

Isilon

Administration and Management

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Welcome to the Isilon Administration and Management course!

Course Overview

Description	This course provides detailed information for administering EMC Isilon scale-out Storage Platform. The course prepares students to perform Isilon storage administration. Topics include the configuration of basic and advanced SMB, NFS client access; FTP, HTTP configurations; data protection/replication in single and multi-cluster implementations, archive deployment, Snapshots and Replication, SNMP integration, data protection, analytics; and support and monitoring.
Audience	This course is intended for EMC, partner and customer personnel involved with the administration of EMC Isilon NAS solutions.
Prerequisites	Prior to attending this course, you should have attended the following course(s): ■ Isilon Fundamentals (e-learning)

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Course Overview



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This course provides detailed information for administering EMC Isilon scale-out Storage Platform. The course prepares students to perform Isilon storage administration. Topics include the configuration of basic and advanced SMB and NFS client access; HTTP configurations, data protection/replication in single and multi-cluster implementations, archive deployment, Snapshots and Replication, SNMP integration, analytics, and support and monitoring.

Introductions

- Name
- Job
- Support and storage experience
- Reason for taking this class

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Course Overview



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This slide introduces the instructor and students.

Logistics

- Breaks
- Lunch
- Restroom locations
- Lab setup

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Course Overview



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This slide discusses the logistical aspects of the class.

EMC Proven Professional Exam

- E20-559 Isilon Solutions Specialist Exam for Storage Administrators
- 60 questions
- 90 minutes to complete
- Practice test available

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Course Overview



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The E20-559 Isilon Solutions Specialist Exam for Storage Administrators is part of the Proven Professional program. The exam consists of 60 questions and the applicant will have 90 minutes to complete the exam. The exam is available through Pearson Vue testing centers.

Agenda

	Modules/Lessons	Labs
Day 1	• Course Overview • Introductions • Logistics • Overview • Agenda • Module 1: Intro to Isilon Clusters • Storage concepts • Isilon nodes and disk types • Intro to OneFS • Access to the cluster • Auditing in OneFS • Module 2: Data Protection and Layout • Data protection • Sub pools, data layout, and actual protection • Isilon cluster caching system	• Lab 1 • Set up initial cluster [video] • Access the Isilon cluster • Manage administration roles and privileges • Lab 2 • Examine data protection settings • Configure data protection at directory and file levels

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Course Overview



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This slide reviews the agenda for day one.

Agenda

	Modules/Lessons	Labs
Day 2	- Module 3: Networking - External connection to enterprise network - DNS and multi-tenancy - SmartConnect and connectivity to cluster - Module 4: Access Management - Access control - Directory service integration - Hadoop - Object storage in OneFS	- Lab 3 - Configure SmartConnect - Create and test DNS records - Lab 4 - Synchronize NTP services - Connect to LDAP and AD domains - Create access zones

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Course Overview



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This slide reviews the agenda for day two.

Agenda

Modules/Lessons

Labs

Day 3

Module 5: User Authentication and File Access

- User identity mapping
- User access to file data
- SMB shares
- NFS exports

Module 6: Storage Administration

- SmartPools and file pools
- SmartQuotas
- SnapshotIQ

Lab 5

- Verify on-disk identities for users and groups
- Configure, modify, and test permissions and ACL policies
- Configure SMB shares and NFS exports
- Configure and connect using HTTP protocol

Lab 6

- Validate SmartPools settings
- Verify and create file pools policies
- Configure SmartQuota directory
- Work with SnapshotIQ



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Course Overview



This slide reviews the agenda for day three.

Agenda

Modules/Lessons

Labs

Day 4

- Module 6 (cont'd): Storage Administration
 - SyncIQ
 - SmartDedupe
- Module 7: Job Engine
 - Job engine architecture
 - Priorities and impact policies
 - Jobs in OneFS
 - Job management

- Lab 6 (cont'd)
 - Modify SyncIQ policy
 - Use SmartDedupe to consolidate disk space
- Lab 7
 - Check job status
 - Create and modify job impact policies and priorities



This slide reviews the agenda for day four.

Agenda

Modules/Lessons

Labs

Day 5

- Module 8: Monitoring
 - Working with system events
 - Using InsightIQ
 - Statistics from the command-line
 - SNMP monitoring overview
 - ESRS overview

- Lab 8
 - Install InsightIQ
 - View performance and file system reports
 - Use `isi statistics` command



This slide reviews the agenda for day five.

Module 1: Intro to Isilon Clusters

Upon completion of this module, you should be able to:

- Define and differentiate storage types
- Describe physical build-out of Isilon
- Create Isilon cluster
- Implement role-based access control
- Explain auditing functionality in OneFS



Module 1: Intro to Isilon Clusters

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Upon completion of this module, you will be able to define and differentiate storage types, describe physical build-out of Isilon, create an Isilon cluster, implement role-based access control, and explain auditing functionality in OneFS.

Lesson 1: Storage Concepts

Upon completion of this lesson, you will be able to:

- Compare and contrast traditional and clustered NAS
- Describe Isilon OneFS operating system
- Define Big Data
- Explain scale-out Data Lakes



Upon completion of this lesson, you will be able to compare and contrast traditional and clustered NAS, describe the Isilon OneFS operating system, define Big Data, and explain scale-out Data Lakes.

NAS Solution Types

- Traditional NAS
 - Independent boxes on network
 - Additional storage add new server
 - Creates NAS sprawl
- Clustered NAS
 - All NAS boxes belong to unified cluster
 - Single point of management



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Module 1: Intro to Isilon Clusters

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Isilon clusters are a network attached storage (NAS) solution. NAS began as independent appliances on the network that were tuned for storage performance. If more storage was needed, you could add another independent NAS box to the network. These independent NAS boxes are also referred to as traditional NAS. However, as more boxes are added to the network, you can end up with NAS sprawl where data is scattered across the network with no single management framework.

Another implementation of NAS is called clustered NAS. In clustered NAS solutions, all NAS boxes belong to a unified cluster that has a single point of management for all. But not all clustered NAS solutions are the same. Some vendors choose to overlay a management interface so that you can manage independent NAS boxes. This gives you a unified management interface, but doesn't actually unify the file system. While this approach does ease the management overhead of traditional NAS, it still does not scale well.

Traditional NAS vs. Isilon



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Module 1: Intro to Isilon Clusters

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Isilon delivers next-generation storage technology. Isilon is not like traditional NAS storage systems. Traditional storage systems take a scale-up approach. Scale-up storage is the traditional architecture that is dominant in the enterprise space and is characterized by extremely high performance, high availability single systems that have a fixed capacity ceiling. In scale-up storage, each filer head connects to all sets of disks. Data is striped into RAID sets of disk drives (8-16), which leads to separate LUNs, volumes, and file systems.

The head/controller can be active/active with both heads accessing the disks or active/passive with one waiting in case the other fails. The heads contain the memory and processor functions. Scale is achieved by adding shelves of disks, or buying a new head/controller.

In a scale-up architecture, vendors put an overarching layer of software that enables a central management point for all of the filer heads and disks. Each still has a separate file system. While many commercially available and proprietary clustered storage options are in use today with a variety of configurations, most rely on industry standard server nodes with a clustered storage operating system that manages the cluster as a unified whole.

Isilon chose to bypass traditional NAS to create a storage system—from the ground up—that is one file system and volume. Each node adds resources (processing power, memory, and disk space) to the cluster. Nodes are peers that work together and stripe data across the entire cluster—not just individual nodes. Isilon takes a scale-out approach.

Scale-up vs. Scale-out Architecture

Scale-up Architecture

- Usually restricted to 16-128 TB
- Block-based storage
- Scale by adding components
- Protects group at same level



Scale-out Architecture

- Upper limits of 50PB+
- File-based storage
- Scale by adding servers
 - Servers have all components
 - Quick scalability!
- Granular file protection



In traditional NAS scale-up solution, the file system, volume manager, and the implementation of RAID are all separate entities. Each entity is abstracted from the other. The functions of each are clearly defined and separate. In a scale-up, solution you have controllers that provide the computational throughput, connected to trays of disks. The disks are then carved up into RAID GROUPS and into LUNs. If you need additional processing, you can add an additional controller, which can run Active/Active or Active/Passive. If you need additional disk, you can add another disk array. To administer this type of cluster, there is an overarching management console that allows for single seat administration. Each of these components are added individually and may have an upper limit of 16TB although some solutions may be higher, but usually not more than about 128TB with current technology. This type of solution is great for specific types of workflows, especially those applications that require block-level access.

In a Scale-out solution, the computational throughput, the disk and disk protection, and the over-arching management are combined and exist within a single node or server. OneFS creates a single file system for the cluster that performs the duties of the volume manager and applies protection to the cluster as a whole. There is no partitioning, and no need for volume creation. Because all information is shared among nodes, the entire file system is accessible by clients connecting to any node in the cluster. Because all nodes in the cluster are peers, the Isilon clustered storage system also does not have any master or slave nodes. All data is striped across all nodes in the cluster. As nodes are added, the file system grows dynamically and content is redistributed. Each Isilon storage node contains globally coherent RAM, meaning that as a cluster becomes larger, it also becomes faster. Each time a node is added, the cluster's concurrent performance scales linearly.



In traditional NAS systems the file system, volume manager, and the implementation of RAID are all separate entities. Each entity is abstracted from the other. The file system is responsible for the higher level functions of authentication, authorization. The volume manager controls the layout of the data while RAID controls the protection of the data (data protection). The functions of each are clearly defined and separate.

OneFS is not only the operating system but also the underlying file system that drives and stores data. OneFS creates a single file system for the cluster that also performs the duties of the volume manager and applies protection to the cluster as a whole. There is no partitioning, and no need for volume creation. Because all information is shared among nodes, the entire file system is accessible by clients connecting to any node in the cluster. Because all nodes in the cluster are peers, the Isilon clustered storage system also does not have any master or slave nodes. All data is striped across all nodes in the cluster.

As nodes are added, the file system grows dynamically and content is redistributed. Each Isilon storage node contains globally coherent RAM, meaning that, as a cluster becomes larger, it also becomes faster. Each time a node is added, the cluster's concurrent performance scales linearly.

OneFS Architecture

- Distributed clustered file system
 - Spans across nodes
 - Grows dynamically
 - Achieves data redundancy by striping
- Runs on all nodes in cluster
- Shares information between nodes
- Optimizes processes and applications



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Module 1: Intro to Isilon Clusters

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The key to Isilon's storage cluster solutions is the architecture of OneFS, which is a distributed cluster file system. This means that a single file system spans across every node in a storage cluster and, as nodes are added, that file system automatically redistributes content across the entire node. Data redundancy is accomplished by striping data across the nodes instead of the disks so that redundancy and performance are increased. For the purposes of data striping, you can consider each node as an individual device.

There is no single master device that controls the cluster. Each node is a peer that shares the workload and acts as a controller for incoming data requests independently, preventing bottlenecks caused by multiple simultaneous requests. This also prevents outages caused by hardware failures because there is no single controlling interface to the cluster.

OneFS Architecture (cont'd)



In an enterprise network environment, clients connected to the enterprise network can connect to the resources stored on an Isilon cluster using standard file access protocols. Each node in an Isilon cluster is also connected to a back-end InfiniBand network that enables each to coordinate and continually adjust to the changing needs of the cluster as a whole.

Scale-out NAS and Big Data?

- Big Data defined:
 - Collection of data so large, diverse, and fast-changing that it's difficult for traditional technology to efficiently process and manage
- Scale-out compatibility:
 - Ability to add disk, memory, CPU on demand
 - Single, production data repository
 - Separates analytics from storage



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Module 1: Intro to Isilon Clusters

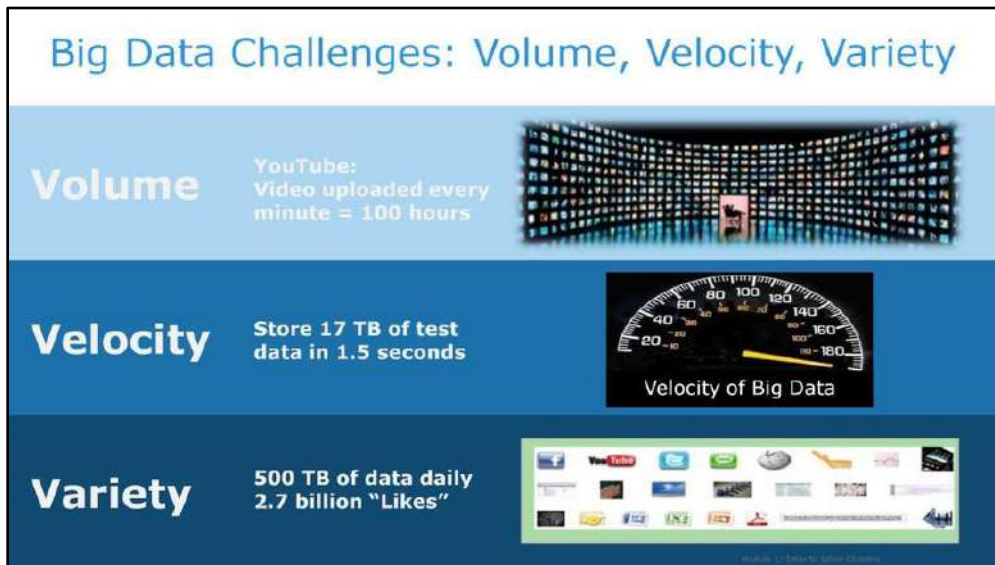


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The term Big Data is being used across the technology industry but what exactly is Big Data? Big Data is defined as any collection of *data* sets so large, diverse, and fast changing that it is difficult for traditional technology to efficiently process and manage. What exactly makes computer data, big data?

The storage industry says that Big Data is digital data having too much volume...velocity...or variety, to be stored traditionally. To make sure the three V's of Big Data are perfectly clear, let's consider some examples.

Why does scale-out NAS blend so well with a Big Data workflow? One of the first reasons is due to the every growing and changing nature of Big Data: on demand storage. With Isilon an administrator can add terabytes of data in seconds and thus allows them to dynamically grow their repository in terms of disk, memory and CPU. Add to this that with Isilon, the cluster functions as a single repository of data and so there is no need to move production data from individual silos into the cluster for analytics. Analytics can be run in real-time on production data. Lastly, by separating compute or analytic server from the storage, there are fewer and smaller analytic servers that need to be in the Big Data compute environment. There is more information on Big Data and analytics later in this course.



What do we mean by volume? Consider any global website that works at scale. YouTube's press page says YouTube ingests 100 hours of video every minute. That is one example of Big Data volume.

What's an example of velocity? Machine-generated workflows produce massive volumes of data. For example, the longest stage of designing a computer chip is physical verification, where the chip design is tested in every way to see not only if it works, but also if it works fast enough. Each time researchers fire up a test on a graphics chip prototype, sensors generate many terabytes of data per second. Storing terabytes of data in seconds is an example of Big Data velocity.

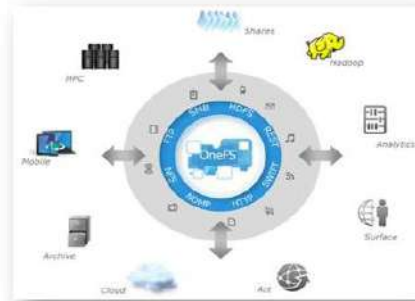
Perhaps the best example of variety is the world's migration to social media. On a platform such as Facebook, people post all kinds of file formats: text, photos, video, polls, and more. According to a CNET article from June 2012, Facebook was taking in more than 500 terabytes of data per day, including 2.7 billion Likes and 300 million photos. Every day. That many kinds of data at that scale represents Big Data variety.

The "Three Vs" – volume, velocity, and variety – often arrive together. When they combine, administrators truly feel the need for high performance, higher capacity storage. The three Vs generate the challenges of managing Big Data.

Growing data has also forced an evolution in storage architecture over the years. Growing data has also forced an evolution in storage architecture over the years due to the amount of data that needs to be maintained, sometimes for years on end. Isilon is a Big Data solution because it can handle the volume, velocity, and variety that defines the fundamentals of Big Data.

Scale-out Data Lake

- Stores data from multiple sources (file shares/web apps, device and cloud)
- Enables data access for variety of uses
- Scales to meet demand of growth
- Provides tiering ability
- Maintains simplicity, even at petabyte scale
- EMC Isilon Scale-out Data Lake whitepaper:
 - <http://www.emc.com/collateral/white-papers/h13172-isilon-scale-out-data-lake-wp.pdf>



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Module 1: Intro to Isilon Clusters

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A scale-out data lake is a large storage system where enterprises can consolidate vast amounts of their data from other solutions or locations, into a single store—a data lake. The data can be secured and analysis performed, insights surfaced, and actions. Enterprises can then eliminate the cost of having silos or “island” of information spread across their enterprises. The scale-out data lake further enhances this paradigm by providing scaling capabilities in terms of capacity, performance, security and protection.

For additional information, see the EMC whitepaper *The EMC Isilon Scale-out Data Lake* located at the following URL:

<http://www.emc.com/collateral/white-papers/h13172-isilon-scale-out-data-lake-wp.pdf>

Lesson 1: Summary

Having completed this lesson, you are now able to:

- Compare and contrast traditional and clustered NAS
- Describe Isilon OneFS operating system
- Define Big Data
- Explain scale-out Data Lakes

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Module 1: Intro to Isilon Clusters



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Having completed this lesson, you are now able to compare and contrast traditional and clustered NAS, describe the Isilon OneFS operating system, define Big Data, and explain scale-out Data Lakes.

Lesson 2: Isilon Hardware

Upon completion of this lesson, you will be able to:

- Differentiate Isilon nodes types
- Characterize target workflows per node
- Illustrate internode communications



Module 1: Intro to Isilon Clusters

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Upon completion of this lesson, you will be able to differentiate Isilon node types, characterize target workflows per node, and illustrate internode communications.

Isilon Nodes

- Uses nodes instead of controllers or filers
- Nodes are peers
- Node height: 1U, 2U, or 4U
- Each node is a server with:
 - CPU
 - Memory (RAM)
 - Disks
 - Local flash drive (OS)
 - Storage disks
 - NVRAM



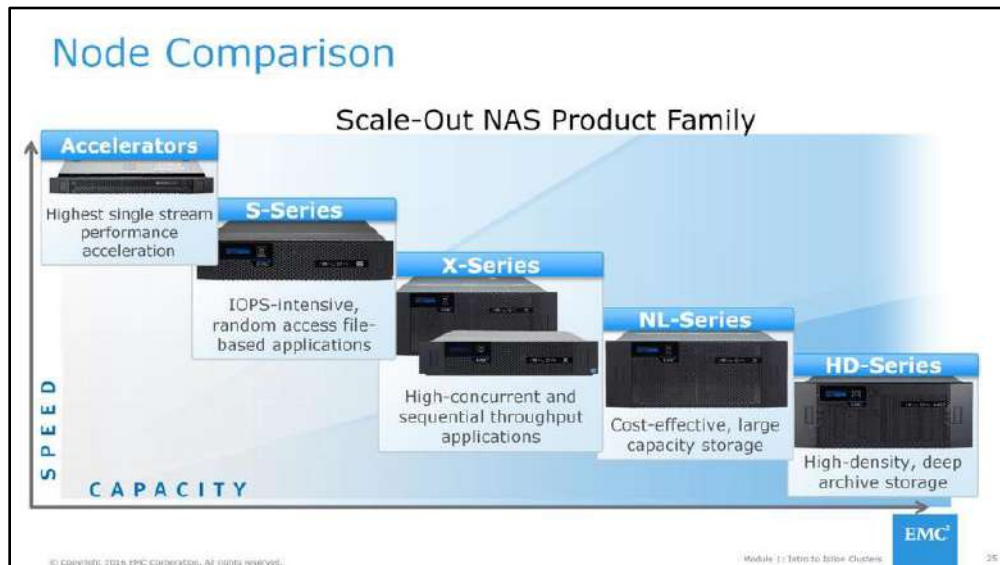
Module 1: Intro to Isilon Clusters

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The basic building block of an Isilon NAS cluster is a node. The Isilon nodes provide the hardware-base on which the OneFS operating system executes.

Architecturally, every Isilon node is a peer to every other Isilon node in a cluster, allowing any node in the cluster the ability to handle a data request. The nodes are equals within the cluster and no one node acts as *the* controller or *the* filer. Instead, the OneFS operating system unites all the nodes into a globally coherent pool of memory, CPU, and capacity. As each new node is added to a cluster, it increases the aggregate disk, cache, CPU, and network capacity of the cluster as a whole.

All nodes have two mirrored local flash drives that store the local operating system, or OS, as well as drives for client storage. All storage nodes have a built-in NVRAM cache that is either battery backed-up or that performs a vault to flash memory in the event of a power failure. The *vault to flash* is similar to the “vault” concept in the VNX and VMAX. If you lose power, the batteries give you enough power to take all the pending writes in memory (NVRAM in the case of an Isilon) and place them into a special area of storage from which they can be retrieved after power is restored.



The EMC Isilon product family consists of five node series: A-Series, S-Series, X-Series, NL-Series, and the HD-Series.

- **A-Series:** A performance accelerator is used when additional disk is not needed but performance enhancements are required. Ideal for streaming large data sets, extremely fast low latency concurrent reads. A Backup accelerator is used to offload backup jobs and connects directly to a tape or virtual tape library.
- **S-Series:** The S-Series is for ultra-performance primary storage and is designed for high-transactional and IO-intensive tier 1 workflows.
- **X-Series:** The X-Series strikes a balance between large capacity and high-performance storage. X-Series nodes are best for high-throughput and high-concurrency tier 2 workflows and also for larger files with fewer users.
- **NL-Series:** The NL-Series is designed to provide a cost-effective solution for tier 3 workflows, such as nearline storage and data archiving. It is ideal for nearline archiving and for disk-based backups.
- **HD-Series:** The HD-Series is the high-density, deep archival platform. This platform is used for archival data that must be retained for long, if not indefinite, periods of time.

How Nodes Form a Cluster

- Minimum of 3 like-type nodes form a cluster
- Maximum 144 nodes in single cluster
- Combine different node types for:
 - Different work flows
 - Tiering/archival
 - Application requirements



Module 1: Intro to Isilon Clusters

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All clusters must start with a minimum of three like-type or identical nodes. This means that when starting a new cluster, you must purchase three identical nodes (i.e., three S-Series nodes, three X-Series nodes, or three NL-Series nodes). You cannot purchase one single S-Series node, one X-Series node, and one NL-Series node, and then combine them to form a three-node cluster.

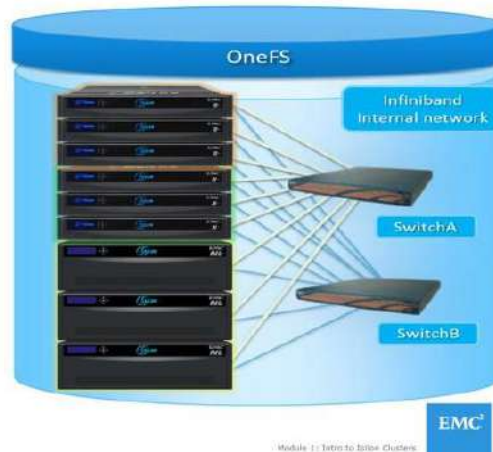
All nodes must *initially* be purchased in groups of three due to the way that OneFS protects the data. You can buy three S-Series nodes, three X-Series nodes and three NL-Series nodes, and combine them into a single cluster. If you accidentally bought three S-Series nodes and two X-Series nodes, you could still form a cluster but only the three S-Series nodes would be writeable. The two X-Series nodes would add memory and processing to the cluster but would sit in a read-only mode until a third X-Series node was joined. Once the third X-Series node was joined, the three X-nodes would automatically become writable and add their storage capacity to the whole of the cluster.

When the minimum of three like-types nodes is met, you can buy in any denomination of nodes of that type. For example, you might start out with a 3-node cluster of X-Series nodes and then purchase one single X-Series node, or 18 more X-Series nodes; again, once the three node minimum is met, any number or type of nodes can be added.

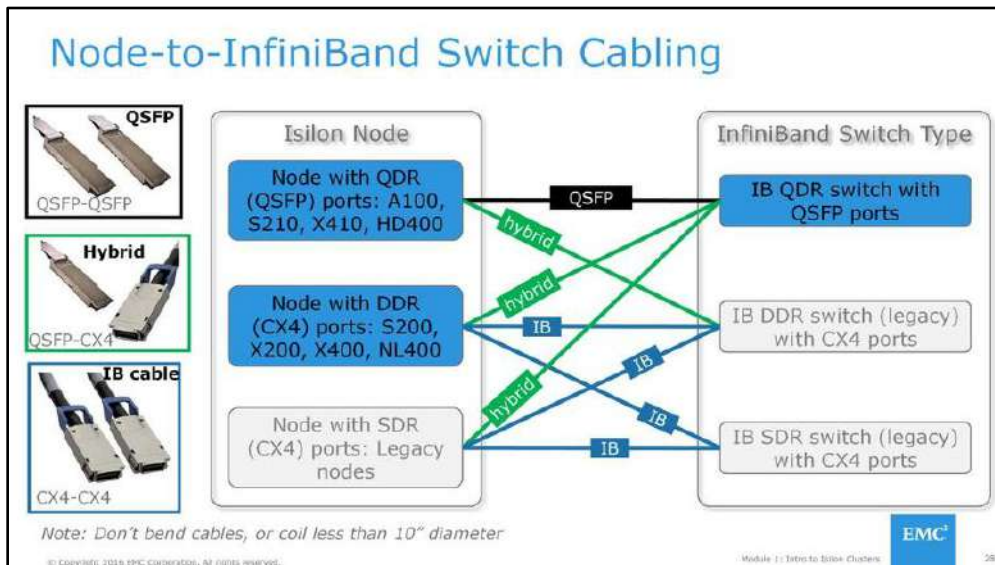
As of this publication, clusters can scale up to a maximum of 144 nodes and access 36.8 TB of global system memory.

InfiniBand Internal Network

- Dual InfiniBand switches
 - Provides fast internode communication
 - Allows failover if one switch fails
 - Isilon-only traffic on back-end or internal network
 - Do not daisy-chain switches
- Switches available in QDR, DDR



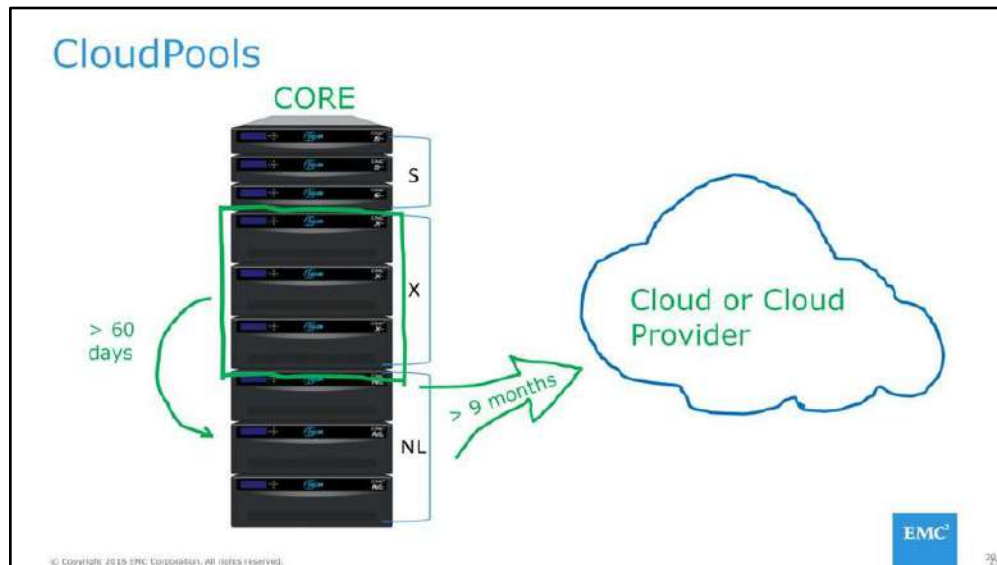
An Isilon cluster uses separate internal and external networks for back-end and front-end connectivity. For the internal network, the nodes in an Isilon cluster are connected by a technology called InfiniBand. An Isilon cluster uses InfiniBand for intra-cluster data and messages. InfiniBand is a point-to-point microsecond-latency interconnect that is available in 20 Gb/sec Double Data Rate (DDR), and 40 Gb/sec Quad Data Rate (QDR) models of switches. InfiniBand delivers the extreme low latency that is needed for the cluster nodes to function as one cluster. Using a switched star topology, each node in the cluster is one hop away from any other node. EMC Isilon recommends that you avoid using the internal network for any purpose other than intra-cluster communication. An Isilon cluster can be configured to use redundant InfiniBand switches for the internal interconnect. You need to procure a switch that is large enough to accommodate all the nodes in the cluster and allow for growth. If you fill up all the ports on the back-end switches, you will need to buy larger switches as it is absolutely not supported to 'daisy chain' the back-end switches.



Connection from the nodes to the internal InfiniBand network now comes in copper or optical, depending on the node type. You should use care when handling InfiniBand cables as bending or mishandling them can result in damaged and unusable cables. Initially, implementation engineers would use the 'hand through the hole' measurement to ensure that the cables were not coiled too tightly (i.e., if your hand can fit through the cable loop, then you're okay); however, it is safer to remember not to coil the cables less than 10 inches in diameter to ensure they do not become damaged. Never bend cables beyond their recommended bend radius. You should consult the recommendation of your cable manufacturer.

Shown in this diagram is the cable type for connecting nodes to the InfiniBand switch. The pictures show the three types of cables, each come in varying lengths. The QSFP (Quad Small Form-factor Pluggable) cable has connectors to allow connection to a QDR switch's QSFP port. Nodes with QSFP ports are the A100, S210, X410, and HD400. Use a hybrid QSFP-CX4 cable to connect nodes that have QSFP ports to DDR InfiniBand switches with CX4 ports. You can also connect DDR nodes to a QDR switch using a hybrid cable. Note that for legacy nodes and legacy InfiniBand switches, a CX4-to-CX4 IB cable is used.

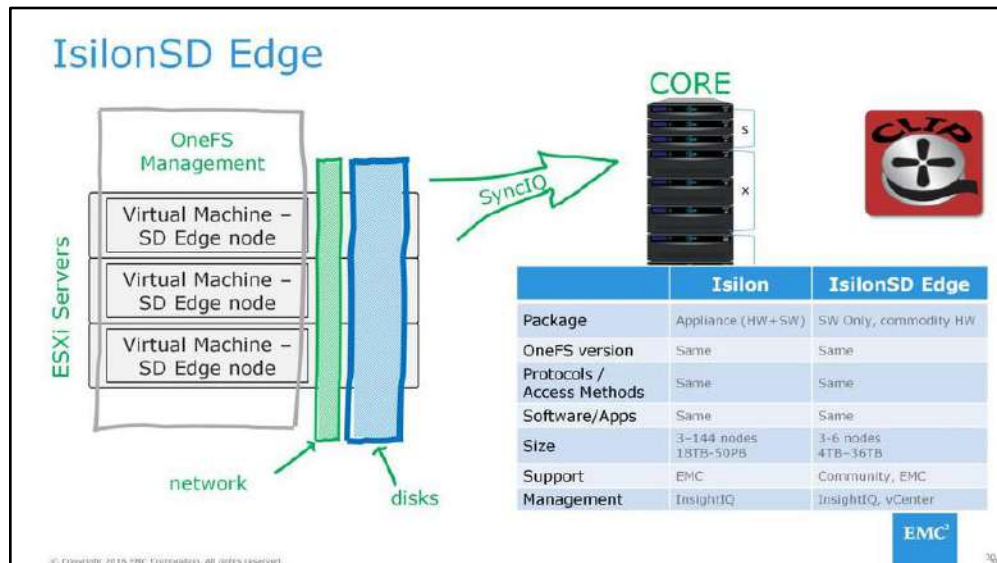
When using optical, you need a QSFP Optical transceiver to plug into the QDR port. The optical cable plugs into the transceivers. For additional information, see the *Isilon Site Preparation and Planning Guide* located on <http://support.emc.com>.



We mentioned data tiering in scale-out Data Lake and how different nodes can be implemented in a tiering solution. Let's take a closer look by examining an edge-to-core-to-cloud solution. We'll start at CloudPools. CloudPools is the feature that extends tiering beyond the enterprise's core and is discussed in detail later in this course. As an example, frequently accessed general purpose file data such as media, documents, presentations, etc. may reside primarily on the X-Series tier as indicated. This data has a policy that moves files that have not been accessed for more than 60 days to the NL-Series tier. We can then have a CloudPools policy that moves files that have not been accessed for more than nine months to the cloud. A user accessing a file that resides on the cloud tier could see slower performance as this is dependent on the cloud choice and actual location of the data.

Essentially what CloudPools does is provide lower TCO for archival-type data by optimizing primary storage with intelligent data placement. CloudPools integrates seamlessly with the cloud. It eliminates management complexity and allows a flexible choice of cloud providers.

Data can also be pulled from the cloud back to the enterprise.



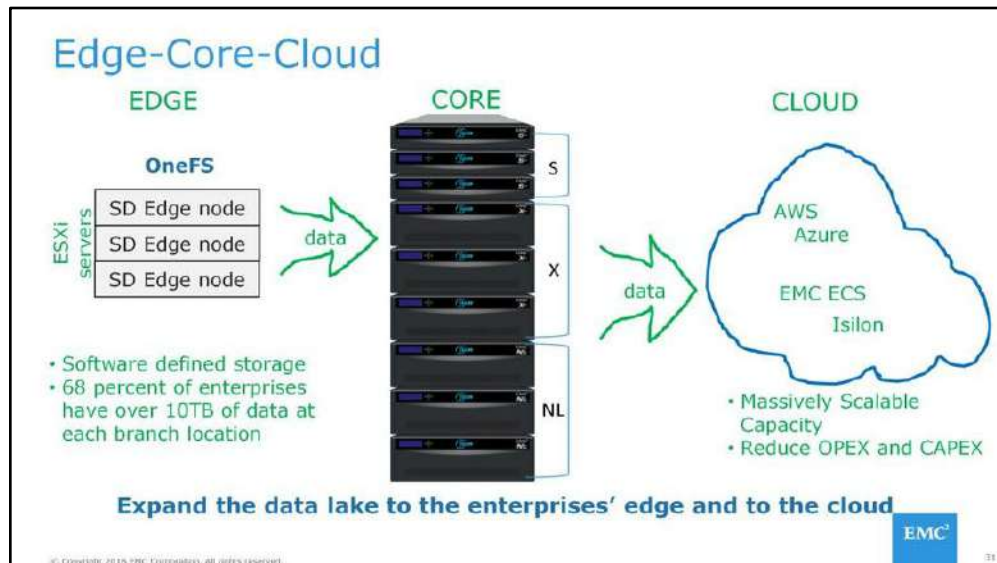
Another component in the edge-to-core-to-cloud solution is Isilon SD Edge. This is a software defined scale-out NAS running OneFS and leveraging the OneFS protocols and access methods, and enterprise grade features. For our design we are especially interested in using SyncIQ to consolidate data to the core. Replicating the data may eliminate the need for backups at the edge sites. SyncIQ are covered in greater detail later in this course. The table compares SD Edge with Isilon. The notable differences are SD Edge scaling to 36 TB and a cluster can have from 3 to 6 nodes.

SD Edge addresses the common challenges the customer face when trying to manage remote offices. Most notably the solution is installed on a virtual environment on commodity hardware, eliminates disparate islands of storage, adds data protection, and simplifies management. In the solution, SD Edge can help consolidate data under the "core" data center. It's simple, agile and cost efficient, ideal for remote locations with limited IT resources. It can be managed with standard VMware tools, removing much of the management complexity.

The IsilonSD Edge Foundation Edition is a free download for non-production use and has EMC Community only support.

IsilonSD Edge video:

https://www.youtube.com/watch?v=BgNzHRZMmo4&list=PLbssOJyyvHuXZ_3JKT5ugbuHPQqZm3e5f&index=1



Here we can use IsilonSD Edge with CloudPools to form an edge-to-core-to cloud solution. SD Edge is the edge component and CloudPools is the cloud mechanism. At a high level, this expands the data lake beyond the data center. First is the ability to consolidate and replicate remote location data in a remote office/branch office (ROBO) type solution. Second is the use of a public or private cloud to tier data out of the "core" platforms.

In the diagram, the branch office is employing commodity servers with VMware ESXi and SD Edge running on them. This is a software defined solution. As many as 68 percent of enterprises have over 10TB of data at each branch location. Data moves from the edge locations to the core. CloudPools allow data to expand beyond the core and into the cloud. Cloud vendors such as Amazon Web Services and Microsoft Azure are supported as well as EMC Elastic Cloud Storage and even Isilon storage. The overall concept of CloudPools is to move old and inactive data to more cost efficient storage, taking advantage of massively scalable storage and reducing the enterprises' OPEX and CAPEX. In doing so, we expand the data lake to the enterprise's edge and to the cloud.

Lesson 2: Summary

Having completed this lesson, you are now able to:

- Differentiate Isilon nodes
- Characterize target workflows per node
- Illustrate internode communications

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Module 1: Intro to Isilon Clusters



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Having completed this lesson, you are now able to differentiate Isilon nodes, characterize target workflows per node, and illustrate internode communications.

Lesson 3: OneFS Management

Upon completion of this lesson, you will be able to:

- Create cluster and add node
- Differentiate between administrative interfaces
- Explain `isi` command structure



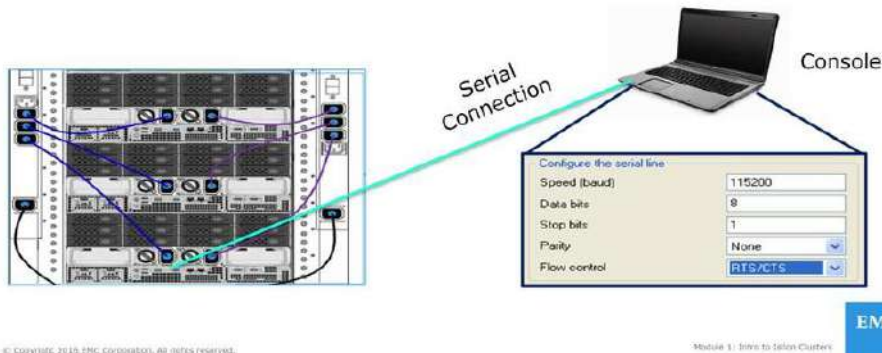
Module 1: Intro to Isilon Clusters

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Upon completion of this lesson, you will be able to create a cluster and add a node, differentiate between administrative interfaces, and explain **isi** command structure.

Creating a Cluster – Serial Connection

- Adding first cluster node
- Must use terminal emulator, such as PuTTY



To initially configure an Isilon cluster, the CLI must be accessed by establishing a serial connection to the node designated as node 1. The serial console gives you serial access when you can't or don't want to use the network. Other reasons for accessing using a serial connection may be for troubleshooting, site rules, a network outage, etc.

The serial port is usually a male DB9 connector. This port is called the management port. Connect a serial null modem cable between a serial port of a local computer, such as a laptop, and the management port on the node designated as node 1. As most laptops today no longer have serial connections, you might need to use a USB-to-serial converter.

On the local computer, launch a serial terminal emulator, such as PuTTY. Configure the terminal emulator utility to use the following settings:

- Transfer rate = 115,200 bps
- Data bits = 8
- Parity = none
- Stop bits = 1
- Flow control = hardware

Either a command prompt or a Configuration Wizard prompt will appear. The command prompt displays the cluster name, a dash (-), a node number, and either an hash (#) symbol or a percent (%) sign. If you log in as the **root** user, it will be a # symbol. If you log in as another user, it will be a % symbol. For example, **Cluster-1#** or **Cluster-1%**. This prompt is the typical prompt found on most UNIX and Linux systems.

OneFS Configuration Wizard

- Log on to node you want join
- Configuration Wizard or command prompt appears
 - Visible on new nodes only

```
Welcome to the Isilon IQ configuration wizard.
Copyright (c) 2001-2016 EMC Corporation. All Rights Reserved.
Enter 'help' at any prompt for additional information on that step.
Enter 'back' at any prompt to return to previous step.
Enter 'quit' at any prompt to discard all changes.
Enter 'manual' at any prompt to switch to manual mode (console).

Node build: Isilon OneFS v8.0.0.0 B_8_0_0_037(RELEASE)
Node serial number: SU200-000NZZ-3YEF

Select an option:
I 11 Create a new cluster
I 21 Join an existing cluster
I 31 Exit wizard and configure manually
I 41 Reboot into SmartLock Compliance mode

Wizard >>> █
```

Option 1
Use to create
new cluster

Option 2
Use to join
nodes 2-144

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Module 1: Intro to Isilon Clusters

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When a node is first powered on or reformatted, the Configuration Wizard automatically starts. If the Configuration Wizard starts, the prompt displays as shown above. There are four options listed:

1. Create a new cluster
2. Join an existing cluster
3. Exit wizard and configure manually
4. Reboot into SmartLock Compliance mode

Choosing option 1 creates a new cluster, while option 2 joins the node to an existing cluster. If you choose option 1, the Configuration Wizard steps you through the process of creating a new cluster. If you choose option 2, the Configuration Wizard ends after the node finishes joining the cluster. You can configure the node using the web administration interface or the CLI. After completing the Configuration Wizard, you can access the settings configured in the Configuration Wizard in the CLI Configuration Console.

For more information about the Configuration Wizard, take the *Isilon Initial Configuration* e-learning course.

- New nodes add CPU, memory and possibly capacity
- Join nodes to cluster via front panel, Configuration Wizard, web administration interface, or CLI command `isi devices`
- Nodes automatically assigned node number and IP address
- Reimaged to match cluster OS version
 - Older versions upgraded
 - Newer versions downgraded



```
Welcome to the Intel® IaaS configuration wizard.  
Copyright (c) 2009-2016 EMC Corporation. All Rights Reserved.  
Enter 'help' at any prompt for additional information on each step.  
Enter 'back' at any prompt to return to previous step.  
Enter 'quit' at any prompt to discard all changes.  
Enter 'manual' at any prompt to switch to manual mode (console).  
  
Mode build: [ellen RHEL7 v0.8.0.0.0_0_0_0_0_0_0] (RELEASE)  
Node serial number: 2U2R0-B000ZG-37EY  
  
Select an option:  
[ 1 ] Create a new cluster  
[ 2 ] Join an existing cluster  
[ 3 ] Exit wizard and configure manually  
[ 4 ] Reboot into SmartLock Compliance mode  
  
Wizard >> [ 1 ]
```



If a node attempts to join the cluster with a newer or older OneFS version, the cluster will automatically reimage the node to match the cluster's OneFS version. After this reimage completes, the node finishes the join. A reimage should not take longer than 5 minutes, which brings the total amount of time taken to approximately 10 minutes. For clusters that use a OneFS version prior to 5.5.x, do not join the node to the cluster. First, reimage the node to the same OneFS version as the cluster before joining the node.

To see a video of an Isilon customer adding a new node in one minute, visit YouTube at:
http://www.youtube.com/watch?v=Y1CIWH4T_pY

Node ID vs. Node LNN

- Node ID
 - Cannot be changed
 - Not reused
 - Unique for each new node
- Node LNN
 - Can be changed
 - Reusable
- Some features use LNN, others use

```
# isi config
isi config >>> status advanced
```

Device Health:					
LNN	ID	Cluster IP	GUID	Type	Status
1	7*	1.2.3.1	00074..	Storage	Okay
2	9	1.2.3.2	00074..	Storage	Okay
4	8	1.2.3.4	00074..	Storage	Okay

```
# isi config
isi config >>> lnnset
```

LNN	Device ID	Cluster IP
1	7	1.2.3.1
2	9	1.2.3.2
4	8	1.2.3.4



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Module 1: Intro to Isilon Cluster

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Nodes are identified by two different numbers: Node ID and LNN.

The **isi config >> status advanced** command checks and verifies a node's LNN and the Node ID.

The **isi config >> lnnset** command changes the node's logical node number, or LNN.

Node ID is sometimes referred to as devid, short for device ID. When a node joins a cluster, it is assigned a unique node ID number, for example, ID1. Node ID numbers are never repeated or duplicated in a cluster, and they never change. If a node is replaced with a new node, the new node is assigned a new node ID by the cluster. Because each node ID is always unique, when seen in cluster logs, individual nodes are easily identified. If a node is removed from the cluster and rejoined, the node is assigned a new Node ID.

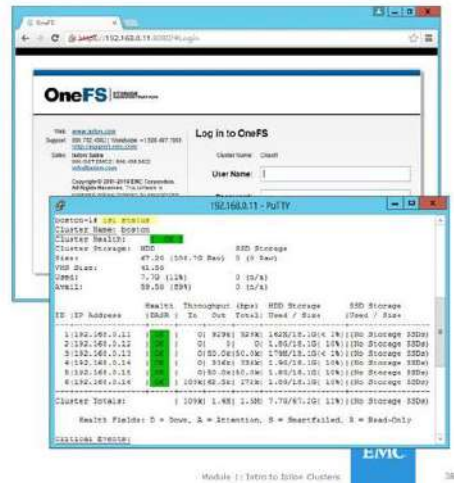
A node's LNN is based on the order a node is joined to the cluster. You can change an LNN in the configuration console for a cluster. To open this console, at the command-line interface, type **isi config**, and then press **ENTER**. At the configuration console prompt, type **lnnset <OldNode#> <NewNode#>**. The LNN of a node displays in the output of the **isi status** command. In logs, the LNN displays with the name of the node, for example: clustername-1.

Another use from the **isi config** console is the version command. This displays the version details of OneFS installed on the cluster.

The information gathered in this way can be useful, not only in interpreting what is happening on a cluster, but also in communication with Technical Support if you have a complex issue.

OneFS Management Tools

- Cluster management options
 - Web administration interface
 - Command-line interface (CLI)
 - Front panel of node
 - Platform Application Programming Interface (PAPI)
- Capabilities vary by interface
 - Some “web administration interface only” functionality
 - Some “CLI only” functionality



You have four options for managing the cluster. You can use the web administration interface, the command-line interface (CLI), or Platform Application Programming Interface (PAPI). PAPI is also referred to as the OneFS application programming interface if you are in the Administration Guide. Management capabilities vary based on which interface you use.

The web administration is robust, but if you're willing to dive into the CLI, you can do a bit more. Some management functionality is only available from the web administration interface. Conversely, sometimes the CLI offers a function, or a detail of a function, that's not available in the web administration interface.

The LCD screen has five buttons used for basic administration tasks, such as adding the node to a cluster, checking node or drive status, etc. Note that Accelerator nodes don't have an LCD screen.

The Platform Application Programming Interface, or PAPI, is a scriptable tool for addressing the cluster, and that it is secured by the same permissions that drive everything else, including role-based access control (RBAC).

Web Administration Interface Overview

- Graphical user interface
 - Browser-based
 - HTTPS
 - Port 8080
 - Connect to any node in cluster
- Use **root**, **admin**, or account with privileges
- Times out after 4 hours



Browser support: Chrome, Internet Explorer, Mozilla Firefox, Apple Safari

The web administration interface is a graphical interface you can use to manage your Isilon cluster.

The web administration interface requires that at least one IP address is configured on one of the external Ethernet ports on one of the nodes. The Ethernet port IP address is either configured manually or by using the Configuration Wizard.

To access the web administration interface from another computer, use an internet browser to connect to port 8080.

Log in using the **root** account, **admin** account, or an account that's the member of a role that has the `ISI_PRIV_LOGIN_PAPI` privilege. After you open the web administration interface, there is a four-hour login timeout.

Interfaces Adapt to User Privileges

Gray tabs = no privileges

The image displays two screenshots of the OneFS web administration interface. The top screenshot shows the interface for a user with 'System Admin and Security Admin' roles, where all navigation tabs (Dashboard, Cluster Management, File System, Data Protection, Access, Protocols) are active and highlighted in blue. The bottom screenshot shows the interface for a user with limited privileges, where the 'File System' and 'Protocols' tabs are grayed out, indicating no access. Both screenshots include a red circle around the 'File System' and 'Protocols' tabs in the top navigation bar. To the right of each screenshot is a text box explaining the user's role and privileges. The OneFS logo and version information (8.0.0) are visible in the top left and right of the screenshots. The EMC logo is in the bottom right corner.

User with System Admin and Security Admin Roles = Full Control

User in a role with limited privileges

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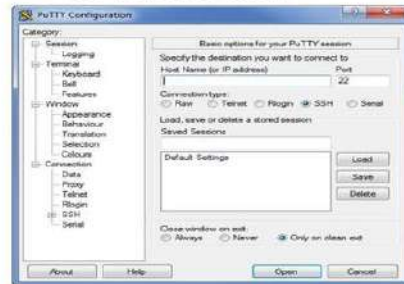
The ability to access certain tabs and features depend on the privileges of the account used to login and are a part of the RBAC function, which is covered in detail later in the module. Once a user has been assigned to a role, all administrative interfaces, including the web administration interface, recognize the privileges of the logged in user.

If you log in as the **root** account, you have full access to all the tabs and licensed features of OneFS, however, if you log in with an account that does not have full privileges, you will see that certain tabs and features are grayed out and you are unable to access or change the settings on these tabs.

Notice in the screenshot on the slide that this user only has privileges to NFS and SMB. The navigation for all other areas are greyed out and unavailable to this user.

CLI Overview

- OneFS CLI accessibility
 - Direct serial connection
 - SSH connections
 - Via Ethernet address
 - CLI access and uses:
 - Controlled by administrator role
 - Permissions dependent
 - Viewing or configuring system
 - Configuration Console
 - Supported management
 - `isi` commands
- Text-based command interface
 - SSH client-based (e.g. PuTTY)
 - No dedicated client



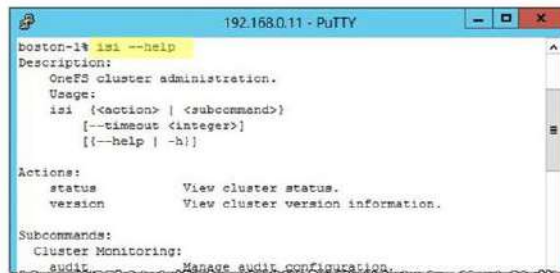
To access the CLI out-of-band, a serial cable is used to connect to the serial port on the back of each node. CLI can also be accessed in-band once an external IP address has been configured for the cluster. Both ways are done using a terminal emulation application, such as PuTTY.

As with the web administration interface, you can delegate responsibilities to your staff, and limit the management options available to them in the CLI. Access to the interface changes based on the administrator's assigned privileges.

The CLI can also be used to view and change configuration settings for individual nodes and the cluster. The CLI is a text-based command interface. You can access the CLI using any SSH client, such as PuTTY. As with the web administration interface, you can connect your preferred SSH client to any node in the cluster to do administration work. Because Isilon is built upon FreeBSD, many UNIX-based command, such as **grep**, **ls**, **cat**, etc., work via the CLI. There are also Isilon-specific commands known as **isi** (pronounced "izzy") commands that are specifically designed to manage OneFS. There is a CLI Reference guide available at <http://support.emc.com> that will provide you with a rich, in-depth listing of all customer-facing commands and their usage.

CLI Capabilities

- Executed in UNIX shell environment
 - OneFS commands
 - Scriptable interface
 - Many UNIX commands available
- Command capabilities
 - Customizable
 - Options / switches / flags
 - Not all administration functionality available



```
boston-14 isi --help
Description:
  OneFS cluster administration.
Usage:
  isi {<action> | <subcommand>}
  [--timeout <integer>]
  [{--help | -h}]

Actions:
  status      View cluster status.
  version     View cluster version information.

Subcommands:
  cluster Monitoring:
    audit      Manage audit configuration.
```



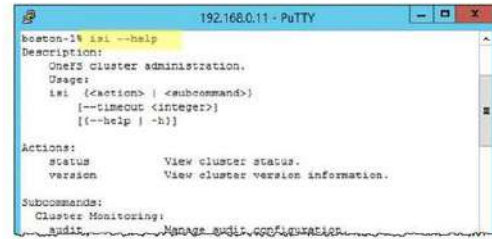
OneFS is built upon FreeBSD UNIX. Every node runs OneFS, including the many FreeBSD kernel and system utilities. Commands in OneFS are executed in a UNIX shell environment. The default shell is zsh. OneFS commands are code built on top of the UNIX environment and are specific to OneFS management. The UNIX shell environment use in OneFS allows scripting and execution of many of the original UNIX commands. Precautions should be taken when writing scripts and cron jobs within OneFS. Certain guidelines and procedures should be followed to appropriately implement the scripts so as to not interfere with regular cluster operations. Access to the CLI is performed either through a serial console or using SSH connections and an SSH client of your choice. PuTTY is a popular, free SSH client available for use.

The CLI command use includes the capability to customize the base command with the use of options, also known as switches and flags. A single command with multiple options result in many different permutations, and each combination results in different actions performed. Understanding the options available for the commands is essential to proper command use. Improper use of a command using the wrong command can be potentially dangerous to the cluster, the node, or to customer data. Commands can also be used together in compound command structures combining UNIX commands with customer facing and internal commands to customize command use even further.

The CLI can be used to do many things, including running the Configuration Console, which comprises all of the settings that were configured during the initial installation via the Configuration Wizard. The CLI can also be used to view and change configuration settings for individual nodes and the cluster.

OneFS CLI Command Usage

- **isi** command
 - View command structure
`man isi --help`
 - Available subcommands and actions
`isi <command> --help`
 - Arguments for specific action
`isi <command> <subcommand> <action> --help`
- Command reference available
 - Different references for different OneFS versions



```
192.168.0.11 - PuTTY
bocton-14 isi --help
Description:
  OneFS cluster administration.
Usage:
  isi {<action> | <subcommand>}
  [--timeout <integer>]
  [--help | -h]

Actions:
  status      View cluster status.
  version     View cluster version information.

Subcommands:
  Cluster Monitoring:
  audit       Manage audit configuration.
```

The *CLI Administration Guide* is available and provides an alphabetical list of **isi** commands that can run to configure, monitor, and manage an Isilon clustered storage system and the individual nodes in a cluster. The **man isi** or **isi --help** command is probably the most important command for a new administrator. It provides an explanation of the many **isi** commands available. You can also view a basic description of any command and its available options by typing its name followed by the **-h** option at the command-line: **<command> -h**.

To view more detailed information at the command-line, refer to the **isi man** page: **man isi <command>** or the Isilon OneFS Command Line Reference for your version of OneFS.

PAPI Overview



- Scriptable tool
 - HTTPS interface to Isilon
 - Separate from web administration interface
 - Sometimes called OneFS API
- Commands not PAPI-aware
 - `isi config`
 - `isi get`
 - `isi set`
 - `isi services`

OneFS API Reference: https://support.emc.com/docu66301_OneFS-8.0.0-API-Reference.pdf?language=en_US

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Module 1: Intro to Isilon Clusters



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PAPI is a scriptable interface for managing the cluster, and that it is secured by the same permissions that drive everything else, including RBAC. PAPI runs through HTTPS, so that all PAPI communications are encrypted, and OneFS applies authentication and RBAC controls to PAPI commands to ensure that only authorized commands are executed. PAPI conforms to the principles of the Representation State Transfer (REST) architecture. One of the chief benefits of PAPI is that it is easy to script, enabling customers to easily automate their storage administration.

An understanding of HTTP/1.1 (RFC 2616) is required to use the API. Whenever possible, HTTP/1.1 defines the standards of operation for PAPI. For more information, see the OneFS Platform API Reference. PAPI commands are structured like URLs, and can be directly executed in a browser provided that the browser supports authentication. For example:

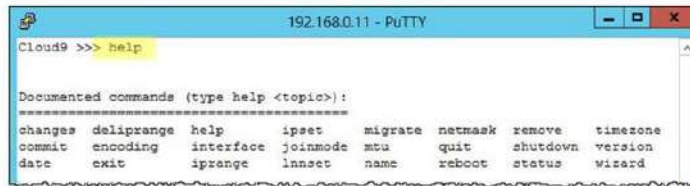
<https://isilon.example.com:8080/platform/3/snapshot/snapshots>

PAPI commands include a PAPI version (in the example, the **3** after the **platform**) so that PAPI scripts are more robust when a cluster's OneFS is upgraded. If the upgrade introduces a new version of PAPI, some backwards compatibility ensures that there is a grace period for old scripts to be rewritten.

Some commands are not PAPI-aware, meaning that RBAC roles will not apply. These commands are internal, low-level commands that are available to administrators through the CLI.

Configuration Console

- Edit settings created with Configuration Wizard
 - `isi config`
- Common commands
 - `shutdown`
 - `status`
 - `name`
- Opens submenu where:
 - Only commands from `isi config` submenu are available
 - Other `isi` commands not available
 - Console changes menu prompt to `>>>`



The **isi config** command opens the Configuration Console where node and cluster settings can be configured. The Configuration Console contains settings that are configured during the Configuration Wizard that ran when the cluster was first created. After you make all the necessary configuration updates to the cluster, they are saved and you are prompted to reboot the cluster as needed. The **changes** command displays a list of changes to the cluster configuration that are entered into the Configuration Console, but have not been applied to the system yet. For example, **joinmode [<mode>]** displays the current cluster add node setting, when executed without any argument.

Sets the cluster add node setting, when appended with one of the following arguments:

- **manual:** Configures the cluster to add new nodes in a separate, manually executed process.
- **secure:** Configures the cluster to disallow any new node from joining the cluster externally. It also makes some other aspects of the operation more secure.

When in the **isi config** console, other Isilon configuration commands are unavailable and only **isi config** commands are valid. You must type **exit** to get back to the default CLI.

Recommended Proper Cluster Shutdown

- Shutdown using CLI or web administration interface
- Proper shutdown use `isi config`
 - Entire cluster:
 - `shutdown all`
 - Single node:
 - `shutdown 3`
- Never use UNIX `shutdown -p` command to shutdown cluster
 - NVRAM not flushed properly
 - Potential data corruption

```

192.168.0.11 - PuTTY
boston-18 isi config
Welcome to the Isilon IQ configuration console.
Copyright (c) 2001-2016 EMC Corporation. All Rights Reserved.
Enter 'help' to see list of available commands.
Enter 'help <command>' to see help for a specific command.
Enter 'quit' at any prompt to discard changes and exit.

Node build: Isilon OneFS v5.0.0.0 B_E_0_037 (RELEASE)
Node serial number: JF100-000N11-0VEF

boston >>> shutdown all

!! You are about to shutdown the entire cluster
Are you sure you wish to continue? [no]
>>> y

boston >>> shutdown 6

!! You are about to shutdown the following nodes: 6
Are you sure you wish to continue? [no]
>>> y
  
```

An administrator can restart or shut down the cluster via the web administration interface or the CLI.

The procedure from the web administration interface:

1. Go to **Cluster Management > Hardware Configuration > Shutdown & Reboot Controls**.
2. Optional: In the **Shut Down or Reboot This Cluster** section, select the action that you want to perform.
 - To shut down the cluster, click **Shut down**, and then click **Submit**.
 - To stop the cluster and then restart it, click **Reboot**, and then click **Submit**.

The procedure from the CLI:

1. Run the **isi config** command.
2. The command-line prompt changes to indicate that you are in the **isi config** subsystem.
 - To restart a single node or all nodes on the cluster, run the **reboot** command.
 - To restart only a single node by specifying the logical node number (lnn):
reboot 6
 - To shut down a single node or all nodes on the cluster, run the **shutdown** command.
 - To shut down all nodes on the cluster, run **shutdown all**.

Do not shut down Isilon nodes the same way that you would shut down UNIX computers; the UNIX **shutdown -p** command, **halt** command, or **reboot** command should never be used to shutdown clusters. This may result in NVRAM not being flushed properly. These native UNIX commands do not elegantly interact with the cluster's code, because the OneFS cluster file system is built as a separate layer on top of UNIX. The file system can think the node is still mounted when it is not connected, and some services can be left with incomplete operations and be left in a hung state.

Graceful Shutdown Internal Steps

- What happens when you properly shutdown a node?
 - Nodes gracefully releases client connections
 - Clients in dynamic pools seamlessly failover to another node
 - NFSv3
 - Clients in static pools have to reconnect
 - SMB/NFSv4/HDFS clients
 - NVRAM contains data not yet been written
 - Uncommitted writes in NVRAM journal flushed to disk
 - Depending on amount of data completion time may vary
- Manually flush journal for a single node by running `isi flush`
 - To flush all nodes, run `isi_for_array -s isi_flush`



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Nodes and clusters often require proper shutdown. When a node is properly shut down, clients gracefully release their connections to the node and all writes are properly flushed from the NVRAM journal. Dynamic client, such as NFSv3, failover seamlessly to another node. Static clients, such as SMB, NFSv4 and HDFS, disconnect from the current node and reconnect to a different node. The NVRAM journal is flushed to disk after all clients are disconnected from the node. Data must be written to disk in order to ensure file system integrity and verify no data is lost.

There may be times when you want to manually flush journals on nodes. This may be to test the journal itself, or because of a performance testing step or a number of other reasons, such as an abundance of caution prior to applying a shutdown command to the cluster. If you want to manually flush writes stored in the node journal to the file system, you can run the **`isi_for_array -s isi_flush`** command. Output similar to the following appears:

```
mycluster-4# isi_for_array -s isi_flush
mycluster-1: Flushing cache...
mycluster-1: Cache flushing complete.
mycluster-2: Flushing cache...
mycluster-2: Cache flushing complete.
mycluster-3: Flushing cache...
mycluster-3: Cache flushing complete.
mycluster-4: Flushing cache...
mycluster-4: Cache flushing complete.
mycluster-4#
```

If a node fails to flush its data, you receive output similar to the following below, where node 1 and node 2 fail their flush command:

```
mycluster-4# isi_for_array -s isi_flush
mycluster-1: Flushing cache...
vinvalbuf: flush failed, 1 clean and 0 dirty bufs remaining
mycluster-2: Flushing cache...
fsync: giving up on dirty
```

Run the **`isi_for_array -s isi_flush`** command again. If any node fails to flush, contact EMC Technical Support. All nodes must successfully flush before proceeding to the next step.

Lesson 3: Summary

Having completed this lesson, you are now able to:

- Create cluster and add node
- Differentiate between administrative interfaces
- Explain `isi` command structure



Having completed this lesson, you are now able to create a cluster and add a node, differentiate between administrative interfaces, and explain **isi** command structure.

Lesson 4: Administration Roles

Upon completion of this lesson, you will be able to:

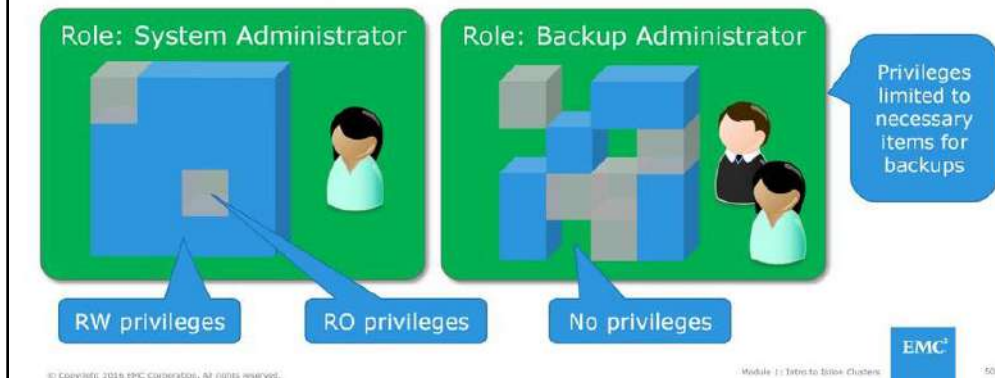
- Describe RBAC
- Establish built-in roles and privileges
- Understand benefits of RBAC
- Manage RBAC



Upon completion of this lesson, you will be able to describe role-based administration control, or RBAC, establish built-in roles and privileges, understand benefits of RBAC, and manage RBAC.

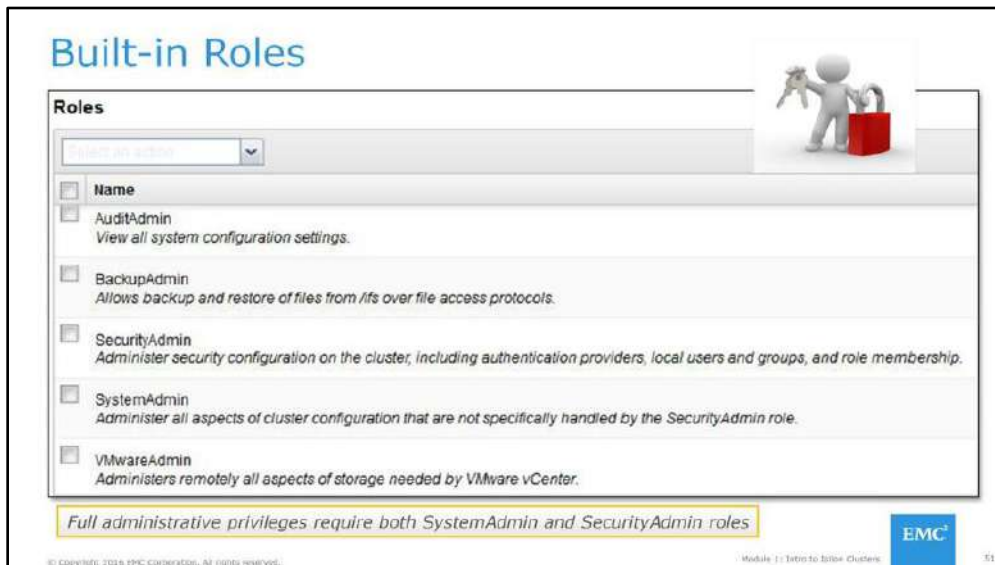
Role-based Access Control Overview

- Increases cluster security
- Separates administrative access from data access



Role-based administration defines the ability to perform specific administrative functions to a specific privilege. The graphic highlights two roles, each role has different privileges assigned. A user can be assigned to more than one role and will then have the combined privileges of those roles. Shown is the individual assigned the System Administrator role is also assigned the Backup Administrator role.

Role-based access enables you to separate out some administrative privileges and assign only those that a user needs to perform their job or specific tasks. As shown, the individual assigned the Backup Administrator role is not given, nor does the individual need, all administrative privileges to just perform a subset of administrative tasks. This makes access to the configuration of the cluster much more restrictive.



In OneFS, there are five built-in roles that have a predefined set of privileges that cannot be modified. These pre-defined roles are listed below and on the slide.

- **AuditAdmin:** Provides read-only access to configurations and settings. It is a useful role for IT and support engineers who must collect system configuration details to investigate a customer issue.
- **BackupAdmin:** Provides permission for backing up and restoring files. This allow you to circumvent the traditional file access checks, the same way that the **root** account has the privileges to circumvent the file access checks; this is all that BackupAdmin allows you do to. You cannot use the backup and restore privileges to change any of the configuration options as you can when logged in as the **root** user.
- **SecurityAdmin:** Provides the ability to manage authentication to the cluster. The ability to create roles and elevate privileges makes this the most trusted role. The SecurityAdmin role does not have permissions for administering other aspects of the system, such as SMB and NFS settings, quotas, or snapshots.
- **SystemAdmin:** Provides all administrative functionality not exclusively defined under the SecurityAdmin role. Members of this role have all of the privileges necessary to administer the entire OneFS cluster.
- **VmwareAdmin:** Provides all administrative functionality required by the vCenter server to effectively utilize the storage cluster. Members of this role have access to the web administration interface and read-write access to a variety of cluster options.

Assign users to both the SystemAdmin and the SecurityAdmin roles to provide full administration privileges to an account. By default, the **root** and **admin** users are members of both of these roles.

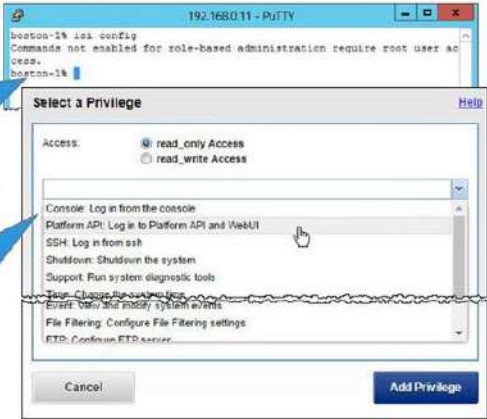
Privileges Overview

Privileges can:

- Grant/restrict access to features
- Read or full control access

Privileges needed to access:

- Web administration interface
- Secure shell (SSH)
- Console (serial port access)



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Roles both simplify administrative access to the cluster, by limiting the operations users can perform, and protects the system and customer data from those who do not require access. A role is made up of the privileges (read_only or read_write) that can be performed on an object. OneFS offers both built-in and custom roles. The graphic shows creating a custom role that's allowing SSH and web administration Read/Write access. Additional privileges can be added.

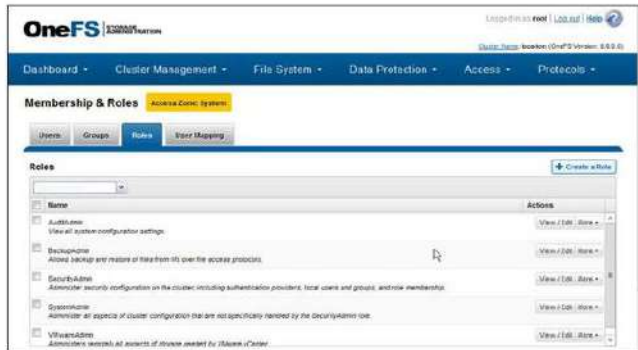
With the implementation of role-based administration, access to configuration protocols is now more restricted. Users must be added to a privileged role in order for them to access the cluster using the web administration interface, the platform API, or SSH. Previously, anyone who could authenticate to the cluster could login using SSH. Now, the privilege needed to access the cluster using SSH access is not given automatically, and administrative users must be added to a role with the SSH login privilege in order to connect using that protocol.

Accounts for **root** and **admin** user exist on the cluster. The **root** account has full control through the CLI and web administration interface whereas the admin account only has access through web administration interface and no privileges into file explorer.

Review the Isilon OneFS administration guides for more information about all the privileges.

RBAC Management

Access → Membership & Roles → Roles

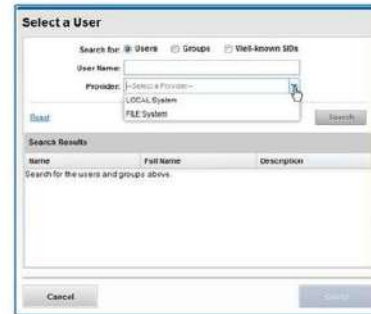


Using the web administration interface, you can create roles, add privileges, and assign members.

The video clip shows the navigation from the **Dashboard** by clicking the **Access** menu, and then selecting the **Roles** tab. Here you can create custom roles or edit the built-in roles to assign users or alter privileges.

Best Practices for RBAC

- Perform in-depth needs-based security review
- Define role-holders
 - Who needs access, what to access, why is access needed
- Assign min set of needed privileges
- Failsafe root account
- Audit review
- Should not exceed 200 roles



Some best practices for assigning users to roles is to first perform an in-depth needs-based security review. Once individuals are identified, their roles are defined based on the job requirements. It's a matter of who needs what access and why. Assign users to roles that contain the minimum set of necessary privileges. For most purposes, the default permission policy settings, system access zone, and built-in roles are sufficient. If not, custom roles can be created.

A failsafe root account and password should be generated and distributed among a quorum of responsible corporate officers. Add an audit review process to ensure the roles are used, not abused, sufficient, and membership up to date.

Exceeding 200 roles could impact cluster performance. Troubleshooting guidance can be found in the Administration – Role Based Access Control (RBAC) / Command Line Interface (CLI) guide.

Lesson 4: Summary

Having completed this lesson, you are now able to:

- Describe RBAC
- Establish built-in roles and privileges
- Understand benefits of RBAC
- Manage RBAC



Having completed this lesson, you are now able to describe RBAC, establish built-in roles and privileges, understand benefits of RBAC, and manage RBAC.

Lesson 5: Establishing Audit Capabilities

Upon completion of this lesson, you will be able to:

- Describe Isilon's auditing implementation
- Explain types of auditing
- Illustrate auditing workflow
- Identify audit log locations

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Upon completion of this lesson, you will be able to describe Isilon's auditing implementation, explain the types of auditing, illustrate the auditing workflow, and identify audit log locations.

What is Auditing?

- Auditing is defined as:
 - Ability to log specific activity on cluster
 - Configuration changes
 - Client protocol activity
- Audit log files sent to external 3rd party audit applications
- Required for regulatory and organizational compliance
 - HIPAA
 - SOX
 - Governmental



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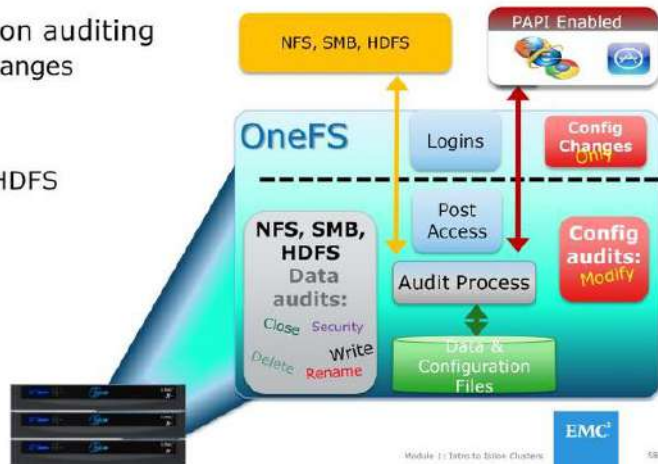
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Auditing is the ability to log specific activities on the cluster. The two activities included are the ability to audit any configuration changes and to audit the client protocol activity. Client protocol activity includes access to the cluster and any actions performed in regards to the data on the cluster such as read, modify, delete, rename, logon, and logoff. The audit system also provides the capability to make the audit logs available to third party audit applications for review and reporting. Audit capabilities are required to meet regulatory and organizational compliance mandates. These include HIPAA, SOX, governmental agency, and other requirements.

Protocol Auditing

- System configuration auditing
 - Configuration changes
 - PAPI enabled
- Protocol auditing
 - NFS, SMB, and HDFS
 - Close
 - Delete
 - Rename
 - Write
 - Set_security



The auditing capabilities in OneFS include: monitoring pre-access configuration changes (cluster login failures/success) and post-access (protocol and configuration) changes to the cluster. Cluster configuration changes are pre and post access and tracking any change is a critical aspect to regulatory compliance. Only the configuration changes made through PAPI are logged. The other post access activity logs what the NFS and SMB client did in regards to the data on the cluster. Auditing provides the capability to track if the data was accessed, modified, created, and deleted.

Audit Capabilities

- Can audit on per-zone basis
 - Ability to select event type per zone
 - Can choose what specific items are forwarded:
 - Close, create, delete, logoff, etc.
- Select specific events to log
 - i.e., Default set of protocol event in System zone but audit only successful attempts to delete files in HR-Zone
- Events logged on node where activity initiated
 - Stored: `/ifs/.ifsvar/audit/logs`
 - Logs roll over when 1GB size reached



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System configuration auditing tracks and records all configuration events that are handled by the OneFS API through the CLI. When you enable system configuration auditing, no additional configuration is required. System configuration auditing events are stored in the config audit topic directories.

Protocol auditing tracks and stores activity performed through SMB, NFS, and HDFS protocol connections. You can enable and configure protocol auditing for one or more access zones in a cluster. If you enable protocol auditing for an access zone, file-access events through the SMB, NFS, and HDFS protocols are recorded in the protocol audit topic directories. You can specify which events to log in each access zone. For example, you might want to audit the default set of protocol events in the System access zone but audit only successful attempts to delete files in a different access zone. The audit events are logged on the individual nodes where the SMB, NFS, or HDFS client initiated the activity. The events are then stored in a binary file under **`/ifs/.ifsvar/audit/logs`**. The logs automatically roll over to a new file after the size reaches 1 GB.

Syslog Forwarding

- Syslog
 - Protocol conveys notification messages
- Config events logged to:
 - /var/log/audit_config.log
 - All data for all zones logged
 - Config events not forwarded to CEE (common Event Enabler)
- Syslog configured for audit_protocol
 - All protocol events per node forwarded to:
 - /var/log/audit_protocol.log



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In OneFS, if the configuration audit topic is selected then, by default, all data regardless of the zone, is logged in the **audit_config.log**, which is the **/var/log** directory. This is configurable and can be changed. If the protocol audit topic is selected, customers have some options as to what exactly they can forward. They can chose the zone they want to audit using the **isi zone zones modify <zonenumber>** command and they can select the events within the zone they want to forward. For example, a customer may be only be interested in successful delete attempts on the System zone. Syslog is configured with an identity of **audit_protocol**. By default, all protocol events are forwarded to the **audit_protocol.log** file that is saved to the **/var/log/**directory, regardless of the zone in which they originated. A CEE (common event enabler) enables third-party auditing applications to collect and analyze protocol auditing logs.

Configuration Audit Changes

- Configuration auditing
 - Enabled through CLI or web administration interface
- Commands to enable/disable:
 - `isi audit settings modify --config-auditing-enabled {true | false}`
 - `isi audit settings modify --config-syslog-enabled {true | false}`
- Config locations logged to:
 - `/var/log/audit_config.log`
- CEE default port 12228



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Configuration auditing is enabled only through the CLI. You use the **isi audit settings modify** command to enable auditing. To enable configuration auditing you add the **--config-auditing-enabled true** option and to enable syslog auditing you add the **--config-syslog-enabled true** option. Both PAPI and web administration interface configuration changes to be logged to the **audit_config.log** file that is located in the **/var/log** directory. To disable configuration auditing, run the same command you used to enable it, but change the value to **false** at the end of the command.

The CEE servers listen, by default, on port 12228. In order to confirm or to verify what ports OneFS is using to talk to CEE servers, run the **isi audit settings view** command.

Audit Log Compression

- Compression of on disk audit data
- Log files compressed on file roll over
- About 90% savings
- Log files stored on per-node basis
- Logs located:
 - /ifs/.ifsvar/audit/logs/node##/topic/



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OneFS uses an audit log compression algorithm on file roll over. This is on-the-fly compression and decompression of on-disk audit data and is handled transparently to the user. The estimated space savings from this compression is 90%. Audit log files are located in **/ifs/.ifsvar/audit/logs/nodeXXX/topic** directory and are compressed as binary. In previous versions of OneFS, these log files were stored in the same path, but in an uncompressed state.

Best Practices for Auditing

- Audited events consumes resources
 - Only configure zones for events needed by auditing application
- Install and configure 3rd party auditing software prior to enabling OneFS feature



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Module 1: Intro to Isilon Clusters

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Because each audited event consumes system resources, EMC Isilon recommends that you only configure zones for events that are needed by your auditing application.

In addition, Isilon recommends that you install and configure third-party auditing applications before you enable the OneFS auditing feature. Otherwise, the large backlog performed by this feature may cause results to not be updated for a considerable amount of time.

Lesson 5: Summary

Having completed this lesson, you are now able to:

- Describe Isilon's auditing implementation
- Explain types of auditing
- Illustrate auditing workflow
- Identify audit log locations



Having completed this lesson, you are now able to describe Isilon's auditing implementation, explain the types of auditing, illustrate the auditing workflow, and identify audit log locations.

Module 1: Summary

Having completed this module, you are now able to:

- Define and differentiate storage types
- Describe physical build-out of Isilon
- Create an Isilon cluster
- Implement role-based access control
- Explain auditing functionality in OneFS



Module 1: Intro to Isilon Cluster

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Having completed this module, you can now define and differentiate storage types, describe physical build-out of Isilon, create an Isilon cluster, implement role-based access control, and explain auditing functionality in OneFS.

Lab 1 - Intro to Isilon Clusters

- Creating cluster and adding nodes
- Validating cluster configuration
- Managing administration roles and privileges



Module 1: Intro to Isilon Clusters

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In this lab, you will first watch a video showing how the initial configuration steps of a cluster. Then you will get hands-on experience by connecting to the cluster, joining nodes to the cluster, validating the cluster configuration using the CLI, and practice managing administrator roles using RBAC.

Lab 1.1 – Task 1: Initial Configuration Video



- Steps to install a node
- System racked, connected, and powered on
- Adding a node



Let's take a moment to look at an initial configuration of a cluster. The demonstration shows the Implementation Engineer using the Configuration Wizard to install a node after the system has been racked, connected, and powered on. Also shown is adding a node once the initial node is installed. Click on the "clip" icon to launch the video.

Module 2: Data Protection and Layout

Upon completion of this module, you should be able to:

- Describe file striping in OneFS
- Identify and configure different Requested Protection levels
- Explain Suggested Protection
- Differentiate data layout for access patterns
- Compare Requested Protection to Actual Protection
- Illustrate caching in OneFS
- Describe file read and write processes

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Module 2: Data Protection and Layout



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Upon completion of this module, you should be able to describe file striping in OneFS, identify and configure different Requested Protection levels, explain Suggested Protection, differentiate data layout for available access patterns, compare Requested Protection to Actual Protection, illustrate caching in OneFS, and describe the file read and write processes.

Lesson 1: Data Protection and Overhead

Upon completion of this lesson, you will be able to:

- Describe stripes and stripe units
- Illustrate layout for Requested Protection
- Differentiate Requested Protection schemes
- Discuss protection overhead for each protection scheme

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Module 2: Data Protection and Layout



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Upon completion of this lesson, you will be able to describe stripes and stripe units, illustrate layout for Requested Protection, differentiate the Requested Protection schemes, and discuss the protection overhead impact for each protection scheme.

Block vs. File Storage

Block Storage	File Storage
• Structured data, generally • SAN technology • Sequence of bytes at fixed length • Single piece of file or whole file • Easy to query • Best for high input/output, low latency (i.e., transaction processing)	• Unstructured data, generally • NAS technology • Discrete unit of information, defined by application or created by user • Only useful as complete file • Too large for database apps, high I/O • Isilon specialized in file-based storage

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Module 2: Data Protection and Layout

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One way of categorizing data storage systems is to describe them as block-based or file-based. Block data is structured data usually found in SAN (storage area network) technology, for example the VNX, whereas file data is unstructured data that is usually associated with NAS (network attached storage) technology, such as Isilon.

A block of data is a sequence of bits or bytes in a fixed length; the length is determined by the file system. Saving a single piece of data requires the operating system, or OS, to break the file into blocks and each block is written to a particular sector (area) of the drive. A single file may require compiling many, many blocks together. Block data is especially useful when working with small bits of information that need to be accessed or written frequently; for example, a large database full of postal codes. Someone querying the database probably wants only some or one of the postal codes, but rarely wants all of them. Block data makes it easy to gather information in partial sets and is particularly adept at handling high volumes of small transactions, such as, stock trading data, which could generate one billion 18k files in only a few hours. Block format is the go-to for flexibility and for when you need intensive speed of input and output operations.

File data is created depending upon the application and protocol being used. Some applications store data as a whole file, which is broken up and sent across the network as packets. All of the data packets are required to reassemble the file. Unlike block where you can grab only one type of postal code, in file storage you would need the whole file content in order for it to be useful. For example, a PDF file is generally not readable unless you have all of it downloaded; having only part of the file will generate an error and not allow the file to be opened. File-based data is organized in chunks too large to work well in a database or in an application that deals with intense amounts of transactions.

Isilon specializes in handling file-based data. Can Isilon do block-based storage? Technically, yes, but if you are looking for a block-based solution there are other EMC products that specialize in block and would best handle that type of workflow.

Data Protection Overview

- Goal: Protect data and data integrity
 - When components fail, data still needs to be available
- Protection calculated per file
 - Files get cut into data stripes
- Stripe units distributed across cluster nodes
 - Written to individual nodes
 - Not to individual drives within same node
 - Allows multiple drives/nodes to fail and data remain accessible
- Node with client connection:
 - Calculates FEC (forward error correcting) stripes
 - Determines write plan



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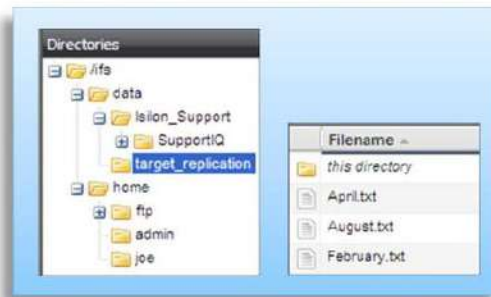
Module 2: Data Protection and Layout

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In OneFS, data protection is calculated on individual files. To calculate protection, individual files are logically broken into 128 KB stripe units. Stripe width is the number of stripe units that you can create before you need to create a protection stripe unit (called FEC). Each file is broken down into smaller 128 KB stripes units, then protection is calculated for the file and protection stripe units are created. The data stripe units and the protection stripe units together form a stripe. Stripe units are then distributed to individual nodes across the cluster. As a result, when a file is needed, multiple nodes in the cluster are able to deliver the data back to the requesting user or application. This dramatically improves overall performance, especially when hundreds, and even thousands, of these requests are made simultaneously from an application. Due to the way in which OneFS applies protection, files that are 128 KB in size or smaller are actually mirrored.

Flexible Data Protection

- FEC
 - Uses Reed-Solomon algorithm
 - Uses software erasure coding
 - Isilon does not use RAID
- Set protection at multiple levels
 - System default
 - Per node pool
 - Per directory
 - Per file

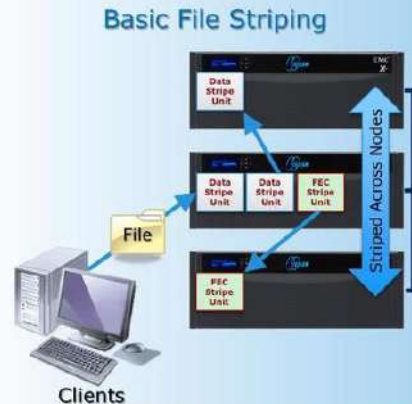


The Isilon system uses the Reed-Solomon algorithm, which is an industry standard method to create error-correcting codes, or ECC, at the file level. EMC Isilon systems do not use hardware or software-based RAID. FEC works much like RAID-5, in that it generates protection data blocks and stores them separately from the data blocks. OneFS can support protection levels of up to $N+4n$. The data can be protected with a $N+4n$ scheme, where up to four drives, nodes or a combination of both can fail without data loss. On an Isilon cluster, you can enable multiple protection levels that allow a cluster to sustain two, three, or four simultaneous failures without resulting in data loss. In OneFS, protection is calculated per individual files and not calculated based on the hardware. OneFS provides the capability to set a file's protection level at multiple levels. The Requested Protection can be set by the default system setting, at the node pool level, per directory, or per individual file.

Data Layout – File Striping

- Data is striped across nodes
 - Data stripe units
 - FEC stripe unit(s)

1. Client saves file to node
2. File divided into data stripe units
3. FEC stripe unit(s) calculated
4. Stripe width assembled
5. Stripe units striped across nodes



OneFS stripes the data stripe units and FEC stripe units across the nodes. Some protection schemes use more than one drive per node. OneFS uses advanced data layout algorithms to determine data layout for maximum efficiency and performance. Data is evenly distributed across nodes in the node pool as it is written. The system can continuously reallocate where the data is stored and make storage space more usable and efficient. Depending on the file size and the stripe width, as the cluster size increases, the system stores large files more efficiently.

Within the cluster, every disk within each node is assigned both a unique GUID (global unique identifier) and logical drive number and is subdivided into 32MB cylinder groups comprised of 8KB blocks. Each cylinder group is responsible for tracking, via a bitmap, whether its blocks are used for data, inodes or other metadata constructs. The combination of node number, logical drive number and block offset comprise a block or inode address and fall under the control of the aptly named Block Allocation Manager (BAM).

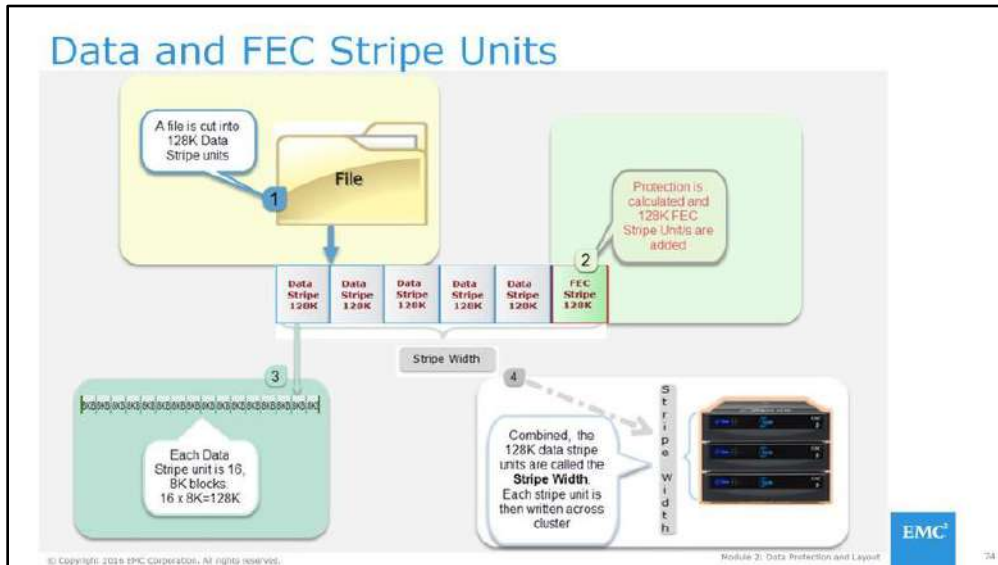
Displayed is a simple example of the write process.

The client saves a file to the node it is connect to.

The file is divided into data stripe units. The data stripe units are assembled into the maximum stripe widths for the file.

FEC stripe unit(s) are calculated to meet the Requested Protection level.

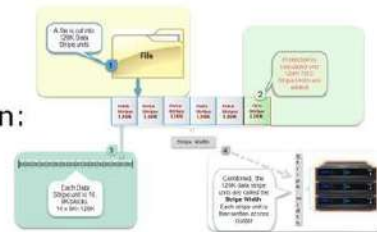
The data and FEC stripe units are striped across nodes.



The data stripe units and protection stripe units are calculated for each file stripe by the Block Allocation Manager (BAM) process. The file data is broken in to 128KB data stripe units consisting of 16 x 8KB blocks per data stripe unit. A single file stripe width can contain up to 16 x 128KB data stripe units for a maximum size of 2MB as the portion of the file's data. A very large file will have thousands of file stripes per file distributed across the node pool. The protection is calculated based on the Requested Protection level for each file stripe using the data stripe units assigned to that file stripe. The BAM process calculates 128KB FEC stripe units to meet the Requested Protection level for each file stripe. The higher the desired protection level, the more FEC stripes units are calculated.

File Stripes and File Stripe Width

- Each file divided into 128K *data stripe units*
 - 16, 8KB blocks per data stripe unit
 - Maximum 16 data stripe units per file stripe
 - $16 \times 128\text{KB} = 2\text{MB}$
- Each stripe consists of:
 - Data stripes
 - Protection (FEC) stripes
- File *stripe width* (size) varies based on:
 - File size
 - Number of nodes in node pool
 - Requested Protection level



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Files written to Isilon are divided in the file stripes. File stripe is a descriptive term and is referred to by different names such as; stripes, protection stripes, or data stripes. File stripes are portions of a file that are contained in a single data and protection band distributed across nodes on the cluster. Each file stripe contains both data stripe units and protection stripe units. The file stripe width or size of the stripe varies based on the file size, the number of nodes in the node pool, and the Requested Protection level to be applied the file. The number of file stripes can range from a single stripe to thousands of stripes per file.

The data stripe units and protection stripe units are calculated for each file stripe by the Block Allocation Manager (BAM) process. The file data is broken in to 128KB data stripe units consisting of 16 x 8KB blocks per data stripe unit. A single file stripe width can contain up to 16 x 128KB data stripe units for a maximum size of 2MB as the portion of the file's data. A very large file will have thousands of file stripes per file distributed across the node pool. The protection is calculated based on the Requested Protection level for each file stripe using the data stripe units assigned to that file stripe. The BAM process calculates 128KB FEC stripe units to meet the Requested Protection level for each file stripe. The higher the desired protection level, the more FEC stripes units are calculated.

Mirrored Data Protection

- Protected as mirror copies
 - 2x to 8x mirror settings
 - # indicates total number of data copies
 - Original *plus* 1–7 duplicate copies
 - Default for metadata and system files
- Explicitly set
 - Protection options to select mirroring
 - Use with predominantly small files
- Actual Protection applied for other Requested Protection levels
 - When small files
 - When node pool smaller than required for Requested Protection



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Mirrored data protection is exactly what the description would indicate. The protection blocks are copies of the original set of data blocks. OneFS includes the capability to use 2X to 8X mirrored protection. The number indicates the total number of data copies to be stored on the cluster. The original data blocks plus one to seven duplicate copies. In addition to protecting file data, mirroring is used to protect the file's metadata and some system files that exist under /ifs in hidden directories.

Mirroring can be explicitly set as the Requested Protection level in all available locations. One particular use case is where the system is used to only store small files. A file of 128KB or less is considered a small file. Some workflows store millions of 1KB to 4KB files. Explicitly setting the Requested Protection to mirroring can save fractions of a second per file and reduce the write ingest time for the files.

Mirroring is set as the Actual Protection on a file even though another Requested Protection level is specified under certain conditions. If the files are small, the FEC protection for the file results in a mirroring. The number of mirrored copies is determined by the loss protection requirements of the Requested Protection. Mirroring is also used if the node pool is not large enough to support the Requested Protection level. As an example, if there are 5 nodes in a node pool and N+3n is the Requested Protection, the file data is saved at the 4X mirror level as the Actual Protection.

N+Mn Protection Levels

N + Mn

N = Number of data stripe units
Mn = Number of simultaneous failures tolerated
 (Equals number of protection (FEC) stripe units)

Protection Level	Drive Failures	Node Failures	Minimum Node Pool Size	Maximum Stripe Width
N+1n	1	1	3 nodes - (2 data + 1 FEC)	17 - (16 data + 1 FEC)
N+2n	2	2	5 nodes - (3 data + 2 FEC)	18 - (16 data + 2 FEC)
N+3n	3	3	7 nodes - (4 data + 3 FEC)	19 - (16 data + 3 FEC)
N+4n	4	4	9 nodes - (5 data + 4 FEC)	20 - (16 data + 4 FEC)



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N+Mn illustrates the primary protection level in OneFS. N represents the number of data stripe units and Mn represents the number of simultaneous drive or node failures that can be tolerated without data loss. M also represents the number of protection or FEC stripe units created and added to the protection stripe to meet the failure tolerance requirements. The available N+Mn Requested Protection levels are +1n, +2n, +3n, and +4n.

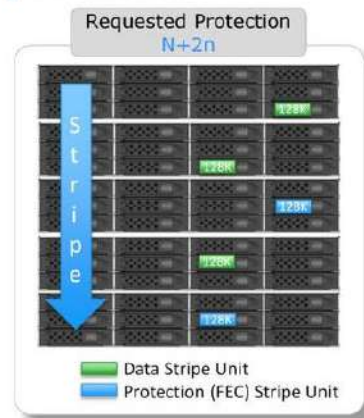
N must be greater than M to gain benefit from the data protection. Referring to the chart, the minimum number of nodes required in the node pool for each Requested Protection level are displayed, three nodes for N+1n, five nodes for N+2n, 7 nodes for N+3n, and 9 nodes for N+4n. If N equals M, the protection overhead is 50 percent. If N is less than M, the protection results in a level of FEC calculated mirroring.

The drives in each node are separated into related sub pools. The sub pools are created across the nodes within the same node pool. The sub pools create additional drive failure isolation zones for the node pool. The number of sustainable drive failures are per sub pool on separate nodes. Multiple drive failures on a single node are equivalent to a single node failure. The drive loss protection level is applied per sub pool.

With N+Mn protection, only one stripe unit is located on a single node. Each stripe unit is written to a single drive on the node. Assuming the node pool is large enough, the maximum size of the file stripe width is 16 data stripe units plus the protection stripe units for the Requested Protection level. The maximum stripe width per N+Mn protection level is displayed.

FEC Protection - Single Drive per Node

- One data stripe unit or FEC stripe unit per node
- **N** = # data stripe units in file stripe
- **M** = Simultaneous failures per value
 - Number of failed drives
 - Number of failed nodes
 - Combination of drive(s) and node(s)



As mentioned previously, some protection schemes utilize a single drive per node per protection stripe. As displayed in the graphic, only a single data stripe unit or a single FEC stripe unit are written to each node. These Requested Protection levels are referred to as $N+M$ or $N+Mn$. In the OneFS web administration interface and command-line interface, the syntax is represented as $+Mn$. M represents the number of simultaneous drive failures on separate nodes that can be tolerated at one time. It also represents the number of simultaneous node failures at one time. A combination of both drive failures on separate nodes and node failures is also possible.

Examples: N+Mn Protection Stripe

+1n	+2n	+3n	+4n	N+Mn Level
Data	Data	Data	Data	Node 1
Data	Data	Data	Data	Node 2
FEC	Data	Data	Data	Node 3
	FEC	Data	Data	Node 4
	FEC	FEC	Data	Node 5
		FEC	FEC	Node 6
		FEC	FEC	Node 7
			FEC	Node 8
			FEC	Node 9

- Requested Protection level for each N+Mn
 - Data and FEC combination
 - Minimum number of nodes illustrated

Data stripe units and FEC stripe units can be placed on any node, in any order in the node pool

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Module 2: Data Protection and Layout

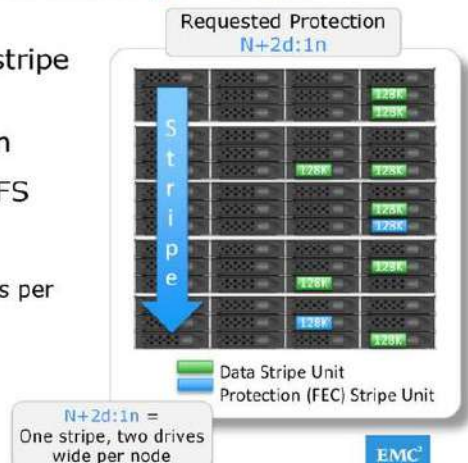
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In the chart is an illustration of each requested N+Mn Requested Protection level over the minimum number of required nodes for each level. The data stripe units and protection stripe units can be placed on any node in the node pool and in any order. The number of data stripe units is dependent on the size of the file and the size of the node pool up to the maximum stripe width. As illustrated, N+1n has one FEC stripe unit per protection stripe, N+2n has two, N+3n has three, and N+4n has four.

N+2n and N+3n are the two most widely used Requested Protection levels for larger nodes pools, node pools with around 15 nodes or more. The ability to sustain both drive or node loss drives the use when possible.

FEC Protection - Multiple Drives per Node

- Multiple data stripe unit or FEC stripe unit per node
- Referred to as **N+M:B** protection
- Represented as **+Md:Bn** in OneFS
 - **Md** = Number of tolerated simultaneous drive failures
 - **M** = Number of FEC stripes units per file stripe
 - **:** represents an **or** conjunction
 - **Bn** = Number of tolerated simultaneous node failures



The other FEC protection schemes utilize multiple drives per node. The multiple drives contain parts of the same protection stripe. Multiple data stripe units and FEC stripe units are placed on separate drive on each node. This is referred to as $N+M:B$ or $N+Md:Bn$ protection. These protection schemes are represented as $+Md:Bn$ in the OneFS web administration interface and the CLI. The M value represents the number of simultaneous tolerable drive failures on separate nodes without data loss. It also represents the number of FEC stripe units per protection stripe. The $:$ (colon) represents an "or" conjunction. The B value represents the number of tolerated node losses without data loss.

Unlike $N+Mn$, $N+Md:Bn$ has different values for the number of drive loss and node losses tolerated before data loss may occur. When a node loss occurs, multiple stripe units are unavailable from each protection stripe and the tolerable drive loss limit is reached when a node loss occurs.

Displayed is an example of a 1MB file with a Requested Protection of $+2d:1n$. Two stripe units, either data or protection stripe units are placed on separate drives in each node. Two drive on different nodes per sub pool can simultaneously be lost or a single node without the risk of data loss.

N+Md:Bn Protection Levels

- One stripe with multiple stripe units per node
 - Stripes across multiple drives per node
- Lowers protection overhead with smaller number of nodes
 - Simulates larger node pool

N+Md:Bn

N = Number of data stripe units
M = Number of tolerable drive failures
B = Number of tolerable node failures

Protection Level	Drive Failures OR Node Failures	Maximum Stripe Width
N+2d:1n	2	18 - (16 data + 2 FEC)
N+3d:1n	3	19 - (16 data + 3 FEC)
N+4d:1n	4	20 - (16 data + 4 FEC)

Stripe units per node

FEC stripe units per file stripe

N+2d:1n is the OneFS default protection level

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N+Md:Bn utilizes multiple drives per node as part of the same data stripe with multiple stripe units per node. N+Md:Bn protection lowers the protection overhead by increasing the size of the protection stripe. N+Md:Bn simulates a larger node pool by utilizing the multiple drives per node. The single protection stripe spans the nodes and each of the included drives on each node. The supported N+Md:Bn protections are N+2d:1n, N+3d:1n, and N+4d:1n. N+2d:1n is the default node pool Requested Protection level in OneFS. M is the number the number of stripe units or drives per node, and the number of FEC stripe units per protection stripe. The same maximum of 16 data stripe units per stripe is applied to each protection stripe. The maximum stripe width for each Requested Protection level is displayed in the chart.

Examples: N+Md:Bn Protection Stripe

Drives per node = # FEC stripe units

+3d:1n			+2d:1n		N+Md:Bn Level
FEC	Data	Data	Data	Data	
Data	FEC	Data	FEC	Data	
Data	Data	FEC	Data	FEC	
3	2	1	2	1	
					Node 1
					Node 2
					Node 3
					Drive

Data stripe units and FEC stripe units can be placed on any node, in any order in node pool

+4d:1n				N+Md:Bn Level
Data	FEC	Data	Data	
Data	Data	FEC	Data	
FEC	Data	Data	FEC	
4	3	2	1	
				Node 1
				Node 2
				Node 3
				Drive

Displayed are examples for the available N+Md:Bn Requested Protection levels. The data stripe units and FEC stripe units can be placed on any node in the node pool in any order. As displayed, N+2d:1n contains 2 FEC stripe units, and has 2 stripe units per node. N+3d:1n contains 3 FEC stripe units, and has 3 stripe units per node. N+4d:1n contains 4 FEC stripe units, and has 4 stripe units per node.

N+2d:1n is the default Requested Protection in OneFS and is an acceptable protection level for smaller nodes pools and node pools with smaller drive sizes.

N+3d:1n and N+4d:1 are most effective with larger file sizes on smaller node pools. Smaller files are mirrored when these protection levels are requested.

FEC Protection – Advanced Protection Levels

- Advanced form of $N+M_d:B_n$
 - **M** = Number of FEC stripe units per protection stripe
 - **Two** stripe units per node
 - Number of drives per node set (2)
 - Does not equal number of FEC stripe units per stripe
 - Provides greater than 1 node loss protection

Protection Level	Drive Failures	Other Failures	Maximum Data Stripe Width
$N+3d:1n1d$	3	OR 1 node + 1 drive	18 - (15 data + 3 FEC)
$N+4d:2n$	4	2 nodes	20 - (16 data + 4 FEC)



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In addition to the previous $N+M_d:B_n$ there are two advanced forms of Requested Protection. M represents the number of FEC stripe units per protection stripe. However, the number of drives per node and the number of stripe units per node is set at two. The number stripe units per node does not equal the number of FEC stripe units per protection stripe. The benefit to the advanced $N+M_d:B_n$ protection levels are they provide a higher level of node loss protection. Besides the drive loss protection, the node loss protection is increased. The available Requested Protection levels $N+3d:1n1d$ and $N+4d:2n$.

$N+3d:1n1d$ includes three FEC stripe units per protection stripe, and provides protection for three simultaneous drive losses, or one node and one drive loss. The higher protection provides the extra safety during data rebuilds associated with the larger drive sizes of 4TB and 6TB. The maximum number of data stripe units is 15 and not 16 when using $N+3d:1n1d$ Requested Protection.

$N+4d:2n$ includes four FEC stripe units per stripe, and provides protection for four simultaneous drive losses, or two simultaneous node failures.

Examples: Advanced N+Md:Bn Protection Stripe

Drives per node $\neq$ # FEC stripe units

N+Md:Bn Level	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Drive
3 FEC Stripe Units, 2 Drives per Node +3d:1n1d	Data	Data	FEC	Data	Data	Data	1
	Data	FEC	Data	FEC	Data	Data	2
4 FEC Stripe Units, 2 Drives per Node +4d:2n	Data	Data	FEC	Data	FEC	Data	1
	Data	FEC	Data	FEC	Data	Data	2

Data stripe units and FEC stripe units can be placed on any node in any order in the node pool

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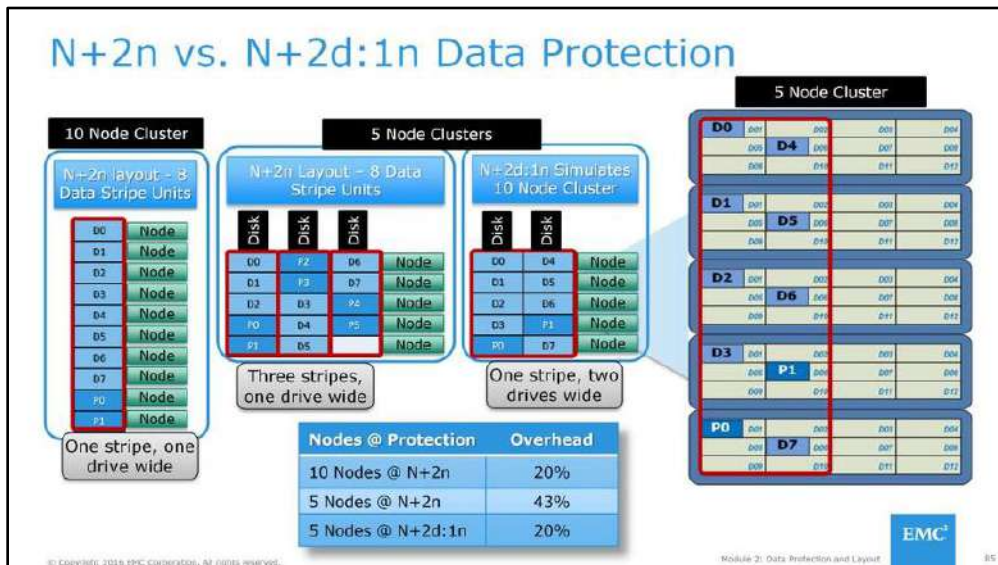
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Displayed are examples of the advanced N+Md:Bn protection schemes. Two drives per node per protection stripe. The number of FEC stripe units does not equal the number of drives used for the protection stripe. Even if one node is lost, there is still a greater level of protection available. Like other Requested Protection levels, the data stripe units and FEC stripe units can be placed on any node in the node pool and on any drive.

N+3d:1n1d is the minimum protection for node pools containing 6TB drives. The extra protection is required to maintain MTDL during the time required to rebuild data from a failed drive.

The use of N+4d:2n is expected to increase especially for smaller to middle sized node pools as larger drives are introduced.



Another illustration from the previous example to assist in clarifying N+2:1 even better.

There are 8 data stripe units to write in a stripe (8 x 128K) ~ 1 MB file. The desired protection includes the ability to sustain the loss of two hard drives.

If there is a 10 node cluster, 2 FEC stripe units would be calculated on the 8 data stripe units using an N+2 protection level. The protection overhead in this case is 20 percent.

However there is only a 5 node cluster to write to. Using N+2 protection, the 1 MB file would be placed into 3 separate data stripes, each with 2 protection stripe units. A total of 6 protection stripe units are required to deliver the Requested Protection level for the 8 data stripe units. The protection overhead is 43 percent. Using N+2:1 protection the same 1 MB file requires 1 data stripe, 2 drives per node wide per node and only 2 protection stripe units. The 10 stripe units are written to 2 different drives per node. The protection overhead is the same as the 10 node cluster at 20 percent.

Protection Overhead

Number of Nodes	OneFS 7.2.0 +	N+1n	N+2n	N+2d:1n	N+3n	N+3d:1n	N+3d:1n:1d	N+4n	N+4d:1n	N+4d:2n
	OneFS 7.1.1 & Older	N+1	N+2	N+2:1	N+3	N+3:1	n/a	N+4	N+4:1	N+4:2
3	3	2+1 (33%)	3x (67%)	4+2 (33%)	3x (67%)	6+3 (33%)	3+3 (50%)	3x (67%)	8+4 (33%)	3x (67%)
4	4	3+1 (25%)	2+2 (50%)	6+2 (25%)	4x (75%)	9+3 (25%)	5+3 (38%)	4x (75%)	12+4 (25%)	4+4 (50%)
5	5	4+1 (20%)	3+2 (40%)	8+2 (20%)	4x (75%)	12+3 (20%)	7+3 (30%)	5x (80%)	16+4 (20%)	6+4 (40%)
6	6	5+1 (17%)	4+2 (33%)	10+2 (17%)	3+3 (50%)	15+3 (17%)	9+3 (25%)	5x (80%)	16+4 (20%)	8+4 (33%)
7	7	6+1 (14%)	5+2 (29%)	12+2 (14%)	4+3 (43%)	16+3 (16%)	11+3 (21%)	5x (80%)	16+4 (20%)	10+4 (29%)
8	8	7+1 (13%)	6+2 (25%)	14+2 (13%)	5+3 (38%)	16+3 (16%)	13+3 (19%)	4+4 (50%)	16+4 (20%)	12+4 (25%)
9	9	8+1 (11%)	7+2 (22%)	16+2 (11%)	6+3 (33%)	16+3 (16%)	15+3 (17%)	5+4 (44%)	16+4 (20%)	14+4 (22%)
10	10	9+1 (10%)	8+2 (20%)	16+2 (11%)	7+3 (30%)	16+3 (16%)	15+3 (17%)	6+4 (40%)	16+4 (20%)	16+4 (20%)
11	11	10+1 (9%)	9+2 (18%)	16+2 (11%)	8+3 (27%)	16+3 (16%)	15+3 (17%)	7+4 (36%)	16+4 (20%)	16+4 (20%)
12	12	11+1 (8%)	10+2 (17%)	16+2 (11%)	9+3 (25%)	16+3 (16%)	15+3 (17%)	8+4 (33%)	16+4 (20%)	16+4 (20%)
13	13	12+1 (8%)	11+2 (15%)	16+2 (11%)	10+3 (23%)	16+3 (16%)	15+3 (17%)	9+4 (31%)	16+4 (20%)	16+4 (20%)
14	14	13+1 (7%)	12+2 (14%)	16+2 (11%)	11+3 (21%)	16+3 (16%)	15+3 (17%)	10+4 (29%)	16+4 (20%)	16+4 (20%)
15	15	14+1 (7%)	13+2 (13%)	16+2 (11%)	12+3 (20%)	16+3 (16%)	15+3 (17%)	11+4 (27%)	16+4 (20%)	16+4 (20%)
16	16	15+1 (6%)	14+2 (13%)	16+2 (11%)	13+3 (19%)	16+3 (16%)	15+3 (17%)	12+4 (25%)	16+4 (20%)	16+4 (20%)
17	17	16+1 (6%)	15+2 (12%)	16+2 (11%)	14+3 (18%)	16+3 (16%)	15+3 (17%)	13+4 (24%)	16+4 (20%)	16+4 (20%)
18	18	16+1 (6%)	16+2 (11%)	16+2 (11%)	15+3 (17%)	16+3 (16%)	15+3 (17%)	14+4 (22%)	16+4 (20%)	16+4 (20%)
19	19	16+1 (6%)	16+2 (11%)	16+2 (11%)	16+3 (16%)	16+3 (16%)	15+3 (17%)	15+4 (21%)	16+4 (20%)	16+4 (20%)
20	20	16+1 (6%)	16+2 (11%)	16+2 (11%)	16+3 (16%)	16+3 (16%)	15+3 (17%)	16+4 (20%)	16+4 (20%)	16+4 (20%)

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The protection overhead for each protection level depends on the file size and the number of nodes in the cluster. The percentage of protection overhead declines as the cluster gets larger. In general, N+1n protection has a protection overhead equal to one node's capacity, N+2n protection has a protection overhead equal to two nodes' capacity, N+3n is equal to three nodes' capacity, and so on.

OneFS also supports optional data mirroring from 2x-8x, allowing from two to eight mirrors of the specified content. Data mirroring requires significant storage overhead and may not always be the best data-protection method. For example, if you enable 3x mirroring, the specified content is explicitly duplicated three times on the cluster; depending on the amount of content being mirrored, this can require a significant amount of capacity.

The table displayed indicates the relative protection overhead associated with each FEC Requested Protection level available in OneFS. Indicators include when the FEC protection would result in mirroring.

Lesson 1: Summary

Having completed this lesson, you can now:

- Describe stripes and stripe units
- Illustrate layout for Requested Protection
- Differentiate Requested Protection schemes
- Discuss protection overhead for each protection scheme

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Having completed this lesson, you can now describe stripes and stripe units, illustrate layout for Requested Protection, differentiate between $N+Mn$, $N+Md:Bn$ and advanced $N+Md:Bn$ Requested Protection schemes, and discuss the protection overhead impact for each protection scheme.

Lesson 2: Protection Management

Upon completion of this lesson, you will be able to:

- Identify Requested Protection configuration areas
- Differentiate between Requested Protection levels
- Modify Requested Protection in web administration interface
- Recognize when node pool protection is below Suggested Protection

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Upon completion of this lesson, you will be able to identify Requested Protection configuration areas, differentiate between levels of Requested Protection configuration, modify the Requested Protection in the web administration interface, and recognize when node pool is below the Suggested Protection.

Data Protection Terminology

Requested Protection	• Determines amount of redundant data on cluster • Default for node pools: N+2:1
Suggested Protection	• Level of protection recommended by OneFS • Can't be modified
Actual Protection	• Level of protection OneFS actually applies to data • Can be more than Requested Protection but never less
Mirrored Protection	• Copies data to multiple locations • Can have 2-8 mirrors

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Module 2: Data Protection and Layout

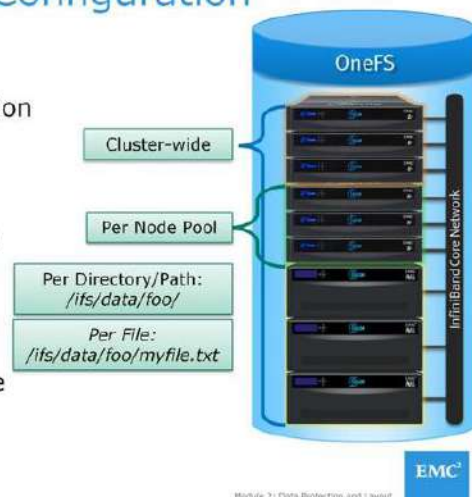


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On the slide are the high-level descriptions used when talking about data protection in OneFS. These are described in further detail in this lesson.

Requested Protection Configuration

- Can be set at multiple levels
 - Cluster-wide default protection
 - Default file pool policy
 - Per storage pool default
 - Per node pool
 - File system specific settings
 - Per directory
 - Per file
- Management options
 - Web administration interface
 - CLI
 - PAPI



Requested Protection configuration is available at multiple levels. Each level is used to control protection for specific reasons. From a cluster-wide setting, the Requested Protection in the default file pool policy is applied to any file or folder that has not been set by another Requested Protection policy. A Requested Protection level is assigned to every node pool. In OneFS, the Requested Protection can be set at the directory or individual file level.

Management of the Requested Protection levels is available using the web administration interface, CLI, or Platform Application Programming Interface (PAPI). Management using web administration interface and CLI management are discussed in this course.

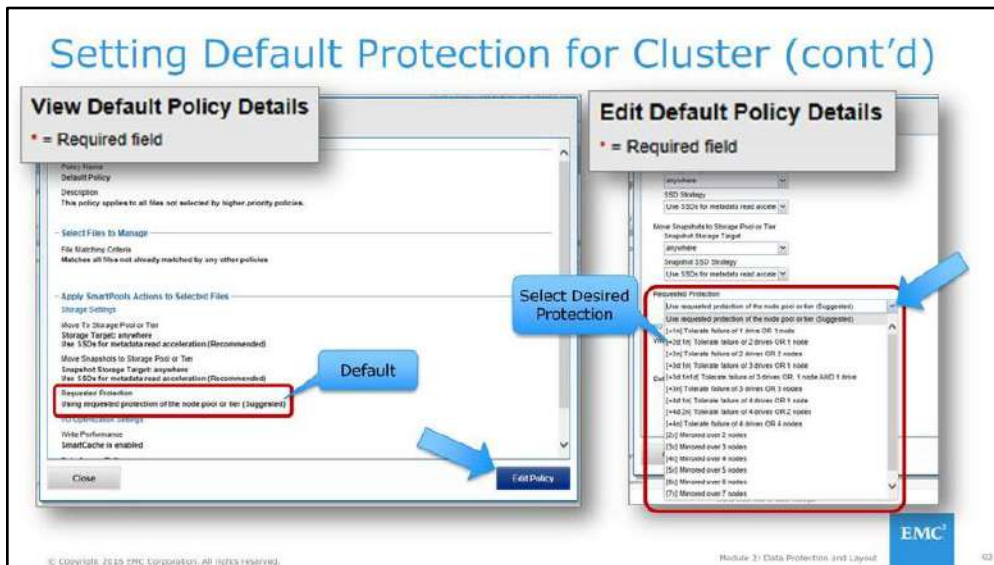
Setting Default Protection for Cluster

- Sets default file pool policy
 - Applied to all files without higher policy
 - Suggested: Use storage pool default

File System → Storage Pools → File Pool Policies



The cluster-wide default data protection setting is made using the default file pool policy. The setting will be applied to any file or directory that does not have a higher priority setting. The default setting is to use the Requested Protection setting for the storage pool. To edit the default setting, navigate to **File System > Storage Pools > File Pool Policies**, and click **View / Edit** on the **Default Policy** line.



The **View Default Policy Details** window is displayed with the current default file pool policy settings. The current protection is displayed under **Requested Protection**. The default protection setting is **Using requested protection of the node pool or tier (Suggested)**.

To change the setting, click **Edit Policy**. The **Edit Default Policy Details** window is displayed. The current settings are changed to drop-down menus.

Click the drop-down arrow to display the available options. After selecting the desired Requested Protection, click **Save**.

The default file pool policies are applied when the **SetProtectPlus** or **SmartPools** job runs.

Storage Pool Protection Setting

- Set per node pool
 - OneFS default = $N+2d:1n$
 - HD400 minimum = $N+3d:1n1d$
- Minimum should meet Suggested Protection

File System → Storage Pools → SmartPools

The screenshot shows the EMC OneFS web administration interface. The navigation path is **File System** → **Storage Pools** → **SmartPools**. The **SmartPools** tab is selected, showing a table of **Tiers & Node Pools**. The table has columns: Name, State, Nodes, Requested Protection, SSD/L3, HDD % Used, SSD % Used, and Actions. The row for **v260_24gb_2gb** is highlighted, and the **Requested Protection** value is **+2d:1n**. A green box labeled **Current Protection** points to the **Requested Protection** column. A red box labeled **4** points to the **View / Edit / More** link in the **Actions** column.

Name	State	Nodes	Requested Protection	SSD/L3	HDD % Used	SSD % Used	Actions
v260_24gb_2gb	Good	1-6	+2d:1n		13.2%	–	View / Edit / More

The default file pool policy protection setting uses the node pool or tier setting. Requested Protection is set per node pool. When a node pool is created, the default Requested Protection applied to the node pool is $+2d:1n$.

The required minimum Requested Protection for an HD400 node pool is $+3d:1n1d$. You are requested to modify the HD400 node pool Requested Protection to meet this minimum. The Requested Protection should meet the minimum Requested Protection level for the node pool configuration. The minimum is based on MTTDL calculations for the number of nodes and the drive configuration in the nodes. If the Requested Protection requires modification, here is where the node pool Requested Protection is modified.

To view and modify the Requested Protection setting for the node pools in the web administration interface, navigate to **File System > Storage Pools > SmartPools**. The current Requested Protection for each node pool is displayed in the **Tiers & Node Pools** section. Click **View / Edit** to modify the settings.

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Automated Settings - File Pool Policies

- Automated policies to manage data
 - Requested Protection
 - Other file settings
 - Location on available storage pool
 - I/O optimization settings
- SmartPools license
 - Licensed – Create custom file pool policies
 - Customized policies for directory and file selection
 - Customized actions for each policy
 - SmartPools and file pool policies covered in later in course
 - No license - Default file pool policy only



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SmartPools file pool policies are used to automate data management including the application of Requested Protection settings to directories and files, the storage pool location, and the I/O optimization settings. In this lesson, we discuss the setting of Requested Protection. SmartPools and file pool policies are discussed in detail in the *Storage Administration* module. A SmartPools license is required to create custom file pool policies. Custom policies can be filtered on many different criteria for each policy including file path or metadata time elements. Without a SmartPools license on the default file pool policy is applied.

Manual Protection Settings

- Manually configure protection for:
 - Directory or subdirectory
 - File
- Not recommended
 - Overrides automated settings
 - Can result in unexpected behavior
 - Requires manual change back to default settings to reinstate
- Web administration interface
 - **File System Explorer**
- Command-line interface
 - `isi set` command

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Manual settings can be used to modify the protection on specific directories or files. The settings can be changed at the directory or subdirectory level. Individual file settings can be manually changed. Best practices recommend against using manual settings, because manual settings can return unexpected results and create management issues as the data and cluster age. Once set manually, the settings either need to be reset to default to use automated file pool policy settings or continue as manually managed settings. Manual settings override file pool policy automated changes. Manual changes are made using file system explorer is used in the web administration interface and the **isi set** command in the CLI.

File System Explorer

File System → File System Explorer → Explorer

The screenshot shows the File System Explorer interface. At the top, there's a breadcrumb trail: File System → File System Explorer → Explorer. Below this is a navigation bar with tabs: Dashboard, Cluster Management, File System, Data Protection, Access, and Protocols. The main content area is titled 'File System Explorer' and shows a directory tree on the left and a file listing on the right. Callouts point to various features: 'Browse or Search for files' points to the search bar; 'Shows you the directory path' points to the path field; 'Allows you to move back a level' points to the 'Parent Directory' link; 'Change Protection settings under View/Edit' points to the 'View/Edit' link in the file listing; and 'Legend defines icons' points to the legend at the bottom.

- View directories and files
- Set directory or file protection
- Requires **root** access

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File system explorer is used to view the directories and files on the cluster. You can also modify the properties of any directory or file. The properties are stored for each file in OneFS. You need to log in as **root** in order to access file system explorer. File system explorer is located under **File System > File System Explorer** in the web administration interface. To navigate to the specific file or directory, expand the directory tree on the left. Once you have located the directory, click the specific directory to view the files and the next level subdirectories. You can also search for a file using the search box or browse directly to a directory/file if you know the path. The properties are displayed on the page directory listings page. To modify the protection level, click **View/Edit**.

Suggested Protection Overview

- Notification for node pool protection vs. MTTDL
 - Visual notification in web administration interface
 - CELOG event, if below Suggested Protection
- Calculates and stores MTTDL for each node pool
 - Uses same calculations as Isilon Sizing Tool
 - Pre-Sales Engineering node pool and cluster sizing tool



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Suggested Protection refers to the visual status (in SmartPools Summary) and CELOG event notification, for node pools that are set below the calculated Suggested Protection level. The Suggested Protection is based on meeting the minimum mean time to data loss, or MTTDL, standard for EMC Isilon node pools. MTTDL is a statistical calculation based on hardware and protection factors that estimate the likelihood of a failure resulting in data loss.

When a new node pool is added to a cluster or the node pool size is modified, the Suggested Protection level is calculated and the MTTDL calculations are compared to a database for each node pool. The calculations use the same logic as the Isilon Sizing Tool, which is an online tool used primarily by EMC Isilon Pre-Sales engineers and business partners. The tool is used to determine appropriate node pool sizing for a customer workflow, and calculates the appropriate Suggested Protection levels based on the node pool size and node configuration.

Why is Suggested Protection Important?

- Data loss is bad!
 - Data at risk when below MTDDL standards
 - Default Requested Protection is +2d:1n
- What occurs in the field
 - Node pool starts small and grows
 - Default setting no longer adequate
 - Protection not increased
- Monitoring and comparing not easy



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So why is Suggested Protection important? Because data loss is bad. This is an obvious statement but it's the underlying reason why the Suggested Protection monitoring feature is important.

When a node pool is below the Mean Time to Data Loss, or MTDDL, standards, the data is at risk. This doesn't mean data loss will occur, it does indicate the data is below the MTDDL standards. Anything that puts data at risk is considered something to be avoided.

The default Requested Protection setting for all new node pools is +2d:1n, which protects the data against either the simultaneous loss of two drives or the loss of a single node. What commonly occurs is a node pool starts small and then grows beyond the configured Requested Protection level. The once adequate +2d:1n Requested Protection level is no longer appropriate, but is never modified to meet the increased MTDDL requirements. The Suggested Protection feature provides a method to monitor and notify users when the Requested Protection level should be changed.

Suggestion Protection Status

- Notified when below Suggested Protection
 - Node pools within Suggested Protection levels not displayed
- Default settings
 - New clusters – enabled, by default
 - Upgraded clusters – disabled, by default
 - Requires field review and customer discussion
 - Enabled through non-customer-facing command

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The Suggested Protection feature notifies the administrator only when the Requested Protection setting is below the suggested level for a node pool. The notification doesn't give the suggested setting and node pools that are within Suggested Protection levels are not displayed. Suggested Protection is part of the SmartPools health status reporting.

By default, the Suggested Protection feature is enabled on new clusters. On clusters upgraded from a version prior to OneFS 7.2, the feature is disabled by default. This is by design because a field review and customer discussion is necessary to mitigate any concerns and to fully explain the Suggested Protection feature before it is turned on. Some customer node pools may be below the Suggested Protection level and, although important to meet MTDL, it is not a critical situation. The discussion consists of the impact on protection overhead, any potential workflow impacts, and an assessment of any risk. After the discussion, the feature can be enabled using a non-customer-facing command. Customers should contact their EMC Isilon account team to arrange a field review.

Suggested Protection Notifications

File System → Storage Pools → Summary

The screenshot shows the 'Storage Pools' summary page in the EMC Veeam web interface. The breadcrumb trail is 'File System → Storage Pools → Summary'. The page has tabs for 'Summary', 'File Pool Policies', 'SmartPools', and 'SmartPools Settings'. The 'Summary' tab is active, showing a 'Status' section with a table of modules and their health.

Module	Status	Message	Action
Policies	Good	All file pool policies are healthy.	
SmartPools	Info	v200_24gb_2gb has a different requested protection from the suggested protection of +2n: 1n.	View Details
SmartPool Settings	Good	All settings are healthy.	
CloudPools	Caution	CloudPools is not licensed.	View Details

A red callout bubble points to the 'SmartPools' row, containing the text: 'If suggested and requested protections are different you'll get an 'informational' FYI'.

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In the web administration interface, Suggested Protection notifications are located under **File System > Storage Pools > Summary** and are included with other storage pool status messages. A node pool below the Suggested Protection level is displayed as a SmartPools module, with an **Info** status, and a message stating **Node pool <node pool name> has a different requested protection from the suggested protection of <level>**.

Displayed is an example of the **v200_24gb_2gb** node pool with a Requested Protection level that is different than the suggested. For this example, the node pool's Requested Protection was configured as +1n to generate the status message. To modify the settings, go to the **SmartPools** tab and click **View/Edit** on the pool.

Lesson 2: Summary

Having completed this lesson, you can now:

- Identify Requested Protection configuration areas
- Differentiate between Requested Protection levels
- Modify Requested Protection in web administration interface
- Recognize when node pool protection is below Suggested Protection

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Having completed this lesson, you can now identify Requested Protection configuration areas, differentiate between Requested Protection levels, modify the Requested Protection in the web administration interface and CLI, and recognize when node pool protection is below Suggested Protection.

Lesson 3: Data Layout and Actual Protection

Upon completion of this lesson, you will be able to:

- Describe drive layout with access pattern
- Identify Requested Protection from Actual Protection
- Illustrate Actual Protection layout

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Upon completion of this lesson, you will be able to explain sub pools and their relationship with data protection, describe drive layout with access pattern, identify Requested Protection from Actual Protection, and illustrate Actual Protection layout.

Data Layout Detail

- Four variables determine data layout:
 - Number of nodes in cluster
 - Protection level applied to node pool/directory/file
 - File size
 - Data access pattern
- Data laid out in most efficient, economical, highest performing way
- Large files distributed maximum to 6 drives per node
 - Striped across the node pool



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There are four variables that combine to determine how data is laid out. This makes the possible outcomes almost unlimited when trying to understand how the system will work. The number of nodes in the cluster affects the data layout because data is laid out vertically across all nodes in the cluster, then number of nodes determines how wide the stripe can be. $N+Mn$ where N is the number of data stripe units and Mn is the protection level. The protection level also affects data layout because you can change the protection level of your data down to the file level, and the protection level of that individual file changes how it will be striped across the cluster. The file size also affects data layout because the system employs different layout options for larger files than for smaller files to maximize efficiency and performance. The disk access pattern modifies both prefetching and data layout settings associated with the node pool. Disk access pattern can be set at a file or directory level so you are not restricted to using only one pattern for the whole cluster.

Ultimately the system's job is to lay data out in the most efficient, economical, highest performing way possible. You can manually define some aspects of how it determines what is best, but the process is designed to be automated.

The maximum number of drives for streaming is six drives per node across the node pool for each file.

Data Access Patterns

- **Concurrency (default)**
 - Best for many clients accessing same data
 - Prefers one drive per node for all protection stripes of a file
 - Moderate prefetching
- **Random**
 - Same layout as Concurrency, but with prefetching turned off
 - Optimized for random access
- **Streaming**
 - Best for single stream workflows
 - Prefers as many drives as possible per file within pool
 - Very aggressive prefetching



Module 2: Data Protection and Layout

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An administrator from the web management or CLI interface can optimize layout decisions made by OneFS to better suit the workflow. The data access pattern influences how a file is written to the drives during the write process.

Concurrency is used to optimize workflows with many concurrent users access the same files. The preference is that each protection stripe for a file is placed on the same drive or drives depending on the Requested Protection level. For example, a larger file with 20 protection stripes, each stripe unit from each protection stripe would prefer to be placed on the same drive in each node. Concurrency is the default data access pattern. Concurrency influences the prefetch caching algorithm to prefetch and cache a reasonable amount of anticipated associated data during a read access.

Streaming is used for large streaming workflow data such as movie or audio files. Streaming prefers to use as many drives as possible, within the given pool, when writing multiple protection stripes for a file. Each file is written to the same sub pool within the node pool. With a streaming data access pattern, the protection stripes are distributed across the 6 drives per node in the node pool. This maximizes the number of active drives per node as the streaming data is retrieved. Streaming also influences the prefetch caching algorithm to be highly aggressive and gather as much associated data as possible.

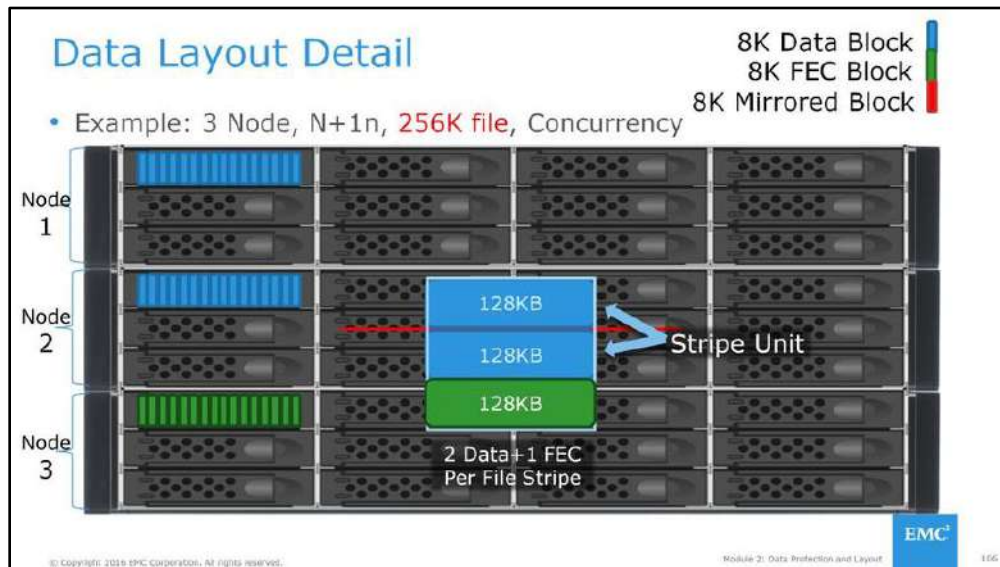
A random access pattern prefers using a single drive per node for all protection stripes for a file just like a concurrency access pattern. With random however, the prefetch caching request is minimal. Most random data does not benefit from prefetching data into cache.

Access can be set from the web administration interface or the CLI. From the CLI, the drive access pattern can be set separately from the data layout pattern.

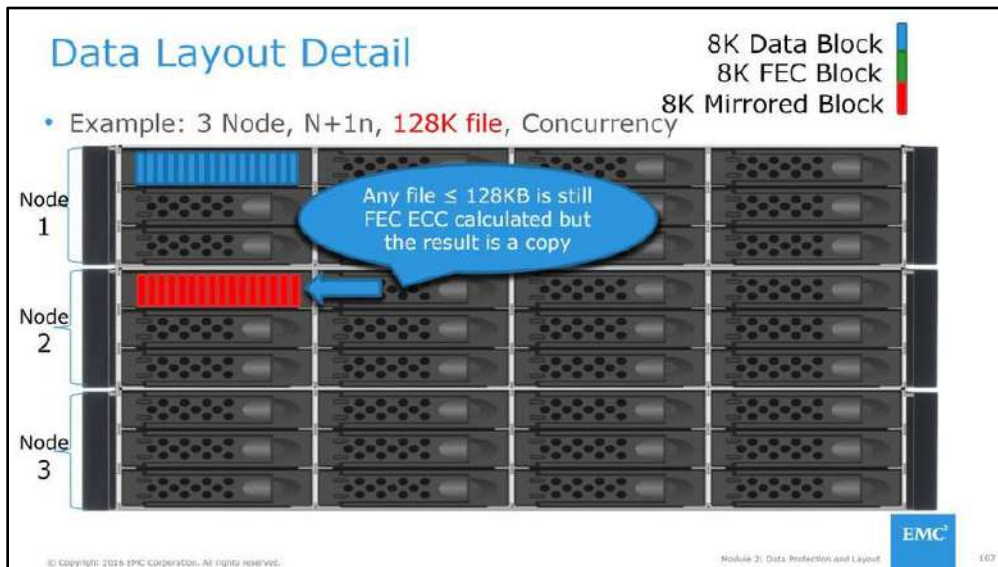
isi set -a <default|streaming|random> -d <#drives> <path/file>

Options:

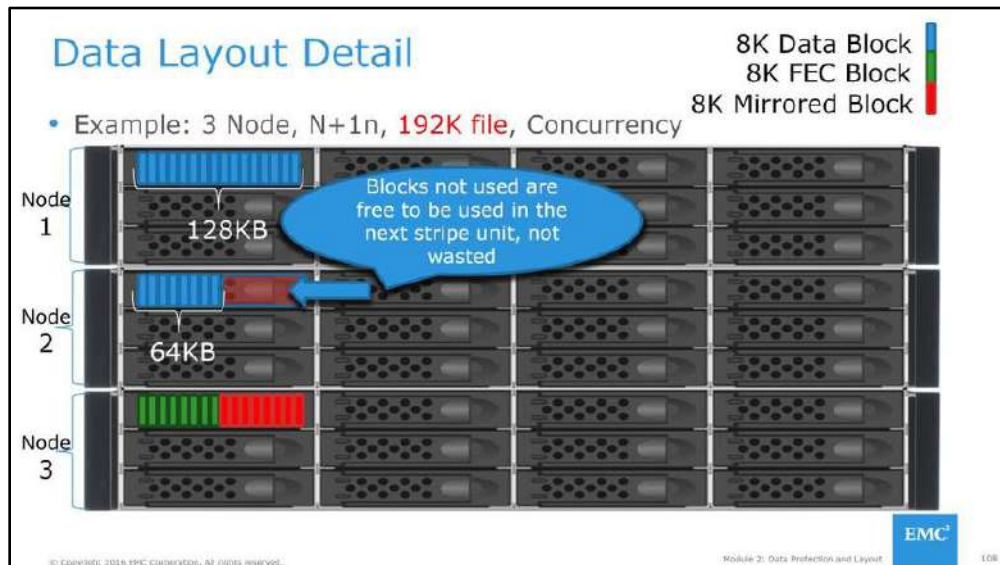
- **-a <value> -<default|streaming|random>** - Specifies the file access pattern optimization setting.
- **-d <@r drives>** - Specifies the minimum number of drives that the file is spread across.
- **-l <value> - <concurrency|streaming|random>** - Specifies the file layout optimization setting. This is equivalent to setting both the **-a** and **-d** flags.



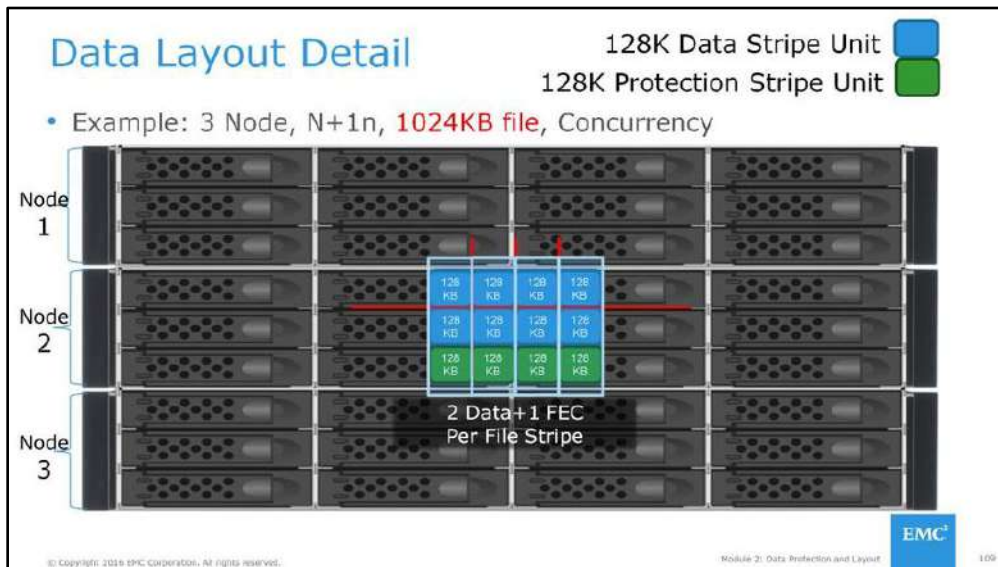
The process of striping spreads all write operations from a client across the nodes of a cluster. The example in this animation demonstrates how a file is broken down into chunks, after which it is striped across disks in the cluster along with forward error correction (FEC). Even though a client is connected to only one node, when that client saves data to the cluster, the write operation occurs in multiple nodes in the cluster. This is also true for read operations. A client is connected to only one node at a time, however when that client requests a file from the cluster, the node to which the client is connected will not have the entire file locally on its drives. The client's node retrieves and rebuilds the file using the back-end InfiniBand network.



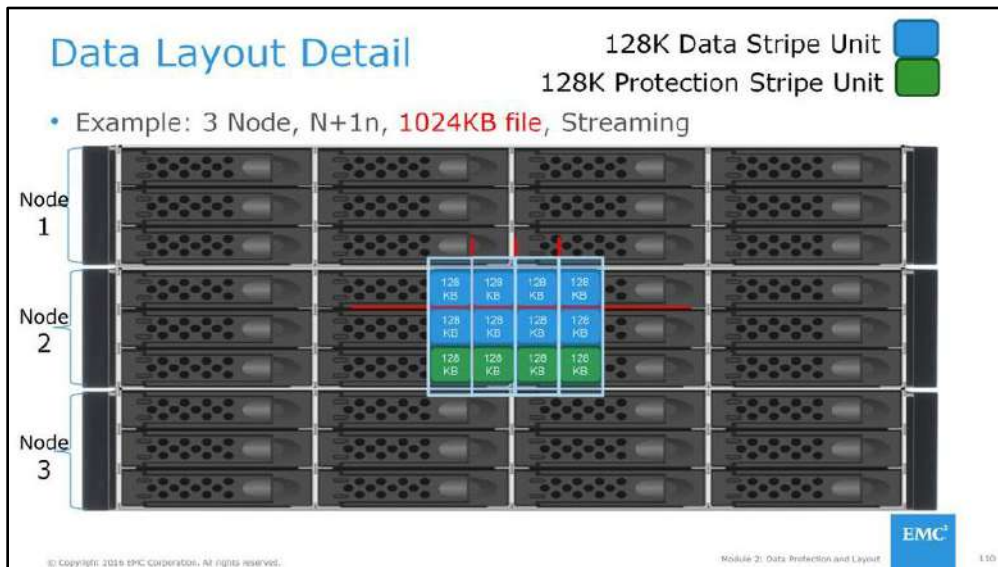
All files 128 KB or less are mirrored. For a protection strategy of N+1 the 128 K file would have a 2X mirroring; the original data and one mirrored copy. We will see how this is applied to different files sizes.



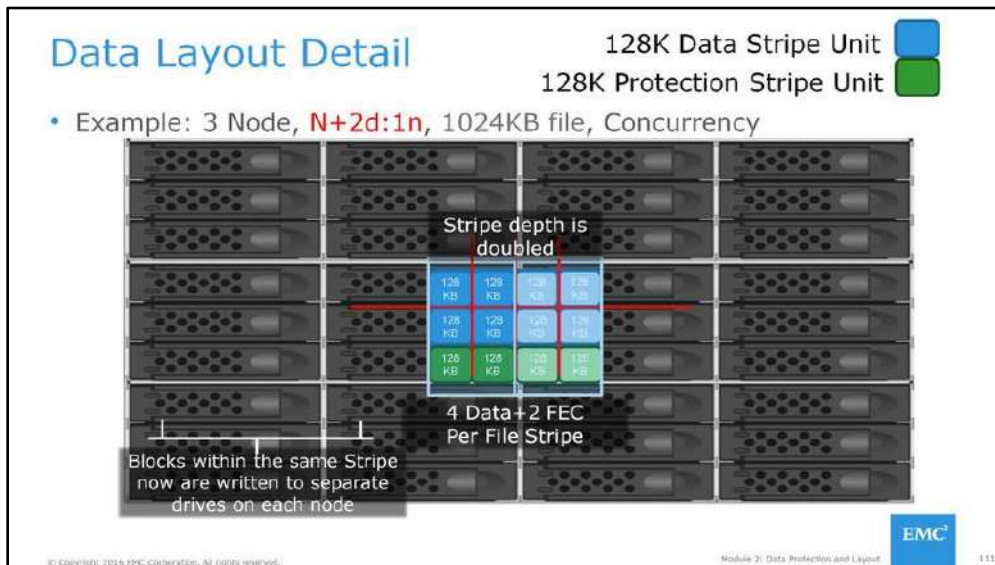
OneFS is designed to withstand multiple simultaneous component failures (currently four) while still affording unfettered access to the entire file system and dataset. Data protection is implemented at the file system level and, as such, is not dependent on any hardware RAID controllers. This provides many benefits, including the ability add new data protection schemes as market conditions or hardware attributes and characteristics evolve. Because protection is applied at the file-level, a OneFS software upgrade is all that's required in order to make new protection and performance schemes available.



This slide further reviews the data layout detail.



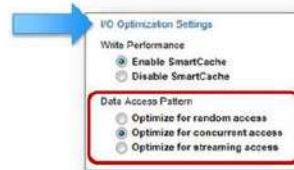
This example shows how the data is striped to different drives using a streaming layout.



OneFS also supports several hybrid protection schemes. These include N+2:1 and N+3:1, which protect against two drive failures or one node failure, and three drive failures or one node failure, respectively. These protection schemes are particularly useful for high density node configurations, where each node contains up to thirty six, multi-terabyte SATA drives. Here, the probability of multiple drives failing far surpasses that of an entire node failure. In the unlikely event that multiple devices have simultaneously failed, such that the file is “beyond its protection level”, OneFS will re-protect everything possible and report errors on the individual files affected to the cluster’s logs.

Data Layout Management

- Management levels
 - Mostly same as Requested Protection
 - Exception: Not set per node pool
 - Default policy
 - SmartPools file pool policies
 - Manual settings
 - File system explorer
 - `isi set` command
- Configured under **I/O Optimization Settings**
 - Data Access Pattern



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Data layout is managed the same way as Requested Protection. The exception is data layout is not set at the node pool level. Settings are available in the default file pool policy, with SmartPools file pool policies, and manually set using either **File System Explorer** in the web administration interface or the **isi set** command in the CLI. The settings are located in the I/O optimization sections under data access pattern.

Edit Default Policy Details

• **Storage Settings**

Copy important data to another volume

Where To Store Data In This Storage Pool

Storage Target: AnyStorage

SMB In-Reply: Use USN for metadata read access

How Frequently To Store Data In This Storage Pool

Storage Target: AnyStorage

Storage Size (in MB): Any

Use USN for metadata read access: ☒

Retention Protection

Use requested protection of the source pool or file (Suggested): ☒

Write Performance

☒ Enable SmartCache

☐ Disable SmartCache

Data Access Pattern

☒ Optimize for random access

☐ Optimize for sequential access

☐ Optimize for streaming access

Cancel

```

[1] <concurrency> | <streaming> | <random>
[2] <strategy> | <name>
[3] <on | off>
[4] <on | off>
[5] <strategy> | <avoid> | <metadata> | <metadata-write>
[6] <data>
[7] <file> | <path> | <lin>

```

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Actual Protection Applied to File

- Meets minimum Requested Protection loss tolerance
 - Except if node pool too small
- May be different than Requested Protection
 - More efficient protection available
 - Small files
 - Smaller than minimum node pool size
- Example: Requested Protection = **+2d:1n**
 - 18 nodes and 2MB file = **+2n** Actual Protection applied
 - 128KB file = **3x** Actual Protection applied

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In OneFS, the Actual Protection applied to a file depends on the Requested Protection level, the size of the file, and the number of nodes in the node pool.

The Actual Protection level is what the cluster actually does. This is not necessarily the same as the Requested Protection level, but here are the rules:

- Actual Protection must meet or exceed the Requested Protection level.
- Actual Protection may change in the interests of efficiency. For example, if you have a Requested Protection of **+2d:1n** and there is a 2MB file and a node pool of at least 18 nodes, the file is actually laid out as **+2n**.
- Actual Protection depends upon file size. If you have a small file of 128KB, the file is actually protected using 3x mirroring, because at that file size the FEC calculation results in mirroring.
- In both cases, the minimum drive loss protection of 2 drives and node loss protection of 1 node are exceeded by the Actual Protection applied to the file.
- The exception to meeting the minimum Requested Protection is if the node pool is too small and unable to support the Requested Protection minimums. For example, a node pool with 3 nodes and set to **+4n** Requested Protection. The maximum supported protection is 3x mirroring in this scenario.

Actual Protection vs. Requested Protection

OneFS 7.2.0 OneFS 7.1.1 & Older		N+1n	N+2n	N+2d:1n	N+3n	N+3d:1n	N+3d:1n1d	N+4n	N+4d:1n	N+4d:2n	Annotations
Default Max Stripe Width		N+1	N+2	N+2:1	N+3	N+3:1	n/a	N+4	N+4:1	N+4:2	
Available Nodes	2	1+1	2x	2+2/2	2x	3+3/3	2x	2x	4+4/4	2x	Mirroring, below minimum size for Requested Protection
	3	2+1	3x	4+2/2	3x	6+3/3	3+3/2	3x	8+4/4	3x	Minimum for Requested Protection
	4	3+1	2+2	6+2/2	4x	9+3/3	5+3/2	4x	12+4/4	4+4/2	
	5	4+1	3+2	8+2/2	4x	12+3/3	7+3/2	5x	16+4/4	6+4/2	
	6	5+1	4+2	10+2/2	3+3	15+3/3	9+3/2	5x	16+4/4	8+4/2	@ Requested Protection
	7	6+1	5+2	12+2/2	4+3	15+3/3	11+3/2	5x	16+4/4	10+4/2	
	8	7+1	6+2	14+2/2	5+3	15+3/3	13+3/2	4+4	16+4/4	12+4/2	
	9	8+1	7+2	16+2/2	6+3	15+3/2	15+3/2	5+4	16+4/4	14+4/2	> than maximum # nodes @ Requested Protection
	10	9+1	8+2	16+2/2	7+3	15+3/2	15+3/2	6+4	16+4/2	16+4/2	
	11	10+1	9+2	16+2/2	8+3	15+3/2	15+3/2	7+4	16+4/2	16+4/2	
	12	11+1	10+2	16+2/2	9+3	15+3/2	15+3/2	8+4	16+4/2	16+4/2	
	13	12+1	11+2	16+2/2	10+3	15+3/2	15+3/2	9+4	16+4/2	16+4/2	
	14	13+1	12+2	16+2/2	11+3	15+3/2	15+3/2	10+4	16+4/2	16+4/2	
	15	14+1	13+2	16+2/2	12+3	15+3/2	15+3/2	11+4	16+4/2	16+4/2	
	16	15+1	14+2	16+2/2	13+3	15+3/2	15+3/2	12+4	16+4/2	16+4/2	
	17	16+1	15+2	16+2/2	14+3	15+3/2	15+3/2	13+4	16+4/2	16+4/2	
	18	16+1	16+2	16+2	15+3	15+3	15+3	14+4	16+4/2	16+4/2	Actual Protection changed from Requested Protection
	19	16+1	16+2	16+2	16+3	16+3	16+3	15+4	16+4/2	16+4/2	
	20	16+1	16+2	16+2	16+3	16+3	16+3	16+4	16+4	16+4	

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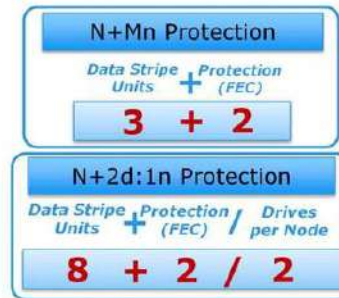
Displayed is a chart indicating the Actual Protection applied to a file according to the number of nodes in the node pool.

- Orange indicates the Actual Protection applied would use mirroring.
- Dark blue indicates files protected at 50% storage overhead, while offering the Requested Protection level.
- White with bold black indicate the Requested Protection is applied in that range.
- White with grey indicates the maximum size of the protection stripe is reached and a subset of the available nodes will be used for the file.
- Burgundy indicates the Actual Protection applied is changed from the Requested Protection while meeting or exceeding the Requested Protection level, for reasons of efficiency.

The chart is provided as a reference. If you see the Actual Protection does not match the Requested Protection level it may have been changed to be more efficient given the file or number of nodes in the node pool.

isi get Output

- Displays
 - Actual Protection setting for file
 - Applied per stripe
 - Manual settings applied
 - Nodes containing file
 - Node pool
 - Metadata location
 - Drives per file stripe
 - And more



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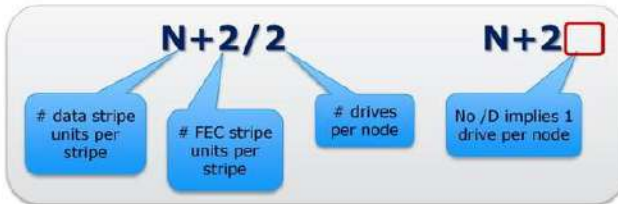
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The calculated file protection is displayed. N+Mn protection displays the number of data stripe units + the number of protection stripe units calculated per data stripe. N+Md:Bn is displayed as the number of data strip units + the number of protection stripe units divided by the number of drives per node. N+2d:1n is displayed as N+2/2, N+3d:1n is displayed as N+3/3, and +3d:1n1d is displayed as N+3/2. Using this nomenclature you can identify the calculated protection and view the protection per stripe in the output.

Actual Protection Nomenclature

- Displayed in `isi get -D` output
 - N = Number of data stripe units per stripe
 - +M = Number of FEC stripe units per stripe
 - /D = Number of drives per node
 - Mirroring displayed as Mx



Requested Protection	Representation in isi get Output
N+1n	N+1
N+2n	N+2
N+2d:1n	N+2/2
N+3n	N+3
N+3d:1n	N+3/3
+3d:1n1d	N+3/2
+4d:1n	N+4/4
+4d:2n	N+4/2

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The Actual Protection nomenclature is represented differently than Requested Protection when viewing the output showing Actual Protection from the **isi get -D** or **isi get -DD** command. The output displays the number of data stripe units plus the number of FEC stripe units divided by the number of disks per node the stripe is written to. The chart displays the representation for the Requested Protection and the Actual Protection. N is replaced in the Actual Protection with the number of data stripe units for each protection stripe. If there is no / in the output, it implies a single drive per node. Mirrored file protection is represented as 2x to 8x in the output.

Finding File Protection – CLI

- View file protection

```
isi get
```

- Useful options

- d – Displays directory attributes
- D – Displays detailed file attributes
- DD – Displays more detailed file attributes

- Syntax

```
isi get <path/filename>
isi get -d <path>
isi get -D <path/filename>
isi get -DD <path/filename>
```

Protection
Information

isi get -DD
Output Sections

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To find the protection setting from the CLI, using the **isi get** command provides detailed file or directory information. The primary options are **-d <path>** for directory settings and **-DD <path>/<filename>** for individual file settings.

The **isi get -DD** output has three primary locations containing file protection, a summary in the header, line item detail settings in the body, and detailed per stripe layout per drive at the bottom. Each of these are explored in more detail using three examples.

isi get Output – No Options

- `isi get <path> or <path/filename>`
 - Outputs all files in directory or single file information
 - Includes Requested Protection and Actual Protection

The diagram illustrates the output of the `isi get` command. It shows two examples of command execution and their corresponding output tables. Annotations point to specific parts of the output:

- Requested Protection:** Points to the `POLICY` column in the first table.
- Manually Set:** Points to the `LEVEL` column in the first table.
- Actual Protection:** Points to the `PERFORMANCE` column in the first table.
- Access Pattern:** Points to the `COAL` column in the first table.
- Path Only:** Points to the `FILE` column in the first table.
- Path & File:** Points to the `FILE` column in the second table.
- Min # Drives = 4:** Points to the `LEVEL` column in the second table.

Example 1: Directory Path

```
se-sandbox2-cluster-1# isi get /ifs/data/training
```

POLICY	LEVEL	PERFORMANCE	COAL	FILE
default	3k	concurrency on		./
+31	5k/2	concurrency on		./
default	4+2/2	concurrency on		Different file size with ls and du command.txt
+2	3k	concurrency off		Dir_Protect/
+1	2+1	random	off	Isilon-Supportability-and-Compatibility-Guide-.pdf
default	4+2/2	concurrency on		OneFS-7.1-CLI-Administration-Guide.pdf
default	4+3/3	concurrency on		OneFS-7.1-Event-Reference.pdf
default	4+2/2	concurrency on		OneFS-7.1-Web-Administration-Guide.pdf
default	4+2/2	concurrency on		Role_Based_Access_OneFS_7.pdf
default	4+2/2	concurrency on		Thumbs.db
default	4+2/2	concurrency on		selog-events-to-anmp.py

Example 2: Specific File

```
se-sandbox2-cluster-1# isi get /ifs/data/training/isilon-synciq-wp.pdf
```

POLICY	LEVEL	PERFORMANCE	COAL	FILE
default	4+2/2	concurrency/64	on	isilon-synciq-wp.pdf

The **isi get** command can be used to display protection settings on an entire directory path or a specific file without any options. The **POLICY** or Requested Protection policy, the **LEVEL** or Actual Protection, the **PERFORMANCE** or data access pattern are displayed for each file. Use with a directory path only displays the properties for every file and subdirectory under the specified directory path. Use with the path and directory specified displays the properties for the specific file. In the example, several files are manually set for protection, one file is manually set for data access pattern a random, and one file has a minimum drive requirement set as part of the data access pattern.

Video of Data Protection



- Data protection overhead
- Protection management
- Data layout options



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Let's take a moment to review Isilon's data protection. The video reviews the concepts covered in Lessons 1 through 3. Click on the "clip" icon to launch the video or go to this link: <https://www.youtube.com/watch?v=drmNedzzH34&feature=youtu.be>

Lesson 3: Summary

Having completed this lesson, you can now:

- Describe drive layout with access pattern
- Identify Requested Protection from Actual Protection
- Illustrate Actual Protection layout

Having completed this lesson, you can now explain sub pools and their relationship with data protection, describe drive layout with access pattern, identify Requested Protection from Actual Protection, and illustrate Actual Protection layout.

Lesson 4: Caching

Upon completion of this lesson, you will be able to:

- Describe different caching in OneFS
- Illustrate read cache process
- Differentiate between asynchronous and synchronous write processes
- Define Endurant cache

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Upon completion of this lesson, you will be able to describe different caching in OneFS, illustrate the read cache process, differentiate between an asynchronous write and synchronous write process, and define the Endurant cache.

Cluster Caching Basics

- Each storage node has at least 12GB of RAM
 - Up to 256GB
- Portion of RAM used for read and write cache
 - No need to tune
- Cache pool used collaboratively across cluster
- Some SSD used as cache
 - Some metadata cached



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There are several methods that Isilon clusters uses for caching. Each storage node contains standard DRAM (between 12GB and 256GB, although older nodes may have less) and this memory is primarily used to cache data that is on that particular storage node and is actively being accessed by clients connected to that node. Each node also contributes to and has access to a cluster-wide cache that is globally accessible and coherent across all nodes. A portion of the DRAM is dynamically allocated and adjusted as read and write cache as needed. Each node communicates with the cache contained on every other node and extracts any available cached file data as needed. Some node pools use SSDs as a specialized cache. The use of SSDs for cache is optional but enabled by default.

Caching in OneFS

- Caching maintains copy of metadata and/or data in location other than primary storage location
 - Used to accelerate access
 - Cache is temporary
 - Organized into levels
 - Each level serves specific purpose
- Cache levels indicate
 - Immediacy of information
 - Relative latency
 - Difference in storage methodology
 - Duration
 - Emptying (flushing) methodology



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Module 2: Data Protection and Layout

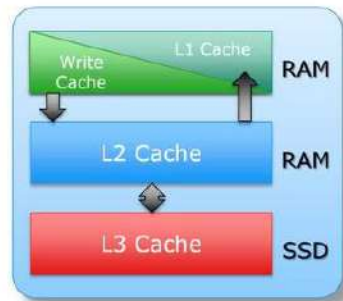
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What is caching? Caching maintains a copy of metadata and or the user data blocks in a location other than primary storage. The copy is used to accelerate access to the data by placing the copy on a medium with faster access than the drives. Because cache is a copy of the metadata and user data, any data contained in cache is temporary and can be discarded when no longer needed. Cache in OneFS is divided into levels and each level serves a specific purpose in read and write transactions.

The cache levels provide a guidance to the immediacy of information from a client-side transaction perspective, the relative latency or time to retrieve or write information, and indicates how the cache is refreshed, how long the data is available and how the data is emptied or flushed from cache.

Caching in OneFS (cont'd)

- Available caching
 - Client-side cache
 - L1 cache
 - Write cache
 - Storage / node-side cache
 - L2 cache
 - L3 cache option



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Caching in OneFS consist of the client-side L1 cache and write coalescer, and L2 storage and node-side cache. Both L1 cache and L2 cache are managed and maintained in RAM. However, OneFS also has the capability to use SSDs as L3 cache. As displayed, L3 cache interacts the L2 cache and is contained on SSDs. Each cache has its own specialized purpose and work together to provide performance improvements across the entire cluster.

L1 Cache - Client-side Cache

- Connected client transaction specific
- Level 1 cache
 - All blocks pertain to immediate read requests
 - Very fast
 - Stored in RAM
 - Information flushed after read transaction
- Write cache / write coalescer
 - Same as L1 cache pertaining to write transactions
 - Optimizes write operations to disk
 - Flushed after successful write transaction



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Level 1, or L1, cache is the client-side cache. It is the immediate buffer on the node connected to the client and is involved in any immediate client data transaction. In OneFS, L1 cache specifically refers to read transaction requests, or when a client requests data from the cluster. L1 cache collects the requested data from the L2 cache of the nodes that contain the data. L1 cache is stored in a segmented area of the node's RAM and as a result is very fast. Following a successful read transaction, the data in L1 cache is flushed or emptied to provide space for other transactions.

Related to L1 cache is the write cache or the write coalescer that buffers write transactions from the client to be written to the cluster. The write coalescer collects the write blocks performs the additional process of optimizing the write to disk. The write cache is flushed after successful write transactions.

In OneFS, the two similar caches are distinguished based on their read or write functionality. Client-side caching includes both the in and out client transaction buffers.

L2 Cache - Storage / Node-side Cache

- Level 2 cache
 - Stores blocks from previous read and write operations
 - Stores anticipated blocks for read request
 - Read ahead cache, prefetching
 - Very fast - Stored in RAM
 - Information flushed by age when full - least recently used (LRU)
 - Does not survive node reboot
- Node-specific, not client-connection-specific
 - Only data blocks from specific node
 - Interacts with node's drive subsystem and L3 cache
 - HDDs and SSDs
 - Read or write operations



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Level 2, or L2, cache is the storage side or node-side buffer. L2 cache stores blocks from previous read and write transactions, buffers write transactions to be written to disk and prefetches anticipated blocks for read requests, sometimes referred to as read ahead caching. L2 cache is also contained in the node's RAM and is very fast and available to serve L1 cache read requests and take data handoffs from the write coalescer. For write transactions, L2 cache works in conjunction with the NVRAM journaling process to insure protected committed writes. L2 cache is flushed by the age of the data as L2 cache becomes full.

L2 cache is node specific. L2 cache interacts with the data contained on the specific node. The interactions between the drive subsystem, the HDDs and the SSDs on the node go through the L2 cache for all read and write transactions. L2 cache on any node communicates as requested by the L1 cache and write coalescers from any other node.

L3 Cache - Additional Storage / Node-side Cache

- Provides additional level of storage / node-side cache
- Uses SSDs as **read** cache
 - Slower than RAM, faster than HDD access
- Extension of L2 cache
- Caches blocks from HDD on same node
 - Caches local node data only
 - Automatically selects blocks to cache
 - Caches metadata and user random data blocks
 - Cached blocks are durable
 - Stored on SSD media
 - Not lost during reboot
 - Flushed based on first in / first out (FIFO)
 - As filled, oldest blocks flushed first



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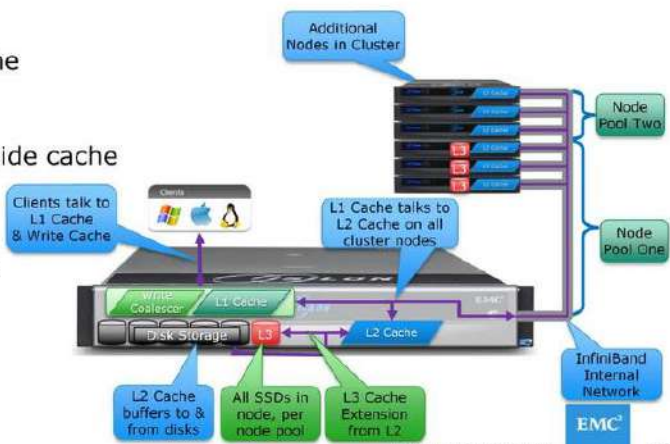
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Level 3, or L3, cache provides an additional level of storage node-side cache utilizing the node's SSDs as read cache. SSD access is slower than access to RAM and is relatively slower than L2 cache but significantly faster than access to data on HDDs. L3 cache is an extension of the L2 read cache functionality. Because SSDs are larger than RAM, SSDs can store significantly more cached metadata and user data blocks than RAM. Like L2 cache, L3 cache is node specific and only caches data associated with the specific node. Advanced algorithms are used to determine the metadata and user data blocks cached in L3. Because the cache is on SSD and not in RAM, unlike L2 cache, L3 cached data is durable and survives a node reboot without requiring repopulating. When L3 cache becomes full and new metadata or user data blocks are loaded into L3 cache, the oldest existing blocks are flushed from L3 cache. L3 cache should always be filled with blocks being rotated as node use requires.

OneFS Caching Levels

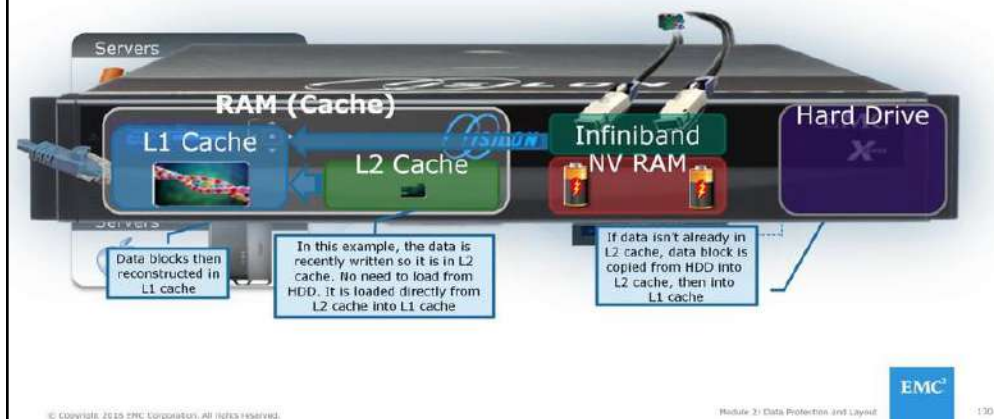
- OneFS cache
 - Client-side cache
 - L1 cache
 - Write cache
 - Storage/node-side cache
 - L2 cache
 - L3 cache
- Accelerator nodes
 - L1 cache only



Displayed in a diagram of a seven node cluster divided into two node pools with a detailed view of one of the nodes. Illustrated are the clients connected to the L1 cache and the write coalescer. The L1 cache is connected to the L2 cache on all of the other nodes and within the same node. The connection to other nodes occurs over the InfiniBand internal network when data contained on those nodes is required for read or write. The L2 cache on the node connects to the disk storage on the same node. The L3 cache is connected to the L2 cache and serves as a read only buffer. L3 cache is spread across all of the SSDs in the same node and enabled per node pool.

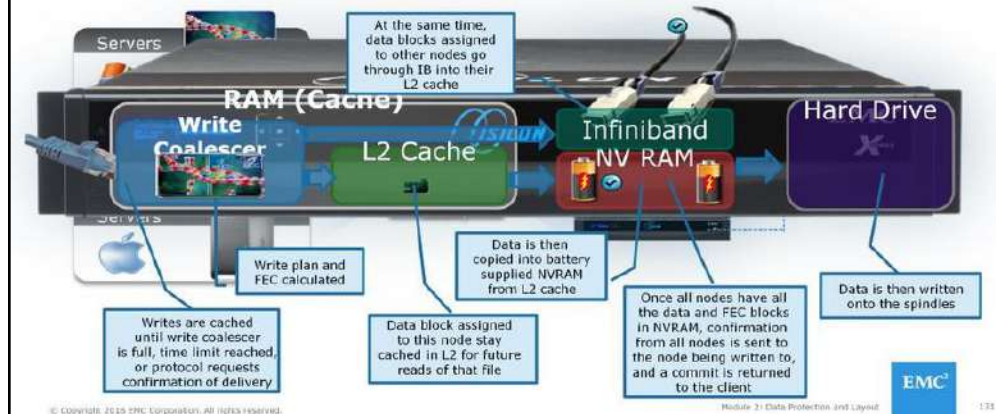
Accelerator nodes do not allocate memory for level 2 cache. This is because accelerator nodes are not writing any data to their local disks, so there are no blocks to cache. Instead accelerator nodes use all their memory for level 1 cache to service their clients. Cache is used differently in the accelerator nodes. Because an accelerator has no local disk drives storing file system data, its entire read cache is L1 cache, however by definition all the data handled by an accelerator is remote data. The cache aging routine in the accelerator cache is LRU-based, as opposed to the dropbehind used in storage node L1 cache. This is because the size of the accelerator's L1 cache is larger, and the data in it is much more likely to be requested again, so it is not immediately removed from cache upon use. In a cluster consisting of storage and accelerator nodes, the primary performance advantage of accelerators is in being able to serve more clients, and potentially hold a client's working set entirely in cache.

The Anatomy of a Read



When a client requests a file, the node to which the client is connected uses the **isi get** command to determine where the blocks that comprise the file are located. The first file inode is loaded and the file blocks are read from disk on all other nodes. If the data isn't already in the L2 cache, data blocks are copied in the L2. The blocks are sent from other nodes by InfiniBand. If the data was already in L2 Cache, we don't need to load it from the hard disks—we just wait for the data blocks from the other nodes to arrive. Otherwise, the node gets the data load from the local hard disks, and then the file is reconstructed in L1 cache and sent to the client.

The Anatomy of an Asynchronous Write



When a client requests that a file be written to the cluster, the node to which the client is connected is the node that receives and processes the file. That node creates a write plan for the file including calculating FEC. Data blocks assigned to the node are written to the NVRAM of that node. Data blocks assigned to other nodes travel through the InfiniBand network to their L2 cache, and then to their NVRAM. Once all nodes have all the data and FEC blocks in NVRAM a commit is returned to the client. Data block(s) assigned to this node stay cached in L2 for future reads of that file. Data is then written onto the spindles.

The layout decisions are made by the BAM on the node that initiated a particular write operation. The BAM makes the decision on where best to write the data blocks to ensure the file is properly protected. To do this, the BSW generates a write plan, which comprises all the steps required to safely write the new data blocks across the protection group. Once complete, the BSW then executes this write plan and guaranty its successful completion. OneFS will not write files at less than the desired protection level, although the BAM will attempt to use an equivalent mirrored layout if there is an insufficient stripe width to support a particular FEC protection level.

What is Endurant Cache?

- Endurant Cache provides:
 - Ingest, staging, and protection of stable synchronous writes
 - Lowers synchronous write latency by improving time to acknowledgement (ACK) to client
 - Enhanced node efficiency in writing synchronous files to disk
- Endurant Cache beneficial workflows:
 - Improve NFS synchronous write performance
 - Improve VMware write performance
 - Improve small block random writes



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So what is Endurant Cache?

Endurant Cache, or EC, is only for synchronous writes or writes that require a stable write acknowledgement be returned to the client. EC provides Ingest and staging of stable synchronous writes. EC manages the incoming write blocks and stages them to stable battery backed NVRAM. Insuring the integrity of the write. EC also provides Stable synchronous write loss protection by creating multiple mirrored copies of the data, further guaranteeing protection from single node and often multiple node catastrophic failures.

The EC process lowers the latency associated with synchronous writes by reducing the “time to acknowledge” back to the client. The process removes the Read-Modify-Write operations from the acknowledgement latency path.

The other major improvement in over all node efficiency with synchronous writes comes from utilizing the Write Coalescer’s full capabilities to optimize writes to disk.

Endurant Cache was specifically developed to improve NFS synchronous write performance and write performance to VMware VMFS and NFS datastore.

A use case for EC is anywhere that VMFS is in use, with the goal of improving the stability of storage in cases where writes might be interrupted by outages.

What Does Endurant Cache Do?



- **Ingest and Staging of Stable Synchronous Writes**
 - **Ingest** - Client sends data block(s) to node's Write Coalescer with sync write ACK request
 - **Staged & Stable** - At requested ACK point, EC Logwriter process mirrors data block(s) to EC Log files in NVRAM
 - ACK returned to client
 - Standard async write process followed
 - **Completed** - EC Log files cleared when standard async write process is stable
- **When is ACK returned to client?**
 - When protocol sends request
 - Could be every block or larger group of blocks
 - 4K, 8K, 32K, 64K or larger



Module 2: Data Protection and Layout

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So what does the Endurant Cache process do?

The Endurant Cache, or EC, ingests and stages stable synchronous writes. Ingests the write into the cluster – The client sends the data block or blocks to the node's Write Coalescer with a synchronous write acknowledgement, or ACK, request. The point of the ACK request varies depending on the application, and the form of the ACK request also varies based on the client protocol. EC manages how the write request comes into the system.

Stages and stabilizes the write – At the point the ACK request is made by the client protocol, the EC Logwriter process mirrors the data block or blocks in the Write Coalescer to the EC log files in NVRAM where the write is now protected and considered stable. This process is very similar to many block storage systems.

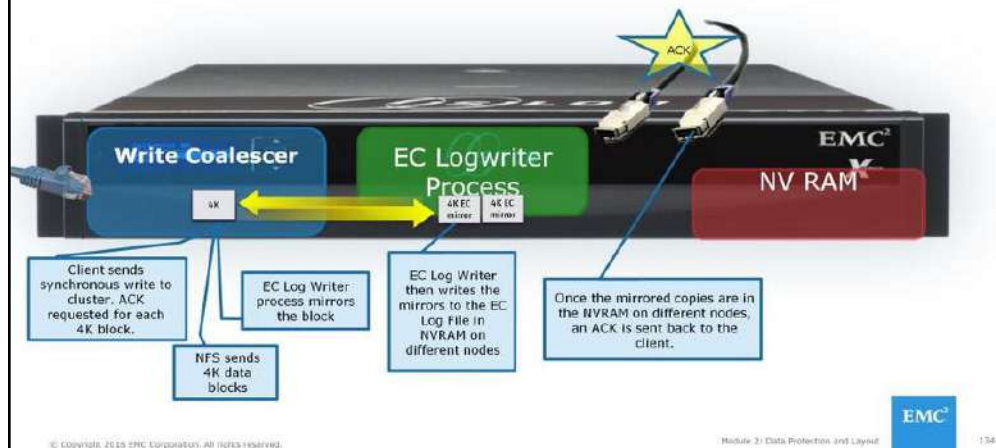
Once stable, the acknowledgement or ACK is now returned to the client. At this point the client considers the write process complete. The latency or delay time is measured from the start of the process to the return of the acknowledgement to the client.

From this point forward, our standard asynchronous write process is followed. We let the Write Coalescer manage the write in the most efficient and economical manner according to the Block Allocation Manager, or BAM, and the BAM Safe Write or BSW path processes.

The write is completed – Once the standard asynchronous write process is stable with copies of the different blocks on each of the involved nodes' L2 cache and NVRAM, the EC Log File copies are de-allocated using the Fast Invalid Path process from NVRAM. The write is always secure throughout the process. Finally the write to the hard disks is completed and the file copies in NVRAM are de-allocated. Copies of the writes in L2 cache remain in L2 cache until flushed through one of the normal processes.

How is it determined when the acknowledgement is returned to the client? The answer is like many with technology, it depends. It depends on the application and its interaction with the protocol. Applications are designed to receive acknowledgements at specific block size points. It also depends upon the protocol and when the protocol makes the request to the storage system, usually at the behest of the application. So for some applications and protocols the acknowledgement request could be as little as every 4K or 8K block sent, or it could be at different incremental sizes, or it could be after an entire file write has been completed.

The Anatomy of a Synchronous Write



Let's look at an example of a new file synchronous write – and diagram how the write process occurs in OneFS with Endurant Cache.

In this example, we are attached to an NFS client, sending 4KB blocks writing a 512KB file with a simple Acknowledgement to be returned after the entire file is written and assuming an N+1 protection level. First, a client sends a file to the cluster requesting a synchronous write acknowledgement. The client begins the write process by sending 4KB data blocks. The blocks are received into the node's Write Coalescer; which is a logical separation of the node's RAM similar to but distinct from L1 and L2 Cache. Once the entire file has been received into the Write Coalescer, the Endurant Cache (EC) LogWriter Process writes mirrored copies of the data blocks (with some log file-specific information added) in parallel to the EC Log Files, which reside in the NVRAM. The protection level of the mirrored EC Log Files is based on the Drive Loss Protection Level assigned to the data file to be written; the number of mirrored copies equals 2X, 3X, 4X, or 5X.

Once the data copies are received into the EC Log Files, a stable write exists and the Write Acknowledgement is sent back to the client, indicating that a stable write of the file has occurred. The client assumes the write is completed and can close out the write cycle with its application or process.

The Write Coalescer then processes the file just like a non-EC asynchronous write at this point. The Write Coalescer fills and is flushed as needed in an asynchronous write fashion, also sometimes referred to as a lazy write, according to the Block Allocation Manager (BAM) and the BAM Safe Write (BSW) path processes. The file is divided into 128-K Data Stripe Units (DSUs); Protection is calculated and FEC Stripe Units (FSUs) are created; the write plan is then determined (Disk Pool, Disk Drives, Blocks on Drives). The 128-K DSUs and FSUs are written to their corresponding nodes' L2 Cache and NVRAM. Then the EC Log Files are cleared from NVRAM. The 128-K DSUs and FSUs are then written to physical disk from L2 Cache. Once written to physical disk, the DSU and FSU copies created during the asynchronous write are de-allocated from NVRAM, but remain in L2 Cache until flushed to make room for more recently accessed data.

The write process is now complete. The acknowledgement was returned to the client prior to the majority of the latency-intensive Read-Modify-Write operations, enabling us to gain all of the benefit of the Write Coalescer efficiencies while maintaining a secure stable write.

L3 Cache Settings

- L3 cache enabled on new node pool, by default
 - Global setting to enable L3 cache
- Enable per node pool
 - On or off configuration setting
 - No visible settings to change
 - Consumes all SSD in node pool
 - Cannot coexist with other SSD strategies
 - Within same node pool
 - No metadata read acceleration
 - No metadata read/write acceleration
 - No data on SSD
 - SSDs cannot participate in GNA



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Module 2: Data Protection and Layout

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L3 cache is enabled by default for all new node pools added to a OneFS 7.1.1 cluster. New node pools containing SSDs are automatically enabled. A global setting is provided in the web administration interface to change the default behavior. Each node pool can be enabled or disabled separately. L3 cache is either on or off and no other visible configuration settings are available.

L3 cache consumes all SSD in the node pool when enabled. L3 cache cannot coexist with other SSD strategies on the same node pool; no metadata read acceleration, no metadata read/write acceleration, and no data on SSD. SSDs in an L3 cache enabled node pool cannot participate as space used for GNA either.

L3 effectively acts as an extension of L2 cache with respect to reads and writes on a node, and the process of reading or writing, with the exception of the larger available cache, is substantially unchanged.

Every HD400 node comes with an SSD so as to be able to use L3 cache to improve its performance. This illustrates how much of a difference a larger cache can make when managing the kinds of huge capacity that such nodes contain.

Lesson 4: Summary

Having completed this lesson, you can now:

- Describe different caching in OneFS
- Illustrate read cache process
- Differentiate between asynchronous and synchronous write processes
- Define Endurant cache

Having completed this lesson, you can now describe different caching in OneFS, illustrate the read cache process, differentiate between an asynchronous write and synchronous write process, and define the Endurant cache.

Module 2: Summary

Having completed this module, you should be able to:

- Describe file striping in OneFS
- Identify and configure different Requested Protection levels
- Explain Suggested Protection
- Differentiate data layout for available access patterns
- Compare Requested Protection to Actual Protection
- Illustrate caching in OneFS
- Describe file read and write processes

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Module 2: Data Protection and Layout



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Having completed this module, you should be able to describe file striping in OneFS, identify and configure different Requested Protection levels, explain Suggested Protection, differentiate data layout for available access patterns, compare Requested Protection to Actual Protection, illustrate caching in OneFS, and describe the file read and write processes.

Lab 2 - Data Protection and File Layout

- Verifying data protection and file layout
- Configuring data protection settings

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Module 2: Data Protection and Layout



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In these labs, you'll practice how to calculate and configure protection levels for your cluster at the directory level and the file level.

Module 3: Networking

Upon completion of this module, you will be able to:

- Identify front-end network properties
- Define NIC aggregation options
- Connect to external IP network
- Differentiate between Basic and Advanced SmartConnect features
- Configure name resolution for cluster

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Module 3: Networking



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Upon completion of this module, you will be able to identify the front-end network properties, define the NIC aggregation options, connect to the external IP network, differentiate between Basic and Advanced SmartConnect features, and configure name resolution for the cluster.

Lesson 1: Connecting to the Network

Upon completion of this lesson, you will be able to:

- Identify properties of front-end NICs
- Examine NIC aggregation
- Establish parameters for configuration choices
- Differentiate SBR and default routing in OneFS

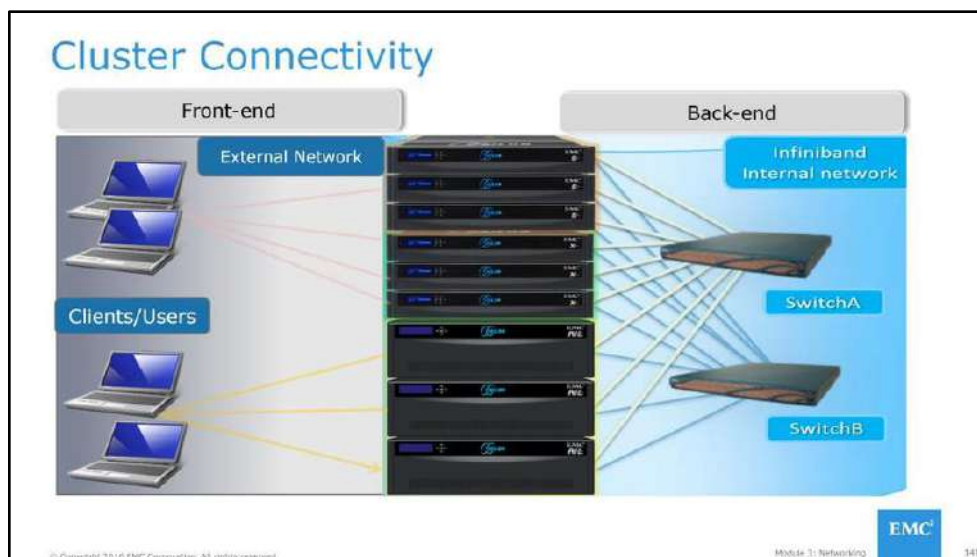
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Module 3: Networking



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Upon completion of this lesson, you will be able to identify properties of front-end NICs, examine NIC aggregation, establish parameters for configuration choices, and differentiate SBR and default routing in OneFS.



Ask the 'Big Picture' questions and do the research to determine the types of workflow in the environment, what your SLAs are, VLAN support, and determine your available IP ranges.

What does our application workflow look like?

- Do we need direct client connections to performance tier?
- What protocols will I need to support?
- What are service level agreements with client departments?
- Do we need VLAN support?
- Will we need NIC aggregation?
- What IP Ranges are available for use?
- Do we have multiple ranges?
- Will we have limited IP addresses per range?

Networking Refresher

- Node may service requests from any front-end port
- No dedicated 'controllers'
- File data accessible from all nodes via all protocols
- Nodes communicate internally
- Clients connect to different nodes based on performance needs



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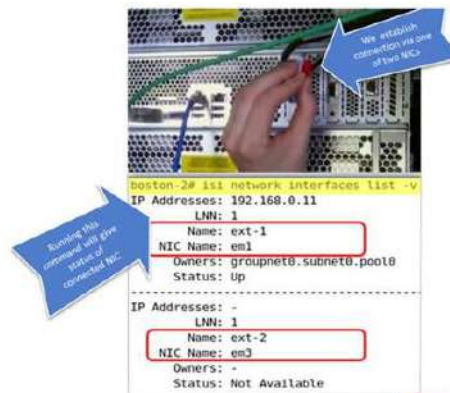
Module 3: Networking

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Using what we have learned so far in the course, keep in mind the following when considering our questions and introducing the front-end hardware: Clients can access their files via a node in the cluster because the nodes communicate with each other via the InfiniBand back-end to locate and move data. Any node may service requests from any front-end port. There are no dedicated 'controllers'. File data is accessible from all nodes via all protocols. Nodes communicate internally. Clients can connect to different nodes based on performance needs.

Network Interface Options

- Front-end or External network faces clients
 - Up to 4 front-end ports
 - Ports 1 GigE or 10 GigE
- Front-end ports have Interface name and NIC name
 - Interfaces labelled:
 - ext-1, ext-2
 - 10gige-1, 10gige-2
 - NIC Names:
 - em0, em1, em2, etc.



Isilon nodes can have up to four front-end or external networking adapters depending on how the customer configured the nodes. The external adapters are labelled ext-1, ext-2, ext-3, ext-4, 10gige-1, 10 gige-2 and can consist of 1 GigE or 10 GigE ports depending on the configuration of the node. A client, can connect to the cluster on any of the four interface depending on how the administrator has configured the cluster. There are no dedicated controllers or filers through which all clients connect to the cluster. Each front-end adapter on any node can answer client-based requests or administrator function calls. It is good practice to verify each external adapter can be reached by: **ping**, by the web administrator interface, and by connecting to a share, for example: \\192.168.0.27\\sales or \\10.10.10.17\\finance from clients on the network.

Using the **isi network ifaces list -v** command, you can see both the interface name and its associated NIC name. For example, **ext-1** would be an interface name and **em1** would be a NIC name. NIC names are required if you want to do a **tcpdump** and may be required for additional command syntax. It is important to understand that the Ethernet ports can be identified by more than one name.

Link Aggregation

- Aggregated on same NIC, across ports, into single logical connection
- Provides improved throughput and redundancy
 - Both NICs used for client I/O but two channels not “bonded” to single 2/20 Gigabit link

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Link aggregation, also known as NIC aggregation, is an optional IP address pool feature that allows you to combine the bandwidth of a single node's physical network interface cards into a single logical connection for improved network throughput and redundancy. For example, if a node has two physical Gigabit Ethernet (GigE) interfaces on the external network, both are logically combined to act as one interface. You cannot NIC aggregate mixed interface types, meaning that a 10 GigE must be combined with another 10 GigE, and not with a 1 GigE.

The link aggregation mode determines how traffic is balanced and routed among aggregated network interfaces. The aggregation mode is selected on a per-pool basis and applies to all aggregated network interfaces in the IP address pool. OneFS supports dynamic and static aggregation modes. A dynamic aggregation mode enables nodes with aggregated interfaces to communicate with the switch so that the switch can use an analogous aggregation mode. Static modes do not facilitate communication between nodes and the switch.

Link Aggregation Types

Aggregation Type	Description
Round-robin	Distributes IP addresses sequentially across all nodes in cluster.
Failover	Examines number of existing connections on each node and distributes to nodes with fewer connections.
LACP	Evaluates file system per node and distributes to balance throughput consumption.
FEC	Examines CPU load on each node and distributes IP addresses to balance CPU workload.

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OneFS provides support for the following link aggregation modes:

Round-robin: Static aggregation mode that rotates connections through the nodes in a first-in, first-out sequence, handling all processes without priority. Balances outbound traffic across all active ports in the aggregated link and accepts inbound traffic on any port. **Note:** This method is not recommended if your EMC Isilon cluster is using TCP/IP workloads.

Active/Passive Failover: Static aggregation mode that switches to the next active interface when the primary interface becomes unavailable. The primary interface handles traffic until there is an interruption in communication. At that point, one of the secondary interfaces will take over the work of the primary.

Link Aggregation Control Protocol (LACP): Dynamic aggregation mode that supports the IEEE 802.3ad Link Aggregation Control Protocol (LACP). You can configure LACP at the switch level, which allows the node to negotiate interface aggregation with the switch. LACP balances outgoing traffic across the interfaces based on hashed protocol header information that includes the source and destination address and the VLAN tag, if available. This option is the default aggregation mode. LACP allows a network device to negotiate and identify any LACP enabled devices and create a link. This is performed by sending packets to the partnered LACP enable device. LACP monitors the link status and will fail traffic over if a link has failed. LACP balances outgoing traffic across the active ports based on hashed protocol header information and accepts incoming traffic from any active port. Isilon is passive in the LACP conversation and listens to the switch to dictate the conversation parameters. Fast EtherChannel balances outgoing traffic across the active ports based on hashed protocol header information and accepts incoming traffic from any active port. The hash includes the Ethernet source and destination address, and, if available, the VLAN tag, and the IPv4/IPv6 source and destination address.

Loadbalance (FEC): Static aggregation method that accepts all incoming traffic and balances outgoing traffic over aggregated interfaces based on hashed protocol header information that includes source and destination addresses.

Link Aggregation Requirements

- One link aggregation type per aggregation pair
 - One port can not be Round-robin and other FEC
 - Both Round-robin or both FEC
- Primarily used for NIC failover purposes
- Enable NIC aggregation on cluster before enabling on switch
 - Per Isilon Administration Guide recommendations

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When planning link aggregation, remember that pools that use the *same* aggregated interface cannot have *different* aggregation modes. For example, if they are using the *same* two external interfaces, you cannot select **LACP** for one pool and **Round-robin** for the other pool. You must select the same aggregation method for all participating devices. A node's external interfaces cannot be used by an IP address pool in both an aggregated configuration and as individual interfaces. You must remove a node's individual interfaces from all pools before configuring an aggregated NIC. You must enable NIC aggregation on cluster before enabling on switch in order to allow communication continuation. Doing it on the switch first may stop communication from the switch to the cluster and result in unexpected downtime.

Best Practices for Link Aggregation

- Each NIC serves separate stream
- Do NOT mix aggregated and single interface in same pool 
 - Results in intermittency on single interfaces
- You cannot aggregate across nodes

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OneFS uses link aggregation primarily for NIC failover purposes. Both NICs are used for client I/O, but the two channels are not bonded into a single 2 Gigabit link. Each NIC is serving a separate stream or conversation between the cluster and a single client. You will need to remove any single interfaces if they are a part of the aggregate interface - they cannot co-exist. In general, it is best practices not to mix agg and non-agg interfaces in the same pool. Such a configuration will result in intermittency on the single connection.

Also, the aggregated NICs must reside on the same node. You cannot aggregate a NIC from node1 and a NIC from node2. Link aggregation provides improved network throughput and physical network redundancy.

NIC and LNI Mapping Differences

- LNI numbers correspond to physical position of NIC on back of node

Logical Network Interface (LNI)	Aggregated LNI
ext-1 ext-2	ext-agg = ext-1 + ext-2
ext-1 ext-2 ext-3 ext-4	ext-agg = ext-1 + ext-2 ext-agg-2 = ext-3 + ext-4 ext-agg-3 = ext-3 + ext-4 + ext-1 + ext-2
ext-1 ext-2 10gige-1 10gige-2	ext-agg = ext-1 + ext-2 10gige-agg-1 = 10gige-1 + 10gige-2

Table Reference: From the OneFS Web Administration Guide

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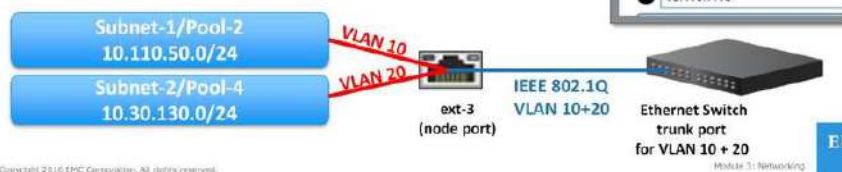
LNI (logical network interface) numbering corresponds to the physical positioning of the NIC ports as found on the back of the node. LNI mappings are numbered from left to right starting in the back of the node.

Remember that aggregated LNIs are listed in the interface in the order in which they are created. NIC names correspond to the network interface name as shown in command-line interface tools, such as **ifconfig** and **netstat**. You can run these commands to verify the output shown in the chart. Up to three VLANs can be configured per network interface. For additional information and to see the chart on the slide, see the OneFS Administration Guide of the appropriate version of your cluster.

If you want to do link aggregation and join together multiple interfaces, then you must use one of the ext-agg interfaces. Link aggregation is configured on a node by node basis and aggregated links cannot span across multiple nodes. If you use the ext-agg interfaces, then you cannot use its associated individual interfaces. For example, if on node 1, you aggregate ext-1 and ext-2, you must use the ext-agg interface and cannot use the individual ext-1 and ext-2 interfaces.

Cluster VLAN Support

- Optional for external subnets
- Physical network port carries network traffic of multiple subnets
- Tags Ethernet frames with VLAN information
- Configuration
 - VLAN Tagging
 - VLAN ID



Virtual LAN (VLAN) tagging is an optional front-end network subnet setting that enables a cluster to participate in multiple virtual networks.

A VLAN is a group of hosts that communicate as though they are connected to the same local area network regardless of their physical location. Enabling the Isilon cluster to participate in a VLAN provides the following advantages:

- Multiple cluster subnets are supported without multiple network switches
- Security and privacy is increased because network traffic across one VLAN is not visible to another VLAN

Ethernet interfaces can be configured as either access ports or trunk ports. An access port can have only one VLAN configured on the interface; it can carry traffic for only one VLAN. A trunk port can have two or more VLANs configured on the interface; it can carry traffic for several VLANs simultaneously.

To correctly deliver the traffic on a trunk port with several VLANs, the device uses the IEEE 802.1Q encapsulation (tagging) method that uses a tag that is inserted into the frame header. This tag carries information about the specific VLAN to which the frame and packet belong. This method enables packets that are encapsulated for several different VLANs to traverse the same port and maintain traffic separation between the VLANs. The encapsulated VLAN tag also enables the trunk to move traffic end-to-end through the network on the same VLAN.

VLAN tags are set on the cluster side as the VLAN ID setting. The switch port needs to be configured for that VLAN ID and configured as a trunk port if multiple VLANs are configured for the external physical port of a cluster node.

Note: An Ethernet interface can function as either an access port or a trunk port; it cannot function as both port types simultaneously. Configuring a VLAN requires advanced knowledge of how to configure network switches to enable this option. Consult your network administrator and switch documentation before configuring a cluster for a VLAN.

Default Routing Complexity

- Asymmetric routing
 - One path from source to target but different path back
- No ability to prefer 10 GigE interface over 1 GigE
- Highest priority gateway has lowest number
 - i.e., number 1 is highest priority, number 20 would be lowest priority
- Lower priority gateways used only if highest priority gateway unreachable



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Routing is the process of determining how to get IP packets from a source to a destination. When responding to client computers, OneFS IP routing attempts to find a matching route, starting with the most specific match. If no specific match is found, IP routing uses the default route (if there is one). There is only one active default outbound route on any particular node at any one time.

Asymmetric Routing means that packets might take one path from source to target, but a completely different path to get back. UDP supports this, but TCP does not; this means that most protocols will not work properly. Asymmetric Routing often causes issues with SyncIQ, when dedicated WAN links for data replication are present. It also has the potential to reduce client I/O for customers that are unaware of how routing works.

In the graphic on the slide we see seven subnets created on the cluster. Only one gateway is created per subnet, however, each of the gateways has a priority. OneFS will always use the highest-priority gateway that is operational, regardless of where the traffic originated. This means that all traffic leaving the cluster leaves through the highest priority gateway (lowest number). In the slide, that would be Network 2's gateway because it has the lowest number/highest priority. If we know all the subnets that are in Network2 or Network3, etc., this approach might work, but we will have to define static routes on the cluster for those subnets.

Another challenge prior to OneFS 7.2 is that there is no way to prefer a 10 GigE interface over a 1 GigE and so, if both a 1 GigE and a 10 GigE were in the same subnet, although traffic might *arrive* on the 10 GigE network, it might *go out* the 1 GigE interfaces. This is called Asymmetric Routing.

OneFS only uses the highest priority gateway configured in *all* of its subnets, falling back to a lower priority one only if the highest priority one is unreachable.

What is Source-based Routing (SBR)?

- Routes data based on packet source IP
 - Global/cluster-wide setting
- Not enabled by default
 - Enabled via CLI only 
- SBR takes priority over static routes
 - SBR only supports IPv4 protocol
- Functionally equivalent to Packet-Reflect on Celerra or VNX

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SBR mitigates how previous versions of OneFS only used the highest priority gateway. Source-based routing ensures that outgoing client traffic (from the cluster) is directed through the gateway of the source subnet.

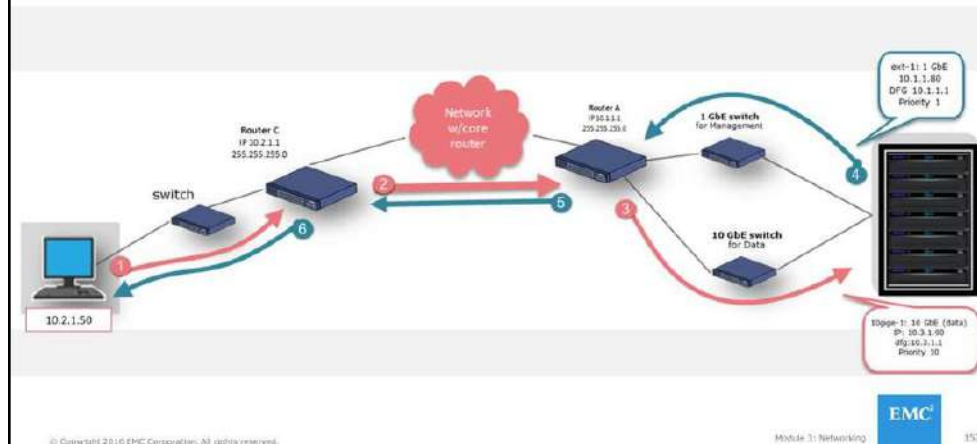
If enabled, source-based routing is applied across the entire cluster. It automatically scans your network configuration and creates rules that enforces client traffic to be sent through the gateway of the source subnet. Outgoing packets are routed via their source IP address. If you make modifications to your network configuration, SBR adjusts its rules. SBR is configured as a cluster wide setting that is enabled via the CLI.

SBR rules take priority over static routes. If static routes are configured in any pools, they may conflict with SBR. SBR only supports the IPv4 protocol.

SBR was developed to be enabled or disabled as seamlessly as possible. SBR configures itself automatically based on the network settings of the cluster. When enabled, whether during the day or at night, packets leaving the cluster will be routed differently. How this affects a customer depends on their network setup but this feature is designed to be as seamlessly as possible when enabled or disabled.

For those who are familiar with the concept of Packet-Reflect on an EMC Celerra or VNX, this is functionally equivalent to that feature that allows traffic that comes in, from an IP on a physical interface with a specific VLAN tag, to go out that same interface to the same IP, with the same VLAN tag.

Default Routing - No SBR



In the above slide, the client must send a packet to the Isilon cluster at IP address 10.3.1.90.

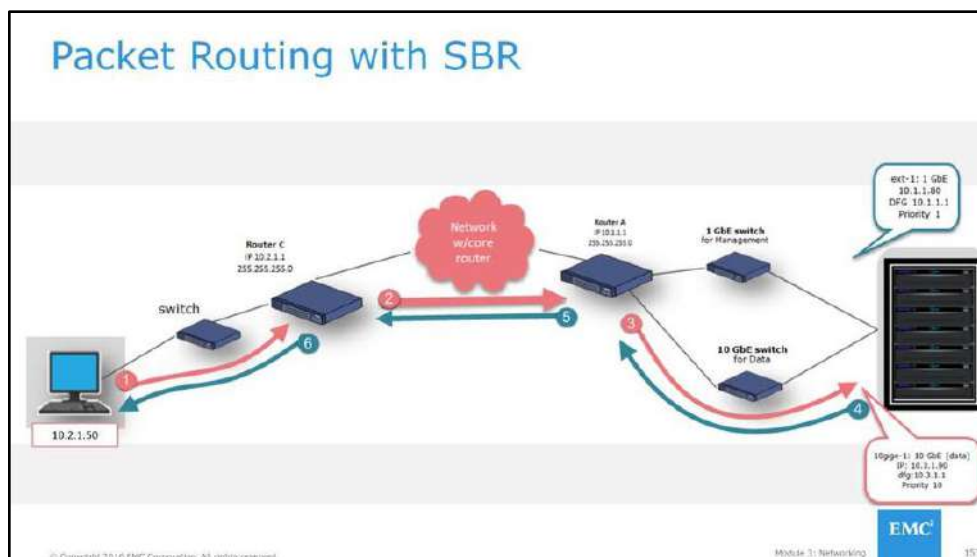
1. The client determines that the destination IP address is not local and it does not have a static route defined for that address. The client sends the packet to its default gateway, Router C, for further processing.
2. Router C receives the packet from the client and examines the packet's destination IP address and determines that it has a route to the destination through the router at 10.1.1.1, Router A.
3. Router A receives the packet on its external interface and determines that it has a direct connection to the destination IP address, 10.3.1.90. Router A sends the packet directly to 10.3.1.90 using its internal interface on the 10GbE switch.
4. Isilon must send a response packet to client and determines that the destination IP address, 10.2.1.50, is not local and that it does not have a static route defined for that address.

OneFS determines which gateway to send the response packet to based on its default gateways' priority numbers. Gateways with lower priority numbers have precedence over gateways with higher priority numbers.

OneFS has two default gateways: 10.1.1.1 with a priority of 1 and 10.3.1.1 with a priority of 10.

OneFS chooses the gateway with priority 1: 10.1.1.1.

OneFS sends the packet to gateway 10.1.1.1 through the 1 GbE interface, not the 10 GbE interface.



Instead of relying on the destination IP to route, the SBR feature on Isilon creates a dynamic forwarding rule. The system makes note of the client's IP and the subnet on the Isilon that the packet arrived. It then creates a reverse rule so packets going to that IP will always be forwarded to the default gateway on for that subnet. As an example, if you have a subnet of 10.3.1.x with a gateway of 10.3.1.1, whenever a packet arrives at the cluster destined for any IP in the 10.3.1.x subnet, a rule will be made to send return packets to the gateway 10.3.1.1 regardless of what is in the routing table or gateway priorities. The way it is currently implemented it also bypasses any static routes that you may have configured.

In the above slide, the client must send a packet to the Isilon cluster at IP address 10.3.1.90.

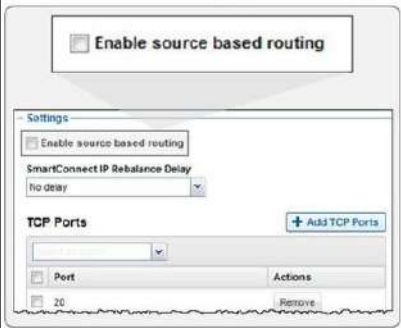
1. The client determines that the destination IP address is not local and it does not have a static route defined for that address. The client sends the packet to its default gateway, Router C, for further processing.
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4. Isilon must send a response packet to the client: OneFS sends the packet to gateway 10.3.1.1 through the 10 GbE interface that received the packet.

For additional information see:

<https://community.emc.com/community/products/isilon/blog/2014/11/28/routing-and-isilon-how-to-get-from-a-to-b-and-back-again>

Enabling SBR

Web Administration Interface



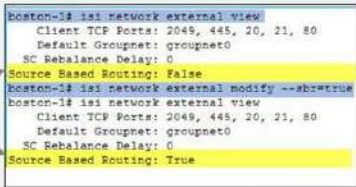
The screenshot shows the 'Web Administration Interface' with a toggle for 'Enable source based routing' set to 'On'. Below it, the 'Settings' section also has 'Enable source based routing' checked. There is a 'SmartConnect IP Rebalance Delay' dropdown set to 'no delay' and a 'TCP Ports' section with a table containing port 20.

CLI

- Enable source-based routing
 - `isi network external modify --sbr=true`
- Disable source-based routing
 - `isi network external modify --sbr=false`
- Determine if cluster is running SBR
 - `isi network external view`

SBR not enabled

SBR enabled



```

Boston-1# isi network external view
Client TCP Ports: 2049, 445, 20, 21, 80
Default Groupnet: groupnet0
SC Rebalance Delay: 0
Source Based Routing: False
Boston-1# isi network external modify --sbr=true
Boston-1# isi network external view
Client TCP Ports: 2049, 445, 20, 21, 80
Default Groupnet: groupnet0
SC Rebalance Delay: 0
Source Based Routing: True
            
```

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You can enable SBR from the CLI or the web administration interface. Shown are the options on the **Cluster Management > Network Configuration** page. In the **Settings** section is the option to enable or disable SBR.

Using the CLI, SBR can be enabled or disabled by running the **isi network external modify** command as shown on the screen. There are no additional options for the command.

To view if SBR is enabled on a cluster, you can run the **isi networks eternal view** command. In the output, if SBR is *not* enabled on the cluster, **Source Based Routing** is **False**. If SBR *is* enabled, **Source Based Routing** is **True**.

What is NANON?

- NANON is: **Not All Nodes On Network**
 - Node without cabled front-end NIC
 - Data still accessible anywhere on cluster
 - Sometimes referred to as NENON (E=Every)
- Best practice is temporary solution
 - Not enough ports and waiting for new switch
 - Proof of concept testing
 - Can run in production but not common

Front-End
Switch



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It is critical that this slide be presented in this manner:

Isilon clusters can get big; very big. Up to 50PB as of the publication of this course. At a certain point most customers are expanding their clusters, not because they need more front-end IO, but because they need more capacity. Imagine a 15-node X400 cluster, with 2x10Gbe links per node. The total potential bandwidth at that point is $2 \times 10 \times 15 = 300\text{Gbps}$, or 37.5GBps. In most cases adding more nodes at this point is going to be done for capacity and aggregated cache/CPU/disk spindle count reasons, rather than front-end IO. As a result, some customers choose to stop connecting additional nodes to the front-end network, because the cost of Network switches and optics cannot be justified.

This decision has pros:

- Lower network cost
- Non-network connected nodes can have maintenance performed at any time, as long as enough nodes are online to meet protection criteria, so patches, firmware updates, etc., are never disruptive to clients on these nodes.

This decision has cons:

- Cons will be discussed on the next slide, to explain why generally this is not an advisable configuration.

NANON Considerations and Best Practices

- Certain OneFS features require all nodes connected
 - Quota notifications
 - NTP/SMB Time
 - Anti-virus – ICAP
 - SNMP services
 - ESRS*
 - Lowest Node ID should always be connected
- Best Practice
 - Connect all nodes to front-end switches
 - At least one interface in static SmartConnect zone



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There are, however, certain features, like anti-virus, that require all the nodes that access files to have IP addresses that can reach the ICAP (Internet control adaptation protocol) server. Additionally, the lowest LNN (logical node number) should always be connected as there are cluster wide notifications that go out via the LNN. If using SMB, it is recommended to have ALL NODES connected to the network as the LNN needs to communicate notifications, SupportIQ information, ESRS, and log files out of the cluster, as well as ensure there are no clock skew or time issues. The recommended best practices would be to ensure that all nodes are wired to the network and possess an IP address. Quota notifications won't work with a NANON cluster. If this is required, please contact technical support for assistance.

*The most recent guidance is that ESRS will work without all nodes able to directly communicate with the ESRS Gateway, however, requests must be proxied through non-connected nodes, and as such this approach is not recommended.

The logic behind the Best Practice stipulating a static SmartConnect zone is that when registering nodes with the ESRS gateway, a static IP must be associated with each node. A Dynamic SmartConnect zone is not an appropriate fit for this, because the IP addresses could easily move to other nodes.

Lesson 1: Summary

Having completed this lesson, you are now able to:

- Identify properties of front-end NICs
- Examine NIC aggregation
- Establish parameters for configuration choices
- Differentiate SBR and default routing in OneFS

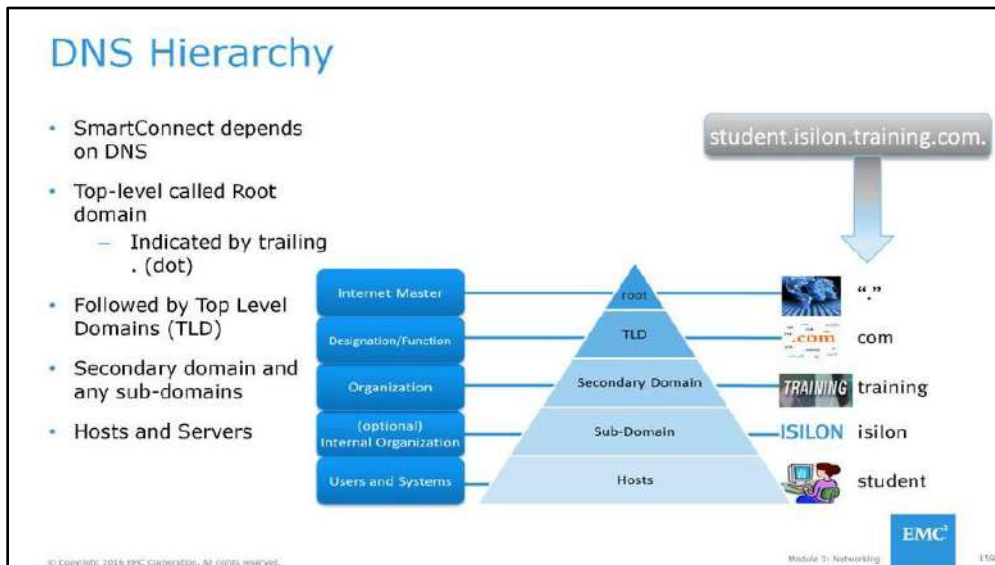
Having completed this lesson, you are now able to identify properties of front-end NICs, examine NIC aggregation, establish parameters for configuration choices, and differentiate SBR and default routing in OneFS.

Lesson 2: Name Resolution

Upon completion of this lesson, you will be able to:

- Understand name resolution process
- Identify host and name server records
- Explain use of FQDN

Upon completion of this lesson, you will be able to understand name resolution process, identify host and name server records, and explain use of FQDN.



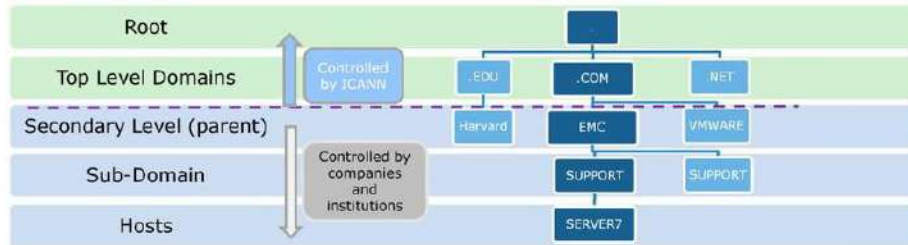
The Domain Name System, or DNS, is a hierarchical distributed database. The names in a DNS hierarchy form a tree, which is called the DNS namespace.

There are a set of protocols specific to DNS to allow for name resolution, more specifically, a Fully Qualified Domain Name, or FQDN, to IP Address resolution.

- The top-level of the DNS architecture is called the **root** domain and it represented by a single "." dot.
- Below the **root** domain are the **Top Level Domains**, or TLDs. These domains are used to represent companies, educational facilities, non-profits, and country codes: *.com, *.edu, *.org, *.us, *.uk, *.ca, etc., and are managed by a Name Registration Authority.
- The **Secondary Domain** represents the unique name of the company or entity, such as EMC, Isilon, Harvard, MIT, etc.
- The last record in the tree is the **Hosts** record, which indicates an individual computer or server. Domain names are managed under a hierarchy headed by the Internet Assigned Numbers Authority (IANA), which manages the top of the DNS tree by administrating the data in the root nameservers.

Fully Qualified Domain Name (FQDN)

- FQDN tells us location of object in DNS tree
- Provides name to IP Address resolution
 - Easy way to remember internet locations
 - Server7.support.emc.com. A/AAAA 192.168.2.31



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A Fully Qualified Domain Name, or FQDN, is the DNS name of an object in the DNS hierarchy. A DNS resolver query must resolve a FQDN to its IP address so that a connection can be made across the network or the internet. If a computer cannot resolve a name or FQDN to an IP address, the computer cannot make a connection, establish a session or exchange information.

An example of a FQDN looks like this:

Server7.support.emc.com.

Reading from left to right, a FQDN starts with the most specific information, in this case, the local computer/server named *server7*, then the delegated domain or sub-domain *support*, followed by the secondary or parent domain *EMC*, and lastly, the Top Level Domain, which is *.com*.

In DNS, a FQDN will have an associated HOST or A record (AAAA if using IPv6) mapped to it so that the server can return the corresponding IP address.

Student-04.isilon.training.com A 192.168.0.31

Secondary domains are controlled by companies, educational institutions, etc., where as the responsibility of management of most top-level domains is delegated to specific organizations by the Internet Corporation for Assigned Names and Numbers or ICANN, which contains a department called the Internet Assigned Numbers Authority (IANA). For more details, see the IANA website at <http://www.iana.org>.

DNS Host Record: A or AAAA Record

- Hostname is unique identifier: computer name
 - For example, host name may be:
 - www, mail, ftp, webmail, laptop-c78, Thor, student-04
- **A record:** Maps hostname to 32-bit IPv4 address
 - Server7.support.emc.com A 192.168.15.12
 - Host name: Server7
 - Domain name: Support.emc.com
 - FQDN: Server7.support.emc.com
- **AAAA record:** Maps hostname to 128-bit IPv6 address
 - ipv6_host1.support.emc.com AAAA 4321:0:1:2:3:4:567:89ab
 - Host name: ipv6_host1
 - Domain name: support.emc.com

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An A-record maps the hostname to a specific IP address to which the user would be sent for each domain or subdomain. It is simple name-to-IP resolution.

For example, a server by the name of server7 would have an A record that mapped the hostname *server7* to the IP address assigned to it:

Server7.support.emc.com A 192.168.15.12

- *Server7* is the hostname
- *Support.emc.com* is the domain name
- *Server7.support.emc.com* is the FQDN

Provides an easy way to remember internet locations. May not remember IP *192.168.251.189* but it's easier to remember *www.isilon.training.com*, with *www* being the hostname.

In IPv6, the difference is the IP address, not the FQDN. Where IPv4 contains four octets of bits (4 * 8 bits= 32 bits) IPv6 has hexadecimal digits (0 1 2 3 4 5 6 7 8 9 a b c d e f) separated by colons. The allocation of IPv6 addresses and their format is more complex than that of IPv4, so in an IPv6 environment you should remember to use the AAAA record in DNS, and consult with the network administrator to ensure that you are representing the IPv6 addresses correctly.

Name Server Record (NS)

- Identifies authoritative name servers for zone or domain
 - Used to delegate name resolution for subdomains
- Examples of subdomains for Mycompany.com:
 - Seattle.Mycompany.com
 - Boston.Mycompany.com
- Example:
 - Delegate control of subdomain Seattle.Mycompany.com to name server called SrvNS that sits in Seattle

Seattle.Mycompany.com	NS	SrvNS.Mycompany.com
SrvNS.Mycompany.com A	192.168.0.100	

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The Name Server Record, or NS Records, indicate which name servers are authoritative for the zone or domain. NS Records are used by companies that want to divide their domain into subdomains. Subdomains indicate that you are delegating a portion of your domain name to a different group of name servers. You create NS records to point the name of this delegated subdomain to different name servers.

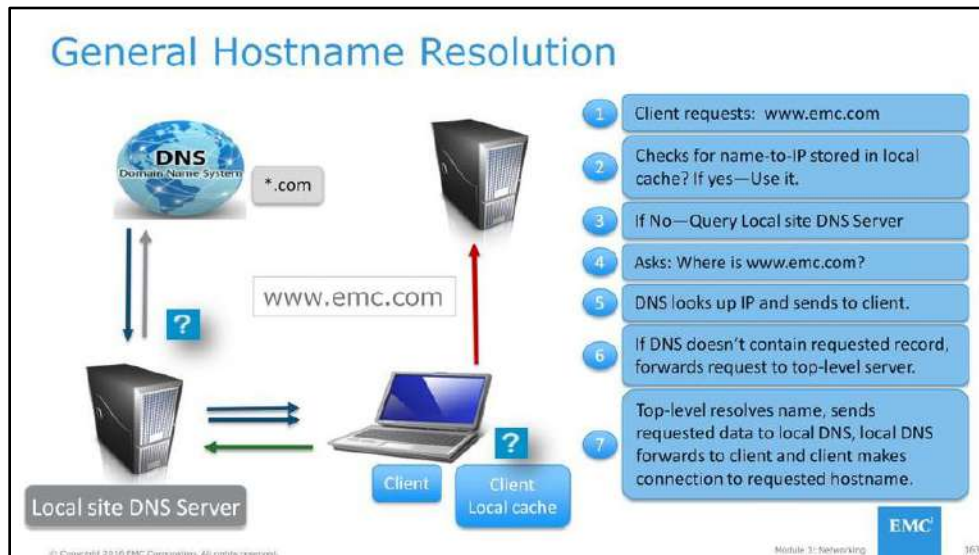
For example, say you have a domain called **Mycompany.com** and you want all DNS Lookups for **Seattle.Mycompany.com** to go to a server located in Seattle. You would create an NS record that maps **Seattle.Mycompany.com** to the Name Server in Seattle with a hostname of SIP thus the mapping looks like:

Seattle.Mycompany.com	NS	SrvNS.Mycompany.com
------------------------------	-----------	----------------------------

This states that anyone looking to resolve cluster.isilon.training.com should go and query the NS called sip.isilon.training.com. You would then have an A record that maps the hostname of sip.isilon.training.com to the IP address, as follows:

SrvNS.Mycompany.com	A	192.168.0.100
----------------------------	----------	----------------------

Now anyone looking for **Seattle.Mycompany.com** will be redirected to **SrvNS.Mycompany.com** and **SrvNS.Mycompany.com** can be found at 192.168.0.100.



DNS Name Resolution and Resolvers

When a client needs to resolve a Fully Qualified Domain Name (FQDN) it follows the following steps:

1. The client will look in its local cache to see if it has already done a lookup for that host or FQDN. If it has, it will cache the hosts *resource record* also known as an A or AAAA (quad A) record, and will use the name-to-IP mapping that sits in its local cache.
- 2-3-4. If there is not an entry in the local cache, the computer will make a call to the DNS server configured within the operating system. This request is called a resolver or resolver query. The request asks the DNS, "Do you know the IP address of `www.emc.com`?"
- 5-6. The DNS server that receives the request will check its local zones to see if they contain a zone for `isilon.training.com`. If it has a copy of the zone (all of the DNS entries for a particular secondary domain), it will query the zone for the hosts A or AAAA record and return the host-to-IP mapping to the client. An A record:
emc.com A 192.168.0.31
7. The DNS server returns the IP to the client, who caches the information, and then attempts to make a connection directly to the IP address provided by the DNS server.

Lesson 2: Summary

Having completed this lesson, you are now able to :

- Understand name resolution process
- Identify host and name server records
- Explain use of FQDN

Having completed this lesson, you are now able to understand name resolution process, identify host and name server records, and explain use of FQDN.

Lesson 3: DNS and Multi-tenancy

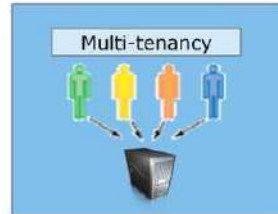
Upon completion of this lesson, you will be able to:

- Define multi-tenancy
- Establish network hierarchy
- Identify groupnet function
- Review networking Best Practices

Upon completion of this lesson, you will be able to define multi-tenancy, establish network hierarchy, identify groupnet function, and review networking best practices.

Single vs. Multi-tenancy

- Single tenancy
 - Single instance of software and infrastructure serves single customer
- Multi-tenancy
 - Single instance of software and infrastructure serves multiple customers
 - Each customer called *tenant*



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In the computer realm, multi-tenancy is defined as the ability to host multiple customers in a single cloud, application or storage device. Each customer in that environment is called a tenant.

With OneFS, multi-tenancy refers to the ability of an Isilon cluster to simultaneously handle more than one set of networking configurations.

Domain name **resolvers** are the names given to computers, commonly located with Internet Service Providers (ISPs) or institutional networks that are used to respond to a user request to resolve a domain name.

Network Hierarchy

- Groupnet = top tier of network hierarchy
- Enables different networking properties for name resolution
 - For multi-tenancy
- Contains:
 - Subnets
 - DNS client settings

The screenshot shows the EMC Isilon Network Hierarchy interface. On the left, a table lists network objects. A red arrow labeled "DNS Settings" points to a settings panel on the right. Another red arrow labeled "Top level Network object" points to the "groupnet0" entry in the "Networks" table.

Networks	Detail	Type	Description
groupnet0	DNS Servers: 192.168.0.201	Default	Initial groupnet
subnet0	NET 192.168.0.0/24	IPv4	Initial ext-1 subnet

Settings

- Name: groupnet0
- Description: Initial groupnet

DNS Settings

- DNS Servers: 192.168.0.201
- DNS Search Suffixes: barrlocal
- DNS Resolver Rotate: Disabled
- Server-side DNS Search: Enabled
- DNS Cache: Enabled

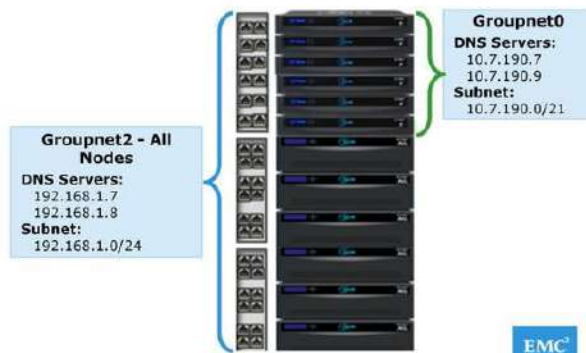
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Groupnets reside at the top tier of the networking hierarchy and are the configuration level for managing multiple tenants on your external network. DNS client settings, such as nameservers and a DNS search list, are properties of the groupnet. You can create a separate groupnet for each DNS namespace that you want to use to enable portions of the Isilon cluster to have different networking properties for name resolution. Each groupnet maintains its own DNS cache, which is enabled by default. A groupnet is a container that includes subnets, IP address pools, and provisioning rules. Groupnets can contain one or more subnets, and every subnet is assigned to a single groupnet. Each EMC Isilon cluster contains a default groupnet named groupnet0 that contains an initial subnet named subnet0, an initial IP address pool named pool0, and an initial provisioning rule named rule0.

Multi-tenancy for OneFS Networking

- Simultaneously handles more than one set of networking configurations
- Example:
 - Groupnets can include all cluster nodes (**Groupnet2**)
 - Or just specific nodes (**Groupnet0**)

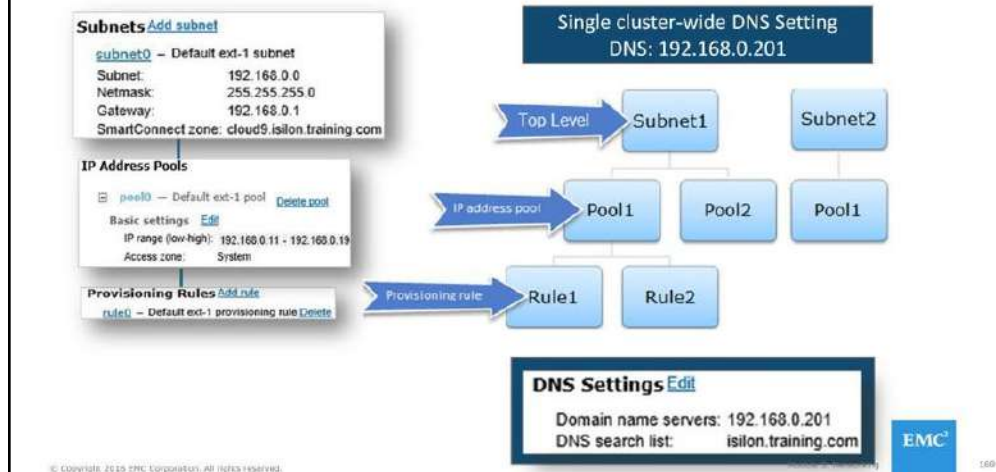


In OneFS 8.0, *Multi-tenancy* refers to the ability of a OneFS cluster to simultaneously handle more than one set of networking configurations. *Multi-Tenant Resolver*, or *MTDNS*, refers to the subset of that feature pertaining specifically to hostname resolution against DNS name servers. These features have now been made available to customers in OneFS 8.0. Each tenant on the cluster can have its own network settings.

On the slide, we see that this cluster has the ability to connect to two separate external network configurations: the 10.7.190.x network and the 192.168.1.x network. Both of these networks are separate from each other and have their own DNS servers, which Isilon can now identify and resolve.

Prior to OneFS 8.0, only one set of DNS servers could be defined on the cluster: This was a global cluster setting. Now in OneFS 8.0, Isilon is able to host multiple networks with multiple DNS servers using a new object called a *groupnet*. Groupnets will be discussed over the course of the next few slides.

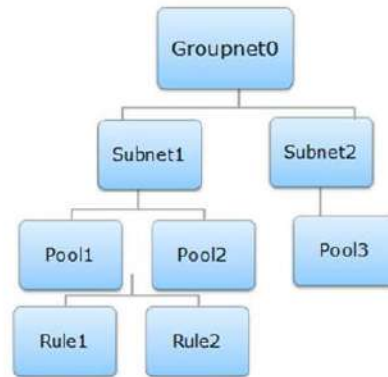
OneFS 7.2.x and Prior: Network Hierarchy



In OneFS 7.2.x and prior versions, a subnet was the highest level of the network configuration. All networking settings were configured below the subnet level where an administrator would configure the SmartConnect Zone name, the IP address pools, the access zones associated with those pools, and any provisioning rules that might need to be created. There is a single cluster-wide DNS setting associated with the cluster and it was not able to accommodate any DNS servers that existed on separate networks. This is an issue with multi-tenancy support when you may be hosting companies or departments that sit on completely different networks in disparate locations.

OneFS 8.0: Network Hierarchy

- Groupnet object is top-level network configuration object
 - All network settings exist below this object
- *Groupnet0* is default groupnet name
 - Groupnet0 identical to OneFS 7.2 configuration
 - Access zone and authentication providers must exist within *one and only one groupnet*
 - Must live in same groupnet to associate with one another
- DNS settings are per Groupnet
 - Create another Groupnet *only* if separate DNS settings required 



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A new networking object is introduced in OneFS 8.0 as part of the multi-tenant feature. Groupnets are how the cluster communicates with the world. If the cluster needs to talk to another customer's authentication domain, your cluster needs to know how to find that domain and requires a DNS setting to know how to route out to that domain.

Groupnets store all subnet settings, they are the top-level object and all objects live underneath Groupnet0. OneFS 8.0 groupnets can now contain individual DNS settings that were one single global entry in previous versions.

After upgrade, administrators will see a Groupnet0 object; this is no different from what a customer had prior to the upgrade, with the whole cluster pointing at the same DNS settings. Groupnet0 is the default groupnet.

Conceptually it would be appropriate to think of groupnets as a *networking tenant*. Different groupnets allow portions of the cluster to have different networking properties for name resolution.

Additional groupnets should be created only in the event that a customer requires a unique set of DNS settings.

Groupnet Interaction with Access Zones

- Access Zones and Authentication providers:
 - May exist within only one groupnet
 - Must live in same groupnet to associate with one another
 - Providers (AD, Krb5, LDAP, NIS) use groupnet DNS for hostname resolution
- Example:
 - Active Directory provider CLOUD9 must exist within same groupnet as Zone1 in order to be added to Zone1 auth provider list

```
isi zone zones modify -zone=zone1 -add-auth-provider=ads:CLOUD9
```

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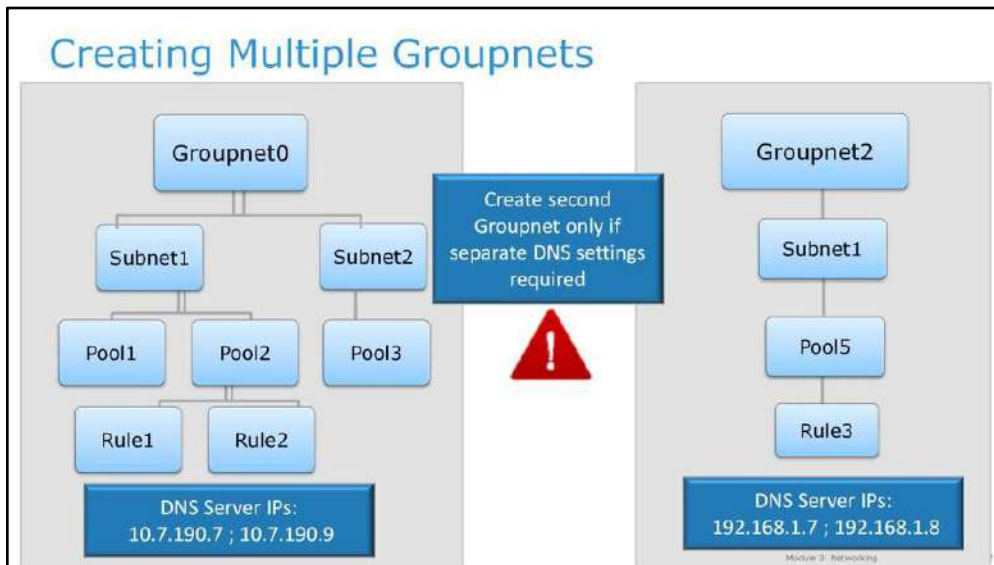
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Because groupnets are the top networking configuration object, they have a close relationship with access zones and the authentication providers. The groupnet defines the external DNS settings for remote domains and authentication providers so the external authentication providers will have an extra parameter that defines the groupnet in which they exist. Access Zones and authentication providers must exist within one and only one groupnet. When the cluster joins an Active Directory server, the cluster must know which network to use for external communication with this external AD domain. Because of this, if you have a groupnet, both the access zone and authentication provider must exist within same groupnet or you will see an error indicating that this is not the case. Access Zones and authentication providers must exist within one and only one groupnet.

Authentication providers and access zones must exist in the same groupnet to be associated with one another. Active Directory provider CLOUD9 must exist in within the same groupnet as Zone1 in order to be added to Zone1's auth provider list. The **isi zone zones modify -zone=zone1 -add-auth-provider=ads:CLOUD9** command would associate Zone1 with the AD provider called CLOUD9.



Having multiple groupnets on the cluster means that you are configuring access to completely separate and different networks. You *only need to configure another groupnet* if separate DNS settings are required, otherwise the cluster will run perfectly well under the default **Groupnet0** groupnet.

If necessary, you can have a different groupnet for every access zone, although you do not need one. Because you can have up to fifty access zones, that allows for up to fifty groupnets.

Configuring Groupnets

Create Networking and Access Zone Environment	<code>isi network groupnets create <id> --dns-servers=<ip></code>	Create groupnet
	<code>isi zone zones create <name> <path> --groupnet=<groupnet name></code>	Create zone
	<code>isi network subnets create <id> <addr-family> {ipv4 ipv6} <prefix-len></code>	Create subnet
	<code>isi network pools create <id> --access-zone=<zone name></code>	Create pool
Create Authentication Providers	<code>isi auth ads create <name> <user> --groupnet=<groupnet name></code>	
Associate Providers with Zone	<code>isi zone zones modify <name> --auth-providers=<list of auth providers></code>	

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When creating a groupnet with access zones and providers in the same zone, you have to create them in the proper order.

First you create the groupnet.

`isi network groupnets create <id> --dns-servers=<ip>`

Then you create the access zone and tell it which groupnet you want to associate it with.

`isi zone zones create <name> <path> --groupnet=<groupnet name>`

Once that is done, you then create the networking information; subnets and pools.

`isi network subnets create <id> <addr-family> {ipv4 | ipv6} <prefix-len>`

`isi network pools create <id> --access-zone=<zone name>`

You must create the access zone *after* the groupnet because when you create the networking/pool you must point the point at the access zone.

Then you add your provider(s) and point it/them to the groupnet.

`isi auth ads create <name> <user> --groupnet=<groupnet name>`

Finally you associate your authentication providers with your zone.

`isi zone zones modify <name> --auth-providers=<list of auth providers>`

Best Practices for DNS and Multi-tenancy

- DNS settings are per Groupnet 
 - Create another Groupnet *only* if separate DNS settings required
- Follow proper build order:
 - Create in this order:
 1. Groupnet
 2. Access zone
 3. Subnet
 4. Pool
 - Then create Authentication provider
 - Finally associate Provider to access zone

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There is no need to create multiple groupnets unless there is a need for two separate set of DNS settings. Groupnets are an option for those clusters that will be hosting multiple companies, departments, or clients that require their own DNS settings. Follow the proper creation order to eliminate frustration. You cannot create these out of order because they are dependencies.

Lesson 3: Summary

Having completed this lesson, you are now able to:

- Define multi-tenancy
- Establish network hierarchy
- Identify groupnet function
- Review networking Best Practices

Having completed this lesson, you are now able to define multi-tenancy, establish network hierarchy, identify groupnet function, and review networking best practices.

Lesson 4: Client Connectivity

Upon completion of this lesson, you will be able to:

- Describe SmartConnect benefits
- Identify required DNS settings
- Understand client connectivity using SmartConnect
- Evaluate SmartConnect Best Practices

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Module 3: Networking

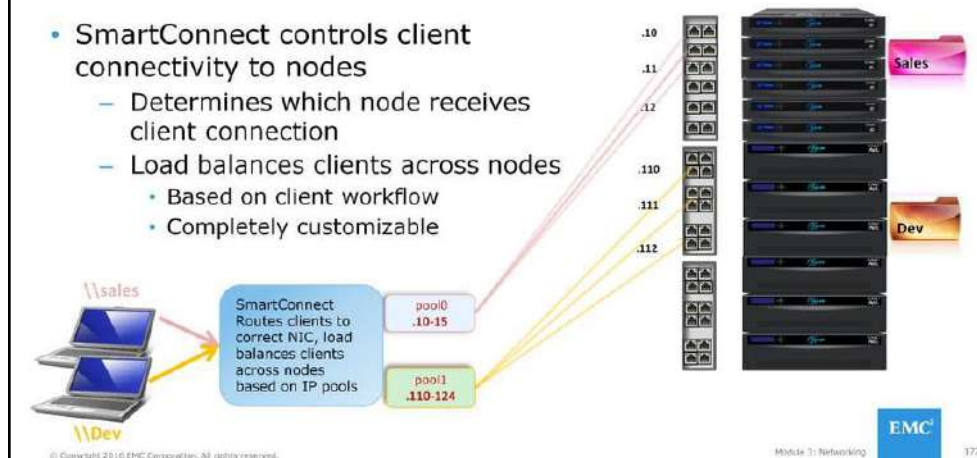


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Upon completion of this lesson, you will be able to describe SmartConnect benefits, identify required DNS settings, understand client connectivity using SmartConnect, and evaluate SmartConnect Best Practices.

Client Access: The Big Picture

- SmartConnect controls client connectivity to nodes
 - Determines which node receives client connection
 - Load balances clients across nodes
 - Based on client workflow
 - Completely customizable



Isilon has many different components and an Isilon cluster can be as simple or as complex as an individual's environment. Knowing how all of the internal features interact is integral to troubleshooting and explaining how the cluster works. Often times access zones and SmartConnect are misunderstood or believed to be the same type of client routing feature but in fact they are distinctly different and dependent on one another. SmartConnect is a client load balancing feature that allows segmenting of the nodes by performance, department or subnet. SmartConnect deals with getting the clients from their devices to the correct front-end interface on the cluster. That is the key, the CORRECT front-end interface for their job function/segment/department. Once the client is at the front-end interface, the associated access zone then authenticates the client against the proper directory service; whether that is external like LDAP and AD or internal to the cluster like the local or file providers. Access zones do not dictate which front-end interface the client connects to, it only determines what directory will be queried to verify authentication and what shares that the client will be able to view. Once authenticated to the cluster, mode bits and ACLs (access control lists) dictate the files, folders and directories that can be accessed by this client. Remember, when the client is authenticated Isilon generates an access token for that user. The access token contains all the permissions and rights that the user has. When a user attempts to access a directory the access token will be checked to verify if they have the necessary rights.

As a best practice, the number of access zones should not exceed 50. The maximum number of access zones has yet to be established.

What Does SmartConnect Do For Us?

- Provides name resolution for cluster
 - Clients connect using friendly name
 - Utilizes existing DNS infrastructure
 - Cluster resolves user-friendly name to specific node IP addresses
- Balances clients across nodes
 - Provides load balancing
 - Keeps all clients from connecting to single node
 - Configurable to best suit need

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SmartConnect zones allow a granular control of where a connection is directed. An administrator can segment the cluster by workflow allowing specific interfaces within a node to support different groups of users. SmartConnect is a client connection balancing management feature (module) that enables client connections to be balanced across all or selected nodes in an Isilon cluster. It does this by providing a single virtual host name for clients to connect to, which simplifies connection mapping.

SmartConnect enables client connections to the storage cluster using a single host name or however many host names a company needs. It provides load balancing and dynamic NFS failover and failback of client connections across storage nodes to provide optimal utilization of the cluster resources. SmartConnect eliminates the need to install client side drivers, enabling administrators to manage large numbers of clients in the event of a system failure.

SmartConnect provides name resolution for the cluster. The cluster appears as a single network element to a client system. Both cluster and client performance can be enhanced when connections are more evenly distributed.

SmartConnect simplifies client connection management. Based on user configurable policies, SmartConnect Advanced applies intelligent algorithms (e.g., CPU utilization, aggregate throughput, connection count or Round-robin) and distributes clients across the cluster to optimize client performance. SmartConnect can be configured into multiple zones that can be used to ensure different levels of service for different groups of clients. All of this is transparent to the end-user.

SmartConnect can remove nodes that have gone offline from the request queue, and prevent new clients from attempting to connect to a node that is not available. In addition, SmartConnect can be configured so new nodes are automatically added to the connection balancing pool.

SmartConnect Licensing

SmartConnect Basic (unlicensed)	SmartConnect Advanced (licensed)
• Static IP allocation • Multiple subnets – Single pool per subnet – Single SC DNS zone per subnet • One balancing option: – Round-robin	• Dynamic allocation – NFSv3 failover – Plus static IP allocation • Multiple subnets – Multiple pools per subnet – Multiple SC zone names per subnet • Four balancing options: – Round-robin, Connection count, Throughput, and CPU usage – Can set Rebalance policy

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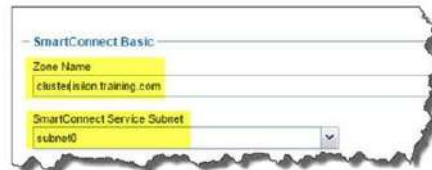
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In traditional NAS scale-up solution, the file system, volume manager, and the implementation of RAID are all separate entities. Each entity is abstracted from the other. The functions of each are clearly defined and separate. In a scale-up, solution you have controllers that provide the computational throughput, connected to trays of disks. The disks are then carved up into RAID GROUPS and into LUNs. If you need additional processing, you can add an additional controller, which can run Active/Active or Active/Passive. If you need additional disk, you can add another disk array. To administer this type of cluster, there is an overarching management console that allows for single seat administration. Each of these components are added individually and may have an upper limit of 16TB although some solutions may be higher. This type of solution is great for specific types of workflows, especially those applications that require block-level access.

In a Scale-out solution, the computational throughput, the disk and disk protection, and the over-arching management are combined and exist within a single node or server. OneFS creates a single file system for the cluster that performs the duties of the volume manager and applies protection to the cluster as a whole. There is no partitioning, and no need for volume creation. Because all information is shared among nodes, the entire file system is accessible by clients connecting to any node in the cluster. Because all nodes in the cluster are peers, the Isilon clustered storage system also does not have any master or slave nodes. All data is striped across all nodes in the cluster. As nodes are added, the file system grows dynamically and content is redistributed. Each Isilon storage node contains globally coherent RAM, meaning that as a cluster becomes larger, it also becomes faster. Each time a node is added, the cluster's concurrent performance scales linearly.

SmartConnect Components

- SmartConnect service IP (SIP)
 - IP address pulled out of subnet
 - Never used in pool
 - Interfaces with DNS server
- SmartConnect Zone name
 - One name per pool
 - Friendly name for users
 - **sales**.isilon.training.com
 - `\\sales`
 - **cluster**.isilon.training.com
 - `\\cluster`
- DNS:
 - Add NS delegation record for SmartConnect Zone
 - Add A or AAAA record for SmartConnect Service IP



SmartConnect Basic

Zone Name
cluster.isilon.training.com

SmartConnect Service Subnet
subnet10

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The SmartConnect Service IP (SSIP or SIP) is one IP address that is pulled out of the subnet. This IP address will never be put into one of the pools, the same way you would not put a static server IP address into a DHCP scope. The SIP is a virtual IP within the Isilon configuration, it is not bound to any of the external interfaces. It resides on the node with the lowest logical number. If that node goes down, the SIP would seamlessly move to the next lowest logical node number. For example, if you had a 5 node cluster and the SIP was answering DNS queries from node 1, if node 1 went down, the SIP would move to node 2 and node 2 would start answering the DNS queries. The SmartConnect zone name is a *friendly* fully-qualified domain name (FQDN) that users can type to access the cluster.

SmartConnect Service IP

- SIP answers DNS queries sent to cluster
 - Can have one SIP per subnet and multiple SIPs per cluster
- SIP lives on node with lowest array ID

MTU

☒ 1500 (Standard Frames)

☐ 9000 (Jumbo Frames)

☐ Custom

SmartConnect Service IP

192.168.0.100

— Advanced Settings —

☐ Allow VLAN Tagging

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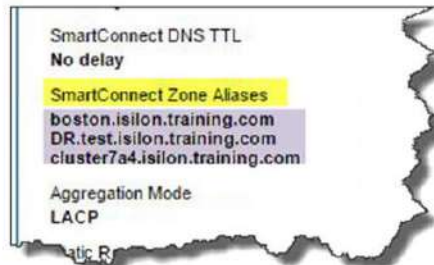


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The SmartConnect service IP answers queries from DNS. There can be multiple SIPs per cluster and they will reside on the node with the lowest array ID for their node pool. If the cluster is very large and contains multiple node pools with multiple subnets, the SIP for each subnet resides on the node with the lowest array ID for that subnet. If you know the IP address of the SIP and wish to know just the zone name, you can use **isi_for_array ifconfig -a | grep <IP of SIP>** and it will show you just the zone that the SIP is residing within.

SmartConnect Zone Aliases

- Consolidate legacy servers when name cannot change
- Allows quick view of all DNS names on cluster
- Requires Service Principal Name (SPN) in Active Directory



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SmartConnect zone alias are a useful tool to use when you are consolidating legacy servers to the Isilon cluster and are required to keep the original server names used by the clients. SmartConnect zone aliases enable you to easily view all the DNS names that a cluster answers for.

This approach requires you to create Service Principal Name (SPN) records in Active Directory or in MIT Kerberos for the SmartConnect zone names, as a component of the cluster's machine account. To create the SPN records, use the CLI **isi auth** command after you add the zone alias, similar to the following:

```
isi auth ads spn check --domain=<domain.com> --repair
```

Cluster Name Resolution Steps

1. Create Host (A) record
2. Create NS record
3. Configure subnet
 1. Specify SSIP
4. Configure pool
 1. Name SmartConnect zone
 2. Select connection policy
 3. Specify SmartConnect service subnet

The screenshot shows the SmartConnect configuration window. Under the 'SmartConnect Basic' tab, the 'Zone Name' is set to 'cloud9.isilon.training.com' and the 'SmartConnect Service Subnet' is 'subnet0'. Under the 'SmartConnect Advanced' tab, the 'Client Connection Balancing Policy' is 'Round-robin', the 'IP Failover Policy' is 'Round-robin', the 'Rebalance Policy' is 'Automatic', and the 'Allocation Method' is 'Static'.

```
cloud9.isilon.training.com NS sip.isilon.training.com
sip.isilon.training.com   A 192.168.0.100
```

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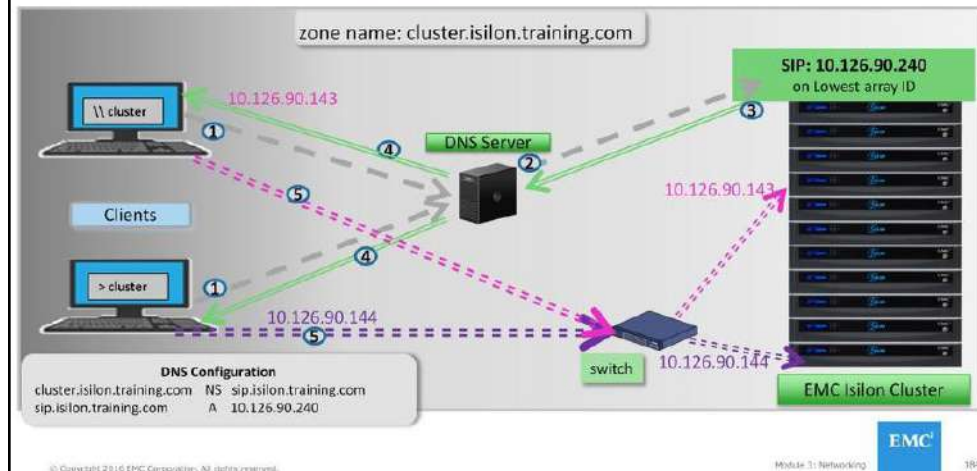
To configure SmartConnect, you must also create records on your DNS server. If the clients on your network use DNS for name resolution, you must configure the network DNS server to forward cluster name resolution requests to the SmartConnect service on the cluster. You can configure SmartConnect name resolution on a BIND server or a Microsoft DNS server. Both types of DNS server require a new name server, or NS, record be added to the existing authoritative DNS zone to which the cluster belongs. In the Microsoft Windows DNS Management Console, an NS record is called a New Delegation. On a BIND server, the NS record must be added to the parent zone (in BIND 9, the "IN" is optional). The NS record must contain the FQDN that you want to create for the cluster and the name you want the client name resolution requests to point to. In addition to an NS record, an A record (for IPv4 subnets) or AAAA record (for IPv6 subnets) that contains the SIP of the cluster must also be created.

In this example, *cluster.isilon.com* is the name you want your clients to use when connecting to the cluster.

- cluster.isilon.com. IN NS ssip.isilon.com.
- ssip.isilon.com. IN A 10.10.10.10

A single SmartConnect zone does not support both IP versions, but you can create a zone for each IP version and give them duplicate names. So, you can have an IPv4 subnet and IP address pool with the zone name *test.mycompany.com* and you can also define IPv6 subnet using the same zone name.

Cluster Name Resolution Process



SmartConnect leverages the customer's existing DNS server by providing a layer of intelligence within the OneFS software application. Specifically, all clients are configured to make requests from the resident DNS server using a single DNS host name (i.e., cluster). (1) Because all clients point to a single host name (`cluster.isilon.training.com`), it makes it easy to manage large numbers of clients. (2) The resident DNS server forwards the lookup request for the delegated zone to the delegated zone's server of authority, in this case the SIP address of the cluster. SmartConnect evaluates the environment and determines which node (single IP address) the client should connect to, based on the configured policies. (3) It then returns this information to the DNS server, (4) which, in turn, returns it to the client. (5) The client then connects to the appropriate cluster node using the desired protocol.

Best Practices for DNS Delegation

- Delegate to (A) record - also called host record
 - Allows failover of entire cluster by changing just one DNS record
- Use one (NS) name server record per SmartConnect zone
 - Permits failover of single SmartConnect zone
 - Failover of one zone does not affect other zones
 - Great for disaster recovery testing
- Do not use single delegation for entire cluster
 - Requires failover of entire cluster and not specific zone

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This section describes best practices for DNS delegation for Isilon clusters.

Delegate to address (A) records, not to IP addresses.

The SmartConnect service IP on an Isilon cluster must be created in DNS as an address (A) record, also called a host entry. An A record maps a URL such as `www.emc.com` to its corresponding IP address. Delegating to an A record means that if you ever need to failover the entire cluster, you can do so by changing just one DNS A record. All other name server delegations can be left alone. In many enterprises, it is easier to have an A record updated than to update a name server record, because of the perceived complexity of the process.

Use one name server record for each SmartConnect zone name or alias.

Isilon recommend creating one delegation for each SmartConnect zone name or for each SmartConnect zone alias on a cluster. This method permits failover of only a portion of the cluster's workflow—one SmartConnect zone—without affecting any other zones. This method is useful for scenarios such as testing disaster recovery failover and moving workflows between data centers.

Isilon does not recommend creating a single delegation for each cluster and then creating the SmartConnect zones as sub records of that delegation. Although using this method would enable Isilon administrators to change, create, or modify their SmartConnect zones and zone names as needed without involving a DNS team, this method causes failover operations to involve the entire cluster and affects the entire workflow, not just the affected SmartConnect zone.

Lesson 4: Summary

Having completed this lesson, you are now able to:

- Describe SmartConnect benefits
- Identify required DNS settings
- Understand client connectivity using SmartConnect
- Evaluate SmartConnect Best Practices

Having completed this lesson, you are now able to describe SmartConnect benefits, identify required DNS settings, understand client connectivity using SmartConnect, and evaluate SmartConnect Best Practices.

Lesson 5: Configuration Options

Upon completion of this lesson, you will be able to:

- Identify load balancing options
- Explain uses of multiple zones
- Differentiate static and dynamic pools

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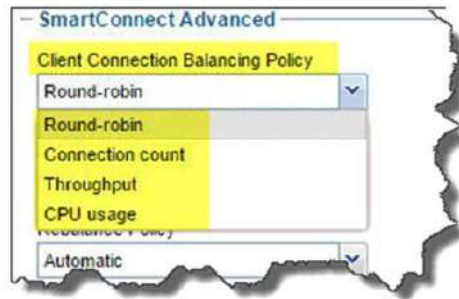


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Upon completion of this lesson, you will be able to identify load balancing options, explain uses of multiple zones, and differentiate static and dynamic pools.

Load Balancing Options

- SmartConnect load balance based on Connection Policy
- If SmartConnect licensed:
 - Round-robin
 - Connection count
 - Throughput
 - CPU usage
- If SmartConnect not licensed:
 - Round-robin only



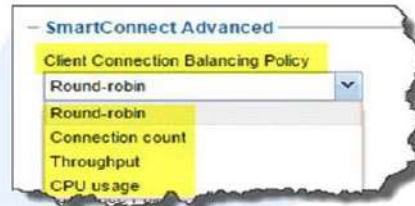
SmartConnect will load balance client connections across the front-end ports based on what the administrator has determined to be the best choice for their cluster. The options are different depending on whether SmartConnect is licensed or not. If a cluster is licensed the administrator has four options to load balance: **Round-robin**, **Connection count**, **Throughput**, and **CPU usage**.

If the cluster does not have SmartConnect licensed, it will load balance by Round-robin only.

The next slide goes into detail about each of the four client load balancing options.

Balancing Policies

- Client Connection Balancing Policies > Administrator defined
- Selection depends on type of workflow



Options	Description
Round Robin	Selects the next available node on a rotating basis, and is the default policy if no other policy is selected.
Connection Count	Determines the number of open TCP connections on each available node to optimize the cluster usage.
Network Throughput	Sets the overall average throughput volume on each available node to optimize the cluster usage.
CPU Usage	Examines average CPU utilization on each available node to optimize the cluster usage.

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Connection Policies are based on what the administrator decides is best for their workflow.

If the setting is **Round-robin**, as a very basic example, the first client that connects will go to node 1, the second to node 2, the third to node 3, etc.

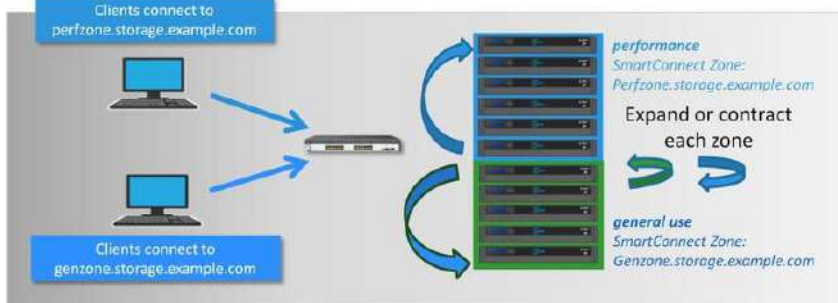
The second option for client load balancing is **Connection count**. Because OneFS is aware of what goes on with all of the nodes, the SIP can load balance by sending clients to the nodes with the least amount of client connections. If one node has seven clients connecting and another has only four, then the SIP will send the next client connection to the node with only four connections.

The **Throughput** policy allows the cluster to load balance based on the current network throughput per node, thus sending the next client connection to the node with the least network throughput.

Lastly, **CPU usage** sends the client connections to the node with the least CPU utilization at the time the client connects. This helps spread the load across the nodes and does not over burden any one node. The **Connection count** policy directs new connections to nodes that have fewer existing connections in an attempt to balance the number of connections to each node. Connection count data is collected every 10 seconds. The **Throughput** policy directs new connections to nodes that have lower external network throughput. Network throughput data is collected every 10 seconds. The **CPU usage** policy looks at the processor load on each node and directs new connections to nodes with lower CPU utilization in an attempt to balance the workload across the cluster nodes. CPU statistics are collected every 10 seconds.

Multiple SmartConnect Zones per Subnet

- Different client network performance needs
- Different SmartConnect zone names for different clients



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Because each SmartConnect zone is managed as an independent SmartConnect environment, they can have different attributes, such as the client connection policy. For environments with very different workloads, this provides flexibility in how cluster resources are allocated. Clients use one DNS name to connect to the performance zone and another to connect to the general use nodes. The performance zone could use CPU Utilization as the basis for distributing client connections, while the general use zone could use **Round-robin** or **Connection count**, which will optimize the allocation of cluster resources based on client requirements and workloads.

A customer can create a subnet and/or pool to be used by a high compute farm to give a higher level of performance. This is the performance zone shown above. A second subnet and/or pool is created with a different zone name for general use, often desktops, that do not need as high-level of performance. This is the general use zone. Each group connects to a different name and gets different levels of performance. This way, no matter what the desktop users are doing, it does not affect the performance to the cluster. Because it is still one cluster, when the data is generated from the cluster, it is immediately available to the desktop users.

Isilon does not support dynamic failover for SMB, hence the use of static allocation of IP addresses for SMB when using SmartConnect.

IP Address Pools

- Default IP Pool called: `pool0`
- Controls connectivity
- Contain multiple IP ranges
 - Either single IP or contiguous range
- Specific interfaces on nodes can be members of pool
- Create multiple IP pools within single subnet with SmartConnect Advanced license

Networks	Detail
groupnet0	DNS Servers: 192.168.0.201
subnet0	NET 192.168.0.0/24
pool0	IP Ranges: 192.168.0.11 - 192.168.0.16
rule0	Node Type: any, Interface: ext-1
pool1	IP Ranges: 192.168.0.45 - 192.168.0.50

SmartConnect Basic

Zone Name
finance.isilon.training.com

SmartConnect Service Subnet
subnet0

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IP address pools partition a cluster's external network interfaces into groups or pools of IP address ranges in a subnet, enabling you to customize how users connect to your cluster. Pools control connectivity into the cluster by allowing different functional groups, such as sales, RND, marketing, etc., access into different nodes. This is very important in those clusters that have different node types.

Perhaps a client with a 9-node cluster containing three S-Series nodes, three X-Series nodes, and three NL-Series nodes wants their Research team to connect directly to the S-Series nodes to utilize a variety of high I/O applications. The administrators can then have the Sales and Marketing users connect to the front-end of the X-Series nodes to access their files. This segmentation will keep the Sales and Marketing users from using bandwidth on the Research department's S-Series nodes. An administrator can also create a pool for connectivity into the NL-Series nodes for anyone who may be doing once a month patent research that does not require high performance or daily access.

The first external IP subnet was configured during the initialization of the cluster. The initial default subnet, `subnet0`, is always an IPv4 subnet. Additional subnets can be configured as IPv4 or IPv6 subnets. The first external IP address pool is also configured during the initialization of the cluster. The initial default IP address pool, `pool0`, was created within `subnet0`. It holds an IP address range and a physical port association.

Additional IP address pools can be created within subnets and associated with a node, a group of nodes, or network interface card, or NIC, ports.

Later in this course, we will describe how IP address pools help with providing different classes of service to different categories of users, such as Engineering and Sales.

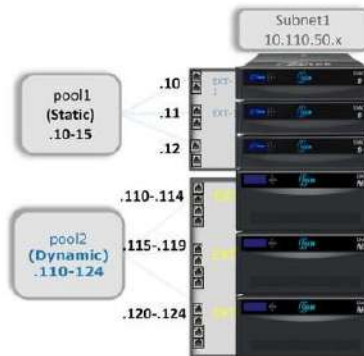
Static vs. Dynamic Pools

- **Static pools**

- Allocates one IP address to each NIC in pool
 - Best for SMB clients and NFSv4
 - Best for use with SyncIQ, SNMP, NDMP
- IP address does not failover if node is lost
 - SMB client must reconnect

- **Dynamic pools**

- Allocates multiple IP addresses per NIC
- Best for NFSv2 and NFSv3 clients
- IP addresses fail over to other nodes in event of node failure



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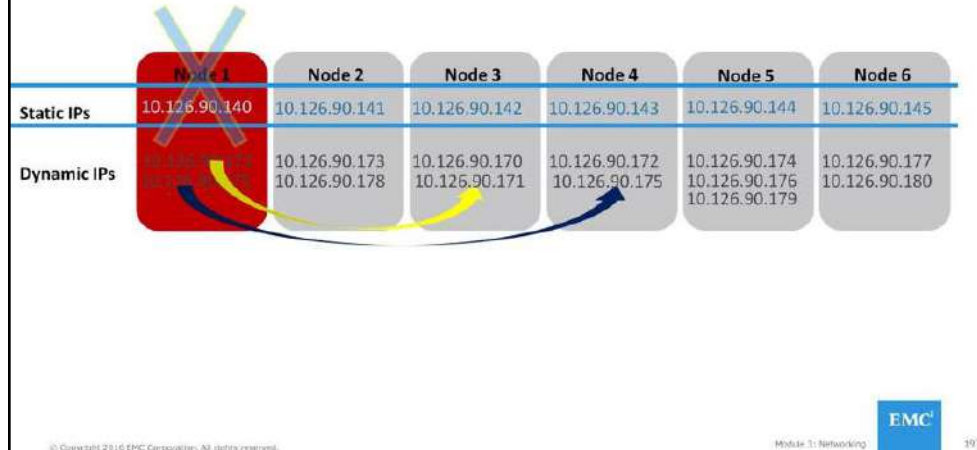
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When configuring IP address pools on the cluster, an administrator can choose either static pools or dynamic pools.

A static pool is a range of IP addresses that allocates only one IP address at a time. Like most computers and servers, a single IP address would be allocated from the pool to the chosen NIC. In the event there are more IP addresses than nodes, as in the above slide where we have three nodes but five IP addresses in the pool, the additional IP addresses will wait to be assigned in the event another node is added to the pool. If another node is added to the static pool then the next IP address from the range (in this case .13) will be assigned. Static pools are best used for SMB clients because of the stateful nature of the SMB protocol. When an SMB client establishes a connection with the cluster the session or "state" information is negotiated and stored on the server or node. If the node goes offline the state information goes with it and the SMB client would have to reestablish a connection to the cluster. SmartConnect is intelligent enough to hand out the IP address of an active node when the SMB client reconnects.

Dynamic pools are best used for NFS clients. Dynamic pools assign out all the IP addresses in their range to the NICs on the cluster. You can identify a Dynamic range by the way the IP addresses present in the interface as .110-.114 or .115-.199 instead of a single IP address like .10. Due to the nature of the NFS protocol being a state-less protocol, in that the session or "state" information is maintained on the client side, if a node goes down, the IP address that the client is connected to will failover (or move) to another node in the cluster. For example, if a Linux client were connected to .110 in our slide and we lost that node, the .110, .111, .112, .113 and .114 IP addresses would be distributed equally to the remaining two nodes in that pool and the Linux client would seamlessly failover to one of the active nodes. The client would not know that their original node had failed.

Dynamic IP Address Allocation & NFS Failover



This is an example illustrating how NFS failover and failback works. In this six-node Isilon cluster, an IP address pool provides a single static node IP (10.126.90.140-145) to an interface in each cluster node. Another pool of dynamic IPs (NFS failover IPs) has been created and distributed across the cluster (10.126.90.170 – 180).

When Node 1 in the Isilon cluster goes offline, the NFS failover IPs (and connected clients) associated with Node 1 failover to the remaining nodes based on the configured IP failover policy (**Round-robin**, **Connection count**, **Throughput**, or **CPU usage**). The static node IP for Node 1 is no longer available.

If a node with client connections established goes offline, the behavior is protocol-specific. NFSv3 automatically re-establishes an IP connection as part of NFS failover. In other words, if the IP address gets moved off an interface because that interface went down, the TCP connection is reset. NFSv3 re-establishes the connection with the IP on the new interface and retries the last NFS operation. However, SMBv1 and v2 protocols are stateful. So when an IP is moved to an interface on a different node, the connection is broken because the state is lost. NFSv4 is stateful (just like SMB) and like SMB does not benefit from NFS failover.

Note: A best practice for all non-NFSv3 connections is to set the IP allocation method to static. Other protocols such as SMB and HTTP have built-in mechanisms to help the client recover gracefully after a connection is unexpectedly disconnected.

Example: Static Pool

```
Cloud9-1: isi network pools view groupnet0.subnet0.pool0
ID: groupnet0.subnet0.pool0
Groupnet: groupnet0
Subnet: subnet0
Name: pool0
Rules: rule0
Access Zone: System
Allocation Method: static
Aggregation Mode: lacp
SC Suspended Nodes: -
Description: Initial ext-1 pool
Ifaces: 1:ext-1, 2:ext-1, 3:ext-1
IP Ranges: 192.168.0.11-192.168.0.19
Rebalance Policy: auto
SC Auto Unsuspend Delay: 0
SC Connect Policy: round_robin
SC Zone: -
SC DNS Zone Aliases: -
SC Failover Policy: round_robin
SC Subnet: -
SC Ttl: 0
Static Routes: -
```

CLI: `isi network pools view <poolname>`

```
- SmartConnect Advanced
Client Connection Balancing Policy
Round-robin
IP Failover Policy
Round-robin
Rebalance Policy
Automatic
Allocation Method
Static
```

WebUI

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The licensed version of SmartConnect allows multiple IP address pools per subnet. Thus, multiple SmartConnect zones with different policies can be created within a subnet, as well.

In this example, the subnet is named subnet0. The SIP is set and subnet0 has two IP address pools – pool0 and belze-pool.

Pool0 has an IP range of 10.126.90.140-149. The SmartConnect settings show the zone name is cluster.isilon.training.com, the connection policy is Round-robin, and the IP allocation method is static.

Each pool member (ext-1 of each node) has one IP address from the IP range. You can see that not all IP addresses in this pool are used. More might be used when more cluster nodes are added, and their interfaces become members of this pool.

Note: Select static as the IP allocation method to assign IP addresses as member interfaces are added to the IP pool. As members are added to the pool, this method allocates the next unused IP address from the pool to each new member. After an IP address is allocated, the pool member keeps the address indefinitely unless:

- The member interface is removed from the network pool.
- The member node is removed from the cluster.

Example: Dynamic Pool

- Second IP address pool in subnet subnet0
- SmartConnect Settings
- Pool Members and IP address range

IP allocation: **Dynamic**
Each IP address pool member has multiple addresses from IP range

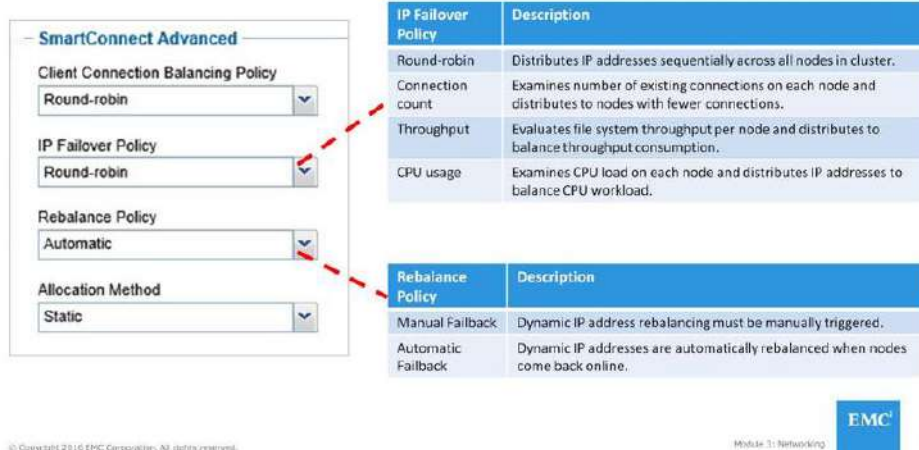
```
Total: 2
Cloud9-1# iscsi network pools view groupnet0.subnet0.NFS-pool
ID: groupnet0.subnet0.NFS-pool
Groupnet: groupnet0
Subnet: subnet0
Name: NFS-pool
Rules: -
Access Zone: System
Allocation Method: dynamic
Aggregation Mode: lacp
SC Suspended Nodes: -
Description: NFS clients
Ifaces: 1:ext-2, 2:ext-2, 3:ext-2
IP Ranges: 192.168.0.150-192.168.0.170
Rebalance Policy: auto
SC Auto Unsuspend Delay: 0
SC Connect Policy: round_robin
SC Zone: nfs.isilon.training.com
SC DNS Zone Aliases: -
SC Failover Policy: round_robin
SC Subnet: subnet0
SC Ttl: 0
Static Routes: -
Cloud9-1#
```

For the second pool in the same subnet, the IP allocation method is set to dynamic. Dynamic IP allocation is only available with SmartConnect Advanced (licensed) and is currently only recommended for use with NFSv3. Dynamic IP allocation ensures that all available IP addresses in the IP address pool are assigned to member interfaces when the pool is created. Dynamic IP allocation allows clients to connect to any IP address in the pool and receive a response. If a node or an interface becomes unavailable, its IP addresses are automatically moved to other member interfaces in the IP address pool.

Note that Dynamic IP allocation has the following advantages:

- It enables NFS failover, which provides continuous NFS service on a cluster even if a node becomes unavailable.
- It provides high availability because the IP address is available to clients at all times.

IP Failover and Rebalance Policies



The screenshot shows the SmartConnect Advanced configuration interface. On the left, there are four dropdown menus: Client Connection Balancing Policy (Round-robin), IP Failover Policy (Round-robin), Rebalance Policy (Automatic), and Allocation Method (Static). Red dashed arrows point from the IP Failover Policy and Rebalance Policy dropdowns to two tables on the right.

IP Failover Policy	Description
Round-robin	Distributes IP addresses sequentially across all nodes in cluster.
Connection count	Examines number of existing connections on each node and distributes to nodes with fewer connections.
Throughput	Evaluates file system throughput per node and distributes to balance throughput consumption.
CPU usage	Examines CPU load on each node and distributes IP addresses to balance CPU workload.

Rebalance Policy	Description
Manual Failback	Dynamic IP address rebalancing must be manually triggered.
Automatic Failback	Dynamic IP addresses are automatically rebalanced when nodes come back online.

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IP rebalancing and IP failover are features of SmartConnect Advanced. The rebalance policy determines how IP addresses are redistributed when node interface members for a given IP address pool become available again after a period of unavailability. The rebalance policy could be:

- **Manual Failback** – IP address rebalancing is done manually from the CLI using **isi network pools rebalance-ips**. This causes all dynamic IP addresses to rebalance within their respective subnet.
- **Automatic Failback** – The policy automatically redistributes the IP addresses. This is triggered by a change to either the cluster membership, external network configuration or a member network interface.

Lesson 5: Summary

Having completed this lesson, you are now able to:

- Identify load balancing options
- Explain uses of multiple zones
- Differentiate static and dynamic pools

Having completed this lesson, you are now able to identify load balancing options, explain uses of multiple zones, and differentiate static and dynamic pools.

Module 3: Summary

Having completed this module, you are now able to:

- Identify front-end network properties
- Define NIC aggregation options
- Connect to external IP network
- Differentiate between Basic and Advanced SmartConnect features
- Configure name resolution for cluster



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Having completed this module, you are now able to identify the front-end network properties, define the NIC aggregation options, connect to the external IP network, differentiate between Basic and Advanced SmartConnect features, and configure name resolution for the cluster.

Lab 3: Networking

- Configuring and testing SmartConnect
- Creating DNS records

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Module 3: Networking



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In these labs, you'll configure SmartConnect and then test the configuration. You will also create DNS records.

Module 4: Access Management

Upon completion of this module, you will be able to:

- Identify best practices for access zones
- Describe File Filtering
- Explain authentication structure
- Detail Directory Service configuration
- Establish benefits of using Isilon with Hadoop
- Understand Isilon implementation of Swift

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Module 4: Access Management



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Upon completion of this module, you will be able to identify best practices for access zones, describe File Filtering, explain authentication structure, detail Directory Service configuration, establish benefits of using Isilon with Hadoop, and understand Isilon implementation of Swift.

Lesson 1: Establishing Access Management

Upon completion of this module, you will be able to:

- Identify access zone functions
- Configure groups and users for an access zone
- Define importance of System access zone
- Implement access zones in OneFS
- Describe File Filtering

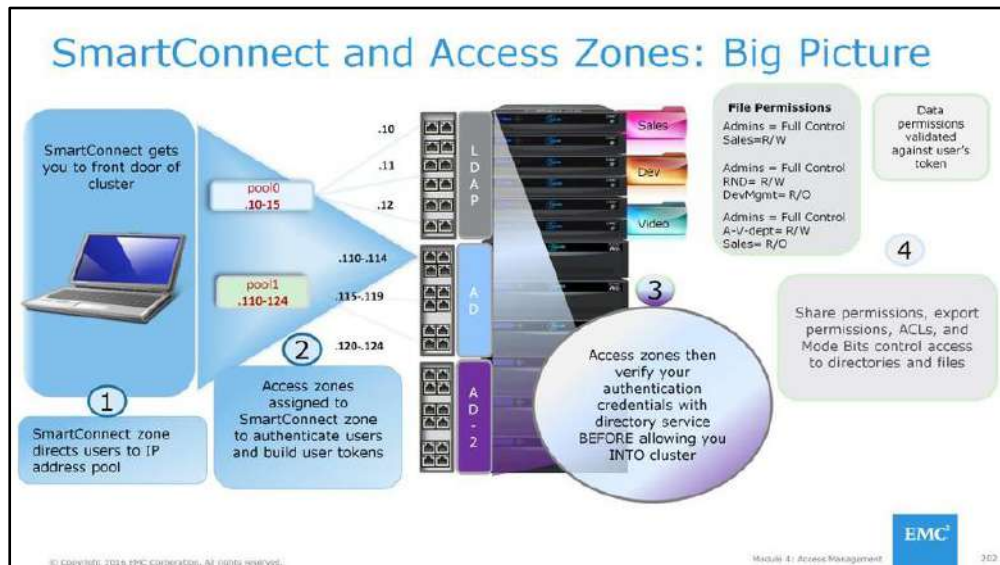
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Upon completion of this module, you will be able to identify access zone functions, configure groups and users for an access zone, define importance of System access zone, implement access zones in OneFS, and describe File Filtering.



Isilon has many different components and an Isilon cluster can be as simple or as complex as an individual's environment. Knowing how all of the internal features interact is integral to troubleshooting and explaining how the cluster works.

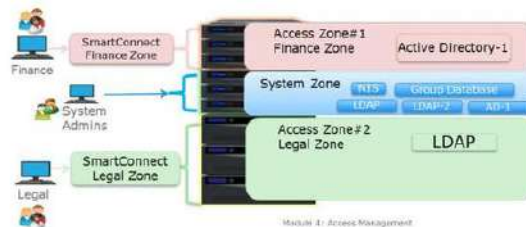
Often times access zones and SmartConnect are misunderstood or believed to be the same type of client routing feature but in fact they are distinctly different and dependent on one another.

- (1) SmartConnect is a client load balancing feature that allows segmenting of the nodes by performance, department or subnet. SmartConnect deals with getting the clients from their devices to the correct front-end interface on the cluster. That is the key, the CORRECT front-end interface for their job function/segment/department.
- (2 & 3) Once the client is at the front-end interface, the associated access zone then authenticates the client against the proper directory service; whether that is external, like LDAP and AD, or internal to the cluster, like the local or file providers. Access zones do not dictate which front-end interface the client connects to, it only determines what *directory* will be queried to verify authentication and what shares that the client will be able to view.
- (4) Once authenticated to the cluster, mode bits and access control lists, or ACLs, dictate the files, folders and directories that can be accessed by this client. Remember, when the client is authenticated, Isilon generates an access token for that user. The access token contains all the permissions and rights that the user has. When a user attempts to access a directory the access token will be checked to verify if they have the necessary rights.

In OneFS 7.0.x, the maximum number of supported access zones is five. As of OneFS 7.1.1, the maximum number of supported access zones is 20.

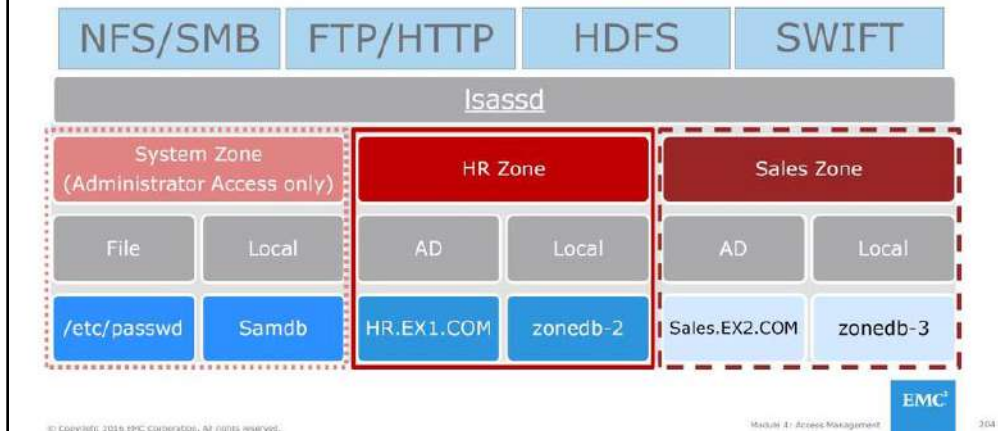
Access Zone Overview

- Partitions cluster into virtual containers
- Isolates data
- Limits who can log into which nodes
- Each zone handles individual client authentication
- Associated with one groupnet
- Controls who can access data
- 50 zone limit



Although the default view of an EMC Isilon cluster is that of one physical machine, you can partition a cluster into multiple virtual containers called access zones. Access zones allow you to isolate data and control who can access data in each zone. Access zones support configuration settings for authentication and identity management services on a cluster, so you can configure authentication providers and provision protocol directories, such as SMB shares and NFS exports, on a zone-by-zone basis. When you create an access zone, a local provider is automatically created, which allows you to configure each access zone with a list of local users and groups. You can also authenticate through a different authentication provider in each access zone.

Access Zone Architecture



The default access zone within the cluster is called the System access zone. The example in this slide displays two additional zones that have been created: an HR access zone and a Sales access zone. Configuration of access zones – or any other configuration of the cluster for that matter – is only supported when an administrator is connected through the System access zone.

Each access zone has their own authentication providers (File, Local, Active Directory, or LDAP) configured. Multiple instances of the same provider can occur in different access zones though doing this is not a best practice.

System Access Zone Overview

- System zone is built-in zone
- RBAC available only in System zone
 - Admins must log into System to manage cluster and other zones
 - References groupnet0, by default
 - Configuration of non-System access zones is not permitted via SSH, OneFS API, or web administration interface
 - Use MMC to create and delete SMB shares in non-System zones

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A cluster includes a built-in access zone named System where you manage all aspects of a cluster and other access zones. By default, all cluster IP addresses connect to the System zone. Role-based access, which primarily allows configuration actions, is available through only the System zone. All administrators, including those given privileges by a role, must connect to the System zone to configure a cluster. The System zone is automatically configured to reference the default groupnet on the cluster, which is groupnet0.

Multiple Access Zones

- Directs incoming clients to zone based on IP Pool
 - Restricts authentication
 - Reduces visible/available SMB shares and NFS exports
 - Can audit individual access zone
- Create Zone Based Directory Tree
 - /ifs/clustername/zone-name
 - OneFS 8.0: Zones can have shared base directory
 - Allows data sharing between zones
 - Share base directories only if required
 - Zones with shared data should also share authentication providers



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To control data access, you associate the access zone with a groupnet, which is a top-level networking container that manages DNS client connection settings and contains subnets and IP address pools. When you create an access zone, you must specify a groupnet. If a groupnet is not specified, the access zone will reference the default groupnet. Multiple access zones can reference a single groupnet. You can direct incoming connections to the access zone through a specific IP address pool in the groupnet. Associating an access zone with an IP address pool restricts authentication to the associated access zone and reduces the number of available and accessible SMB shares and NFS exports. An advantage to multiple access zones is the ability to configure audit protocol access for individual access zones. You can modify the default list of successful and failed protocol audit events and then generate reports through a third-party tool for an individual access zone. You can configure access zones to have a shared base directory, allowing the access zones to share data. Access zones that share a base directory should also share authentication providers. Configuration management of a non-System access zone is not permitted through SSH, the OneFS API, or the web administration interface. However, you can create and delete SMB shares in an access zone through the Microsoft Management Console (MMC).

A base directory defines the file system tree exposed by an access zone. The access zone cannot grant access to any files outside of the base directory. You must assign a base directory to each access zone. Base directories restrict path options for several features such as SMB shares, NFS exports, the HDFS root directory, and the local provider home directory template. The base directory of the default System access zone is **/ifs** and cannot be modified. To achieve data isolation within an access zone, EMC recommends creating a unique base directory path that is not identical to or does not overlap another base directory, with the exception of the System access zone. For example, do not specify **/ifs/data/hr** as the base directory for both the zone2 and zone3 access zones, or if **/ifs/data/hr** is assigned to zone2, do not assign **/ifs/data/hr/personnel** to zone3.

OneFS supports overlapping data between access zones for cases where your workflows require shared data; however, this adds complexity to the access zone configuration that might lead to future issues with client access. For the best results from overlapping data between access zones, EMC recommends that the access zones also share the same authentication providers. Shared providers ensures that users have consistent identity information when accessing the same data through different access zones.

Do I Need Multiple Access Zones?

- Need to perform storage consolidation
- Multiple authentication providers: LDAP1, LDAP2, LDAP3
- Different untrusted AD domains
 - Cannot have different untrusted AD domains in single zone
- Different untrusted AD forests with overlapping directory structures
- Limit SMB share visibility/accessibility
- Limit client access based on access zone/SmartConnect IP pool
- Want to audit specific zones



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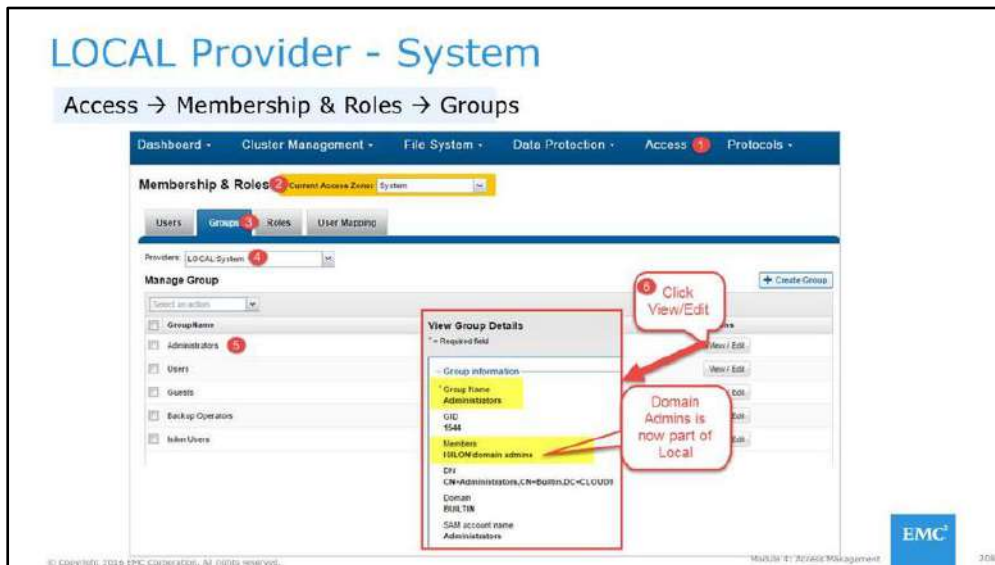
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OneFS enables you to configure multiple authentication providers on a per-zone basis. In other words, it's possible for an Isilon cluster to have more than one instance of LDAP, NIS, File, Local, and Active Directory providers.

Access zones provide a means to limit data access to specific directory structures by access zone and SmartConnect zone/IP address pool. Each access zone can be configured with its own authentication providers, zone aware protocols, such as SMB, FTP, and HTTP, and associated SmartConnect IP address pools. An access zone becomes an independent point for authentication and access to the cluster. Only one Active Directory provider can be configured per access zone. If you connect the cluster to multiple AD environments (untrusted) only one of these AD providers can exist in a zone at one time. Each access zone may also have relationships to the System access zone. This is particularly useful for storage consolidation, for example, when merging multiple storage filers that are potentially joined to different untrusted Active Directory forests and have overlapping directory structures. SMB shares that are bound to an access zone are only visible/accessible to users connecting to the SmartConnect zone/IP address pool to which the access zone is aligned. SMB authentication and access can be assigned to any specific access zone. Here's an example of separate namespaces for SMB/NFS:

- A number of SmartConnect zones are created, such as **finance.emc.com**, **hr.emc.com**. Each of those SmartConnect zones can be aligned to an access zone.
- Users connecting to **\\hr.emc.com** would only see hr shares.
- Users connecting to **\\finance.emc.com** would only see finance shares.
- Having multiple zones allow you to audit specific zones without needing to audit the entire cluster.



When joining the Isilon cluster to an AD domain, the Isilon cluster is treated as a resource.

If the System access zone is set to its defaults, the Domain Admins and Domain Users groups from the AD domain are automatically added to the cluster's local Administrators and Users groups, respectively. Besides the existing local groups, more groups can be created and groups can be edited or deleted. For each access zone, a local provider is automatically created.

It's important to note that, by default, the cluster's local Users group also contains the AD domain group: Authenticated Users. This group enables all users that have authenticated to the AD domain to have access rights (Authenticated Users excludes the Guest and anonymous users; this is how it differs from the group Everyone) to cluster resources. They must also have permissions to read or modify these resources.

The local group can be edited so that only specific users or groups from the AD domain are selected, and would thus have access using the access zone. Another access zone could be created having the same AD provider, but uses separate shares, a different IP address pool (maybe supporting 10 GigE), and different domain users and groups as members of a local group. Note that unlike UNIX groups, local groups can include built-in groups, and global Active Directory groups as members. Local groups can also include users from other providers. Netgroups are not supported in the local provider.

Authentication Sources and Access Zones

- Authentication providers do not belong to one specific zone
 - Visible by zones, not owned by them
- Only join AD providers not in same forest (untrusted forest)
 - Trusts within same forest managed by AD
- Overlapping UIDs not checked
 - Shared UIDs in same zone can potentially cause UID/GID conflicts

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There are three things to know about joining multiple authentication sources through access zones. First, the joined authentication sources do not belong to any zone, instead they are seen by zones; meaning that the zone does not own the authentication source. This allows other zones to also include an authentication source that may already be in use by an existing zone. For example, if you have Zone-A with providers LDAP-1, AD-1 and Zone-B with NIS, not allowing authentication sources to belong to a zone means that the administrator can then create Zone-C with the LDAP-1 provider that was used in Zone-A. Second, when joining AD domains, only join those that are not in the same forest. Trusts within the same forest are managed by AD, and joining them could allow unwanted authentication between zones. Finally, there is no built-in check for overlapping UIDs. So when two users in the same zone - but from different authentication sources - share the same UID, this can cause access issues; additional details on this topic will be covered in the next module.

Best Practices for Access Zones

Best Practice	Detail
Create unique base directory	Achieves data isolation. Base directories can overlap only if workflows share data
System zone for Admin access only	Reserve for configuration access
Create zones to isolate data for different clients	Do not do this if workflow requires shared data
One authentication provider per zone	Simplifies administration
Avoid overlapping UID/GID ranges for providers in same zone	Potential for UID/GID conflicts if overlap in same zone



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You can avoid configuration problems on the EMC Isilon cluster when creating access zones by following best practices guidelines.

Best practice details:

1. Create unique base directories. To achieve data isolation, the base directory path of each access zone should be unique and should not overlap or be nested inside the base directory of another access zone. Overlapping is allowed, but should only be used if your workflows require shared data.
2. Separate the function of the System zone from other access zones. Reserve the System zone for configuration access, and create additional zones for data access. Move current data out of the System zone and into a new access zone.
3. Create access zones to isolate data access for different clients or users: Do not create access zones if a workflow requires data sharing between different classes of clients or users.
4. Assign only one authentication provider of each type to each access zone: An access zone is limited to a single Active Directory provider; however, OneFS allows multiple LDAP, NIS, and file authentication providers in each access zone. It is recommended that you assign only one type of each provider per access zone in order to simplify administration.
5. Avoid overlapping UID or GID ranges for authentication providers in the same access zone: The potential for zone access conflicts is slight but possible if overlapping UIDs/GIDs are present in the same access zone.

File Filtering Overview

- Deny or allow access to files based on file extensions
 - Configured per access zone
 - Custom extension list(s) to:
 - Explicitly deny
 - Explicitly allow
- Multi-protocol feature support
 - SMB, NFS, HDFS and Swift
 - Contains additional SMB support
 - SMB default share
 - Per SMB share level granularity
- No license required



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File filtering enables administrators to deny or allow file access on the cluster based on the file extension. Both the ability to write new files to the cluster or access existing files on the cluster are controlled by file filtering. *Explicitly deny* lists are used to block only the extensions in the list. *Explicitly allow* list permits access to files only with the listed file extensions. There is no limit or pre-defined list of extensions. Customers can create custom extension lists based on their specific needs and requirements. The top-level of file filtering is setup on a per access zone and controls all access zone aware protocols such as SMB, NFS, HDFS and Swift. Any client on any access zone aware protocol is limited by the file filtering rules. At a lower level, file filtering is configurable for the SMB default share, and is configurable as part of any individual SMB share setup. File filtering is included with OneFS 8.0 and no license is required.

Management of Existing Cluster Files

- File management for existing files with filtered extensions
 - Denied file access focus
 - Files filtered according to deny or allow list
- File access
 - No user access to files already existing on cluster
 - No read, modify, or delete capabilities
 - Administrator can access (read) or delete files
 - Access zone or SMB share rules applied
 - No restrictions for direct administrator access



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What happens if you enable file filtering on an existing cluster? The file extensions are used to determine access to the files. Users will not be able to access any file with a denied extension. The extension can be denied through the denied extensions list, or because the extension was not included as part of the allowed extensions list. Administrators can still access existing files. Administrators can read the files or delete the files. Modifying or updating a file is not allowed. If a user or administrator accesses the cluster through an access zone or SMB share without file filtering applied, files are fully available to the user or administrator. How the file filtering rule is applied to the file determines where the file filtering occurs. Administrators with direct access to the cluster can manipulate the files. File filters are applied only when accessed using the four protocols.

File Filtering Configuration

View SMB Share Details
* = Required field

Settings

Name
finance

Description
Finance files

Path
ifs/data/finance

Home Directory Provisioning
Path variables will not be expanded
Directories will not be automatically created

Content Availability
Disabled

Users and Groups
Everyone is not a root with full control

File Filter

File Filter Extensions
Enabled

File Extensions
Deny writes for list of file extensions

File Filtering Extensions
mp3

- File Filtering configuration at three levels:
 - Zone level settings
 - isi file-filter
 - Default share settings
 - isi smb settings shares
 - Specific share setting
 - isi smb shares modify
 - Must have
ISI_PRIV_FILE_FILTER
privilege



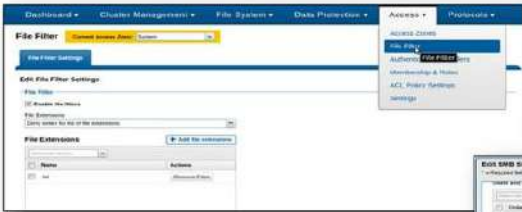
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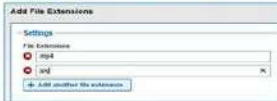
You can configure file filtering at three separate levels within the cluster: At the access zone level, on the default SMB share, and on specific SMB shares. If you are using RBAC to delegate control of this task, you must ensure that the user has the ISI_PRIV_FILE_FILTER privilege.

Per Access Zone Filtering via WebUI

Access → File Filter → File Filter Settings



Extensions must start with a period (.) no wildcards






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In order to configure an entire access zone to be used with File Filtering, you need to navigate to **Access → File Filter → File Filter Settings**, enter the extension of the file, and click submit. The file extension window will not allow the use of wildcards or special characters so add the extension using just the (.) period and extension, such as *.mp3, *.doc, *.jpg.

Why Add File Filtering?

- Common customer request
- Some reasons or uses for file filtering
 - Enforces organization's policies
 - Meets compliance requirements
 - e.g., Email compliance - No *.pst files saved
 - Limits large size file content to share
 - e.g., *.mov or *.mp4 files
 - Avoids potential copy write infringement issues
 - e.g., *.mp3 files
 - Isolates clusters for only specific file use
 - Application specific cluster use, only specific file extensions allowed

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Customers commonly request file filtering and in OneFS 8.0 it can now be delivered. Some of the reasons for file filtering include the capability to enforce organizations policies. With all of the compliance considerations today, organizations struggle to meet many of the requirements.

For example, many organizations are required to make all email available for litigation purposes. To help make sure email is not stored longer than desired, they may not want to allow *.pst files to be stored on the cluster by the users. Some reasons are practical, cluster space costs money. Organization plan storage space increases based on their work. They may not want typically large files, such as video files, to be stored on the cluster, so they can filter *.mov or *.mp4 file extension from being stored. An organizational legal issue is copy write infringement. Many users store their *.mp3 files on the cluster and open a potential issue for copy write infringement for the organization. Another requested use is to limit a cluster for only a specific application with its unique set of file extensions. File filtering with explicit allow list of extensions can help limit the cluster for its singular intended purpose.

Lesson 1: Summary

Having completed this lesson, you are now able to:

- Identify access zone functions
- Configure groups and users for an access zone
- Define importance of System access zone
- Implement access zones in OneFS
- Describe File Filtering

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Having completed this lesson, you are now able to identify access zone functions, configure groups and users for an access zone, define importance of System access zone, implement access zones in OneFS, and describe File Filtering.

Lesson 2: Directory Service Integration

Upon completion of this module, you will be able to:

- Explain authentication structure
- Detail Directory Service configuration
 - Microsoft Active Directory (AD)
 - Lightweight Directory Access Protocol (LDAP)
 - Network Information Service (NIS)
- Understand Local and file sources

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Upon completion of this module, you will be able to explain the authentication structure, detail the Directory Service configuration, Microsoft Active Directory, or AD, Lightweight Directory Access Protocol, or LDAP, Network Information Service, or NIS, understand Local and file sources, and describe access zone role in authentication.

NTP Use in Cluster

- Cluster time sets cluster's date and time settings
 - If multiple NTP servers defined, first server on list used first
 - After NTP server established, can't manually set date and time
 - Internally cluster nodes use NTP to coordinate time settings
- **Risk:** Conflict between SMB time and cluster external NTP
 - If cluster drifts from SMB time, authentication fails
- **Best Practices**
 - NTP only for cluster and AD forest
 - If cluster has 3 nodes, only one node gets external NTP information
 - If cluster is large, first 3 nodes act as chimer



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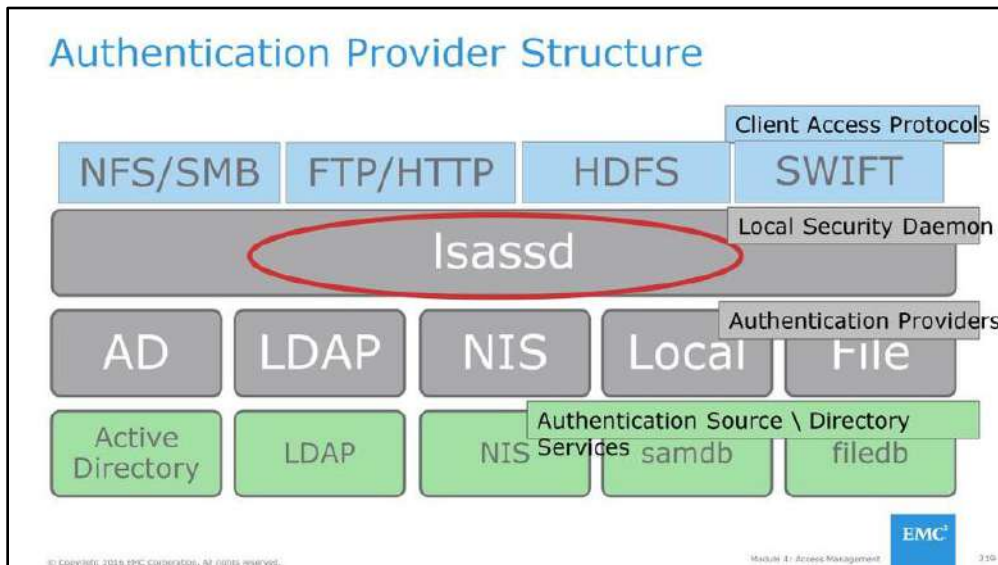
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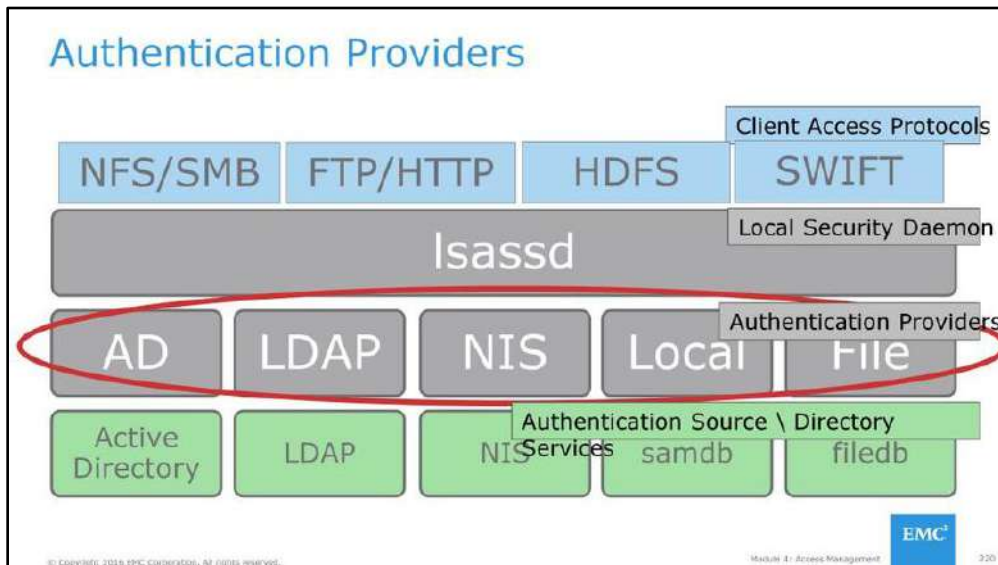
The **Cluster Time** property sets the cluster's date and time settings, either manually or by synchronizing with an NTP server. There may be multiple NTP servers defined. The first NTP server on the list is used first, with any additional servers used only if a failure occurs. After an NTP server is established, setting the date or time manually is not allowed. After a cluster is joined to an AD domain, adding a new NTP server can cause time synchronization issues. The NTP server will take precedence over the SMB time synchronization with AD and overrides the domain time settings on the cluster.

SMB time is enabled by default and is used to maintain time synchronization between the AD domain time source and the cluster. Nodes use NTP between themselves to maintain cluster time. When the cluster is joined to an AD domain, the cluster must stay in sync with the time on the domain controller otherwise authentication may fail if the AD time and cluster time have more than a five minute differential. AD and SMB keep the time on the nodes in sync with the domain controller. The best case support recommendation is to not use SMB time and only use NTP if possible on both the cluster and the AD domain controller. The NTP source on the cluster should be the same source as the AD domain controller's NTP source. If SMB time must be used, then NTP should be disabled on the cluster and only use SMB time.

Only one node on the cluster should be setup to coordinate NTP for the cluster. This NTP coordinator node is called the chimer node. The configuration of the chimer node is by excluding all other nodes by their node number using the **isi_ntp_config add exclude node# node# node#** command. The list excludes nodes using their node numbers separated by a space. The node which was not excluded acts as the NTP chimer node and may be any node you choose on the cluster.



The lsassd, which is pronounced "L-sass-D", is the cluster's authentication daemon. It resides between the access protocols and the lower level services providers. The lsassd daemon mediates between the authentication protocols used by clients and the authentication providers in the third row, that check their data repositories, represented on the bottom row, to determine user identity and subsequent access to files.



Authentication providers support the task of authentication and identity management by verifying users' credentials before allowing them to access the cluster. The authentication providers handle communication with authentication sources. These sources can be external, such as Active Directory (AD), Lightweight Directory Access Protocol (LDAP), and Network Information Service (NIS). The authentication source can also be located locally on the cluster or in password files that are stored on the cluster. Authentication information for local users on the cluster is stored in **/ifs/.ifsvvar/sam.db**.

OneFS supports the use of more than one concurrent authentication source.

Under FTP and HTTP, the Isilon cluster supports Anonymous mode, which allows users to access files without providing any credentials and User mode, which requires users to authenticate to a configured authentication source.

LDAP Directory Service Intro

- Simple directory service
 - Authenticates users and groups accessing cluster
 - Supported by Windows and Linux clients
- Features supported in cluster
 - Netgroups supported
 - Uses base DN's
 - Idapsam schema supported
 - Allows NTLM to authenticate over SMB
 - Multiple LDAP instances can be created



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LDAP can be used in mixed environments and is widely supported. It is often used as a meta-directory that sits between other directory systems and translates between them, acting as a sort of bridge directory service to allow users to access resources between disparate directory services or as a single sign-on resource. It does not offer advanced features that exist in other directory services such as Active Directory. A netgroup, is a set of systems that reside in a variety of different locations, that are grouped together and used for permission checking. For example, a UNIX computer on the 5th floor, six UNIX computers on the 9th floor, and 12 UNIX computers in the building next door, all combined into one netgroup.

Within LDAP, each entry has a set of attributes and each attribute has a name and one or more values associated with it that is similar to the directory structure in AD. Each entry consists of a distinguished name, or DN, which also contains a relative distinguished name (RDN). The base DN is also known as a search DN because a given base DN is used as the starting point for any directory search. The top-level names almost always mimic DNS names; for example, the top-level Isilon domain would be dc=isilon,dc=com for Isilon.com.

You can configure Isilon clusters to use LDAP to authenticate clients using credentials stored in an LDAP repository.

The LDAP provider in an Isilon cluster supports the following features:

- Users, groups, and netgroups
- Configurable LDAP schemas. For example, the Idapsam schema allows NTLM authentication over the SMB protocol for users with Windows-like attributes.
- Simple bind authentication (with or without SSL)
- Redundancy and load balancing across servers with identical directory data
- Multiple LDAP provider instances for accessing servers with different user data
- Encrypted passwords

Enabling LDAP

- Add an LDAP Provider
 - Need one LDAP provider per set of LDAP users
 - Web administration interface
 - **Add an LDAP Provider** page
 - CLI
 - `isi auth ldap`
 - Can connect to more than one LDAP server
- Required information
 - Base distinguished name
 - Port number
 - LDAP server (host name or IP address)



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To enable the LDAP service, you must configure a base distinguished name (base DN), a port number, and at least one LDAP server. Before connecting to an LDAP server you should decide which optional customizable parameters you want to use. You can enable the LDAP service using the web administration interface or the CLI. LDAP commands for the cluster begin with **isi auth config ldap**. To display a list of these commands, run the **isi auth config ldap list** command at the CLI.

If there are any issues while configuring or running the LDAP service, there are a few commands that can be used to help troubleshoot. Often issues involve either misconfigured base DN's or connecting to the LDAP server. The **ldapsearch** command can be used to run queries against an LDAP server to verify whether the configured base DN is correct and the **tcpdump** command can be used to verify that the cluster is communicating with the assigned LDAP server.

Note: AD and LDAP both use TCP port 389. Even though both services can be installed on one Microsoft server, the cluster can only communicate with one of services if they are both installed on the same server.

Add LDAP Provider

Access → Authentication Providers → LDAP → Add an LDAP Provider

The screenshot displays the EMC web administration interface. The top navigation bar includes 'Dashboard', 'Cluster Management', 'File System', 'Data Protection', 'Access', and 'Protocols'. The 'Access' menu is expanded, showing 'Access Zones', 'File Filter', 'Authentication Providers', and 'Membership & Roles'. The 'Authentication Providers' menu item is selected. Below the navigation bar, the 'Authentication Providers' section is visible, with tabs for 'Active Directory', 'LDAP', 'NIS', 'Local Provider', 'File Provider', 'Kerberos Provider', and 'Kerberos Provider'. The 'LDAP' tab is selected. Below the tabs, the 'Manage LDAP Providers' section is shown, featuring a table with columns for 'Provider Name' and 'Status'. The table contains one entry, 'Isilon LDAP', with a green status icon. A '+ Add an LDAP provider' button is located to the right of the table. Red numbered circles (1, 2, 3, 4) indicate the sequence of steps: 1. Click 'Access' in the top navigation bar. 2. Click 'Authentication Providers' in the expanded 'Access' menu. 3. Click 'LDAP' in the 'Authentication Providers' tabs. 4. Click '+ Add an LDAP provider' button.

Dashboard ▾ Cluster Management ▾ File System ▾ Data Protection ▾ Access ▾ 1 Protocols ▾

Access Zones
File Filter
Authentication Providers 2
Membership & Roles

Active Directory 3 LDAP NIS Local Provider File Provider Kerberos Provider K...

Manage LDAP Providers

+ Add an LDAP provider 4

LDAP Providers

Provider Name	Status
Isilon LDAP	View details Del...

EMC

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To configure the cluster to connect to an LDAP server, in the web administration interface, click **Access**, click **Authentication Providers**, click **LDAP**, and then click **Add an LDAP provider**.

Add LDAP Provider (cont'd)

Add an LDAP Provider
* = Required field

* LDAP Provider Name: 1

* Server URIs: 2

URIs should begin with ldap:// or ldaps:// e.g. ldap://12.34.56.78

☒ Load balance servers 3

* Base Distinguished Name:

Groupnet:

Bind to: 4

Password: 5

[Default Query Settings](#)
[User Query Settings](#)
[Group Query Settings](#)
[Netgroup Query Settings](#)
[Advanced LDAP Settings](#)

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Access → Authentication
Providers → LDAP →
Add an LDAP Provider

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To create a new LDAP provider, type the name of the LDAP provider (1) and list one or more LDAP servers (2). The servers must all support the same set of users. You can optionally choose to load balance between multiple LDAP servers. A base DN is also required (3).

In the **Bind to** field, optionally type the distinguished name of the entry to use to bind to the LDAP server (4). In the **Password** field (5), type the password to use when binding to the LDAP server. Click **Add LDAP Provider** (6).

After the LDAP provider is successfully added, the **Manage LDAP Providers** page should display a green status. This means that the cluster can communicate with the LDAP server(s).

Active Directory Service Intro

- Microsoft directory service
- Enables domain users to access cluster
- Joining cluster to Active Directory (AD) domain requires:
 - Domain name
 - Domain account with rights to join domain
 - Domain account password
 - Organizational Unit (OU) (optional)
 - Determines location of computer account
 - Use LDAP style or UNIX style paths
 - LDAP: OU=Designers,OU=Marketing,DC=corp,DC=isilon,DC=com
 - UNIX: Designers/Marketing



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Active Directory, or AD, is a directory service created by Microsoft that controls access to network resources and that can integrate with Kerberos and DNS technologies. Active Directory can serve many functions, but the primary reason for joining the cluster to an AD domain is to enable domain users to access cluster data. OneFS 8.0 supports AES 128-bit and AES 256-bit encryption for Kerberos.

A cluster that joins a domain becomes a domain resource and acts as a file server. The domain join process can take up to several minutes depending on the complexity of the domain being joined. While joining the domain, the browser window displays the status of the process and confirms when the cluster has successfully joined the AD domain. During the process of joining the domain, a single computer account is created for the entire cluster. If the web administration interface is being used to join the domain, you must enable pop-up windows in the browser.

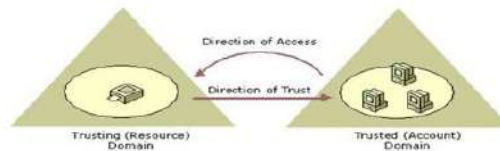
Before joining the domain, complete the following steps:

- NetBIOS requires that computer names be 15 characters or less. Two to four characters are appended to the cluster name you specify to generate a unique name for each node. If the cluster name is more than 11 characters, you can specify a shorter name in the **Machine Name** box in the **Join a Domain** page.
- Obtain the name of the domain to be joined.
- Use an account to join the domain that has the right to create a computer account in that domain.
- Include the name of the OU in which you want to create the cluster's computer account. Otherwise the default OU (Computers) is used.

When a cluster is destined to be used in a multi-mode environment, the cluster connect to the LDAP server first before joining the AD domain, so that proper relationships are established between UNIX and AD identities. Joining AD first and then LDAP will likely create some authentication challenges and permissions issues that will require additional troubleshooting.

Trusts and Pass-through Authentication

- AD provider supports pass through authentication
 - Uses trusts to authenticate users from trusted domain
 - Users from trusted domains can access cluster
 - Cluster is domain resource
- Cluster should belong to only one AD domain within a forest
 - Domains in forest automatically trust each other
 - Can join more than one AD domain *only* if additional domain is untrusted



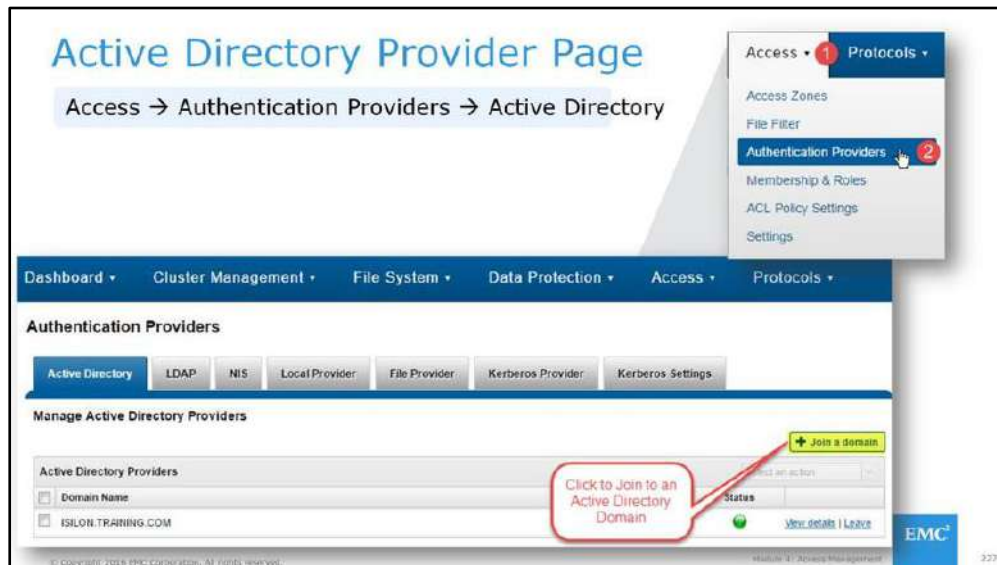
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The AD authentication provider in an Isilon cluster supports domain trusts and NTLM (NT LAN Manager) or Kerberos pass through authentication. This means that a user authenticated to an AD domain can access resources that belong to any other trusted AD domain. Because the cluster is a domain resource, any user that is authenticated to a trusted domain can access the cluster's resources just as members of the cluster's domain can access the cluster's resources. These users must still be given the permission to cluster's resources, but pass through authentication makes it possible to grant trusted users access to the cluster's resources. For this reason, a cluster needs only to belong one Active Directory domain within a forest or among any trusted domains. A cluster should belong to more than one AD domain only to grant cluster access to users from multiple untrusted domains.



To join the cluster to an AD domain, in the web administration interface, click **Access** (1), and then click **Authentication Providers** (2). The different providers are listed on individual tabs.

Click **Active Directory**, and then click **Join a domain**. When a cluster is destined to be used in a multi-mode environment, as a best practice, connect to the LDAP server first, and then join the cluster to the AD domain. This allows the proper relationships to be established between UNIX and AD identities. If the opposite occurs and AD is added before joining an LDAP domain, there will be some authentication challenges and permissions issues, and additional work is needed to remediate these challenges.

Join a Domain

Access → Authentication Providers → Active Directory → Join a domain

Join a Domain
* = Required field

* Domain Name: isilon.training.com

* User: administrator

* Password: *****

Organizational Unit:
If specified, specify OU in the form "OuName" or "OuName1/SubName2"

Machine Account:

Groupnet: grouper0

☐ Enable Secure NFS
Select this option to allow Active Directory authentication for NFS.

[+ Advanced Active Directory Settings](#)



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On the **Join a Domain** page, type the name of the domain you want the cluster to join. Type the user name of the account that has the right to add computer accounts to the domain, and then type the account password. Optionally, if you want to create the computer account in a particular OU, in the **Organizational Unit** field, type the name of the OU and also type the name that you want for the computer account. If you do not type a computer account, the cluster name is used.

The **Enable Secure NFS** checkbox enables users to log in using LDAP credentials, but to do this, Services for NFS must be configured in the AD environment. To finish, click **Join**.

NIS Authentication Provider

- Can integrate cluster into existing NIS infrastructure
- Unsecure:
 - Exposes passwd, group and netgroup maps from NIS server
- Supports hostname lookups
- Isilon does not support NIS+



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NIS provides authentication and uniformity across local area networks. OneFS includes a NIS authentication provider that enables you to integrate the cluster into an existing NIS infrastructure in your network. The NIS provider is used by the Isilon clustered storage system to authenticate users and groups that are accessing the cluster. The NIS provider exposes the passwd, group, and netgroup maps from a NIS server. Hostname lookups are also supported. Multiple servers can be specified for redundancy and load balancing.

NIS is different from NIS+, which Isilon clusters do not support.

Local Authentication Provider

- Supports authentication and lookup for local users and groups
 - Does not include system accounts
 - UNIX netgroups are not supported
 - Can include AD domain users and groups
- Commonly used:
 - In small environments
 - Allows local group to host multiple groups from external providers
 - To manage groups ON the cluster



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The Local provider supports authentication and lookup facilities for local users and groups that have been defined and are maintained locally on the cluster. It does not include system accounts such as root or admin. UNIX netgroups are not supported in the Local provider.

The Local provider can be used in small environments, or in UNIX environments that contain just a few clients that access the cluster, or as part of a larger AD environment. The Local provider plays a large role when the cluster joins an AD domain. Like the local groups that are used within an Active Directory environment, the local groups created on the cluster can include multiple groups from any external provider. These external groups would be added to the cluster local group to assist in managing local groups on the cluster.

File Provider

- Authoritative third-party source of user and group information
 - Formatted as: `/etc/spwd.db` and `/etc/group`
- File providers:
 - `etc/passwd` and `etc/group`
- Recommended location/directory for these files
 - `/ifs/.ifsvar`
 - Ensures all nodes have access to same version of files
- Separate files for users, groups, and netgroups



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OneFS uses **`/etc/spwd.db`** and **`/etc/group`** files for users and groups associated with running and administering the cluster. These files do not include end-user account information; however, you can use the file provider to manage end-user identity information based on the format of these files.

The file provider enables you to provide an authoritative third-party source of user and group information to the cluster. The file provider supports the `spwd.db` format to provide fast access to the data in the **`/etc/master.passwd`** file and the **`/etc/group`** format supported by most UNIX operating systems.

The file provider pulls directly from two files formatted in the same manner as **`/etc/group`** and **`/etc/passwd`**. Updates to the files can be scripted. To ensure that all nodes in the cluster have access to the same version of the file provider files, you should save the files to the **`/ifs/.ifsvar`** directory. The file provider is used by OneFS to support the users `root` and `nobody`.

The file provider is useful in UNIX environments where `passwd`, `group`, and `netgroup` files are synchronized across multiple UNIX servers. OneFS uses standard BSD **`/etc/spwd.db`** and **`/etc/group`** database files as the backing store for the file provider. The `spwd.db` file is generated by running the **`pwd_mkdb`** command-line utility. Updates to the database files can be scripted.

You can specify replacement files for any combination of users, groups, and netgroups.

Note: The built-in System file provider includes services to list, manage, and authenticate against system accounts (for example, `root`, `admin`, and `nobody`). Modifying the System file provider is not recommended.

Lesson 2: Summary

Having completed this lesson, you are now able to:

- Integrate NTP servers
- Explain authentication structure
- Detail Directory Service configuration
 - Microsoft Active Directory (AD)
 - Lightweight Directory Access Protocol (LDAP)
 - Network Information Service (NIS)
- Understand Local and file sources



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Having completed this lesson, you are now able to explain the authentication structure, detail the Directory Service configuration, Microsoft Active Directory, or AD; Lightweight Directory Access Protocol, or LDAP; and Network Information Service, or NIS, understand Local and file sources, and describe access zones role in authentication.

Lesson 3: Demystifying Hadoop

Upon completion of this lesson, you will be able to:

- Explain Hadoop components
- Illustrate Hadoop traditional architecture
- Examine benefits of a Data Lake
- Analyze benefits of using Isilon with Hadoop

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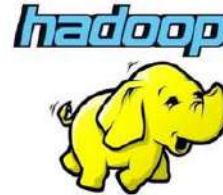


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Upon completion of this lesson, you will be able to explain Hadoop components, illustrate Hadoop traditional architecture, examine benefits of a Data Lake, and analyze benefits of using Isilon with Hadoop.

What is Hadoop?

- Hadoop is:
 - Software library framework
 - Allows for distributed processing of large data sets
 - Analyzes data across groups of computers using simple programming models
- Demystifying Hadoop EMC Video:
 - Explores what Hadoop is, why it's used, and what architecture looks like
 - <https://www.youtube.com/watch?v=xJHv5t8jcM8>



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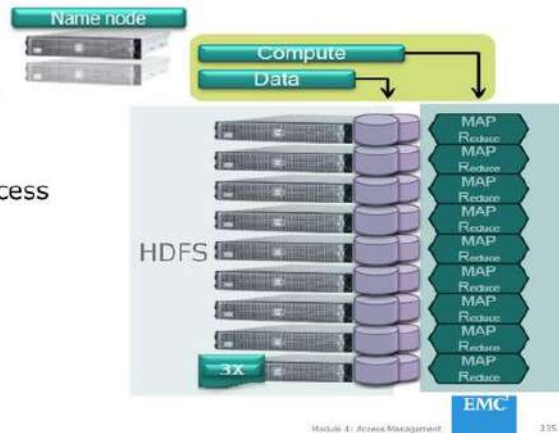
Hadoop is an open source software project that enables the distributed processing of large data sets across clusters of commodity servers. It is designed to scale up from a single server to thousands of servers.

Hadoop clusters can be dynamically scaled up and down based on the available resources and the required services levels. Performance service levels vary widely for processing, queries can take anywhere from a few minutes to multiple days depending on how many nodes and the amount of data requested. Hadoop has emerged as a tool of choice for big data analytics but there are reasons to use it in a typical enterprise environment to analyze existing data to improve processes and performance depending on your business model. We will explore the uses of Hadoop in environments with large data sets and touch upon why Hadoop is also a good fit for corporations that have a lot of data but perhaps aren't traditionally considered a "big data" environment.

Additional information about Hadoop and its origin can be found at:
http://www.sas.com/en_us/insights/big-data/hadoop.html

Components of Conventional Hadoop

- **NameNode:**
 - One Physical Server
 - One secondary
 - Holds Metadata/location data for cluster
- **Compute: Logical**
 - Map/Reduce: logical process
 - Runs on each node
- **Data: Physical**
 - Spread across cluster nodes, 3x mirrored



The **NameNode** holds the location information for every file in the cluster. The file system metadata.

The **Secondary NameNode** is a backup NameNode. This is a passive node that requires the Administrator to intervene to bring it up to primary NameNode.

The **DataNode** server is where the data resides.

The primary resource management is the **Job Tracker**, which manages and assigns work to the **TaskTrackers**.

TaskTracker is a node in the cluster that accepts tasks - Map, Reduce and Shuffle operations from a Job Tracker.

Hadoop Data and Security Overview

- Getting data into Hadoop is separate step
 - Production data resides elsewhere
 - Data is read-only, non-production copy
- Hadoop Admin has R/W access to *all* Hadoop data
 - No way to delegate control to other accounts
 - No built-in data security



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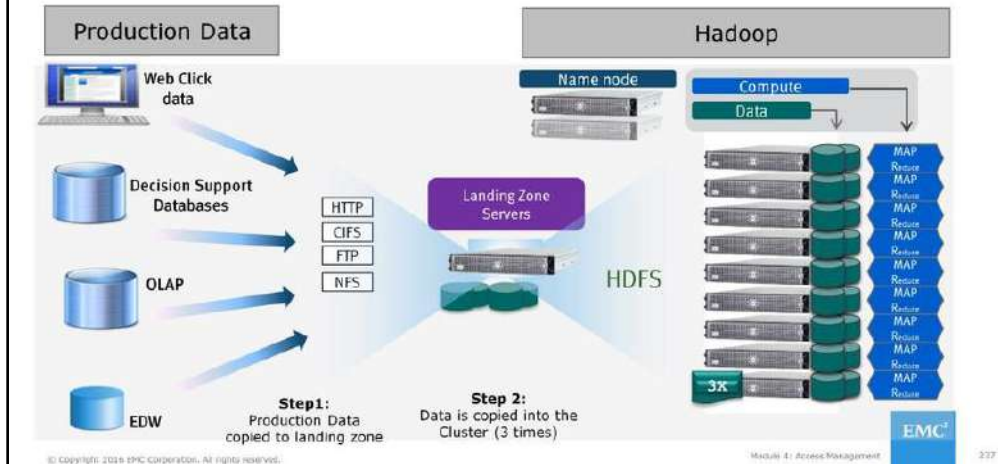
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Populating Hadoop with data can be an exercise in patience. Some distro's and 3rd party utilities can expedite moving data into Hadoop but in a survey done by IDC.

In a traditional Hadoop only environment, we have to remember that the HDFS is a read-only file system. It would be difficult to do analysis on an every changing data set so once the data is on Hadoop, it is read-only. What is the definition of a data silo? According to Wikipedia: *An information silo is a management system incapable of reciprocal operation with other, related information systems.* For example, a bank's management system is considered a silo if it cannot exchange information with other related systems within its own organization, or with the management systems of its customers, vendors, or business partners.

Hadoop, like many open source technologies, such as UNIX and TCP/IP, was not created with security in mind. Hadoop evolved from other open-source Apache projects, directed at building open source web search engines and security was not a primary consideration. There are some security features through the current implementation of Kerberos, the use of firewalls, and basic HDFS permissions. Kerberos is not a mandatory requirement for a Hadoop cluster, making it possible to run entire clusters without deploying any security.

Moving Data into Hadoop



In a traditional Hadoop cluster, the data exists in silos. Production data is maintained on productions server and then copied in some way to a Landing Zone Server, which then imports or ingests the data into Hadoop/HDFS. It is important to note that the data on HDFS is not production data; it is copied from another source, and a process must be in place to update the HDFS data periodically with the production data information.

Data Lakes and Analytics

- Data Lake brings analytics tools to your data
- Makes running analytics more efficient
- Data shared between projects with centralized control & standardized analytics tooling

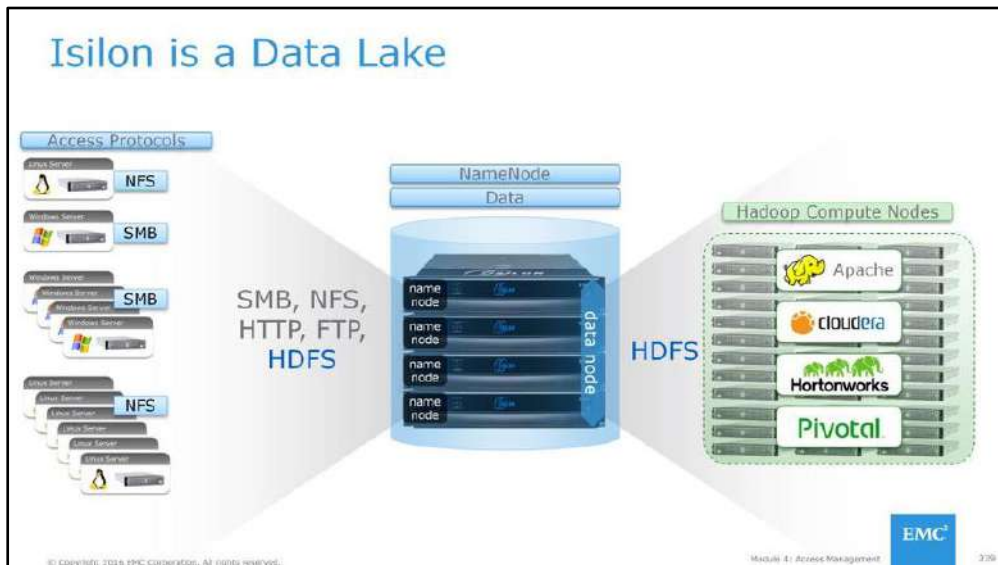


The Data Lake represents a paradigm shift away from the linear data flow model. In Module 1, we introduced the concept of Data Lake which is, most simply, a central data repository that allows you to access and manipulate the data using a variety of clients and protocols. This keeps an IT department from having to manage and maintain a separate storage solution (silo) for each type of data (i.e., SMB, NFS, Hadoop, SQL, etc.). Utilizing Isilon to hold the Hadoop data gives you all of the protection benefits of the Isilon OneFS operating systems. You can select any of the data protection levels that OneFS offers (N+1 through 8x mirroring) giving you both disk and node fault tolerance.

Data Lake based ingest let them capture a wider range of data types than were possible in the past. Data is stored in raw, unprocessed forms to ensure that no information is lost. Massively Parallel Processing and in memory technologies allow data transformation in real-time as data is analyzed. Because the Data Lake brings data sources into a single, shared repository, more tools can be made available on demand to give data scientists and analysts what they need to find insights. The data lake makes it simple to surface those insights in a consistent way to executives and managers so that decisions can be taken quickly and the inclusion of platform as a service makes building 3rd platform applications simple and efficient.

PaaS combined with new approaches like continuous integration and deployment mean that app development cycles can be in the days and weeks rather than months or years. All of which dramatically reduces the time taken from having an idea to identifying insight, taking action and creating value. A Data Lake helps IT and the business run better.

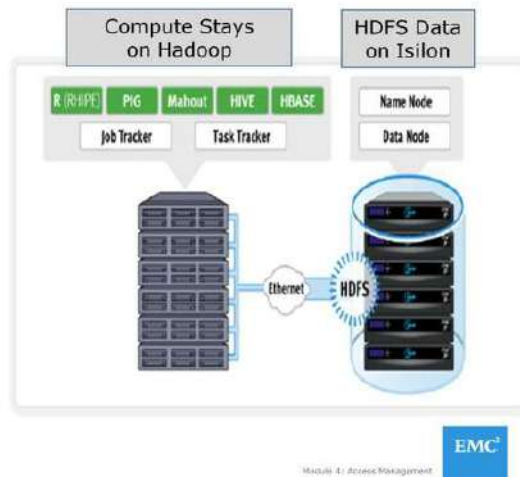
<http://www.emc.com/collateral/white-papers/h13172-isilon-scale-out-data-lake-wp.pdf>



All production data resides on Isilon so there is no need to export it out of your production applications and import it into Isilon the way that you have to with a traditional Hadoop environment. The MapReduce continues to run on dedicated Hadoop compute nodes. Isilon requires this Hadoop front-end to do the data analysis. Isilon simply holds the data so that it can be manipulate, whether by Hadoop or by using various protocols, applications, or clients to access the Hadoop data residing on Isilon.

Isilon with HDFS Data

- Isilon becomes both:
 - NameNode and DataNode
- HDFS data on Isilon
- MapReduce on Hadoop
 - Computations stays on Hadoop
 - JobTracker/TaskTracker



EMC Isilon is the only scale-out NAS platform that provides native support for the Hadoop Distributed File System (HDFS) protocol. Using HDFS as an over-the-wire protocol, you can deploy a powerful, efficient, and flexible data storage and analytics ecosystem. In addition to native integration with HDFS, EMC Isilon storage easily scales to support massively large Hadoop analytics projects. Isilon scale-out NAS also offers unmatched simplicity, efficiency, flexibility, and reliability that you need to maximize the value of your Hadoop data storage and analytics workflow investment. Combine the power of VMware vSphere Data Extension with Isilon scale-out NAS to achieve a comprehensive big data storage and analytics solution that delivers superior value.

The Isilon HDFS implementation is a lightweight protocol layer between OneFS file system and the HDFS clients. This means that files are stored in standard POSIX compatible file system on an Isilon cluster.

Hadoop Advantage Using EMC Isilon

- Data protection:
 - N+1 thru 8x mirroring, Snapshots, SyncIQ, NDMP
- In-place analytics:
 - Run analysis on production data already on Isilon
 - No need to import data or move data around network
- Multi-protocol access to HDFS data:
 - SMB, NFS, FTP, HTTP
- Data security:
 - SEDs, ACLs, Mode Bits, access zones, RBAC
- SEC Compliance using Isilon
- Simple scale and deduplication capability

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Data Protection – Hadoop does 3X mirror for data protection and had no replication capabilities. Isilon supports snapshots, clones, and replication using it's Enterprise features.

No Data Migration – Hadoop requires a landing zone for data to come to before using tools to ingest data to the Hadoop cluster. Isilon allows data on the cluster to be analyzed by Hadoop. Imagine the time it would take to push 100TB across the WAN and wait for it to migrate before any analysis can start. Isilon does in place analytics so no data moves around the network.

Security – Hadoop does not support kerborized authentication it assumes all members of the domain are trusted. Isilon supports integrating with AD or LDAP and give you the ability to safely segment access.

Dedupe – Hadoop natively 3X mirrors files in a cluster, meaning 33% storage efficiency. Isilon is 80% efficient.

Compliance and security – Hadoop has no native encryption. Isilon supports Self Encrypting Drives, using ACLS and ModeBits, access zones, RBAC, and is SEC compliant.

Multi-Distribution Support – Each physical HDFS cluster can only support one distribution of Hadoop...we let you co-mingle physical and virtual versions of any apache standards-based distros you like.

Scale Compute and Storage Independently – Hadoop pairs the storage with the compute o if you need more space, you have to pay for more CPU that may go unused or if you need more compute, you end up with lots of overhead space. We let you scale compute as needed and Isilon for storage as needed; aligning your costs with your requirements.

For additional information on in-place analytics:

<http://www.emc.com/collateral/TechnicalDocument/docu50638.pdf>

Supported Hadoop Versions

		OneFS Compatibility				
Hadoop EcoSystem		7.1.0	7.1.1	7.2.0	7.2.1	8.0.0
Ambari	1.5.1 - 1.6.0			*	*	*
	1.6.1		7.2.0.1	*	*	*
	1.7.0 - 1.7.1			7.2.0.2	*	*
	2.0.0 - 2.0.1			7.2.0.3	7.2.1.1	*
	2.0.2			7.2.0.4	7.2.1.1	*
	2.1.0			7.2.0.4	7.2.1.1	*
	2.1.1				7.2.1.1	*
	2.1.2				7.2.1.2	*
	3.0.0 - 3.3.1				7.2.1.3	8.0.0.1
	3.2.2				7.2.1.3	8.0.0.1
	2.4 (not yet supported)			support coming in future OneFS release	support coming in future OneFS release	

		OneFS Compatibility				
Hadoop EcoSystem		7.1.0	7.1.1	7.2.0	7.2.1	8.0.0
Cloudera	CDH3u2 - CDH3u6	*	*	*	*	*
	CDH4.0 - 4.7	*	*	*	*	*
	CDH5.0 - 5.1			*	*	*
	CDH5.2			7.2.0.1	*	*
	CDH5.3 - 5.4.3			7.2.0.2	*	*
	CDH5.4.4 - 5.7			7.2.0.3	7.2.1.1	*
	CDH5.8					*
	Manager 4.0	*	*	*	*	*
	Manager 5.0 - 5.2			*	*	*
	Manager 5.3 - 5.4			7.2.0.2	*	*
	Manager 5.5			7.2.0.3	7.2.1.1	*
GreenPlum						

KEY • "*" indicates support for that OneFS family. • Exact OneFS version number indicates minimum required version of OneFS. • Blank cell indicates unsupported OneFS version. • Some cells labeled to indicate future support.						
--	--	--	--	--	--	--

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OneFS supports the Hadoop distributions shown on the screen. Where provided, an exact OneFS version number indicates the minimum version of OneFS that is required. For information about how Isilon Scale-out NAS can be used to support a Hadoop data analytics workflow, visit the community information at <https://community.emc.com/docs/DOC-37101>.

Supported Hadoop Versions (cont'd)

		OneFS Compatibility				
Hadoop EcoSystem		7.1.0	7.1.1	7.2.0	7.2.1	8.0.0
Hortonworks	HDP 1.1.1 - 1.3.16	*	*	*	*	*
	HDP 2.0 - 2.1			*	*	*
	HDP 2.2			7.2.0.2	*	*
	HDP 2.3			7.2.0.4	7.2.1.1	*
	HDP 2.4				7.2.1.3	8.0.0.1
	HDP 2.5 (not yet supported)				support coming in future OneFS release	support coming in future OneFS release
IBM	Open Platform 4.0			7.2.0.4	7.2.1.1	*
	Open Platform 4.1				7.2.1.1	*

		OneFS Compatibility				
Hadoop EcoSystem		7.1.0	7.1.1	7.2.0	7.2.1	8.0.0
Hadoop	0.20.209, 0.20.206	*	*	*	*	*
	1.0.0 - 1.8.3	*	*	*	*	*
	1.2.1	*	*	*	*	*
	2.0.3	*	*	*	*	*
	2.2 - 2.4	*	*	*	*	*
	2.5			7.2.0.1	*	*
	2.6			7.2.0.2	*	*
	2.7			7.2.0.4	7.2.1.1	*
	HAAWG 1.1.4	*	*	*	*	*
	HAAWG 1.2.0.6 - 1.2.0.1			*	*	*
Pivotal	HAAWG 1.3 - 1.3.1			7.2.0.3	7.2.1.1	*
	HDB 2.0				7.2.1.3	8.0.0.1
	HD 1.1.1	*	*	*	*	*
	HD 2.0 - 2.1			*	*	*
	HD 3.0 - 3.0.1			7.2.0.3	7.2.1.1	*



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Here is a continuation of the OneFS support for the distributions and products of the Hadoop Distributed File System (HDFS). The source for this information is at <https://community.emc.com/docs/DOC-37101>.

Hadoop and OneFS 8.0

- New web administration interface
 - Full configuration options
- Improved CLI options
 - Controls HDFS settings
 - `isi hdfs`

Authentication type and root directory: Any configuration previously done via CLI now done in web administration interface



New HDFS configuration page in web administration interface

Cable HDFS and change block size

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In OneFS 8.0, the Isilon engineering team made the decision to provide a robust and scalable version of HDFS for this and all future releases. Starting in OneFS 8.0, the HDFS protocol was entirely rewritten in C++ code to increase processing, scalability, a web administration interface, as well as to add additional support for auditing, CloudPools, and SMB file filtering. With this rewrite, OneFS 8.0 has a new foundation, purpose built, to support continued future HDFS innovations.

Let's discuss some of the options on the **Settings** tab:

- The HDFS block size determines how the HDFS service returns data upon read requests from Hadoop compute client. Block size is configurable from 4KB up to 1GB, with a default of 128KB. Setting a larger block size enables nodes to read and write HDFS data in larger blocks.
- The HDFS **Authentication Type** is on a per-access zone basis. The authentication method can be Simple, Kerberos, or both.
- The Ambari client/server framework is a third-party tool that enables you to configure, manage, and monitor a Hadoop cluster through a browser-based interface.

Proxy users for secure impersonation can be created on the **Proxy Users** tab. As an example, you can create an Apache Oozie proxy user to securely impersonates a user called HadoopAdmin, allowing the Oozie user to request that Hadoop jobs be performed by the HadoopAdmin user. Apache Oozie is an application that can automatically schedule, manage, and run Hadoop jobs.

On the **Virtual Racks** tabs, nodes can be preferred along with an associated group of Hadoop compute clients to optimize access to HDFS data.

Lesson 3: Summary

Having completed this lesson, you are now able to:

- Explain Hadoop Components
- Illustrate Hadoop traditional architecture
- Examine benefits of a Data Lake
- Analyze benefits of using Isilon with Hadoop

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Having completed this lesson, you are now able to explain Hadoop components, illustrate Hadoop traditional architecture, examine benefits of a Data Lake, and analyze benefits of using Isilon with Hadoop.

Lesson 4: Object Storage in OneFS

Upon completion of this lesson, you will be able to:

- Identify differences between object and file storage
- Define benefits of object storage
- Describe Isilon implementation of Swift
- Summarize Swift best use cases

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Upon completion of the lesson, you will be able to identify differences between object and file storage, define benefits of object storage, describe Isilon implementation of Swift, and summarize Swift best use cases.

File Storage vs. Object Storage

File Storage	Object Storage
• Hierarchical structure • Manages location of data • Limited metadata • Identifies file type, create/modify date, owner	• Flat hierarchy • Data stored in containers as objects (files) • Not concerned with data location on disk • Rich metadata includes: • Content: what file contains • Context: how often it's accessed, should it be archived, is it active

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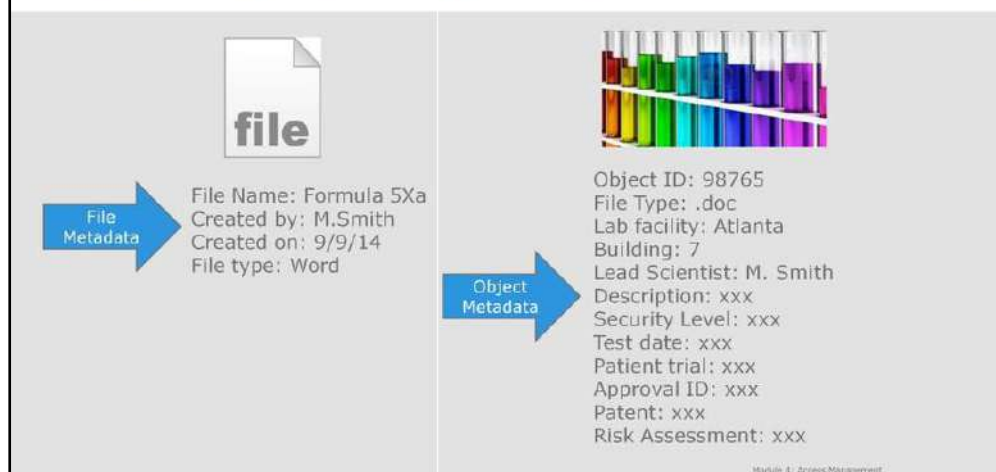
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File storage was developed to deal with a specific set of users who required shared access to a specific set of files. This need led to file access permissions and file locking mechanisms, which allows users to share files and make modifications to files without effecting each other's changes. A file system stores its data in a hierarchy of directories, subdirectories, folders, files. The file system manages the location of the data within the hierarchy; if you want to access a specific file you need to know where to look for the file. Queries to a file system are limited and you might be able to search for a specific type of file *.doc or the name of a file serverfile12*.* but you lack the ability to parse through the files to find out the content contained within them. It is also difficult to determine the context of a file. For example, should it be stored in an archival tier or will this information need to be accessed on a regular basis? It is also hard to determine the content of the data from the limited metadata provided. A document might contain the minutes of a weekly team meeting or it could contain confidential personal performance evaluation data.

Object storage combines the data with a richly populated metadata allowing information to be searched by both what is contained within the file, and how the file should be managed within the system. Instead of a file that tells you the create or modified date, file type and owner, you can have metadata that perhaps tells you the project name, formula results, personnel assigned, location of test and next run date. The rich metadata of an object store allows applications to run analytics against the data. Object storage has a very flat hierarchy and stores its data within containers as individual object. An object storage platform can store billions of objects within its containers and each object can be accessed with a URL. The URL associated with a file allows the file to be located within the container; hence, the path to the physical location of the file on the disk is not required.

Object storage is well suited for workflows with static file data and/or cloud storage.

File vs. Object Storage: Metadata Difference



File systems have metadata that is limited in its depth of information. When accessing a file, you may have the file name, the owner, the create date and the type of file.

In contrast, object-based storage deals with rich, fully-populated metadata allowing for granular description of both the content and the type of storage that it requires, such as archive or regularly accessed.

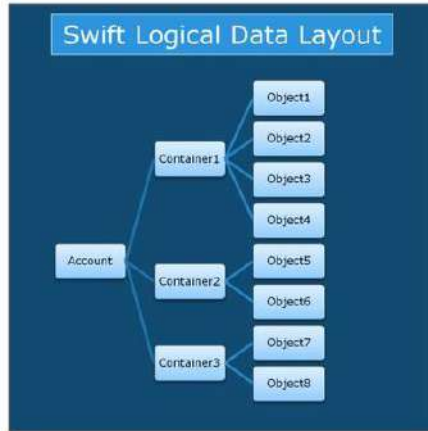
Isilon Swift Implementation

- Hybrid between File and Object storage
- Provides rich metadata with hierarchical file structure
- Integration with OneFS features and multiple protocol access
- Swift API runs as a protocol driver on each node
 - HTTP requests sent to web server on port 28080
 - HTTPS requests proxied via Apache web server on port 8083
 - Ports are not configurable



Isilon Swift is a hybrid between the two storage types, storing Swift metadata as an Alternative Data Stream. It provides the rich metadata of object storage with the hierarchical structure of file system storage. This allows integration with OneFS and the features it supports, such as the other protocols like NFS, SMB, etc., and the data management features, such as deduplication, snapshots, etc. OneFS exposes the Swift API through a Swift protocol driver. An instance of this protocol driver runs on each node in the cluster and handles the API requests. The Swift API is implemented as a set of Representational State Transfer (REST) web services over HTTP or secure HTTP (HTTPS). Because the Swift API is considered as a protocol, content and metadata can be ingested as objects and concurrently accessed through protocols configured on the EMC Isilon cluster. Isilon Swift attempts to provide the best of both worlds; the best of Swift Object-based Storage and the best of EMC Isilon's OneFS. HTTP requests are sent to an internal web server listening on port **28080**. HTTPS requests are proxied through the Apache web server listening on port **8083**.

Accounts, Containers, Objects



Item	Definition
Accounts	Administrative control point
Containers	Organize objects
Objects	Contain user data



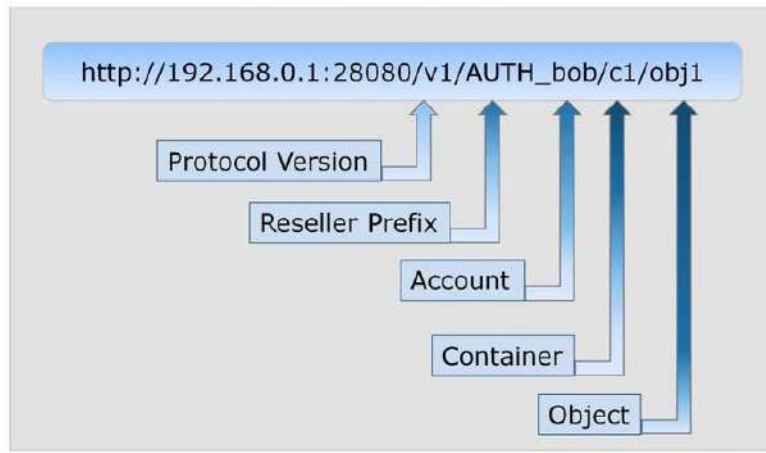
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Let's take a moment and identify accounts, containers, and objects for those who may be unfamiliar with the Swift hierarchy.

Accounts are the administrative control point for containers and objects, Containers organize objects, and Objects contain user data. For users to access objects, they must have an account on the system. An account is the top of the hierarchy.

Swift Terminology: Storage URL



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For those not familiar with Swift terminology, this slide displays what a Swift Storage URL looks like.

The protocol version **/v1** is defined by OpenStack.

The reseller prefix **/AUTH_bob**, where **/AUTH** is a vestige of the OpenStack implementation's internal details that we see leaking into the interface.

The **_bob** portion of the URL is the actual account name that we are using.

The container **/c1** is the container in which an object is stored and the object **/obj1** is the actually object.

Isilon Swift Limitations

- 150 concurrent active connections per node
- Cannot submit PUT request for 0 length object
- Can't delete container unless it's empty
- User name must include FQDN of AD domain



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Isilon Swift supports up to 150 concurrent active connections per EMC Isilon node. When uploading objects or listing containers, the Isilon Swift service can become memory constrained. This will cause a service outage and can affect the client access and performance. To avoid an outage, maintain the Swift Service memory load within 384 MB. Account and container listing requests initiate a full file system walk from the requested entity. Workloads can expect longer response times during the listing operations as the number of containers or objects increase. To prevent response time issues, we recommend that you redistribute or reduce the objects and containers until the response times are within the acceptable limits.

You cannot submit a PUT request to create a zero-length object because PUT is incorrectly interpreted as a pseudo-hierarchical object.

You cannot submit a DELETE request to delete a container if the container is not empty. As a best practice, delete all the objects from the container before deleting the container.

When authenticating with Active Directory (AD) and Isilon Swift, the user name in the X-Auth-User header must include the fully-qualified AD domain name in the form test-name@mydomain.com unless the domain has been configured as the default through the assume-default-domain configuration parameter in the AD provider's configuration.

New in OneFS 8.0

- **Swift Discoverability describes:**
 - Swift storage service that clients connect to
 - Details what service supports
- **Account Support**
 - Allows OneFS to support multi-tenant accounts



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One feature that is very important in a Swift release, especially consumers of the OpenStack protocol, the *Swift Discoverability* describes the Swift storage service that a client is connected to and what that service supports.

Account support allows us to support multi-tenant accounts and get the current Swift define from accounts in home directories to relocate them to be more flexible in how we use them and what we can support.

OneFS 8.0: File System Layout Changes



OneFS 7.2.1 and Prior

- User data in user's home directory
`/ifs/home/<username>/container1/obj1`
- No differentiation between Swift created containers and other containers in user's home directory

OneFS 8.0

- User's data relocated to:
`/<zoneroot>/isi_lwSwift/<useraccount>/container1/obj1`
- All containers and objects in path only created by Swift
- Access to Swift accounts based on identity of specific user



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In OneFS 7.2.1 and earlier, user data was located in the users home directory as shown on the slide, and there was no differentiation between Swift created containers and other containers that were located in the users home directory.

Now, in OneFS 8.0, user data can be found in the `/<zoneroot>/isi_lwSwift` directory, which you can see on the right-hand side of the slide, and all containers and objects in this path are *only* created by Swift. Access to Swift accounts is granted based on the identity and permissions for a specific user.

Containers are assigned to Swift accounts. Objects that store user data reside within containers which are first-level directories below the account directories. Objects are identified by URIs in the form **`http://example.com/v1/account/container/object`**. In this example:

- `example.com` identifies the cluster
- `v1` is the protocol version
- `/account/container/object` is the object storage location

OneFS Swift Provisioning Differences



OneFS 7.2.1 and Prior

- Enabled Swift license and anyone could use
- No administrative control

OneFS 8.0

- Must provision accounts before use:
 1. Enable Swift license
 2. Decide file system user or group ownership
 3. Create account with `isi swift` command
 4. Assign users access to new account
 5. Modify file system permission changes, if necessary



In OneFS 7.2.1 and prior releases, you turned on the Swift license and that all; administrators had no idea what users or accounts were provisioned, who was using the service, nor what they were doing with it. The service was on for anyone to use whether the administrator wanted them to use it or not.

In OneFS 8.0, administrators must provision the accounts before users can use the service, and there are additional steps required in order to get users:

1. Enable Swift license
2. Decide upon file system user or group ownership
3. Create accounts using the **isi swift** command
4. Assign users access to newly created account
5. Make any necessary file system permission changes if you are relocating data into the account.

OneFS 8.0 Swift Upgrade Considerations

- Old accounts deactivated
 - If data existed in users home directory, Admin must:
 - Provision new account
 - Move data from user home directory to new account
 - Swift will not look for data in user's home directory locations
- Something to consider prior to upgrade

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If a customer is using Swift and plans on upgrading to OneFS 8.0, there is some upgrade planning that needs to be done.

Any user currently using Swift will have their old account deactivated as Swift will no longer look in the user's home directory for an account. A plan needs to be put into place to determine which users are using Swift, creating a new account for them under the new Swift path, and then moving the data from their old account into the newly provisioned one.

Swift and OneFS Feature Interoperability

- Works with most major OneFS 8.0 features, including:
 - Access zones
 - SmartConnect
 - Dedupe
 - SyncIQ
 - Server Side Copy
 - etc.
- Not compatible with:
 - Audit filter



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Swift functions with all the major OneFS 8.0 features including, but not limited to, access zones, SmartConnect, Dedupe, SyncIQ, etc.

Currently, the OneFS 8.0 implementation is not compatible with the auditing feature.

Swift Use Cases and Benefits

- Use cases

- Dissimilar protocol storage consolidation
- Automate data collection, storage, and management for data analysis
- Automate data-processing applications

- Benefits

- Secure multi-tenancy
- Leverage cluster features
- Balance work load across all cluster nodes
- Efficient object storage

Ref: Isilon Swift Technical Note

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Listed here are Swift use cases and benefits. Swift enables storage consolidation for applications regardless of protocol, which can help eliminate storage silos. In environments with petabytes of unstructured data, Swift can automate the collection, store and manage the data, such as in a data lake, for later analysis. Swift can be used to automate data-processing applications to store objects on an Isilon cluster and analyze the data with Hadoop through the OneFS HDFS.

Swift benefits include secure multi-tenancy for applications through access zones while uniformly protecting the data with enterprise storage capabilities such as authentication, access control, and identity management. Manage data through enterprise storage features such as deduplication, replication, tiering, performance monitoring, snapshots, and NDMP backups. Swift balances the work load across all of the nodes in a cluster through OneFS SmartConnect and stores object data more efficiently with forward error correction instead of data replication.

Lesson 4: Summary

Upon completion of the lesson, you should be able to:

- Identify differences between object and file storage
- Define benefits of object storage
- Describe Isilon implementation of Swift
- Summarize Swift use cases

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Module 4: Access Management



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After completing this lesson, you should be able to identify differences between object and file storage, define benefits of object storage, describe Isilon implementation of Swift, and summarize Swift use cases.

Module 4: Summary

Having completed this module, you are now able to:

- Identify best practices for access zones
- Describe File Filtering
- Explain authentication structure
- Detail Directory Service configuration
- Establish benefits of using Isilon with Hadoop
- Understand Isilon's implementation of Swift

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Module 4: Access Management



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Having completed this module, you are now able to identify best practices for access zones, describe File Filtering, explain authentication structure, detail Directory Service configuration, establish benefits of using Isilon with Hadoop, and understand Isilon's implementation of Swift.

Lab 4: Access Management

- Synchronizing NTP services
- Connecting to LDAP domain
- Joining Active Directory domain
- Creating access zones



Module 4: Access Management

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In this lab, you'll synchronize NTP services with an Active Directory server, connect to an LDAP domain and Active Directory domain, and create access zones.

Module 5: User Authentication and File Access

Upon completion of this module, you should be able to:

- Understand Isilon identity management
- Explain OneFS permissions mapping
- Configure Isilon protocol access
- Plan OneFS multiprotocol implementations

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Module 5: User Authentication and File Access



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Upon completion of this module, you will know how OneFS deals with user identities, permissions and how protocols afford user access to the cluster.

Lesson 1: Managing User Identity Mapping

Upon completion of this lesson, you should be able to:

- Understand OneFS user identity tracking
- Manage OneFS identity mapping

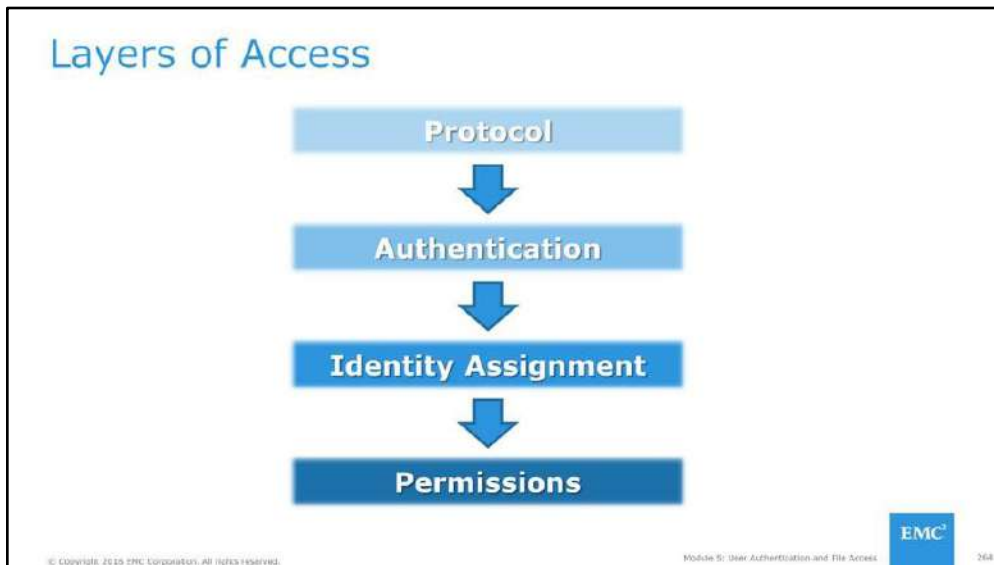


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Module 5: User Authentication and File Access

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Upon completion of this lesson, you will know how OneFS establishes user identities, and how multiple identities are reconciled to provide a consistent user experience.



Interactions with an Isilon cluster have four layers in the process.

The first layer is the protocol layer. This may be Server Message Block, or SMB; Network File System, or NFS; File Transfer Protocol, or FTP; or some other protocol but this is how the cluster is actually reached.

The next layer is authentication. The user has to be identified using some system, such as NIS, local files, or Active Directory.

The third layer is identity assignment. Normally this is straightforward and based on the results of the authentication layer, but there are some cases where identities have to be mediated within the cluster, or where roles are assigned within the cluster based on a user's identity. We will examine some of these details later in this module.

Finally, based on the established connection and authenticated user identity, the file and directory permissions are evaluated to determine whether or not the user is entitled to perform the requested data activities.

Identity Management



Simply put, OneFS's identity management maps the users and groups from separate services in order to provide a single unified identity on a cluster and uniform access control to files and directories, regardless of the incoming protocol. This illustration shows the authentication providers OneFS uses to first verify a user's identity after which users are authorized to access cluster resources. The top layer are access protocols – NFS for UNIX clients, SMB for Windows clients, and FTP and HTTP for all. Between the protocols and the lower level services providers and their associated data repositories is the Isilon Isassd daemon. The Isassd daemon mediates between the authentication protocols used by clients and the authentication providers in the third row, who check their data repositories, represented on the bottom row, to determine user identity and subsequent access to files.

When the cluster receives an authentication request, Isassd searches the configured authentication sources for matches to an incoming identity. If the identity is verified, OneFS generates an access token. This token is not the same as an Active Directory or Kerberos token, but an internal token which reflects the OneFS Identity Management system. When a user attempts to access cluster resources, the system allows or denies access based on matching the identity, user, and group memberships to this same information on the file or folder.

Access Token Generation

Step	Process	Description
1	User identity lookup	Using the initial identity, the user is looked up in all configured authentication providers in the access zone, in the order in which they are listed, until a match is found. The user identity and group list are retrieved from the authenticating provider. Any SIDs, UIDs, or GIDs are added to the initial token. Note An exception to this behavior occurs if the AD provider is configured to call other providers, such as LDAP or NIS.
2	ID mapping	The user's identifiers are associated across directory services. All SIDs are converted to their equivalent UID/GID and vice versa. These ID mappings are also added to the access token.
3	User mapping	Access tokens from other directory services are combined. If the username matches any user mapping rules, the rules are processed in order and the token is updated accordingly.
4	On-disk identity calculation	The default on-disk identity is calculated from the final token and the global setting. These identities are used for newly created files.

Ref: Isilon OneFS 8.0.0 Web Administration Guide
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Module 5: User Authentication and File Access



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Access tokens form the basis of who you are when performing actions on the cluster and supply the primary owner and group identities to use during file creation. For most protocols, the access token is generated from the username or from the authorization data that is retrieved during authentication. Access tokens are also compared against permissions on an object during authorization checks. The access token includes all identity information for the session. OneFS exclusively uses the information in the token when determining if a user has access to a particular resource. The table shows the simplified overview steps of the complex process through which an access token is generated.

Supported Primary Identities

- **UID**
 - UNIX-based user identifier
 - Local, LDAP, file & NIS
- **GID**
 - UNIX-based group identifier
 - Local, LDAP, file & NIS
- **SID**
 - Active Directory-based
 - Users, groups, & other AD objects



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Module 5: User Authentication and File Access

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OneFS supports three primary identity types, each of which can be stored directly on the file system. These identity types are used when creating files, checking file ownership or group membership, and performing file access checks.

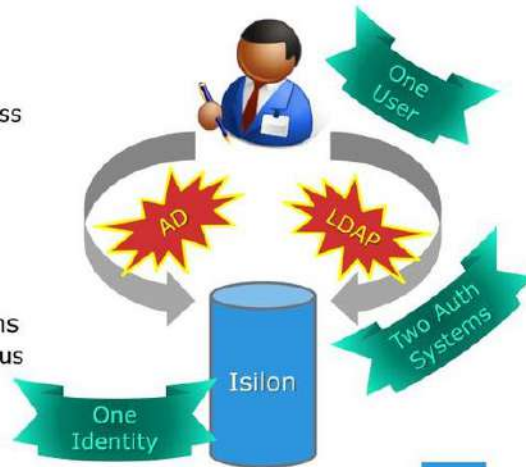
The identity types supported by OneFS are:

- **User identifier, or UID**, is a 32-bit string that uniquely identifies users on the cluster. UIDs are used in UNIX-based systems for identity management.
- **Group identifier, or GID**, for UNIX serves the same purpose for groups that UID does for users.
- **Security identifier, or SID**, is a unique identifier that begins with the domain identifier and ends with a 32-bit relative identifier (RID). Most SIDs take the form S-1-5-21-**<A>**-****-**<C>**-**<RID>**, where **<A>**, ****, and **<C>** are specific to a domain or computer, and **<RID>** denotes the object inside the domain. SID is the primary identifier for users and groups in Active Directory.

The Identity (ID) mapping service maintains relationship information between mapped Windows and UNIX identifiers to provide consistent access control across file sharing protocols within an access zone.

Multiple Identities

- Goal:
 - Treat users uniformly regardless of how they reach cluster
- Integrated environments
 - Corporate mergers
 - Platform migrations
 - Multiplatform environments
- Multiple authentication platforms
 - Homogeneous or heterogeneous
 - Active Directory
 - LDAP, Kerberos, NIS



Although there are multiple ways to authenticate users to the same cluster, the aim is to treat users uniformly regardless of how they reached the cluster. Whether the case is a team of developers who have Windows, Apple, and UNIX operating systems on each desktop, or internal and external sales networks, which are being integrated into a uniform authentication scheme, or two entire corporations which are merging and therefore combining their IT infrastructure, the need is to provide a consistent and uniform mapping of user identities externally to user identities that Isilon uses internally.

This does not apply to a forest of mutually trusting Active Directory servers, because user identification is handled within AD in this scenario so there is no need for the Isilon cluster to perform any disambiguation.

Isilon handles multiple user identities by mapping them internally to unified identities.

ID Mapper Database

- User mapper vs. ID mapper
- Mappings combine different identity types
 - Stored in ID mapper
 - Pair of individual one-way mappings:
 - `UID:10000 -> SID:S-1-5-21-1-2-3-4`
 - Reverse mappings:
 - `SID:S-1-5-21-1-2-3-4 -> UID:10000`
- Built by ID provider
 - Sits between lsassd and other providers
 - Routes authentication requests



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Module 5: User Authentication and File Access

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The User mapper and OneFS ID mapper differ. User mapping provides a way to control permissions by specifying a user's security identifiers, user identifiers, and group identifiers. OneFS uses the identifiers to check file or group ownership. With the user mapping feature, you can apply rules to modify which user identity OneFS uses, add supplemental user identities, and modify a user's group membership. The user mapping service combines a user's identities from different directory services into a single access token and then modifies it according to the rules that you create.

Mappings are stored in a cluster-distributed database called the ID mapper. The ID provider builds the ID mapper based on incoming source and target identity type—UID, GID, or SID. Only authoritative sources are used to build the ID mapper.

Each mapping is stored as a one-way relationship from source to destination. If a mapping is created, or exists, it has to map both ways, and to record these two-way mappings they are presented as two complementary one-way mappings in the database. When an identity request is received, if a mapping already exists between the specified source and the requested type, that mapping is returned.

Identity Mapping Types

- Algorithmic (UNIX SID)
 - Convert UNIX SIDs to UID or GID
 - S-1-22-1-1234 -> 1234
- External
 - Locate on-disk identity type in external directory service
- Manual
 - `isi auth mapping create`
 - `isi auth mapping dump`
 - `isi auth mapping delete`
 - `isi auth mapping token`
- Automatic
 - Generate identity required



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Algorithmic mappings are created by adding a UID or GID to a well-known base SID, resulting in a "UNIX SID." These mappings are not persistently stored in the ID mapper database. For example, if the UNIX SID was S-1-22-1-1234 -> 1234 (with 1234 as the REAL UID), the well-known base SID of S-1-22-1 would be stripped out and the REAL UID of 1234 would be set as the on-disk identity.

External mappings are derived from identity sources outside of OneFS. For example, Active Directory can store a UID or GID along with an SID. When retrieving the SID from AD, the UID/GID is also retrieved and used for mappings on OneFS.

Manual mappings are set explicitly by running the **isi auth mapping** command at the command-line. Manual mappings are stored persistently in the ID mapper database. The **isi auth mapping new** command allocates a mapping between a source persona and a target type (UID, GID, SID, or principal). If a mapping already exists to that type, it will be returned; otherwise, a mapping is created using the current rules. The **isi auth mapping dump** command dumps the kernel mapping database. The **isi auth mapping list** command lists the mappings for UIDs, GIDs or SIDs.

The **isi auth mapping token** command includes options for displaying a user's authentication information by a list of parameters including user name and UID. This allows for detailed examination of identities on OneFS.

Automatic mappings are generated if no other mapping type can be found. In this case, a SID is mapped to a UID or GID out of the default range of 1,000,000-2,000,000. This range is assumed to be otherwise unused and a check is made only to ensure there is no mapping from the given UID before it is used.

Identity Search Order

- Incoming SID needs UID/GID
- Search for following (in order)
 1. If source has UID/GID, use it
 - Look for uidNumber attribute in AD (SFU)
 2. Check existing mappings in ID mapper database
 3. Default name lookup steps
 - Normalize name to lowercase
 - Search for matching name in local, LDAP, and/or NIS or users
 - If successful, create external mapping
 4. Allocate UID or GID



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On new installations and re-imaging, the on-disk identity is set to Native, which is likely to be best identity for a network that has UNIX and Windows clients. If an incoming authentication request comes in, the authentication daemon attempts to find the correct UID/GID to store on disk by checking for the following ID mapping types in this specified order:

1. If the source has a UID/GID, use it. This occurs when incoming requests from AD has Services for NFS or Services for UNIX installed. This service adds an additional attribute to the AD user (**uidNumber** attribute) and group (**gidNumber** attribute) objects. When you configure this service, you identify from where AD will acquire these identifiers.
2. Check if the incoming SID has a mapping in the ID mapper.
3. Try name lookups in available UID/GID sources. This can be a local, or sam.db, lookup, as well as LDAP, and/or NIS directory services. By default, external mappings from name lookups are not written to the ID mapper database.
4. Allocate a UID/GID.

You can configure ID mappings on the **Access** page. To open this page, expand the **Membership & Roles** menu, and then click **User Mapping**. When you configure the settings on this page, the settings are persistent until changed. The settings in here can however have complex implications, so if you are in any doubt as to the implications, the safe option is to talk to Isilon Support staff, and establish what the likely outcome will be.

Secondary Identifiers

- Names in UNIX
 - Unique, case-sensitive namespaces for users and groups
- Names in Windows
 - Single namespace for all objects
 - Not case-sensitive
 - Specifies prefix to target AD domain
 - All names have format similar to email addresses



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Module 5: User Authentication and File Access

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UIDs, GIDs, and SIDs are primary identifiers of identity. Names, such as usernames, are classified as a secondary identifier. This is because different systems such as LDAP and Active Directory may not use the same naming convention to create object names and there are many variations in the way a name can be entered or displayed. Some examples of this include the following:

- UNIX assumes unique case-sensitive namespaces for users and groups. For example, "Name" and "name" can represent different objects.
- Windows provides a single namespace for all objects that is not case-sensitive, but specifies a prefix that targets a specific Active Directory domain. For example domain\username.
- Kerberos and NFSv4 define principals, which requires that all names have a format similar to email addresses. For example name@domain.

As an example, given the username "Petre" and the domain of EMC.COM, the following would be valid names for a single object in Active Directory: Petre, EMC\Petre, and Petre@EMC.COM.

In an Isilon cluster, whenever a name is provided as an identifier, the correct primary identifier of UID, GID, or SID is requested.

The administrator can configure the ID mapping system to record mappings based on names, but it is not the default setting.

On-Disk Identity

- Identifies preferred identity to store on disk (per user/group)
- Determines identity stored in ACLs
 - Can contain SIDs or UID/GIDs
- Three choices:
 - UNIX
 - SID
 - NATIVE



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OneFS uses an on-disk identity store a single identity for users and groups. Using on-disk identities, you can choose whether to have the UNIX or Windows identity stored automatically, or allow the system to determine the correct identity to store. Even though OneFS creates a user token that includes identities from other management systems, OneFS stores an authoritative version of this identity as the preferred on-disk identity. The on-disk identity types are UNIX, SID, and Native. Although you can change the type of on-disk identity, the native identity option is likely to be the best for a network with UNIX and Windows systems. In native mode, OneFS favors setting the UID as the on-disk identity because doing so improves NFS performance. OneFS stores only one type of identifier—either a UID and a GID or a SID—on disk at a time. Therefore, it is important to choose the preferred identity to store on disk because most protocols will require some level of mapping to operate correctly.

On-Disk Identity Types

On-Disk Identity Types

- Assumes mixed environment: locate both ID types
- Default (newly installed cluster)

UNIX-only environment:

- store UID/GIDs
- Search for UID/GID of incoming SID
- Stores SID only if no real UID/GID found

Windows-only environment:

- store SIDs
- Searches for SID of incoming UID/GID

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The available on-disk identity types are UNIX, SID, and Native. This setting is in the web administration interface on the **Access > Settings** page.

- If the **UNIX** on-disk identity type is set, the system always stores the UNIX identifier, if available. During authentication, the system authentication `lsassd` daemon looks up any incoming SIDs in the configured authentication sources. If a UID/GID is found, the SID is converted to either a UID or GID. If a UID/GID does not exist on the cluster, whether it is local to the client or part of an untrusted AD domain, the SID is stored instead. This setting is recommended for NFSv2 and NFSv3, which use UIDs and GIDs exclusively.
- If the **SID** on-disk identity type is set, the system will always store a SID, if available. During the authentication process, `lsassd` searches the configured authentication sources for SIDs to match to an incoming UID or GID. If no SID is found, the UNIX ID is stored on-disk.
- If the **Native** on-disk identity is set, the `lsassd` daemon attempts to locate the correct identity to store on disk by running through each of the ID mapping methods. The preferred object to store is a real UNIX identifier. If a real UNIX identifier is found, it will be used. If a user or group does not have a real UNIX identifier (UID or GID), it will store a real SID. This is the default setting in OneFS 6.5 and later.

If you upgrade from a previous version of OneFS, by default the on-disk-identity is **UNIX**. For new installations or re-imaging, the default on-disk identity type is **Native**.

Lesson 1: Summary

Having completed this lesson, you should now be able to:

- Understand OneFS user identity tracking
- Manage OneFS identity mapping



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Having completed this lesson you should now be able to administer OneFS identity mapping and manage multiprotocol identity conflicts.

Lesson 2: Controlling User Access to File Data

Upon completion of this lesson, you should be able to:

- Differentiate Windows and POSIX permissions
- Understand mapping different permissions
- Relate share permissions to file permissions
- Configure OneFS permissions management

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Upon completion of this lesson, you will understand Windows ACLs and POSIX mode bit permissions differ from each other, as well as how OneFS translates between them when necessary.

Permissions Overview

- Can support NFS and SMB protocols
- Authoritative version stored on disk
- Internal representation also stored
 - No more restrictive than visible permissions
 - Generates synthetic ACL
 - To support additional protocol
- Permissions translation
 - OneFS approximately matches ACLs and Mode Bits
 - Approximated permissions a unique OneFS representation
 - Same file access regardless of connection protocol
 - Access same directories and files with different clients



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Like identities, OneFS also stores permissions on disk. However, storing permissions is more complex than storing identities because each data access protocol uses its own permissions model. To support this, OneFS must not only store an authoritative version of the original permissions for the file sharing protocol that stored the file, but also map the authoritative permissions to a form that is acceptable to the other protocol. OneFS must do so while maintaining the file's security settings and meeting user expectations for access. The result of the transformation preserves the intended security settings on the files and ensures that users and applications can continue to access the files with the same behavior.

To handle cross-protocol file access, OneFS stores an internal representation of the permissions of a file system object, such as a directory or a file. The internal representation, which can contain information from either the POSIX mode bits or the ACLs, is based on RFC 3530, which states that a file's permissions must not make it appear more secure than it really is. The internal representation can be used to generate a synthetic ACL, which approximates the mode bits of a UNIX file for an SMB client. Because OneFS derives the synthetic ACL from mode bits, it can express only as much as permission information as mode bits can and not more.

Because each access protocol can process only its native permissions, OneFS transforms its representation of the permissions into a shape that the access protocol can accept. But because there is no one-to-one mapping between the permissions models of the two protocols, there are some subtle differences in the way the security settings map across protocols. Because the ACL model is richer than the POSIX model, no permissions information is lost when POSIX mode bits are mapped to ACLs. When ACLs are mapped to mode bits, however, ACLs must be approximated as mode bits and some information may be lost.

The rules that Isilon developed were influenced by two documents: RFC 3530, Network File System (NFS) version 4 Protocol, at <http://www.ietf.org/rfc/rfc3530.txt> and a network working group internet draft on mapping between NFSv4 and POSIX draft ACLs at <http://www.citi.umich.edu/projects/nfsv4/rfc/draft-ietf-nfsv4-acl-mapping-03.txt>.

Detailed, updated tables of Isilon's permissions mappings are available from the Isilon support page, as well as from Isilon's Support. If the permissions are not behaving as expected, Isilon's Support staff can help clarify what may be occurring.

Authorization Process

- Access token generated when establishing connection
 - Contains user identity information
 - According to identity mapping process
- Compare access token to file authorization data
 - Can be ACLs or mode bits
 - Identifiers (SID, UID, GID) included
- Optimized for mixed environment
- Access rights enforced consistently across access protocols
- Traditional UNIX permissions by default



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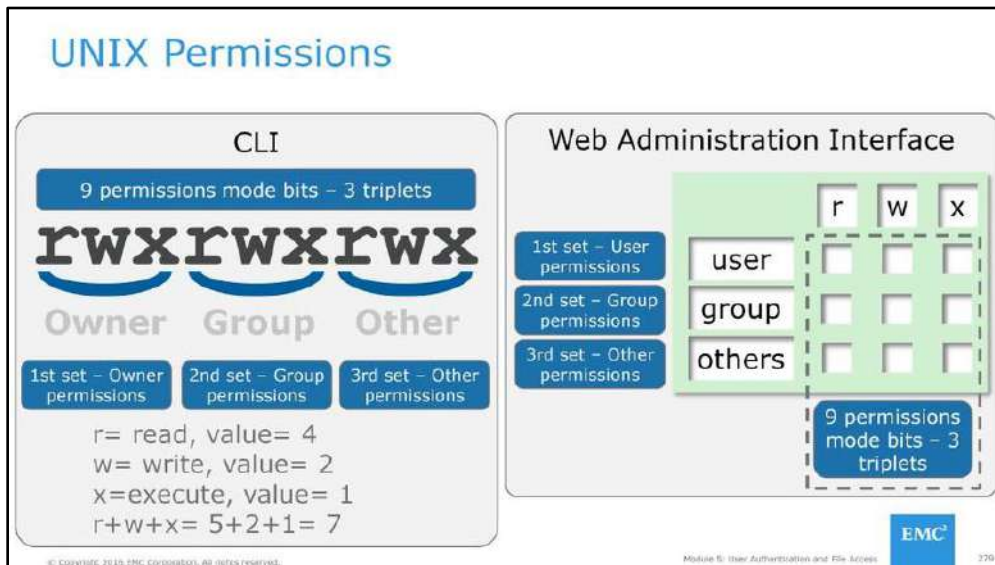
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During file access authorization, OneFS compares the access token presented during the connection with the authorization data found on the file. All user and identity mapping occurs during token generation, so no mapping is performed when evaluating permissions.

OneFS supports two types of authorization data on a file: access control lists (ACLs) and UNIX permissions. Generally, files that are created in a directory that has an ACL or over SMB, receive an ACL. Otherwise, OneFS relies on the POSIX mode bits that define UNIX permissions. In either case, the owner is represented by a UNIX user or group identifier (UID or GID), or by a Windows identifier (SID). A group can be represented only by a GID or SID. Although mode bits are present when a file has an ACL, those bits are provided only for protocol compatibility and are not used for access checks. If required to evaluate a UNIX permission against a file with an ACL, OneFS converts the permissions into the corresponding rights that the caller must possess.

By default, OneFS is configured with the optimal settings for a mixed UNIX and Windows environment. If necessary, you can configure ACL policies to optimize for UNIX or Windows. Regardless of the security model, access rights are enforced consistently across access protocols. A user is granted or denied the same rights to a file when using SMB, or Windows file sharing, as they would when using NFS, or UNIX file sharing. Clusters running OneFS support a set of global policy settings that enable you to customize the default ACL and UNIX permissions settings to best support your environment. By default, OneFS will use ordinary POSIX/UNIX permissions, but those can be replaced by setting an ACL on the file or directory in question.



In a UNIX environment, you modify permissions for users/owners, groups, and others (everyone else who has access to the computer) to allow or deny file and directory access as needed. These permissions are saved in 16 bits, which are called mode bits. You configure permission flags to grant read (r), write (w), and execute (x) permissions to users, groups, and others in the form of permission triplets.

The lower 9 bits are grouped as three 3-bit sets, called triplets, which contain the read (r), write (w), and execute (x) permissions for each class of users (owner, group, other). You set permissions flags to grant permissions to each of these classes.

Assuming the user is not **root**, the class is used to determine if the requested access to the file should be granted or denied. The classes are not cumulative. The first class that is matched is used. Therefore, it is common practice to grant permissions in decreasing order, with the highest permissions given the file's owner and the lowest to users who aren't the owner or the owning group.

The information in the upper 7 bits can also encode what can be done with the file, although it has no bearing on file ownership. An example of such a setting would be the so-called "sticky bit".

OneFS does not support POSIX ACLs, which are different from Windows ACLs.

You can modify the user and group ownership of files and directories, and set permissions for the owner user, owner group, and other users on the system. You can view or modify UNIX permissions in the web administration interface by navigating to the **File System > File System Explorer** page, and selecting the **View/Edit** option for a file or directory. A representation of the **Permissions** section is shown above. You can select or clear the boxes to assign read, write, or execute permissions to the specified account owner (user), group members (group), and anyone (other). To apply setting changes, click **Save Changes**.

OneFS supports the standard UNIX tools for changing permissions, **chmod** and **chown**. The change mode command, **chmod**, can change permissions of files and directories. All options are documented in the man page for **chmod**. It is important to note that changes made using **chmod** can affect Windows ACLs.

The **chown** command is used to change ownership of a file. You must have root user access to change the owner of a file. The basic syntax for **chown** is **chown [-R] newowner filenames**. **Newowner** can be a user or group and can be identified using the account name or ID. The **-R** option applies the new ownership to subdirectories.

Windows Permissions

- Access Control List (ACL)
 - List of Access Control Entries (ACEs)
 - ACE describes:
 - User or group
 - Deny or allow permission
- Rules to remember
 - No Permission = No Access
 - Inherited vs. Explicitly assigned permissions
 - Deny vs. Allow permissions



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In Windows environments, file and directory access rights are defined in Windows Access Control List, or ACL. A Windows ACL is a list of access control entries, or ACEs. Each entry contains a user or group and a permission that allows or denies access to a file or folder. While you can apply permissions for individual users, Windows administrators usually use groups to organize users, then assign permissions to groups instead of individual users. Group memberships can cause a user to have several permissions to a folder or file.

Windows includes many rights that you can assign individually or you can assign a set of rights bundled together as a permission. For example, the Read permission includes the rights to read and execute a file while the Full Control permission assigns all user rights including the right to change ownership and change the assigned permissions of a file or folder.

When working with Windows, you should remember a few important rules that dictate the behavior of Windows permissions. First, if a user has no permission assigned in an ACL, then the user has no access to that file or folder. Second, permissions can be explicitly assigned to a file or folder and they can also be inherited from the parent folder.

By default, when a file or folder is created, it inherits the permissions of the parent folder. If a file or folder is moved, it retains the original permissions. You can view security permissions in the properties of the file or folder in Windows Explorer. If the checkboxes in the **Permissions** dialog are not available (grayed out), those permission are inherited. You can explicitly assign a permission. It is important to remember that explicit permissions override inherited permissions.

The last rule to remember is that Deny permissions take precedence over Allow permissions. However, an inherited Deny permission is overridden by an explicit Allow permission.

Windows-style ACLs

- More complex than mode bits
 - Can express more permission permutations
 - Define more permissions types
 - More granular
- Comprises one or more ACEs for user or group
- Contains set of rights and inheritance bits
- Can deny or grant access for given identity
 - Normally a Windows SID
 - OneFS allows POSIX UIDs and GIDs as identifiers



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ACLs are more complex than mode bits and are also capable of expressing much richer sets of access rules. However, not all POSIX mode bits can be represented by Windows ACLs any more than POSIX mode bits can represent all Windows ACL values. A Windows ACL is composed of one or more access control entries, or ACEs, each representing the security identifier, or SID, of a user or a group as a trustee. Each ACE in the ACL contains its own set of rights that allow or deny access to a file or folder, and can optionally contain inheritance flags to specify that the ACE should be inherited by any child folders and files.

Instead of the standard three permissions available for mode bits, ACLs have 32 bits of fine grained access rights. Of these, the upper 16 bits are general and apply to all object types. The lower 16 bits vary between files and directories but are defined in a compatible way that allows most applications to use the same bits for files and directories. Rights can be used for granting or denying access for a given identity. Access can be blocked to a user explicitly through the use of a deny ACE, or implicitly by ensuring that the user does not directly (or indirectly through a group) appear in an ACE that grants the right in question.

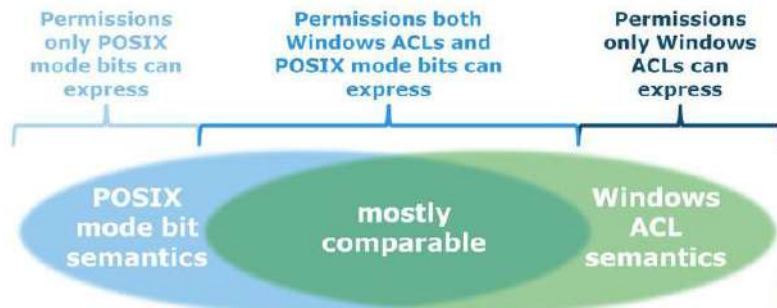
In OneFS, an ACL can contain ACEs with a UID, GID, or SID as the trustee.

On a Windows computer, you can configure ACLs in Windows Explorer.

For OneFS, in the web administration interface, you can change ACLs in the **Access > ACL Policy Settings** page.

Mixed Environments

- NFS and SMB clients
 - NFS exports and UNIX permissions
 - Windows (SMB) shares and Windows permissions (ACLs)



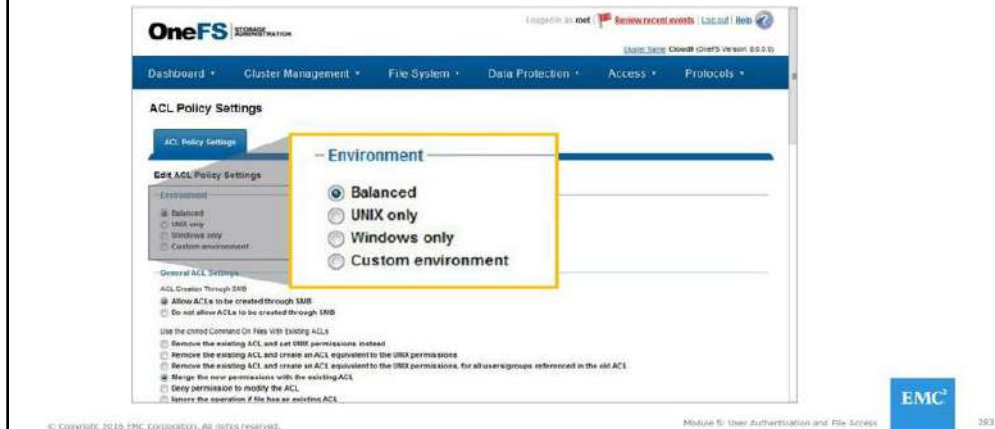
OneFS supports a mixed environment in which NFS exports and SMB shares on the cluster can be configured for the same data. Also, the individual files and folders reached through NFS exports or SMB shares can have UNIX permissions and Windows ACLs assigned. OneFS enables you to choose between ACLs and UNIX permissions. However, no perfect one-to-one mapping exists between the two. The result is multi-protocol access to a data set that contains both Windows ACLs and UNIX permissions.

Both Windows ACLs and standard UNIX permissions can be configured on the cluster. The type used is based on the ACL policies that are set on the file creation method. Generally, files that are created over SMB or within a directory that has an ACL will receive an ACL; otherwise, OneFS relies on the POSIX mode bits that define UNIX permissions. POSIX mode bits are present when a file has an ACL, however these bits are provided only for protocol compatibility and are not used for access checks.

When performing an authorization check, OneFS compares the access token generated during the connection with the authorization data found on the file. If required to evaluate a UNIX permission against a file with an ACL, OneFS converts the permissions into the corresponding rights that the caller must possess.

ACL Policies

Access → ACL Policy Settings



The Isilon cluster includes ACL policies that control how permissions are managed and processed. You can change the Isilon cluster's default ACL settings globally or individually, to best support your environment. These global permissions policies change the behavior of permissions on the system.

To configure the type of authorization to use in your environment:

- Click **Balanced** for cluster permissions to operate in a mixed UNIX and Windows environment. This setting is recommended for most cluster deployments and is the default.
- Click **UNIX only** for cluster permissions to operate with UNIX semantics, as opposed to Windows semantics. This option prevents ACL creation on the system.
- Click **Windows only** for the cluster permissions to operate with Windows semantics, as opposed to UNIX semantics. If you enable this option, the system returns an error on UNIX **chmod** requests.
- Click **Custom environment** to configure individual permission-policy settings.

If you enabled **UNIX only**, **Balanced**, or **Windows only**, the corresponding options in the **Permission Policies** section are automatically enabled or disabled when you click **Submit**.

The cluster's permissions settings are handled uniformly across the entire cluster, rather than by each access zone.

ACL Permission Policy Settings

Access → ACL Policy Settings

General ACL Settings

ACL Creation Through SMB

- ☐ Allow ACLs to be created through SMB
- ☒ Do not allow ACLs to be created through SMB

Use the chmod Command On Files With Existing ACLs

- ☐ Remove the existing ACL and set UNIX permissions instead
- ☐ Remove the existing ACL and create an ACL equivalent to the UNIX permissions
- ☐ Remove the existing ACL and create an ACL equivalent to the UNIX permissions, for all users/groups referenced in the old ACL
- ☒ Merge the new permissions with the existing ACL
- ☐ Deny permission to modify the ACL
- ☐ Ignore the operation if file has an existing ACL

ACLs Created On Directories By the chmod Command

- ☒ Make ACLs inheritable
- ☐ Do not make ACLs inheritable

Use the chown and chgrp Commands On Files With Existing ACLs

- ☒ Modify the owner and/or group and ACL permissions
- ☐ Modify only the owner and/or group
- ☐ Ignore operation if file has an existing ACL

Access Checks (chmod, chown)

- ☒ Allow the file owner and users with WRITE_DAC and WRITE_OWNER permissions to change the mode or owner of the file (Windows mode)
- ☐ Allow only the file owner to change the mode or owner of the file (UNIX mode)

Environment

- ☒ Balanced
- ☐ UNIX only
- ☐ Windows only
- ☐ Custom environment



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Shown here are the settings for the permission policies for each of the environment settings. Remember that these settings cannot be changed if one of the pre-configured policies is chosen.

Select the “clip” icon to see the ACL options for each environment setting.

Synthetic vs. Advanced ACLs

Advanced ACL displays with a + after the mode bits

Synthetic ACL has no +



The image shows two terminal windows from a PuTTY session on 192.168.0.11. The top window shows the command `ls -le studentshare1` and its output, which includes an advanced ACL for `test_document1.txt` with a '+' sign after the permissions. The bottom window shows the command `ls -le software` and its output, which includes a synthetic ACL for `software` without a '+' sign.

```
Cloud9-1# ls -le studentshare1
total 26
-rwxrwxr--+ 1 root wheel 30 May 5 10:22 test_document1.txt
OWNER: user:root
GROUP: group:wheel
CONTROL: dacl_auto_inherited
0: group:Administrators allow inherited file_gen_all,inherited_ace
1: user:root allow inherited file_gen_all,inherited_ace
2: group:Users allow inherited file_gen_read,file_gen_execute,inherited_ace
Cloud9-1#
```

```
Cloud9-1# ls -le software
total 2
-rw-r--r-- 1 root wheel 0 May 12 07:12 swtest
OWNER: user:root
GROUP: group:wheel
SYNTHETIC ACL
0: user:root allow file_gen_read,file_gen_write,std_write_dac
1: group:wheel allow file_gen_read
2: everyone allow file_gen_read
Cloud9-1#
```

When you assign UNIX permissions to a file, no ACLs are stored for that file. However, a Windows system processes only ACLs; Windows does not process UNIX permissions. Therefore, when you view a file's permissions on a Windows system, the Isilon cluster must translate the UNIX permissions into an ACL. In the Isilon cluster, this type of ACL is called a *synthetic ACL*. Synthetic ACLs are not stored anywhere; instead, they are dynamically generated as needed and then they are discarded. Synthetic ACLs are the cluster's translation of UNIX permissions so they can be understood by a Windows client. If a file also has Windows-based ACLs (and not only UNIX permissions), it is considered by OneFS to have advanced ACLs.

If a file has UNIX permissions, you may notice synthetic ACLs when you run the **ls -le** command on the cluster in order to view a file's ACLs. Advanced ACLs display a plus (+) sign when listed using an **ls -l** command.

Managing ACL Permissions

- SSH connection into Isilon cluster
- Options added to several Isilon commands

Command	Output in OneFS
# ls -le	Actual permissions stored on disk; prints ACL from security descriptor
# ls -len	Changes view the owner and group's SID or UID/GID to numerical rather than by name
# ls -lean	Adds hidden directories beginning with dot [.]
# chmod +a	Manipulate ACL entries



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Isilon takes advantage of standard UNIX commands and has enhanced some commands for specific use on Isilon clusters. Using an SSH session to the cluster, the list directory contents (**ls**) command is run to provide file and directory permissions information. Isilon has added specific options to enable reporting on ACLs as well as POSIX mode bits.

The **ls** command options are all designed to be used the long notation format, which is displayed when the **-l** option is used. The long format includes: file mode, number of links, owner name, group name, MAC label, number of bytes in the file, abbreviated month, day-of-month file was last modified, hour file last modified, minute file last modified, and the pathname.

The **-l** option also displays the actual permissions stored on disk. Adding the **-e** option to the **-l** prints the ACLs associated with the file.

The **-n** option, when combined with the **-l** option, displays user and group IDs numerically rather than converting them to a user or group name.

The options are used in combination to report the desired permissions information. Referring to the chart, you can see how adding additional options changes the output.

The **+a** of the **chmod** command mode parses a new ACL entry from the next argument on the command-line and inserts it into the canonical location in the ACL. If the supplied entry refers to an identity already listed, the two entries are combined. The **+a** mode strives to maintain correct canonical form for the ACL, which is local deny, local allow, inherited deny, and inherited allow. By default, **chmod** adds entries to the top of the local deny and local allow lists. Inherited entries can be added by using the **+ai** mode, or specifying the `'inherited_ace'` flag.

Lesson 2: Summary

Having completed this lesson, you should be able to:

- Differentiate Windows and POSIX permissions
- Understand mapping different permissions
- Relate share permissions to file permissions
- Configure OneFS permissions management

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Having completed this lesson you should now understand how POSIX permissions and Windows ACLs differ from each other, and how the ambiguities are resolved in OneFS. You also should know how the exact settings can be managed through the OneFS interface.

Lesson 3: Configuring SMB Shares

Upon completion of this lesson, you should be able to:

- Differentiate SMB functionality from previous versions
- Describe how SMB CA works
- Describe SSC functionality
- Enable SMB sharing
- Configure SMB shares
- Manage automatic creation of home directories

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Upon completion of this lesson, you should be able to differentiate SMB functionality from previous versions, describe how SMB continuous availability, or CA, works, describe SSC functionality, enable SMB sharing, configure SMB shares, and manage automatic creation of home directories.

SMB Failover: OneFS 7.2.1 and prior

- SMB clients connect to single Isilon node
 - During network or node failure SMB clients disconnected
 - Needed to manually reconnect to cluster
- Issues:
 - Noticeable disruption in work
 - Clients aware of disconnection
 - Must manually reconnect to continue working
 - Clients open frequent internal help desk tickets

disruption



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In OneFS 7.2.1 and earlier versions, when an SMB client connects to the cluster, it connects to one single node. In the event that this node goes down or if there is a network interruption between the client and the node, the SMB client would have to reconnect to the cluster manually. This is due in part to the stateful nature of the protocol. This is an issue because it is a noticeable interruption to the clients work. In order to continue working they client must manually reconnect to the share on the cluster. Too many disconnections would also prompt for the clients to open help desk tickets with their local IT department to determine the nature of the interruption/disconnection. Frequent helpdesk tickets divest time from the administrators primary responsibility, administration, and forces time to be spent diagnosing minor disconnection issues.

What is SMB Continuous Availability (CA)?

- SMB Continuous Availability (CA)
 - Clients transparently recover from network or node failure
 - Applies to SMB 3.0 clients:
 - Windows 8, Windows 10, Windows 2012 R2 clients
- Isilon's Non-Disruptive Operations (NDO) initiative
 - Seamlessly moves client from one node to another
 - No manual intervention on client-side
 - Node or network failure transparent to client
 - No disruption to client-side work
 - Supports home directory workflows

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In OneFS 8.0, Isilon offers the continuously available (CA) share option. This allows SMB clients the ability to transparently fail over to another node in the event of a network or node failure. This feature applies to Microsoft Windows 8, Windows 10 and Windows 2012 R2 clients.

This feature is part of Isilon's non-disruptive operation (NDO) initiative to give customers more options for continuous work and less down time. The CA option allows seamless movement from one node to another and no manual intervention required on the client side. This enables a continuous workflow from the client-side with no appearance or disruption to their working time. CA supports home directory workflows, as well.

SMB CA Requirements

- Requirements
 - Service witness protocol (SMB 3.0)
 - Active Directory
 - SmartConnect
- Not enabled by default
 - Must be enabled on share
 - Enabled only upon share creation
- Existing shares
 - Need to share again using CA option



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In SMB 3.0, Microsoft introduced an RPC-based-mechanism that updates the clients to any state change on the SMB servers. This service is called Service Witness Protocol (SWP) and it provides a faster recovery mechanism for SMB 3.0 clients to failover should their server go down. In SMB 1.0 and SMB 2.x, SMB clients use a 'time-out' service using either SMB or TCP. These time-out services must wait for a specific period of time before notifying the client of a server down. These time-outs can take up as much as 30-45 seconds and thus create a high latency that is disruptive to enterprise applications. The SWP requires 'continuously available' file shares and is aware of cluster or scale-out storage. SWP will observe the servers in use and in the event that one is unavailable, will notify the SMB client to release its file handle. This exchange happens within five seconds, thus dramatically decreasing the time from the 30-45 seconds previously used with 'time-outs'.

CA is not available by default on any share on the cluster. CA must be enabled when the share is created. Any existing shares would need to be reshared using the CA option in order to make them highly available. This is something that should be put into an upgrade plan or discussed as an option if the company is interested in using CA for SMB shares.

SMB Server-side Copy

- Server-side Copy offloads specific copy operations to server
 - Copy and paste files without involving client machine
- Benefits:
 - Does not require client machine to copy
 - If file already on server, copy made locally
 - Avoids moving records across network thus:
 - Expedites copy time
 - Considerable performance improvements



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Server-side copy offloads copy operations to the server when the involvement of the client is unnecessary. File data no longer needs to be transferred across the network for copy operations that could have been carried out on the server. Clients making use of server-side copy support, such as Windows Server 2012, can experience considerable performance improvements for file copy operations, like CopyFileEx or "copy-paste" when using Windows Explorer.

Disable SMB Server-side Copy

- Server-side copy enabled by default
 - Global setting: either enabled or disabled
 - Configured via CLI

```
isi smb settings global modify --server-side-copy=no (yes)
```



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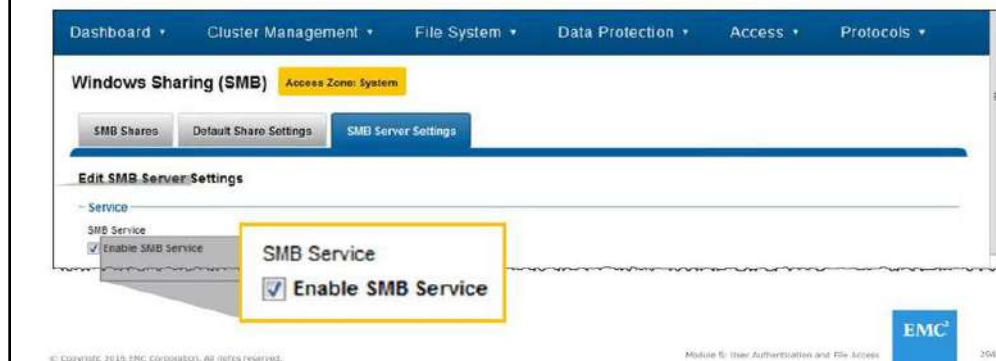
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The server-side copy feature is enabled by default in OneFS 8.0. If the feature is not something that the customer wants enabled, for some very specific and unique reason, then the feature can be disabled via a CLI command.

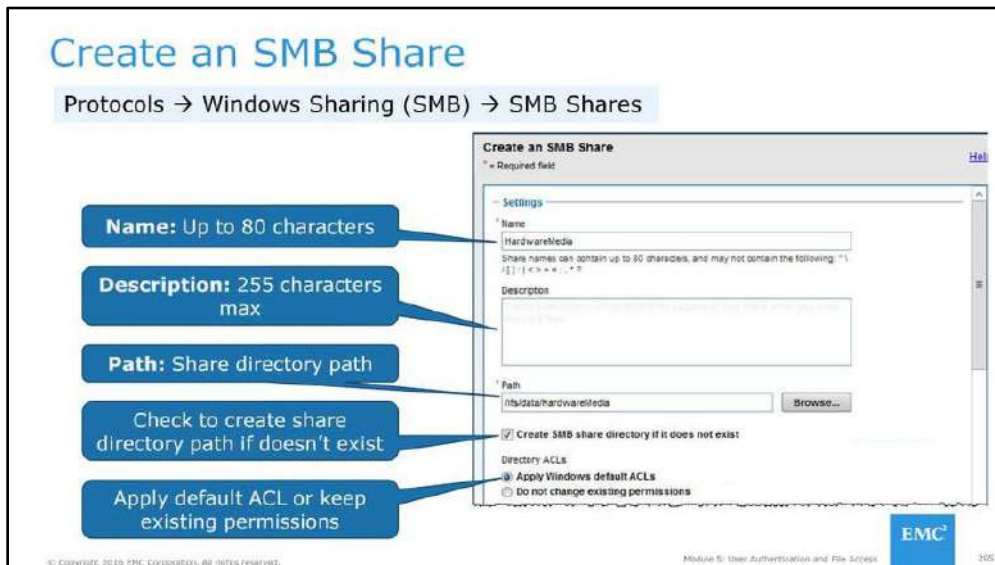
SMB Enabled by Default

- To disable SMB, uncheck box

Protocols → Windows Sharing (SMB) → SMB Server Settings



To enable SMB, in the web administration interface, navigate to the **Protocols > Windows Sharing (SMB)** page, and then select the **SMB Server Settings** tab. The **SMB Server Settings** page contains the global settings that determine how the SMB file sharing service operates. These settings include enabling or disabling support for the SMB service. The SMB service is enabled by default.



Before creating the SMB share, ensure that the drop-down list by the **Windows Sharing (SMB)** title shows the correct **Access Zone**.

- In the **Name** field, type a name for the share. Share names can contain up to 80 characters, and can only contain alphanumeric characters, hyphens, and spaces.
- In the **Description** field, type a comment with basic information about the share you are creating. There is a 255 character limit. A description is optional, but is helpful when managing multiple shares.
- In the **Path** field, type the full path of the share, beginning with **/ifs**, or click **Browse** to locate the share.
- **Create SMB share directory if it does not exist** will create the required directory and then share it if the directory was not already there.
- Also apply the initial **Directory ACLs** settings. These settings can be modified later.
 - To maintain the existing permissions on the shared directory, click the **Do not change existing permissions** option. Caution should be used when applying the default ACL settings as it could overwrite existing permissions in cases where the data has been migrated over onto Isilon. Be aware of what this setting can do prior to implementation.

Directory ACLs

Protocols → Windows Sharing (SMB) → SMB Shares

- Setting cause and effect
- Rule of thumb
 - Set **Do not change existing permissions** if adding a new share to existing directory structure
 - Creating a new share for Windows set **Apply Windows default ACLs**

Directory ACLs

- ☒ Apply Windows default ACLs
- ☐ Do not change existing permissions

Create an SMB Share

* = Required field

Settings

Name: HardwareMedia
Share names can contain up to 80 characters, and may not contain the following: " \ | : < > * . , * ?

Description: HardwareMedia

Path: /ifs/data/hardwareMedia Browse...

☒ Create SMB share directory if it does not exist

Directory ACLs

- ☒ Apply Windows default ACLs
- ☐ Do not change existing permissions

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Let's take a closer look at the **Directory ACLs** setting, and the cause and effect each setting can have. As noted in the previous slide, caution should be taken when applying the default ACL settings.

When a cluster is setup, the default permissions on /ifs may or may not be good for the permissions on your directories. As an example, let's say that /ifs/tmp is an NFS export and you explicitly want the /ifs/tmp mode bit rights set based on UNIX client application requirements. Selecting the **Apply Windows default ACLs** option as shown in the screen capture, overwrites the original ACL which can break the application. Thus, there is risk associated with using **Apply Windows default ACLs** with a currently existing directory.

Conversely, let's say that /ifs/tmp is a new directory created using the CLI in which windows users will create and delete files. When creating the share, if the set **Do not change existing permissions** is set and then users attempt to save files to the share, they would get access denied because "Everyone" only gets Read access. In fact, even as Administrator you would not be able to modify the security tab of the directory to add Windows users because the Mode Bits limit access to only Root.

In summary, a good rule of thumb is as follows:

- If you have an existing directory structure that you want to add a share to, you likely do not want to change the ACL, so select the **Do not change existing permissions** option.
- If you are creating a new share for a new directory you will likely be changing permissions to the ACL to grant Windows users rights to perform operations. Thus, you should set the **Apply Windows default ACLs** option and then once the share is created, go into the Windows Security tab and assign permissions to users as needed.

Create an SMB Share (cont'd)

The screenshot shows the 'Create an SMB Share' wizard with the following settings and callouts:

- Expand path variables such in the share directory path:** Points to the **Allow Variable Expansion** checkbox.
- Automatically create home directories:** Points to the **Enable Home Directories** checkbox.
- Enable SMB CA:** Points to the **Enable continuous availability on the share** checkbox.
- Add, view, or edit users or groups:** Points to the **Users and Groups** section, which includes a table with columns: Order, Account, Run As Root, Permission, and Action.
- Enable file filtering:** Points to the **File Filter** section, which includes a checkbox for **Enable file filters**.
- Show/edit advanced settings:** Points to the **Show Advanced Settings** button.

At the bottom of the wizard are **Cancel** and **Create Share** buttons. The EMC logo is in the bottom right corner.

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The next settings are for **Home Directory Provisioning**, which is covered in the next slide. If needed, apply the **Users and Groups** options. The default permissions configuration is read-only access for the **Everyone** account. Edit or add users or groups to allow users to write to the share. File filtering for the share can be enabled to allow or deny file writes. Also, if needed, apply **Show Advanced Settings**. Any adjustments made to advanced SMB share settings override the default settings for this share only. While it is not recommended, if you need to make changes to the default values themselves, you can make those changes on your **Default Share Settings** tab. The **Advanced Settings** include the CA settings, SMB server settings (behavior of snapshot directories) and the SMB share settings (File and directory permissions settings, performance settings, and security settings).

In the command-line interface, you can create shares using the **isi smb shares create** command. You can also use the **isi smb shares modify** to edit a share and **isi smb shares list** to view the current Windows shares on a cluster.

Home Directory Creation through SMB

* Path

☐ Create SMB share directory if it does not exist

Directory ACLs
☒ Apply Windows default ACLs
☐ Do not change existing permissions

Home Directory Provisioning
☒ Allow Variable Expansion
Include one or more of the following expansion path variables in the share directory path: %U, %L, %D, or %Z
☒ Auto-Create Directories
Create home directories for users when they first access the share path with expansion variables.

Variable	Expansion
%L	Host name of cluster (lowercase)
%D	Netbios domain name
%U	User name, for example student1

- Enabled at SMB share create or modify
- Automatically creates home directories in home share
- No need to create home directories on user-by-user basis
- Maps users automatically to their home directory



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OneFS supports the automatic creation of SMB home directory paths for users. Using variable expansion, user home directories are automatically provisioned. Home directory provisioning enables you to create a single home share that redirects users to their SMB home directories. A new directory is automatically created if one does not already exist. To create a share that automatically redirects users to their home directories, when you create the share in the web management interface, check the **Allow Variable Expansion** box. This automatically expands the **%U** and **%D** in the path to the specified user name and domain name. To automatically a directory for the user if one does not exist, check the **Auto-Create User Directory** box. You may also set the appropriate flags by using the **isi smb** command in the command-line interface.

Set up users to access their home directory by mapping to **//servername/home**. They are automatically redirected to their home directory **/ifs/home/<UserName>**.

The variable **%L** expands to host name of the cluster, in lowercase, **%D** to the Netbios domain name, and **%U** to user name. In this example, expansion variables are used to automatically create a path under which the users will store the home directory files. After the creation, users connecting to this share are automatically redirected to their own home directory according to the used path variables.

The access zone is already implicitly in the directory, because all access for Active Directory is done per access zone and each access zone has its own home directory path. There used to be a variable to make this possible, but it is no longer supported as a variable since the directories are already differentiated by the environment.

Lesson 3: Summary

Having completed this lesson, you should now be able to:

- Differentiate SMB functionality from previous versions
- Describe how SMB CA works
- Describe SSC functionality
- Enable and configure SMB shares
- Manage automatic home directory creation

Having completed this lesson, you should now understand how to differentiate SMB functionality from previous versions, describe how SMB CA works, describe SSC functionality, enable SMB sharing, configure SMB shares, and manage automatic home directory creation.

Lesson 4: Managing NFS Exports

Upon completion of this lesson, you should be able to:

- Explain NFS
- Add NFS exports
- Configure NFS



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Upon completion of this lesson, you should understand the creation of NFS shares, differences between supported versions of NFS, and NFS configuration options.

Network File System (NFS)

- Native to UNIX clients
- Supports NFSv3 and NFSv4
- NFS configuration steps
 - Enable NFS (Default)
 - Create folder on cluster
 - Create NFS export
 - web administration interface
 - `isi nfs export`
 - Create mount point
 - Mount cluster's exported file system on client



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Network File System (NFS) is a protocol that allows a client computer to access files over a network. It is an open standard that is used by UNIX clients. You can configure NFS to allow UNIX clients to address content stored on Isilon clusters. NFS is enabled by default in the cluster; however, you can disable it if it isn't needed. In NFS, sharing is enabled by exporting a directory, which is then imported by clients and made accessible under a mount point. The mount point is the directory that will display files from the server.

The NFS service in an Isilon cluster enables you to create as many NFS exports as needed. To configure NFS, you need to create and manage NFS exports. You can do this through either the web administration interface or the command-line interface.

Isilon supports NFS protocol versions 3 and 4. Kerberos authentication is supported. You can apply individual host rules to each export, or you can specify all hosts, which eliminates the need to create multiple rules for the same host. When multiple exports are created for the same path, the more specific rule takes precedence. For example, 10.10.x subnet has RO (read only) access and 10.10.2.5 has RW (read write) access. In this case, 10.10.2.5 has RW access, even though it is within in the 10.10.x subnet because it is more specific. OneFS can have multiple exports with different rules that apply the same directory.

NFSv4 Failover: OneFS 7.2.1 and Prior

- NFSv4 clients connect to **single** Isilon node
 - If node or network disruption occurs, NFSv4 clients disconnected
 - NFSv4 is stateful protocol
 - State information exists on node, when node offline state is lost
 - IP address assigned to node does not move
 - Clients manually reconnect to share
- Why is this an issue?
 - Noticeable disruption to client work
 - Clients aware of disconnection
 - Must manually reconnect to continue working



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In OneFS 7.2.1 and earlier versions when an NFSv4 client connects to the cluster, it connects to a single node. In the event that this node goes down or if there is a network interruption between the client and the node, the NFSv4 client has to reconnect to the cluster manually. This is due in part to the stateful nature of the protocol. This is an issue because it is a noticeable interruption to the clients work. In order to continue working, they client must manually reconnect to the cluster. Too many disconnections would also prompt for the clients to open help desk tickets with their local IT department to determine the nature of the interruption/disconnection.

What is NFSv4 Continuous Availability?

- Transparent client recovery from node or network failure
- Isilon's non-disruptive operations (NDO) initiative
 - Seamless movement of NFSv4 client between nodes
 - No manual intervention on client side
 - Node or network failure transparent to client
 - No disruption to client side work



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In OneFS 8.0, Isilon offers the continuously available (CA) feature. This option allows NFSv4 clients to transparently fail over to another node in the event of a network or node failure. This feature is part of Isilon's non-disruptive operation initiative to give customers more options for continuous work and less down time. The CA option allows seamless movement from one node to another and no manually intervention on the client side. This enables a continuous workflow from the client side with no appearance or disruption to their working time. CA supports home directory workflows, as well.

NFS CA Requirements

- NFSv4 CA
 - Uses dynamic pools in OneFS 8.0
 - Enabled by default
 - Dynamic pool is automatically CA pool
- Considerations prior to upgrade
 - Determine if NFSv4 using static or dynamic pools
 - For OneFS 7.2.x, static pools are Best Practice with NFSv4
 - Most customers should use static pools with NFSv4
 - If using dynamic pools, be aware after upgrade:
 - CA immediately enabled
 - Customer may see performance hit on these pools



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In OneFS 8.0, NFSv4 CA is enabled by default. This won't affect the 99% of customers who are using NFSv4 with a static IP address pool; however, if a customer is using NFSv4 with a dynamic IP address pool, they will notice a significant drop in the performance of this pool. The best practice is currently to use NFSv4 with a static pool because NFSv4 acts and functions similarly to SMB. In rare instances that a customer decided or was inadvertently told to use a dynamic pool, those customers upgrading to OneFS 8.0 will notice a decrease in the performance of these pools. Planning and reviewing of the current pool types should be done, and the effects explained to those customers prior to upgrading to OneFS 8.0.

Additional NFS Improvements

- In OneFS 7.2.1 and prior:
 - Support up to 1000 exports
- In OneFS 8.0, mountd improvements:
 - Support up to 40K exports
 - Helps meet demand of:
 - Large customers
 - Growing customers
 - NFS startup may take longer on startup
 - Refreshes are immediate once NFS is started
 - Don't need to reload all exports



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Prior to OneFS 8.0 Isilon supported up to 1000 exports, however, many customers required or requested a larger number of exports. With OneFS 8.0, in order to meet the demands of large and growing customers, Isilon now supports up to 40K exports.

Enable NFS

Protocols → UNIX Sharing (NFS) → Global Settings



In the web administration interface, click **Protocols > UNIX Sharing (NFS)**, and then select **Global Settings**. The NFS global settings determine how the NFS file sharing service operates. These settings include enabling support for different versions of NFS. These are the global default settings for all current and future NFS exports. If you change a value in the NFS export settings, that value will change for all NFS exports. Modifying the global default values is not recommended. You can change the settings for individual NFS exports as you create them, or edit the settings for individual exports as needed.

The first step is to make sure that the NFS service is enabled, which is the default. If the NFS service is not needed, then it can be disabled here. The support for NFSv3 is enabled, NFSv4 is disabled by default. If NFSv4 is enabled, the name for the NFSv4 domain needs to be specified in the NFSv4 domain box.

Other configuration steps on the **Global Settings** page are the possibilities to reload the cached NFS exports configuration to ensure any DNS or NIS changes take effect immediately, to customize the user/group mappings, and the security types (UNIX and/or Kerberos), as well as other advanced NFS settings.

Create an NFS Export

Protocols → UNIX Sharing (NFS) → NFS Exports

Setting	Description
Clients	Clients allowed access to the export.
Always Read/Write Clients	Clients allowed read/write access regardless of export's access-restriction setting.
Always Read-Only Clients	Clients allowed read-only access regardless of export's access-restriction setting.
Root Clients	Clients to map as root.

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Go to **Protocols > UNIX Sharing (NFS) > NFS Exports**. Select the access zone from the **Current Access Zone** drop-down list.

Then click the **Create Export** link.

In the **Create an Export** window, you can enter the directory path or paths, or browse to the directory that you want to export. You can add multiple directory paths by clicking **Add another directory path** for each additional path. Optional fields include adding a **Description** for the NFS export using up to 255 characters, and/or specifying the clients that are allowed access via this export. A network host name, an IP address, a subnet, or a netgroup name can be used for reference. For IPv4 addresses, specify in dotted-decimal notation (a.b.c.d). For IPv6 addresses, specify in colon notation. Use one line per entry. The same export settings and rules created here are applied to all the listed directory paths. If no clients are listed in any entries, no client restrictions apply to attempted mounts.

Clients can also be listed by the intended degree of access, so that those clients which should not ever be able to modify data can be listed in **Always Read-Only Clients**, and clients which should be permitted to mount with direct root-level access can be listed in **Root Clients**.

Create an NFS Export (cont'd)

Create an Export
+ Required field

Permissions
☐ Restrict access to read-only
☐ Enable mount access to subdirectories
Allow subdirectories below the paths to be mounted.

Root User Mapping
Use Default Use Custom
Using default value of:
User: nobody
Primary Group: No Primary Group
Secondary Groups: No Secondary Groups

Non-Root User Mapping
Use Default Use Custom
Using default value of:
User mapping is not enabled by default

Failed User Mapping
Use Default Use Custom
Using default value of:
User mapping is not enabled by default

Security Flavors
Use Default Use Custom
UNIX (system)

Advanced Settings
Block Size
Use Default Use Custom
8 KB

Cancel Create Export

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Permissions settings can restrict access to read-only (default is R/W) and enable mount access to subdirectories (allow subdirectories below the path to be mounted). Other export settings are user mappings. The default is **Map root users to nobody** and group is none. Customized mappings can be entered. The default security flavor is UNIX (system). Kerberos security can be set additionally or instead UNIX (system).

The **Advanced Settings** require advanced knowledge. Uninformed changes to these advanced settings could result in operational failures. Make sure you understand the consequences of your changes before saving. Any adjustments made to these settings override the default settings for this export only. While it is not recommended, if you need to make changes to the default values themselves, you can make those changes on your **Export Settings** tab. **Advanced Settings** are performance settings, client compatibility settings, and export behavior settings.

NFSv3 vs. NFSv4

Feature	NFSv3	NFSv4
State	Stateless	Stateful
Environments	UNIX-only	UNIX and Windows (SMB)
Permissions	POSIX mode bits	Supports Windows ACLs
Transport	UDP / TCP	TCP-only
Mount	Exports mounted separately	All exports mounted together as part of pseudo file system
Locking	File locking only (NLM)	File and byte range locking (in NFS protocol)



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NFSv3 does not track state. A client can be redirected to another node, if configured, without interruption to the client. NFSv4 tracks state, including file locks. Automatic failover is not an option in NFSv4.

Because of the advances in the protocol specification, NFSv4 can use Windows Access Control Lists (ACLs). NFSv4 mandates strong authentication. It can be used with or without Kerberos, but NFSv4 drops support for UDP communications, and only uses TCP because of the need for larger packet payloads than UDP will support.

File caching can be delegated to the client: a read delegation implies a guarantee by the server that no other clients are writing to the file, while a write delegation means no other clients are accessing the file at all. NFSv4 adds byte-range locking, moving this function into the protocol; NFSv3 relied on NLM for file locking.

NFSv4 exports are mounted and browseable in a unified hierarchy on a pseudo root (/) directory. This differs from previous versions of NFS.

Lesson 4: Summary

Having completed this lesson, you should now be able to:

- Explain NFS
- Add NFS exports
- Configure NFS

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Module 5: User Authentication and File Access



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Having completed this lesson, you should now understand how the supported NFS versions differ from each other, as well as how to set up exports in each one.

Module 5: Summary

Upon completion of this module, you should be able to:

- Understand Identity Management in Isilon
- Explain Permissions Mapping in OneFS
- Configure protocol access to Isilon
- Plan multiprotocol implementations on OneFS

Having completed this module, you should now know how OneFS deals with user identities, permissions, and how protocols afford user access to the cluster.

Lab 5: User Authentication and File Access

- Viewing on-disk identities
- Modifying permissions
- Configuring SMB shares and home directories
- Managing NFS exports
- Configuring share and export permissions
- Connecting using HTTP protocol



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Module 5: User Authentication and File Access

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In this lab, you'll lookup identities for users and groups, and also look at the identity mappings.

Module 6: Storage Administration

Upon completion of this module, you will be able to:

- Implement SmartPools and file pool policies
- Deploy CloudPools
- Configure SmartQuotas functionality
- Apply SnapshotIQ
- Execute SyncIQ policies and actions
- Accomplish data deduplication

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Module 6: Storage Administration



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After completing this module, you will be able to implement SmartPools and file pool policies, deploy CloudPools, configure SmartQuotas, apply SnapshotIQ, execute SyncIQ policies, and accomplish data deduplication.

Lesson 1: SmartPools and File Pools

Upon completion of this lesson, you will be able to:

- Describe SmartPools functionality
- Explain and configure tiers, node pools, global settings
- Understand node compatibilities
- Understand default policy
- Create file pool policies
- Define SSD strategy use

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Module 6: Storage Administration

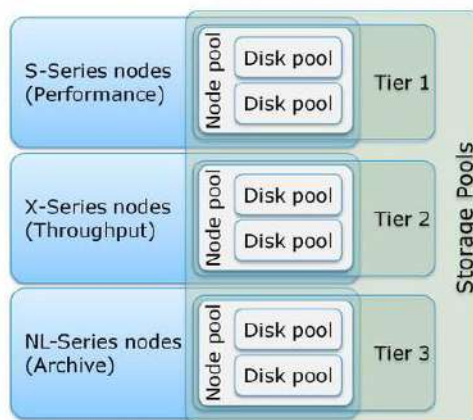


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After completing this lesson, you will be able to describe SmartPools functionality, explain and configure tiers and node pools, configure SmartPools settings, understand node compatibilities, create file pool policies, understand how to apply the default policy, and define using the SSD strategy.

SmartPools and Storage Pools Overview

- Storage pools
- Global cluster settings
- File pools
 - File pool policies
 - File placement
 - Data protection



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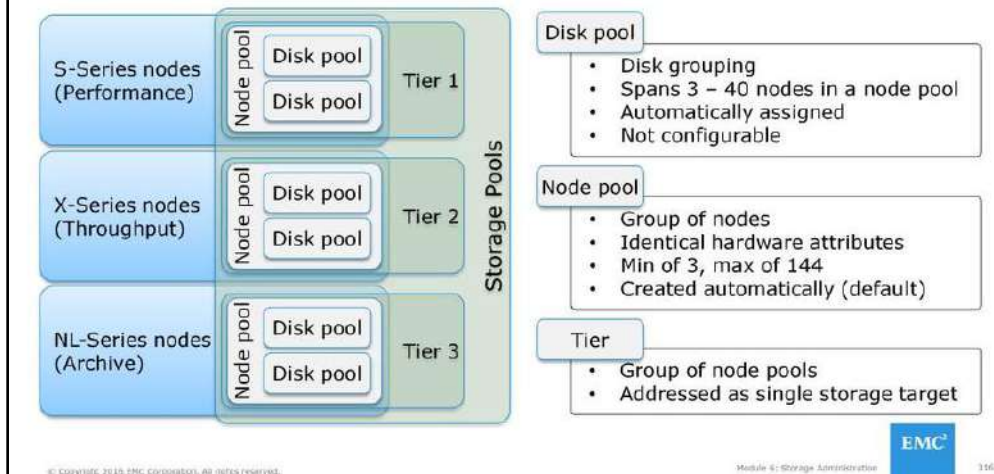
SmartPools is a software module that enables administrators to define and control file management policies within a OneFS cluster. Simply put, with SmartPools data can be segregated based on its business value, putting data on the appropriate tier of storage with the appropriate levels of performance and protection.

Shown here are the building blocks of a storage pool. Storage pools is an abstraction that encompasses disk pools, node pools, and tiers. Storage pools also monitors the health and status of those storage pools at the node pool level. Using storage pools, multiple tiers of Isilon storage nodes (including S-Series, X-Series, NL-Series, and HD-Series) can all co-exist within a single file system, with a single point of management. By using SmartPools, administrators can specify exactly which files they want to live on particular nodes pools and tiers. Node pool membership changes through the addition or removal of nodes to the cluster. Tiers are a grouping of different node pools.

SmartPools manages global settings for the cluster, such as L3 cache enablement status, global namespace acceleration (GNA) enablement, virtual hot spare (VHS) management, global spillover settings, and more. This lesson will cover these settings in detail.

Whereas storage pools define a subset of the cluster's hardware, the file pools are SmartPools logical layer for which file pool policies are applied. File pool policies provide a single point of management to meet performance, requested protection level, space, cost, and other requirements. User created and defined policies are set on the file pools.

Anatomy of Storage Pools



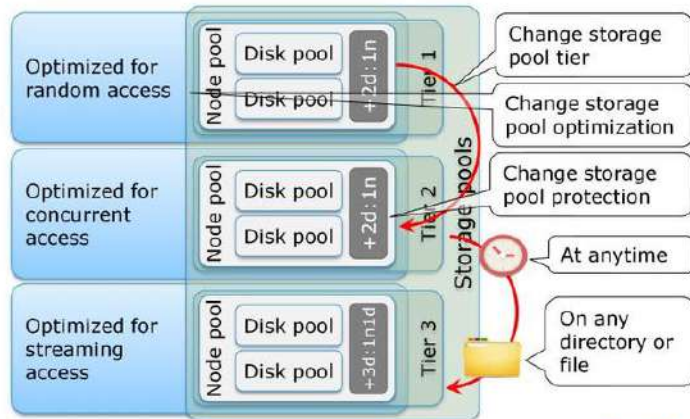
Let's take a look at the storage pools components starting with the smallest unit, disk pools. Similar node drives are automatically provisioned into disk pools with each disk pool representing a separate failure domain. Disk pools span 3 - 40 nodes in a node pool. Data protection stripes or mirrors don't span disk pools, making disk pools the granularity at which files are striped to the cluster. Disk pool configuration is automatic and cannot be configured manually.

A node pool is used to describe a group of similar nodes. There can be from three up to 144 nodes in a single node pool. All the nodes with identical hardware characteristics are automatically grouped in one node pool. A node pool is the lowest granularity of storage space that users manage.

Multiple node pools with similar performance characteristics can be grouped together into a single tier with the licensed version of SmartPools. Multiple tiers can be included in a cluster to meet the business requirements and optimize storage usage.

File Pool Policies

- Automates file management
- Policy driven – filter and action
- File-based not hardware-based
- Applied by SmartPools job



File pool policies are used to determine where data is placed, how it is protected and which other policy settings are applied based on the user-defined and default storage pool policies. File pool policies add the capability to modify the settings at any time, for any file or directory. Files and directories are selected using filters and apply actions to files matching the filter settings. The policies are used to change the storage pool location, requested protection settings, and I/O optimization settings. The management is file-based and not hardware-based. Each file is managed independent of the hardware, and is controlled through the OneFS operating system. The policies are applied in order through the SmartPools job.

SmartPools Licensing

SmartPools Basic (unlicensed)

- Multiple node pools within single tier
- One default file pool policy
- Virtual hot spares

SmartPools Advanced (licensed)

- Multiple node pools
- Multiple tiers
- Multiple file pool policies
- Node pool spillover management



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SmartPools is a licensable software module that provides basic features in an unlicensed state and advanced features after it is licensed. In an unlicensed state, you can create multiple node pools, but only a single tier and only a single file pool. The basic version of SmartPools also supports virtual hot spares, which enable you to reserve space in a node pool that can be used for re-protection of data in the event of a drive failure.

By default, SmartPools basic (unlicensed) is implemented in a cluster. This means that there is one file pool that directs all files in the cluster to one or more node pools in a single tier.

More advanced features are available in SmartPools if you license the software. These advanced features include the ability to create multiple tiers and file pool policies that direct specific files and directories to a specific node pool or a specific tier. Another advanced feature, called disk pool spillover management, enables you to define whether write operations are redirected to another node pool if the target node pool is full. If SmartPools is unlicensed, spillover is automatically enabled.

SmartPools – Unlicensed vs. Licensed

Description	Unlicensed	Licensed
Automatic node pool provisioning	Yes	Yes
Number of tiers	Multiple	Multiple
Separate storage pool targets	No	Yes
Number of file pool policies	One (Default)	Multiple
File pool policy targets	One (Default)	Multiple
File pool policy filters	No	Multiple
Policy-based protection level	One (Default)	Multiple
Metadata acceleration setting	One (Default)	Multiple
I/O optimization	One (Default)	Multiple
Snapshot target	One (Default)	Multiple
Specify spillover target	No	Yes
Global namespace acceleration (GNA)	Yes	Yes



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Referring to the chart, with unlicensed SmartPools, you have a one-tier policy of **anywhere** with all node pools tied to that storage pool target through the default file pool policy. This means that there is one file pool policy that applies that same protection level and I/O optimization settings to all files and folders in the cluster. After purchasing and activating a SmartPools license, the capability to have multiple storage pools containing node pools or tiers with different performance characteristics on the same cluster is enabled. Data can be managed at a granular level through the use of SmartPools file pool policies. Because of the availability to have multiple data target locations, some additional target options are enabled in some global settings. These advanced features include the ability to create multiple storage tiers, multiple file pool policy targets, and multiple file pool policies, each with its own protection, I/O optimization, SSD metadata acceleration, and node pool spillover settings.

Node Compatibility Overview

- Allows combining older and newer nodes from same series
- Provides option to transition slowly
 - Add one node at a time to existing node pools
 - Cost effective
 - Disable node compatibility, when appropriate
- Enable gains from larger node pool
 - Better performance
 - Lower protection overhead for larger files



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The Node Compatibility feature allows you to establish an equivalence association between older and newer class nodes from the same performance series so you can combine them into a single node pool. If no node compatibility is created, nodes can't be merged into the same node pool.

Node compatibility is important for a few reasons. It provides you with the Isilon value to transition slowly to the new hardware over time without a forklift upgrade by allowing you to add one node at a time to an existing node pool. This is more cost effective than adding the three node minimum to start a new node pool with the all new hardware. When a customer has grown the new node counts to sufficient quantities, node compatibility can be disabled on an individual node pool.

Adding nodes to an existing node pool rather than starting a new, smaller node pool enables gains available from larger node pools. Larger files can be striped across a larger number of nodes, and the workload is distributed across more nodes and drives, providing for better performance. Because of having larger protection stripes, fewer FEC protection stripe units are required to protect the files, which results in lower protection overhead and better storage efficiencies.

Node Compatibility Requirements

* New in OneFS 8.0

- Same node class (S2xx, X2xx, X4xx, or NL4xx Series)
- Configuration requirements are same
- Compatible RAM configuration
- Drive configuration
 - Non-L3 cache enabled node pools
 - Same number and capacity HDDs and SSDs, or SED HDDs and SSDs
 - L3 cache required
 - Same HDD capacities
 - Same or different SSD capacities
 - Different drive counts allowed in same node pool *
- Compatibility must be enabled in OneFS

<http://www.emc.com/collateral/hardware/white-papers/h8321-wp-smartpools-storage-tiering.pdf>



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You can enable node compatibility between node pools within the same node series. For example S200 nodes and adding S210 nodes. The supported compatibilities are S200/S210, X200/X210, X400/X410, NL400/NL410. Nodes must meet compatibility requirements, be of the same node class, have identical settings for protection settings, and configuration of SSD strategy or L3 cache, and have compatible RAM capacity. If and SSD strategy is used, all nodes must have the same HDD and SSD configurations. This applies to nodes with SEDs HDDs and SSDs too. Each node must have the same capacity size and quantity of the corresponding drive type. If L3 cache is enabled on the node pool additional options are available. You may still have the same configuration of HDDs and SSDs on all nodes, you may also have nodes with different size SSDs. Added in OneFS 8.0 is the capability to have nodes with different drive counts in the same node pool. This compatibility requires the same size HDDs in all nodes. Compatibilities must be enabled to be applied in OneFS. Node compatibilities can be created before or after you add a new node type to the cluster, and can be disabled or deleted at any time.

RAM Capacity Compatibility

S200	S210	X400	X410	X200	X210	NL400	NL410
24GB	32GB	24GB	32GB	6GB	NA	12GB	NA
48GB	64GB	48GB	64GB	12GB	NA	24GB	24GB
96GB	128GB or 256GB	96GB	128GB	24GB	24GB	48GB	48GB
		192GB	256GB	48GB	48GB		

Combine S200, 96GB nodes with either S210, 128GB or 256GB nodes

Does not require identical RAM capacities

Requires identical RAM capacities

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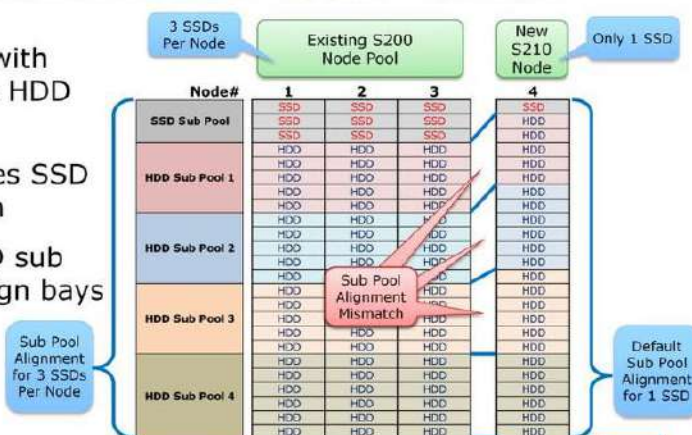
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To enable node compatibility, one requirement is the nodes need to have compatible RAM capacities but the capacity doesn't have to be identical. Shown are the RAM capacities for equivalency. RAM compatibilities are basically the same for both node series except for a slight difference with the higher RAM capacities. The S200 has a maximum of 96GB of RAM and is compatible with S210 nodes with either 128GB or 256GB of RAM. The X400 can have either 96GB or 192GB of RAM. The X400 with 96GB of RAM is compatible with X410 nodes with 128GB of RAM, and the X400 with 192GB RAM is compatible with X410 nodes that have 256GB of RAM.

For the X200/X210 and the NL400/NL410 the RAM amount must be identical. The lower 6GB and 12GB RAM capacities available on the X200 and NL400 do not have compatible RAM configurations available on the X210 and NL410 nodes. An upgrade to the X200 or NL400 node RAM capacity is required before a compatible X210 or NL410 is available.

S200/S210 Different SSD Count - Before

- Add new node with different SSD & HDD counts
- L3 cache enables SSD count mismatch
- Challenge: HDD sub pools do not align bays and counts



Node compatibilities refer to the capability to add dissimilar nodes into the same node pool. Node compatibility is important for a few reasons. It enables a gradual transition to the new hardware without requiring a forklift upgrade. Compatibility allows the mixing of newer and older nodes within the same node series, the nodes with dissimilar sizes of SSDs, and different drive counts. The graphic is used to highlight node compatibility. Shown here is a three node cluster with a pool of SSDs and four sub pools of HDDs. HDD sub pools 1, 2, and 3 contain five drives from each node and sub pool 4 has six. For each compatibility, certain rules exist to enable. The purpose is to enable incremental node pool growth in the same node pool without requiring a minimum of three new nodes resulting in the creation of a new node pool for each node configuration. Shown is adding a fourth node of a different node type, but in the same series. The added node has a different number of SSDs and HDDs. The HDD sizes need to match.

For the customer, this allows a single new node to incrementally be added to a node pool when needed. Larger node pools are more efficient in space utilization and performance especially with larger files. Node compatibilities also enable a node pool to be split into separate node pool when enough newer nodes, or nodes with a similar configuration are present without sacrificing utilization efficiencies or performance.

Adding nodes to an existing node pool rather than starting a new, smaller node pool benefit from gains available from larger node pools. Larger files can be striped across a larger number of nodes, and the workload is distributed across more nodes and drives, providing for better performance. Because of having larger protection stripes, fewer FEC protection stripe units are required to protect the files, which results in lower protection overhead and better storage efficiencies.

S200/S210 Different SSD Count - After

- New node HDD sub pools aligned with existing node pool
- Unaligned HDDs marked as NO_PURPOSE
 - No purpose HDDs not used
 - Must be populated



Using the same graphic, we can illustrate how the different SSD count with the new S210 node is handled. The S210 HDD subpool 1 contains five HDDs, and HDD subpools 2, 3, and 4 each contain six HDDs. This creates a misalignment as shown. The SSD count mismatch is enabled by the SSD compatibility and through the use of L3 cache. Each HDD subpool is directly aligned with the node containing the highest number of SSDs. In this example, there are three SSDs in each of the existing S200 nodes. The unaligned HDDs are marked as NO_PURPOSE in bays 2 and 3. The NO_PURPOSE HDDs are not used by the node for storage, however, the bays must be populated with HDDs. If you add enough similarly configured S210 nodes, you can later remove the SSD count compatibility and the S210 nodes will form their own node pool. When this occurs, the subpools are redistributed and the HDDs are added back as usable capacity in the new node pool.

SSD Count Compatibility Enablement

The screenshot displays the 'Create a Compatibility' dialog box within the EMC VPLEX web administration interface. At the top, a legend identifies icons for Tier, Node Pool, Manual Node Pool, and Unprovisioned Node. Below this, a 'Compatibilities' table lists existing entries, such as 'S210 nodes are SSD compatible'. The 'Create a Compatibility' dialog is open, showing a 'Compatibility Settings' section with the following fields:

- Compatibility Type:** A dropdown menu with 'SSD' selected. A callout points to this field with the text 'Must be enabled'.
- Node Class:** A dropdown menu with 'S210' selected. A callout points to this field with the text 'Must be enabled'.
- SSD Count Compatibility:** A checkbox that is currently checked. A callout points to this checkbox with the text 'Toggle compatibility on or off'.

At the bottom right of the dialog, there are 'Delete' and 'Edit' buttons. The EMC logo is visible in the bottom right corner of the interface.

SSD count compatibility is enabled through the web administration interface or using the command-line interface. In the web administration interface, the checkbox is in the SSD Count Compatibility. You must enable SSD compatibility and node class compatibility as part of the configuration. SSD count compatibility can be toggled on or off by checking or unchecking the box.



In addition to node configuration compatibilities, other compatibility requirements are assessed during the compatibility creation and deletion processes. During the creation of node compatibilities and prior to merging of the node pools, the requested protection level and the L3 cache enablement settings are examined. Both configurations must match before the compatible node pools can be merged together. If the requested protection levels or the L3 cache enablement setting are different, they must be changed to be the same between the compatible node pools.

Displayed is the process to create a node compatibility in the web administration interface. As shown, no node compatibilities have been created prior to you creating the first compatibility.

Click **Create a Compatibility** and the displayed dialogue box opens. Select the node types for the desired compatibility at the top. The preparations to prepare for the node pool merger are displayed. When finished, click **Create a Compatibility** and the confirmation dialogue box appears. You can see the required checkboxes to accept the changes that will be made in the process. Click **Confirm** to proceed. When completed, the new node pool created from the compatibility is displayed in the node pools and tiers containing all of the merged S200 and S210 nodes in this example.

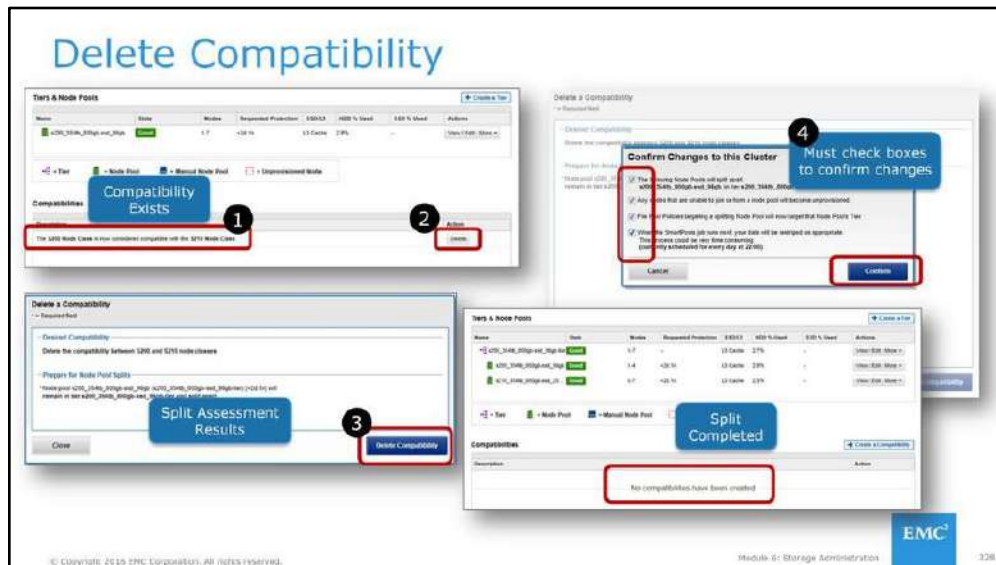
In the CLI, use the command **isi storagepool compatibilities active create** with arguments for the old and new node types. The changes to be made are displayed in the CLI. You must accept the changes by entering **yes**, followed by ENTER to initiate the node compatibility.



In addition to node configuration compatibilities, other compatibility requirements are assessed during the compatibility creation and deletion processes. During the creation of node compatibilities and prior to merging of the node pools, the Requested Protection level and the L3 cache enablement settings are examined. Both configurations must match before the compatible node pools can be merged together. If the Requested Protection levels or the L3 cache enablement setting are different, they must be changed to be the same between the compatible node pools.

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In the CLI, use the command **isi storagepool compatibilities active create** with arguments for the old and new node types. The changes to be made are displayed in the CLI. You must accept the changes by entering **yes**, followed by ENTER to initiate the node compatibility.



When a compatibility is deleted, a split occurs between the different class of node pools and the separate node pools are placed in the same tier. A new tier is created if merged node pool is not already a member of a tier. Each node pool will have the same Requested Protection setting and L3 cache enablement setting as the pre-split compatible node pool. File pool policies are redirected towards the tier and not towards specific node pools.

Displayed is the process to delete a node compatibility in the web administration interface. Click **Delete** next to the desired node compatibility and the displayed dialogue box opens. The preparations to prepare for the node pool split are displayed. When finished, click **Delete Compatibility** and the confirmation dialogue box appears. You can see the required checkboxes to accept the changes that will be made in the process. Click **Confirm** to proceed. When completed, the tier containing the new node pools created from the split is displayed in the node pools and tiers containing a new tier and both node pools as a result of the node pool split, the S200 and S210 nodes in this example. Under **Compatibilities**, the node compatibility is no longer listed.

In the CLI, use the command **isi storagepool compatibilities active delete** with arguments with the compatibility ID number. The changes to be made will be displayed. You must accept the changes by entering **yes**, followed by ENTER to initiate the node compatibility.

Global Namespace Acceleration (GNA)

- GNA accelerates all namespace reads across cluster
- Cluster-wide metadata read acceleration
- Use SSDs to store metadata mirror in different node pools
- Specific requirements:
 - Minimum 1.5% of all disk capacity must be SSD (2% recommended)
 - 20% of nodes must contain SSDs

*L3 cache enable node pools not considered
in of GNA calculations*



Edit SmartPools Settings

Local Storage Settings

- ☒ Increase directory protection to a higher level than its contents
- ☒ Enable global namespace acceleration



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The SmartPools feature allows you to combine different node pools in the same cluster, all in a single file system, and to automatically transfer data among tiers with different performance and capacity characteristics so that data is stored appropriately, based on its value and how it needs to be accessed. GNA enables SSDs to be used for cluster-wide metadata acceleration and use SSDs in one part of the cluster to store metadata for nodes that have no SSDs. The result is that critical SSD resources are maximized to improve performance across a wide range of workflows. Global namespace acceleration can be enabled if 20% or more of the nodes in the cluster contain SSDs and 1.5% or more of the total cluster storage is SSD-based. The recommendation is that at least 2.0% of the total cluster storage is SSD-based before enabling global namespace acceleration. If you go below the 1.5% SSD total cluster space capacity requirement, GNA is automatically disabled and all GNA metadata is disabled. If you SmartFail a node containing SSDs, the SSD total size percentage or node percentage containing SSDs could drop below the minimum requirement and GNA would be disabled.

Any node pool with L3 cache enabled is excluded from GNA space calculations and do not participate in GNA enablement.

Virtual Hot Spare

- Allocate space for data re-protection
 - Percentage of disk space
 - Number of virtual drives in node pool
 - Combination of both
- Space reserved at node pool level
 - Data from lost drives can be repaired within same node pool
 - VHS reservations affect spillover

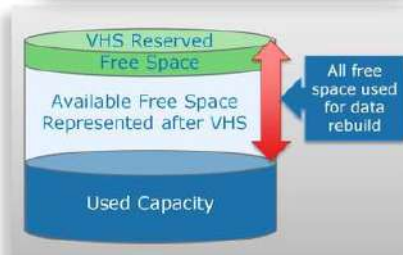
☒ Ignore reserved space when calculating available free space

☒ Deny data writes to reserved disk space

VHS Space Reserved

At least virtual drive(s)

At least % of total storage



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VHS allocation enables you to allocate space to be used for data rebuild in the event of a drive failure. This feature is available with both the licensed and unlicensed SmartPools module. By default, all available free space on a cluster is used to rebuild data. The virtual hot spare option reserves free space for this purpose. VHS provides a mechanism to assure there is always space available and to protect data integrity in the event of overuse of cluster space.

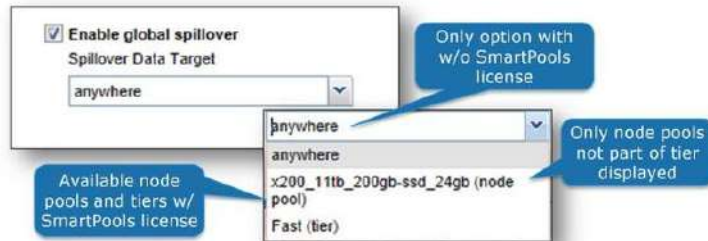
Using the **virtual hot spare (VHS)** option, for example if you specify two virtual drives or 3%, each node pool reserves virtual drive space that is equivalent to two drives or 3% of their total capacity for virtual hot spare, whichever is larger. You can reserve space in node pools across the cluster for this purpose, equivalent to a maximum of four full drives. If you select the option to reduce the amount of available space, free-space calculations exclude the space reserved for the virtual hot spare. The reserved virtual hot spare free space is used for write operations unless you select the option to deny new data writes.

VHS reserved space allocation is defined using these options:

- A minimum number of virtual drives in each node pool (1-4)
- A minimum percentage of total disk space in each node pool (0-20%)
- A combination of minimum virtual drives and total disk space. The larger number of the two settings determines the space allocation, not the sum of the numbers. If you configure both settings, the enforced minimum value satisfies both requirements.

Global Spillover

- Node pool full → data spillover to another node pool or tier
- Requires SmartPools license
- Can be disabled



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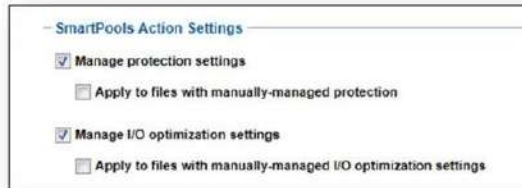
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The **Enable global spillover** and **Spillover Data Target** options configure how OneFS handles a write operation when a node pool is full. Simply put, spillover is node capacity overflow management. With the licensed SmartPools module, a customer can direct data to spillover to a specific node pool or tier group of their choosing. If spillover is not desired, then you can disable spillover so that a file will not move to another node pool.

Virtual hot spare reservations can affect when spillover would occur. If the virtual hot spare reservation is 10 percent of storage pool capacity, spillover occurs if the storage pool is 90 percent full.

SmartPools Action Settings

- Global enablement settings
 - Protection settings
 - I/O Optimization settings
- Enables SmartPools as management for settings
- Override manually-managed settings



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SmartPools Action Settings give you a way to enable or disable managing Requested Protection settings and I/O optimization settings. If the box is unchecked (disabled), then SmartPools will not modify or manage settings on the files. The option to **Apply to files with manually managed protection** provides the ability to override any manually managed Requested Protection setting or I/O optimization. This option can be very useful if manually managed settings were made using file system explorer or the **isi set** command.

Modifying Default File Pool Policy

File Pool Policies

Order	Policy Name	Description	Actions
1	Default Policy	This policy applies to all files not selected by higher-priority policies.	View / Edit / Remove

Edit Default Policy Details

Apply SmartPool Actions to Selected Files

Storage Settings

Move To Storage Pool or Tier
Storage Target:

SSD Strategy

Move Snapshots to Storage Pool or Tier
Snapshot Storage Target:

Snapshot SSD Strategy

Requested Protection

I/O Optimization Settings

Write Performance
☒ Enable SmartCache
☐ Disable SmartCache

Data Access Pattern
☐ Optimize for random access
☒ Optimize for concurrent access
☐ Optimize for streaming access

- Storage location
 - Anywhere
 - Unless licensed
- Data protection
 - Data and snapshot target
 - Protection level
- I/O Optimization
 - SmartCache
 - Data access pattern

The default file pool policy is defined under the default policy. The individual settings in the default file pool policy apply to all files that do not have that setting configured in another file pool policy that you create. You cannot reorder or remove the default file pool policy.

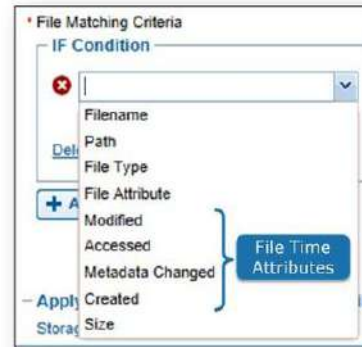
To modify the default file pool policy, click **File System**, click **Storage Pools** and then click the **File Pool Policies** tab. On the **File Pool Policies** page, next to the **default policy**, click **View / Edit**. After finishing the configuration changes, you need to submit and then confirm your changes.

A pool for data and a pool for snapshots can be specified. For data, you can choose any node pool or tier, and the snapshots can either follow the data, or be assigned to a different storage location. You can also apply the cluster's default protection level to the default file pool, or specify a different protection level for the files that are allocated by the default file pool policy.

Under **I/O Optimization Settings**, the **SmartCache** setting is enabled by default. SmartCache can improve performance by prefetching data for read operations. In the **Data access pattern** section, you can choose between **Random**, **Concurrency**, or **Streaming**. Random is the recommended setting for VMDK files. Random access works best for small files (<128 KB) and large files with random access to small blocks. This access pattern turns off prefetching. Concurrency is the default setting. It is the middle ground with moderate prefetching. Use concurrency access for file sets that get a mix of both random and sequential access. Streaming access works best for medium to large files that have sequential reads. This access pattern uses aggressive prefetching to improve overall read throughput.

File Pool Policy Filters

- File name
- Directory location or path
- File type
- File attribute
- File size
- Time attributes
 - Access time (atime)
 - Must enable cluster-wide atime tracking
 - Modify time (mtime)
 - Create time (birthtime)
 - Metadata change time (ctime)



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You can create these filters in the **File Matching Criteria** section when creating or editing a file pool policy. In the **File Matching Criteria** section, click the drop-down list and select the appropriate filter and the appropriate operators. Operators can vary according to the selected filter. Next, you can configure the comparison value, which also varies according to the selected filter and operator.

At least one criterion is required but multiple criteria are allowed. You can add AND statements to a list of criteria. Using AND adds a criterion to the selected criteria block. Files must satisfy each criterion to match the filter. You can configure up to three criteria blocks per file pool policy.

The **Ignore case** box should be selected for files that are saved to the cluster by a Windows client.

File pool policies with path-based policy filters and storage pool location actions are executed during the write of a file matching the path criteria. Path-based policies are first executed when the SmartPools job runs, after that they are executed during the matching file write. File Pool Policies with storage pool location actions and policy filters based on other attributes besides path get written to the node pool with the highest available capacity and then moved, if necessary to match a file pool policy, when the next SmartPools job runs. This ensures that write performance is not sacrificed for initial data placement.

The matching criteria are file name, including the file extension, the directory path, the file type, file attribute, file size, and different file time stamps. The time stamps include the last time accessed, last time modified, create time, and last time the metadata was changed. To use last time accessed or atime, the atime tracking must be enabled on the cluster.

SSD Usage

- Nodes can contain SSDs and HDDs
- Data and metadata can reside on either HDDs or SSDs
 - L3 cache enabled by default
 - Metadata read acceleration SSD strategy
- SSD strategies settings
 - Default file pool policy
 - Using file pool policies (recommended)
 - Manually set (not recommended)
 - File system explorer in web administration interface
 - `isi set` command in CLI



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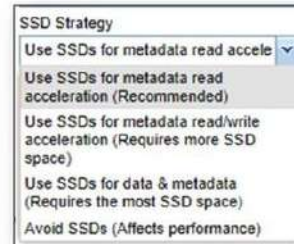
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If a node pool has SSDs, by default, L3 cache is enabled on the node pool. To use the SSDs for other strategies, the L3 cache must first be disabled on the node pool. The metadata read acceleration is the recommended SSD strategy. With metadata read acceleration, OneFS directs one copy of the metadata is directed to SSDs, and the data and remaining metadata copies are directed to reside on HDDs. The benefit of using SSDs for file-system metadata includes faster namespace operations used for file lookups. The settings that control SSD behavior in SmartPools either in the Default File Pool policy or when SmartPools is licensed in the file pool policy settings. Manual setting can be used to enable SSD strategies on specific files and directories, but is not recommended.

SSD Strategy Settings

- Metadata read acceleration
 - One metadata copy on SSDs
 - Data and other metadata copies on HDDs
- Metadata read/write acceleration
 - All metadata copies on SSDs
 - Data on HDDs
- Data on SSDs
 - Data and associated metadata on SSDs
- Avoid SSD
 - Data and all metadata on HDDs

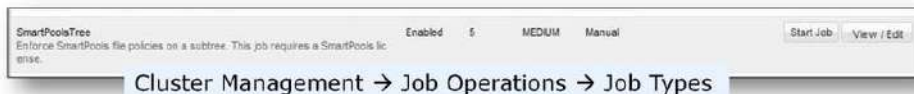


Selecting metadata read acceleration creates one metadata mirror of file metadata on the SSDs and writes the rest of the metadata mirrors plus all user data on HDDs. Selecting metadata read/write acceleration writes all metadata mirrors to SSD. This setting can consume up to 5 or 6 times more SSD space than a metadata read acceleration SSD strategy. Selecting **Use SSDs for data & metadata** writes all of the data and metadata for a file on SSDs. Selecting **Avoid SSDs** writes all associated file data and all metadata mirrors to HDDs only and does not use SSDs.

SSDs are node pool specific and used within only the node pool containing the data. The exception is with global namespace acceleration (GNA). When GNA is enabled if a specific node pool does not contain SSDs, any SSDs within the system that are not used for L3 cache, may be used for metadata read acceleration provided by GNA. If a node pool has SSDs and GNA is enabled, the preference is to use the node pools SSDs first for GNA before using SSDs contained on other node pools.

File Pool Policy Jobs

- Jobs run once a day to enforce file pool policies
- Job Engine job
 - **SetProtectPlus** job - SmartPools unlicensed
 - **SmartPools** job - SmartPools licensed
 - Job Engine restripe category
- **SmartPoolsTree** job
 - Selectively apply SmartPools file pool policies
 - Allows Job Engine to manage resource utilization



File pool policies are applied to the cluster by a job. When SmartPools is unlicensed, the **SetProtectPlus** job applies the default file pool policy. When SmartPools is licensed, the **SmartPools** job processes and applies all file pool policies. By default, the job runs at 22:00 hours every day at a low priority. Policies are checked in order from top to bottom. The **SetProtectPlus** and **SmartPools** jobs are part of the restripe category for the Job Engine. One restripe job at a time is done. Only one restripe job can run at a time. The job results are displayed in the **Cluster Management > Job Operations > Job Reports** or **Job Events**. The Job Engine is discussed in more detail in a later module.

The **SmartPoolsTree** job introduced in OneFS 8.0 is used to apply selective SmartPools file pool policies. It executes the **isi filepool apply** command. Using the Job Engine to control and manage the command allows the Job Engine to manage the resources assigned to the job. This allows for testing file pool policies before applying them to the entire cluster.

SmartPoolsTree Job Settings

The screenshot shows the 'Start a Job' dialog for the 'SmartPoolsTree' job type. The dialog is titled 'Start a Job' and has a section for 'Job Type Details'. The 'Name' field is 'SmartPoolsTree'. The 'Description' is 'Enforce SmartPools file policies on a subtree. This job requires a SmartPools license.' A callout bubble points to the description with the text 'License required'. Below the description is a checkbox for 'Allow Duplicate Jobs'. The 'Priority' is set to '5'. The 'Impact Policy' is set to 'MEDIUM'. The 'Paths' field contains '/ifs' and a 'Browse...' button. A callout bubble points to the 'Paths' field with the text 'Path options Default: /ifs'. Below the 'Paths' field is a button for 'Add another directory path'. There are three checkboxes for execution options: 'Dry run: calculate what would be done', 'Directory only: skip processing of regular files', and 'Policy only: apply policies but skip restriping'. A callout bubble points to these options with the text 'Execution options'. The 'Recurse: process children, recursively' checkbox is checked, and a callout bubble points to it with the text 'Default'. The EMC logo is in the bottom right corner.

The **SmartPoolsTree** job allows file pool policies to be rapidly applied directly to a directory, instead of the entire filesystem. The path and options are part of the job execution settings when starting the **SmartPoolsTree** job. The job can be setup and run using the web administration interface (shown) or the CLI. Web administration interface navigation is **Cluster Management > Job Operations > Job Types**. A SmartPools license is required for the job to be executed. The job enables policies to be applied only to a specific path as needed. There are many options that are useful when dealing with file pool policies. You can use the **Dry run** option to test file pool policies before applying them. You can apply the policy or policies at the directory level only and not process regular files. You can apply only a specific file pool policy. Or you can recursively apply policies to all child directories if desired. And you can use ingest (CLI only) as an alias for directory-only and policy-only options. The options are the same as those available using the previous CLI command.

Policy Template

Policy Templates		
Template Name	Description	Actions
Archive	Move older files to older storage.	View / Use Template
VMware Files	Set VMware files for random-access.	View / Use Template
ExtraProtect	Protect a subset of files at a higher requested protection.	View / Use Template
PoolByPath	Assign files to the performance pool based solely on their path.	View / Use Template

- Opens up partially populated new file pool policy
- Must rename policy
- Can modify and add additional criteria and actions
- Use in web administration interface only
 - View templates only using CLI



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When the template is used, the basic settings are preset to the name of the template along with a brief description. These settings can be changed by the user. A filter is also pre-configured to achieve the specified function, in this case to archive files older than two months. Additional criteria can be configured using the links in the filter box.

You need to decide where to store the archived files and what, if any, changes to make to the protection level. Additionally, you can also change the I/O optimization levels if desired.

You can also use an existing policy as a template by changing the name and any settings you desire, and then saving the policy.

Templates may only be used to create new policies in the web administration interface. In the CLI, the templates provide a guide to creating the CLI text used to create the policy.

SmartPools Configuration

1. Configure existing node pools and tiers
2. Review (modify) Default file pool policy
3. License SmartPools
4. Configure file pool policies
 - a. Select node pools or tiers
 - b. Create / configure file pool policies



Launch
SmartPools
Video



Launch File
Policy
Video



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There are four major steps needed to configure all of the features of SmartPools. The first three steps in configuring SmartPools can be accomplished without licensing SmartPools. By default, unlicensed SmartPools is implemented on a cluster. The default file pool policy only allows an Anywhere storage target. This means that all files in the cluster can be written to any node pool or tier on the cluster. You can have multiple node pools from the same performance series with each assigned to a tier without licensing SmartPools, however, there is no capability to control the location of files to the node pools.

If you decide you need to have multiple tiers or more than one file pool policy, then you must license SmartPools. Additionally, if you want to add nodes pools that are from different series of nodes, then you must license SmartPools. For example, if you have an S-Series node pool, and an NL-Series node pool in the same cluster, you have multiple performance tiers, and a SmartPools is required for the configuration.

Select the clip icon for a short demonstration on the configuration.

Lesson 1: Summary

Having completed this lesson, you can now:

- Describe SmartPools functionality
- Explain and configure tiers, node pools, global settings
- Understand node compatibilities
- Create file pool policies
- Understand default policy
- Define SSD strategy use

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Having completed this lesson, you now know how to describe SmartPools functionality, explain and configure tiers and node pools, configure SmartPools settings, understand node compatibilities, create file pool policies, understand how to apply the default policy, and define using the SSD strategy.

Lesson 2: CloudPools

After completing this lesson, you will be able to:

- Explain CloudPools benefits
- Create and manage CloudPools

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After completing this lesson, you will be able to explain CloudPools benefits and create and manage CloudPools.

CloudPools Overview

- Managed as extension of SmartPools
 - Same policy engine
 - Same administrative and management framework
- Tiering to public, private, hybrid cloud
- Transparent to users and applications
- Extends Data Lake
- Data transmission
 - Compression for network bandwidth optimization
 - Encrypt data sent to cloud



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CloudPools is a licensed addition to SmartPools that allows the definition of another tier of storage the cluster can utilize, the cloud. The SmartPools automated tiering policy engine and framework is used to implement and manage CloudPools, and as with SmartPools, the tiering is transparent to users and applications. CloudPools store connection details on the Isilon cluster and add file pool policies that move archive data out to cloud storage. With CloudPools, an on-premise Isilon data lake can be extended to cloud-scale capacities.

Data moved to the cloud can be compressed for bandwidth optimization. When enabling, files undergo a compression algorithm and then are broken into their 1MB objects for storage, conserving space on the cloud storage resources. Internal performance testing does note a performance penalty for applying compression, and decompressing files on read.

Data encryption can be enabled. Encryption is applied to file data transmitting to the cloud service. Each 128k file block is encrypted using AES256 encryption, then transmitted as an object to the cloud. Internal performance testing notes very little performance penalty for encrypting the data stream. Compression and encryption is enabled on a per-policy basis.

CloudPools Considerations

- Licensed feature
- Cloud services
 - Public cloud - pay for use
 - Private
 - Supported providers
 - EMC Isilon, EMC ECS Appliance, EMC Elastic Cloud Storage Service, Amazon S3, and Microsoft Windows Azure
- Access performance based on cloud connection
- Writes to cloud file cached locally
 - Cache flushed at designated intervals



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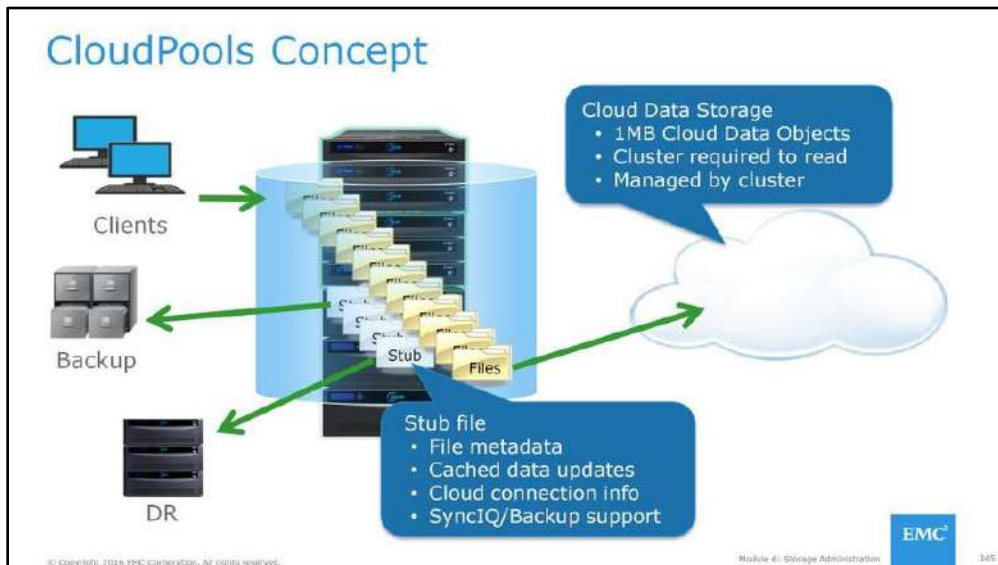
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CloudPools requires SmartPools and CloudPools licenses.

In a public cloud, enterprises may pay only for the capacity they actually use per month, for instance storage of 100TB on a public cloud might be three thousand dollars per month. Once data is stored in the cloud, fees are incurred at a low rate for reading this data, higher for writing or copying of the data and still higher for the removal of that data back to private resources. Pricing varies widely based on performance requirements and other agreements.

Private clouds utilize similar arrays of compute and storage resources, but are offered either within the company network, or connected through a private direct connection rather than the general internet, possibly through a VPN connection. These private object stores may use EMC's ECS or Isilon systems as their base infrastructure and offer a variety of services similar to a public cloud.

When accessing files on the cluster, whether through SMB, NFS, HDFS, SWIFT, etc., files stored in the cloud vs. stored locally on the cluster appear identical. When opening a file stored in the cloud, the cluster makes the appropriate read request to bring the file to view for the client. These read requests will of course incur additional latency dependent on the quality of networking and service connection to the cloud resource, but the client behavior remains the same. Updates to the file are stored in the SmartLink (stub) data cache on the Isilon cluster. At a designated interval, the Isilon cluster will flush cached changes out to the cloud, updating the files. This allows the administrator greater control of cloud storage costs, as writes often incur additional fees.



Shown here is an Isilon cluster with tiering between the nodes. When files are moved to a cloud pool tier, a SmartLink file remains on the cluster (referred to as a “stub” file). The stub files are pointers (contain metadata) to the data moved to the cloud, and any cached data changes not yet written out to the cloud. Stub files have the details for connecting to the appropriate cloud resource for its file. Also, when enabling encryption, the encryption keys become a part of the stub file, further securing cloud data from direct access.

Clients and applications access to data is transparent. So clients simply continue opening files, with a bit longer latency for those files in the cloud. NDMP backups and SyncIQ policies continue as if the data were still in place, save time by just backing up the stub files, or by copying full files as necessary. Additional details for this functionality follows in the SyncIQ section of the training.

Data that is moved to the cloud, is also protected against anyone connecting directly to the cloud. Files are stored in 1MB chunks called Cloud Data Objects that appear unreadable to direct connections. Metadata stored on the Isilon cluster is required to read these files, adding an extra layer of protection to cloud storage.

CloudPools Administration

File System → Storage Pools → CloudPools

Storage Pools

Summary File Pool Policies SmartPools **CloudPools** SmartPools Settings CloudPools Settings

Cloud Storage Accounts

[+ Create a Cloud Storage Account](#)

Name	Type	State	Username	URI	Actions
 RAN dst acct	Isilon RAN	Enabled	root	isilon://10.7.135.67:8080/namespaces/ifs/ran/dst	View / Edit

CloudPools

[+ Create a CloudPool](#)

CloudPools and Accounts	Type	State	Vendor	Description	Actions
 RAN dst CP	Isilon RAN	Enabled	Isilon		View / Edit
 RAN dst acct					

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Once the SmartPools and CloudPools licenses are applied, the web administration interface shows the **CloudPools** tab. Selecting defines the connection details for a cloud service. After a Cloud Storage Account is defined and confirmed, the administrator can define the cloud pool itself. Further additions to the file pool policies allows the definition of a policy that moves data out to the cloud.

Creating Cloud Storage Account

The screenshot shows a 'Create a Cloud Storage Account' dialog box. It has a title bar and a close button. The dialog is divided into sections: 'Connection Settings', 'Cloud Account Information', and a 'Skip SSL certificate validation (not recommended)' checkbox. The 'Connection Settings' section includes a 'Name or Alias' field (containing 'CloudPool_FromData') and a 'Type' dropdown menu. The 'Cloud Account Information' section includes a 'URI' field, a 'User Name' field, and a 'Key' field. Callouts point to these fields: 'Fully Qualified URI for account' points to the 'URI' field, 'Cloud Account Name' points to the 'User Name' field, and 'Account Password' points to the 'Key' field. The 'Type' dropdown menu is open, showing options: 'EMC Isilon', 'EMC ECS Appliance', 'EMC Elastic Cloud Storage Service', 'Microsoft Azure', and 'Amazon S3'. A '+ Create a Cloud Storage Account' button is visible on the left. The bottom of the dialog has 'Cancel' and 'Create Account' buttons. The EMC logo is in the bottom right corner.

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Shown here is the window for creating a cloud storage account. All the fields are required. The **Name or Alias** must be unique to the cluster. The **Type** is the type of cloud account and options are on the drop-down list. The **URI** must use HTTPS and match the URI used to set up the cloud account. The **User Name** must be the name provided to the cloud provider. The **Key** is the account password provided to (or received from) the cloud provider.

Creating CloudPool Storage Target

Create a CloudPool

* Required field

Help

CloudPool Settings

State: **Enabled**

State Details

Name:

Type:

- EMC Isilon
- EMC ECS Appliance
- EMC Elastic Cloud Storage Service
- Microsoft Azure
- Amazon S3

Vendor:

Description:

Accounts in CloudPool

Cancel Create CloudPool

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Once a storage account is created, a CloudPool can be created that is associated or points to the account. Shown here is the window to **Create a CloudPool**. The **Name** must be unique to the cluster. The **Type** is the type of cloud account and the drop-down list has the supported options. The **Vendor** name and **Description** are optional fields. The **Account in CloudPool** is activated after the **Type** is selected and the configured storage accounts will be listed on the drop-down list.

File Pool Policies

CloudPools Settings

- Create SmartPools file pool policy
- Apply CloudPools action settings to policy
- Apply advanced CloudPools settings (customized or default)



SmartPools file pool policies are used to move data from the cluster to the selected CloudPool storage target. When you configure a file pool policy, you have the option to apply CloudPools actions to the selected files. As part of the setting, you select the CloudPool storage target from the available list. You can elect to encrypt the data prior to sending to the specified CloudPool, and you may compress the data before transfer to improve the transfer rate.

File Pool Policies

Advanced CloudPools Settings

Hide Advanced CloudPool Settings

Data Retention Settings

Cloud Data Retention Period

1 week

Incremental Backup Retention Period

5 year

Full Backup Retention Period for NDMP

5 year

Accessibility and Cache Settings

Writeback Frequency

5 hour

Accessibility

Cache archived files locally

Cache Read Ahead

Cache accessed data only

Cache Expiration

1 day

Archiving files with snapshots does not save space on the cluster.

☒ Archive files with snapshots

Setting	Description
Data Retention Settings	Cloud Data Retention Period: Specifies how long cloud objects are retained after a SmartLink file has been deleted. When a SmartLink file is deleted on the local cluster, CloudPools cleans up local resources allocated for the SmartLink files, and also removes the associated cloud objects. This work is performed weekly by the cloud objects garbage collector job. Incremental Backup Retention Period for NDMP Incremental Backup and SyncIQ: Specifies the backup retention period for SmartLink files created with an incremental NDMP backup policy or a SyncIQ policy. This value is only relevant for incremental NDMP backups and SyncIQ replication. Full Backup Retention Period for NDMP Only: Specifies the backup retention period for SmartLink files created with a full NDMP backup policy. This value is only relevant for full NDMP backups.
Accessibility and Cache Settings	Writeback Frequency: Specifies how often SmartLink files edited on the Isilon cluster are written to their associated cloud data objects. Accessibility: Specifies whether or not to cache archived data locally. Local caching enables faster access of cloud data, but reduces space saving on your cluster. Cache Read Ahead: Specifies whether cloud data is fully or partially cached when you access a SmartLink file on the local file system. If the policy specifies partial caching, the system only caches the blocks needed when a file is accessed. If the policy specifies full caching, cloud data is fully cached when the SmartLink file is accessed. Cache Expiration: Specifies the amount of time cloud data that has been cached is retained in the local cache of associated SmartLink files. The system purges the SmartLink cache of data that has not been accessed for the number of days specified.
Archive files with snapshots	Specifies that CloudPools is to archive files that have one or more snapshot versions. The latest versions of such files are archived, and SmartLink files are left in their place on the local cluster. Because the full snapshot versions of files are retained on the cluster, archiving these files does not save space on the cluster.

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A number of default advanced CloudPool options are configured. You may want to modify these setting for the file pool policy based on your requirements. Modification is not be necessary for most workflows. The table is an excerpt from the *Isilon OneFS Version 8.0.0 Web Administration Guide* and provides a description of the advanced fields.

Archive and Recall Files from CLI

```
JSandBox2-2# isi cloud --help
Description:
  Manage CloudPool accounts and pools.
```

```
Usage:
  isi cloud {action} | <subcommand>
  [--timeout <integer>]
  [--help | -h]
```

```
Actions:
  archive      Archive a file.
  recall       Recall a stubbed file.
  restore_obj  Restore the cloud object index for an account.
```

```
Subcommands:
  access      Grant access to CloudPool accounts and FilePool policies.
  accounts    Manage CloudPool accounts.
  jobs        Manage CloudPool jobs.
  pools       Manage Cloud Pools.
  settings    Manage CloudPool top level settings.
```

- Specific file management
- Actions
 - Archive
 - Recall

- Manage CloudPools jobs
 - Separate from Job Engine
- View files in CloudPools job

```
JSandBox2-2# isi cloud jobs --help
Description:
  Manage CloudPool jobs.
```

```
Usage:
  isi cloud jobs {action} | <subcommand>
  [--timeout <integer>]
  [--help | -h]
```

```
Actions:
  cancel       Cancel a CloudPool job.
  create       Create a CloudPool job.
  pause        Pause a CloudPool job.
  resume       Resume a paused CloudPool job.
  list         View a list of CloudPool jobs.
  view         View detailed properties of a CloudPool job.
```

```
Subcommands:
  files        Manage CloudPool jobs.
```



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From the CLI, you have the option to manage specific files. You can archive files to the CloudPool and recall files from the CloudPool using the **isi cloud archive** and **isi cloud recall** commands. The CloudPools job is outside of the Job Engine. Separate commands to manage the CloudPools jobs are provided using the **isi cloud jobs** command. To view the files associated with a specific CloudPools job, use the **isi cloud jobs file** command. More detailed information on these commands is available in the *OneFS 8.0 CLI Administration Guide* available on www.support.emc.com.

isi cloud recall Command



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Files stored in the cloud can be fully recalled using the **isi cloud recall** command. Recall can only be done via the CLI. When recalled, the full file is restored to its original directory, and therefore may still be subject to the same file pool policy that originally archived it and re-archived to the cloud the next time the SmartPools job runs. If this is unintended, the recalled file should be moved to a different, unaffected, directory. The recalled file overwrites the stub file. The command can be executed for an individual file or recursively for all files in a directory path.

CloudPools Limitations

- File recall requires administrative action
 - Limit potential for data removal fees
- File spillover not supported
 - Spillover from CloudPools
 - No statistics automated from cloud on capacity usage
 - No spillover can be triggered
 - Spillover to CloudPools
 - Spillover is temporary measure - risk of file recall fees
- Limited file statistics on data usage details



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In a standard node pool, file pool policies can move data from high performance tiers to storage tiers and back as defined by their access policies. However, data moved to the cloud will remain stored in the cloud unless an administrator explicitly requests data recall to local storage. If a file pool policy change is made that rearranges data on a normal node pool, data will not be pulled from the cloud. Public cloud storage often places the largest fees on data removal from cloud storage, thus file pool policies avoid incurring removal fees by placing this decision in the hands of the administrator.

The connection between a cluster and a cloud pool has limited statistical features. The cluster does not track the data storage used in the cloud. This means file spillover is not supported. Spillover to the cloud again presents the potential for file recall fees. Spillover is designed as a temporary safety net, once the target pool capacity issues are resolved, data would be recalled back to the target node pool.

Additional statistic details, such as the number of stub files on a cluster or how much cache data is stored in stub files and would be written to the cloud on a flush of that cache, is not easily available. Finally, no historical data is tracked on the network usage between the cluster and cloud either in writing traffic or in read requests. These network usage details should be found by referring to the cloud service management system.

Later in this module, SyncIQ's support with CloudPools is discussed.

Lesson 2: Summary

Having completed this lesson, you can now:

- Explain CloudPools benefits
- Create and manage CloudPools

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Having completed this lesson, you now know how to explain CloudPools benefits, create and manage CloudPools, and describe SyncIQ CloudPools support.

Lesson 3: SmartQuotas

Upon completion of this lesson, you will be able to:

- Differentiate types of quotas
- Explain benefits of SmartQuotas
- Understand thin provisioning
- Configure SmartQuotas for directories

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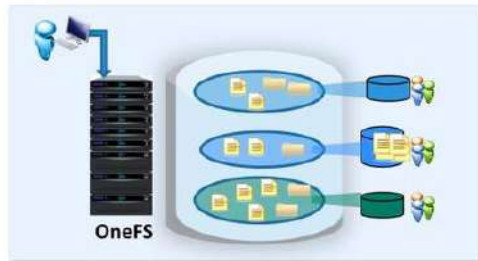


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Upon completion of this lesson, you will be able to differentiate types of quotas, explain benefits of SmartQuotas, understand thin provisioning, and configure SmartQuotas for directories.

SmartQuotas Overview

- Administrative control over storage use
 - Allows limiting, monitoring disk usage
- Notification and reporting options
- Useful for policy enforcement and cost accounting



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SmartQuotas is a software module used to limit, monitor, thin provision, and report disk storage usage at the user, group, and directory levels. Administrators commonly use file system quotas as a method of tracking and limiting the amount of storage that a user, group, or a project is allowed to consume. SmartQuotas can send automated notifications when storage limits are exceeded or approached.

Quotas are a useful way to ensure that a user or department uses only their share of the available space. SmartQuotas are also useful in enforcing an internal chargeback system. SmartQuotas contain flexible reporting options that can help administrators analyze data usage statistics for their Isilon cluster. Both enforcement and accounting quotas are supported, and a variety of notification methods are available.

SmartQuotas allows for thin provisioning, also known as over-provisioning, which allows administrators to assign quotas above the actual cluster size. With thin provisioning, the cluster can be full even while some users or directories are well under their quota limit. Administrators can configure notifications to send alerts when the provisioned storage approaches actual storage maximums enabling additional storage to be purchased as needed.

Usage Limits



Accounting Quotas

- Monitors disk usage
- Analysis
- Future planning
- Threshold subtypes
 - Advisory



Enforcement Quotas

- Monitors and limits disk usage
- Enables notification
- Threshold subtypes
 - Hard
 - Soft



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You can choose to implement accounting quotas or enforcement quotas. Accounting quotas monitor, but do not limit, disk storage. They are useful for auditing, planning, and billing purposes. The results can be viewed in a report. SmartQuotas accounting quotas can be used to:

- Track the amount of disk space that various users or groups use
- Review and analyze reports that can help identify storage usage patterns
- Intelligently plan for capacity expansions and future storage requirements

Enforcement quotas include all of the functionality of accounting quotas, but they also enable the sending of notifications and the limiting of disk storage. Using enforcement quotas, a customer can logically partition a cluster to control or restrict how much storage a user, group, or directory can use. Enforcement quotas support three subtypes and are based on administrator-defined thresholds:

- Hard quotas limit disk usage to a specified amount. Writes are denied after the quota threshold is reached and are only allowed again if the usage falls below the threshold.
- Soft quotas enable an administrator to configure a grace period that starts after the threshold is exceeded. After the grace period expires, the boundary becomes hard, and additional writes are denied. If the usage drops below the threshold, writes are again allowed.
- Advisory quotas do not deny writes to the disk, but they can trigger alerts and notifications after the threshold is reached.

Applying SmartQuota Types

Type	Description	Function
Directory	Quota on directory that includes all files and subdirectories	Limits size of directory structure
User	Quota on user for specific directory	Limits data single user can store in directory
Default User	Quota "template" on all users for specific directory	Limits data all users can store directory
Group	Quota on group for specific directory	Limits data group can store in directory
Default Group	Quota "template" on all groups for specific directory	Limits data all groups can store in directory



Do not enforce quotas on root of file system (/ifs)



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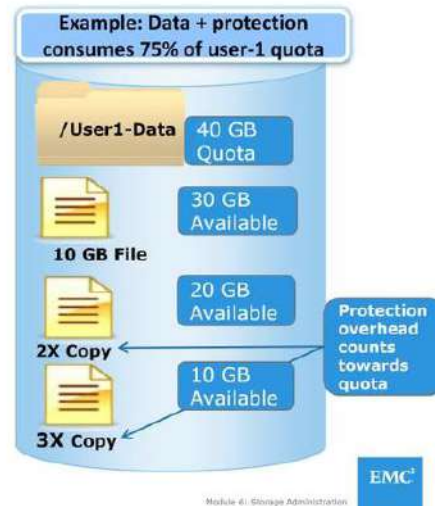
There are five types of quotas that can be configured, which are directory, user, default user, group, and default group.

- Directory quotas are placed on a directory, and apply to all directories and files within that directory, regardless of user or group. Directory quotas are useful for shared folders where a number of users store data, and the concern is that the directory will grow unchecked because no single person is responsible for it.
- User quotas are applied to individual users, and track all data that is written to a specific directory. User quotas enable the administrator to control how much data any individual user stores in a particular directory.
- Default user quotas are applied to all users, unless a user has an explicitly defined quota for that directory. Default user quotas enable the administrator to apply a quota to all users, instead of individual user quotas.
- Group quotas are applied to groups and limit the amount of data that the collective users within a group can write to a directory. Group quotas function in the same way as user quotas, except for a group of people and instead of individual users.
- Default group quotas are applied to all groups, unless a group has an explicitly defined quota for that directory. Default group quotas operate like default user quotas, except on a group basis.

You should not configure any quotas on the root of the file system (**/ifs**), as it could result in significant performance degradation.

Overhead Calculations

- OneFS can include or not included overhead in quota calculations
- Including overhead
 - Limits available disk space
 - Increases future usage calculations
 - Not typically used
- Services that create overhead
 - Disk protection
 - Snapshots



Most quota configurations do not need to include overhead calculations. If you configure overhead settings, do so carefully, because they can significantly affect the amount of disk space that is available to users.

If you include data-protection overhead in a quota usage calculation, disk-usage calculations for the quota subtract any space that is required to accommodate the data-protection settings for that. The options are:

1. **Default:** The default setting is to only track user data, which is just the data that is written by the user. It does not include any data that the user did not directly store on the cluster.
2. **Snapshot Data:** This option tracks both the user data and any associated snapshots. This setting cannot be changed after a quota is defined. To disable snapshot tracking, the quota must be deleted and recreated.
3. **Data Protection Overhead:** This option tracks both the user data and any associated FEC or mirroring overhead. This option can be changed after the quota is defined.
4. **Snapshot Data and Data Protection Overhead:** Tracks all data user, snapshot and overhead with the same restrictions.

For example, consider a user who is restricted by a 40 gigabyte (GB) quota that includes data-protection overhead in its disk-usage calculations. If the cluster is configured with a 2x data-protection level and the user writes a 10 GB file to the cluster, that file actually consumes 20 GB of space: 10 GB for the file and 10 GB for the data-protection overhead. In this example, the user has reached 50% of the 40 GB quota by writing a 10 GB file to the cluster.

Quotas can also be configured to include the space that is consumed by snapshots. A single path can have two quotas applied to it: one without snapshot usage (default) and one with snapshot usage. If snapshots are included in the quota, more files are included in the calculation.

Quotas and Thin Provisioning

- Total quota amount can exceed actual cluster size
- Example:
 - 250 TB cluster with two quotas
 - Quota 1 has 200 terabyte quota
 - Quota 2 has 75 terabyte quota
 - Total space provisioned = 275 terabytes
- Cluster can fill before quotas are reached



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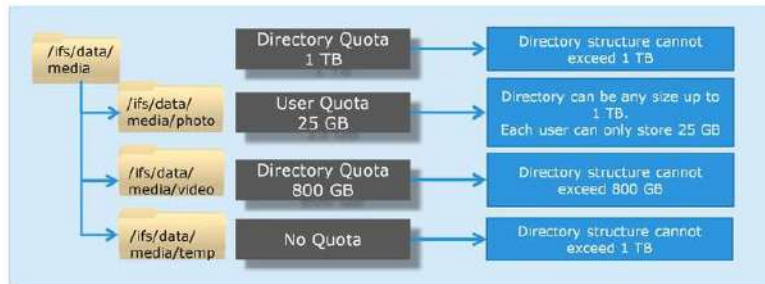
Thin provisioning is a tool that enables an administrator to define quotas that exceed the capacity of the cluster. Doing this accomplishes two things:

1. It allows a smaller initial purchase of capacity/nodes, and the ability to simply add more as needed, promoting a capacity on demand model.
2. It enables the administrator to set larger quotas initially and so that continually increases as users consume their allocated capacity are not needed.

However, thin provisioning requires that cluster capacity use be monitored carefully. With a quota that exceeds the cluster capacity, there is nothing to stop users from consuming all available space, which can result in service outages for all users and services on the cluster.

Nesting Quotas

- Nesting: Multiple quotas within same directory
- Top-level directory quota limits total of combined subdirectories



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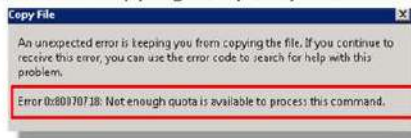
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Nesting quotas refers to having multiple quotas within the same directory structure. In the example shown, all quotas are hard enforced. At the top of the hierarchy, the **/ifs/data/media** folder has a directory quota of 1 TB. Any user can write data into this directory, or the **/ifs/data/media/temp** directory, up to a combined total of 1 TB.

The **/ifs/data/media/photo** directory has a user quota assigned that restricts the total amount any single user can write into this directory to 25 GB. Even though the parent directory (media) is below its quota restriction, a user is restricted within the photo directory. The **/ifs/data/media/video** directory has a directory quota of 800 GB that restricts the capacity of this directory to 800 GB. However, if users place a large amount of data in the **/ifs/data/media/temp** directory, say 500 GB, then only 500 GB of data can be placed in this directory, as the parent directory (**/media**) cannot exceed 1 TB.

Quota Notification Options

- Send email and/or generate cluster event
- Email uses cluster settings
 - Can send to owner (user) or alternate
 - Can use message template
- Email integrates with AD or LDAP email configuration
- Also use mapping to specify addresses



Cluster Event Generated

Event Details	
View Event Help	Queue Event
Create Notification Rule	
Ancestry:	Root event
Instance ID:	4,419
Start time:	2012-11-10 18:16:55
End time:	—
Quoted time:	—
Event type:	500919904
Category:	SmartQuotas events
Message:	SmartQuotas threshold violation on quota denied, domain
Scope:	Cluster
Update count:	0
Event hierarchy:	Coalescing event
Severity:	info
Extrema severity:	info
Value:	0.0
Extreme value:	0.0



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Quota events can generate notifications by email or through a cluster event. The email option sends messages using the default cluster settings. You can specify to send the email to the owner of the event, which is the user that triggered the event, or you can send email to an alternate contact, or both the owner and an alternate. You also have the option to use a customized email message template. If you need to send the email to multiple users, you need to use a distribution list.

If you are using LDAP or Active Directory to authenticate users, the Isilon cluster uses the email settings for the user stored within the directory. If no email information is stored in the directory, or authentication is performed by a Local or NIS provider, you must configure a mapping.

Quota Notifications

- Global notification enabled for quotas, by default
- Customize notifications for each type

Event	Advisory	Soft	Hard
Limit Exceeded	✓	✓	✓
Limit Remains Exceeded	✓	✓	✓
Grace Period Expired		✓	
Write Access Denied		✓	✓



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A default notification is enabled when SmartQuotas is enabled. You can specify different notification parameters for each type of quota (advisory, soft, and hard). You can also set a different notification scheme on individual quotas, which allows you to create a customized notification system. You can establish a default notification scheme for each type of quota, then customize specific notifications as appropriate.

Each type of quota has different events that can trigger a notification, as shown in the table.

- **Limit Exceeded** is triggered when a quota threshold is exceeded for advisory or soft quotas, and when a threshold is reached with a hard quota.
- **Limit Remains Exceeded** generates an alert on a recurring basis while the quota is exceeded.
- **Grace Period Expired** is triggered when a soft quota is exceeded and is not corrected before the grace period has elapsed.
- **Write Access Denied** occurs with a soft quota if the grace period has elapsed or when a hard quota threshold is reached and a user attempts to write data.

Lesson 3: Summary

Having completed this lesson, you are now able to:

- Define types of quotas
- Explain benefits of SmartQuotas
- Understand thin provisioning
- Configure SmartQuotas for directories

Having completed this lesson, you are now able to define types of quotas, explain benefits of SmartQuotas, understand thin provisioning, and configure SmartQuotas for directories.

Lesson 4: SnapshotIQ

Upon completion of this lesson, you will be able to:

- Describe snapshot behavior
- Identify snap types
- Understand how snapshots are saved to disk
- Configure and manage snapshot functionality

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Upon completion of this lesson, you will be able to describe snapshot behavior, identify type of snaps OneFS completes, understand how snapshots are saved to disk, configure and manage snapshot functionality.

Overview of Isilon SnapshotIQ

- Creates logical copy of data
- Useful for data protection
 - User-initiated file restoration
- Taken at directory level
 - Includes subdirectories
 - Not at file system level
- OneFS uses Copy on Write (CoW) and Redirect on Write (RoW) snapshot methods
- Snapshot limits
 - 20,000 total cluster
 - 1,024 per directory
- Licensed feature
 - Activation required
 - Can be disabled
 - License not required for system generated snapshots



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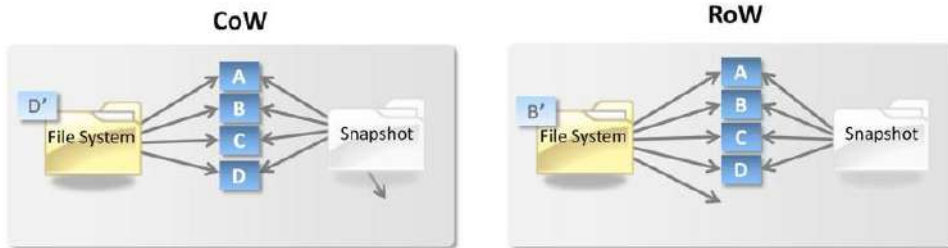
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A OneFS snapshot is a logical pointer to data stored on a cluster at a specific point in time. Snapshots target directories on the cluster, and include all data within that directory, including any subdirectories contained within. This is in contrast to the traditional approach, where snapshots are taken at a file system or volume boundary. Snapshots are more efficient than backing up data to a separate physical storage device in terms of both time and storage utilization.

You can use snapshots to protect data against accidental deletion and modification. If a user modifies a file and later determines that the changes were unnecessary or unwanted, the earlier version of the file can be copied back from the snapshot. Also, because snapshots are available locally, end users can often restore their data without the assistance of a system administrator, saving administrators the time it takes to retrieve the data from another physical location. In addition to using SnapshotIQ as a stand-alone tool for user-initiated file restore, snapshots can also be used for staging content to export, and ensuring that a consistent point-in-time copy of your data is replicated or backed up. To use the SnapshotIQ, you must activate a SnapshotIQ license on the cluster. However, some OneFS operations generate snapshots for internal system use without requiring a SnapshotIQ license. If an application generates a snapshot, and a SnapshotIQ license is not configured, you can still view the snapshot. However, all snapshots generated by OneFS operations are automatically deleted after they are no longer needed. You can disable or enable SnapshotIQ at any time.

SnapshotIQ uses both copy on write (CoW) and redirect on write (RoW) for its differential snapshots. You can configure basic functions for the SnapshotIQ application, including automatically creating or deleting snapshots, and setting the amount of space that is assigned exclusively to snapshot storage. You can configure advanced settings that control user access and directory visibility and configure advanced options for root directory and subdirectory access and visibility for NFS, Windows, and local users. The default is 20,000 snapshots. Snapshots should be set up for separate distinct and unique directories. Do not snapshot the **/ifs** directory. Instead you can create snapshots for the subdirectory structure under the **/ifs** directory. Snapshots only start to consume space when files in the current version of the directory are changed or deleted.

Copy on Write vs. Redirect on Write



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SnapshotIQ captures snapshots of parts of the filesystem (usually defined at the directory level). You can configure basic functions for the SnapshotIQ application, including automatically creating or deleting snapshots, and setting the amount of space that is assigned exclusively to snapshot storage. You can configure advanced settings that control user access and directory visibility and configure advanced options for root directory and subdirectory access and visibility for NFS, Windows, and local users.

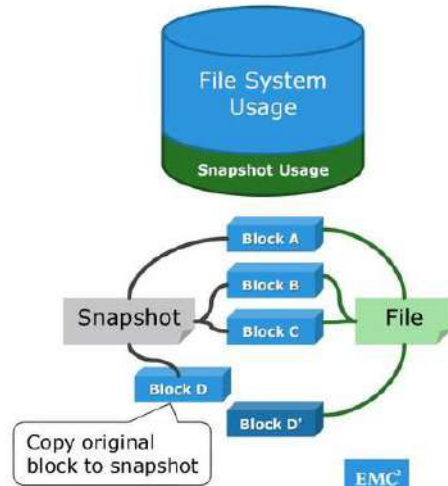
Both CoW (Copy on Write) and RoW (Redirect on Write) are used by OneFS. Both have pros and cons, and OneFS dynamically picks which method to use in order to maximize performance and keep overhead to a minimum. With CoW a new write to HEAD results in the old blocks being copied out to the snapshot version first. Shown here changes are made to "D".

Although this incurs a double write penalty, it results in less fragmentation of the HEAD file, which is better for cache prefetch and related file reading functions. Typically, CoW is most prevalent in OneFS, and is primarily used for small changes, inodes and directories.

Redirect on write (RoW), on the other hand, avoids the double write penalty by writing changes to a snapshot protected file directly to another free area of the file system. However, the flip side to this is increased file fragmentation. Since RoW does not maintain file contiguity because of writing changes to other file system regions, RoW in OneFS is used for more substantial changes such as deletes and large sequential writes.

How Snapshots Work

- Snapshot creation
 - Does not make copy of file
 - Creates additional set of pointers to file
 - Consumes negligible disk space
 - Created almost instantaneously regardless of data amount
- Size grows as data modified
 - Only changed (original) data blocks contained in snapshots
 - Changed data stored in /.snapshots directories
- Snapshot storage
 - Can store snaps on different node pool or tier
 - Automatic expiration/deletion available



Snapshots are created almost instantaneously regardless of the amount of data contained in the snapshot. A snapshot is not a copy of the original data, but only an additional set of pointers to the original data. So, at the time it is created, a snapshot consumes a negligible amount of storage space on the cluster. Snapshots reference or are referenced by the original file. If data is modified on the cluster (**Block D'** in the graphic), only one copy of the changed data is made. Like in the previous example, with CoW the original block (**Block D**) is copied to the snapshot. This allows the snapshot to maintain a pointer to the data that existed at the time that the snapshot was created, even after the data has changed. A snapshot consumes only the space that is necessary to restore the files contained in the snapshot. If the files that a snapshot contains have not been modified, the snapshot consumes no additional storage space on the cluster. The amount of disk space that a snapshot consumes depends on both the amount of data stored by the snapshot and the amount of data the snapshot references from other snapshots. The size of a snapshot reflects the amount of disk space consumed by actual blocks stored in that snapshot. Because snapshots do not consume a set amount of storage space, there is no requirement to pre-allocate space for creating a snapshot. You can choose to store snapshots in the same or a different physical location on the cluster than the original files.

Locating Snapshot Files

- Snapshot location
 - Within each folder being snapped
 - Root of **/ifs**
 - Snapshots in hidden directory called **.snapshot**
- Accessing snapshots
 - Only viewed by users with permissions to that snapshot
 - Examples:
 - `/ifs/data/students/name1/.snapshot`
 - `/ifs/data/students/name2/.snapshot`
 - Name1 can view only their snapshots
 - Name2 can view only their snapshots

```
Cluster-2# cd /ifs/.snapshot
Cluster-2# ls | more
Franklin_Alias
Massachusetts_Alias
ScheduleName_duration_2013-08-28-09-30
```

```
Cluster-2# cd /ifs/data/NA/USA/MA/Franklin/.snapshot
Cluster-2# ls | more
Franklin_Alias
Massachusetts_Alias
ScheduleName_duration_2013-08-28-09-30
```

```
Cluster-2# isi snapshot list | wc -l
1587
```

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Snapshot files can be found in two places. They can be found within the path that is being snapped: i.e., if we are snapping a directory located at **/ifs/data/students/name1**, we would be able to view, thru the cli or a Windows Explorer window (with the view hidden files attribute enabled) the hidden **.snapshot** directory. The path would look like: **/ifs/data/students/name1**. The second location to view the **.snapshot** files is at the root of the **/ifs** directory. From here you can view all the **.snapshots** on the system but users can only open the **.snapshot** directories for which they already have permissions. They would be unable to open or view any **.snapshot** file for any directory to which they did not already have access rights.

There are two paths through which to access snapshots. The first is through the **/ifs/.snapshot** directory. This is a virtual directory that allows you to see all the snaps listed for the entire cluster. Remember the **.** (dot) listed before snapshots (**.snapshots**) makes this a hidden directory. The second way to access your snapshots is to access the **.snapshot** directory in the path in which the snapshot was taken. So if you are snapping **/ifs/data/media**, you can **cd** (change directory) or browse your way to the **/ifs/data/media** path, and you will have access to the **.snapshot** directory for just the snapshots taken on this directory. Because snapshots are a picture of a file or directory at that point in time, the permissions are preserved on snapshots; meaning that if you go and restore a snapshot from 3 months ago, if the owner of that data has left the company, you will need to restore the file and then change/update the permissions. Snapshots are read-only. Snapshots are pointers to a point-in-time in the past. As the data is modified, the changed blocks become owned by the snapshots, and the new blocks are owned by the current version. You cannot go back to the pointers and modify the blocks they point to after the fact. Isilon does provide this functionality in the use of clones or writeable snapshots. Clones can be created on the cluster using the **cp** command and do not require you to license the SnapshotIQ module. See the *OneFS 8.0 Web Administration Guide* for additional information about snapshots or clones. The **isi snapshot list | wc -l** command tells you how many snapshots you currently have on disk.

Snapshot Permissions

- Snapshots preserve point-in-time
- Permissions preserved even if permission or owner changes



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You can take snapshots at any point in the directory tree. Each department or user can have their own snapshot schedule. All snapshots are accessible in the virtual directory **/ifs/.snapshot**. Snapshots are also available in any directory in the path where a snapshot was taken, such as **/ifs/data/music/.snapshot**. Snapshot remembers which .snapshot directory you entered through.

Permissions are preserved at the time of the snapshot. If the permissions or owner of the current file change, it does not affect the permissions or owner of the snapshot version. The snapshot of **/ifs/sales/forecast/dave** can be accessed from **/ifs/.snapshot** or **/ifs/sales/forecast/dave/.snapshot**. Permissions for **../dave** are maintained, and the ability to traverse the **.snapshot** directory matches those permissions.

- Web administration interface
 - SnapshotIQ page
- CLI
 - `isi snapshot`
- Options
 - View existing snapshots
 - List scheduled snapshots
 - Monitor space availability



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To manage SnapshotIQ at the command-line, use the **isi snapshot** command.

isi snapshot settings modify --<options>

Creating Snapshots

- Manual or schedule
 - Includes expiration period
 - System automatically deletes snapshot at end of specified duration
 - Best Practice
 - Run concurrent schedules with different frequencies/expiration to meet protection needs
 - Example:
 - Weekly snapshot, expires after 1 month
 - Daily snapshot, expires after 1 week
 - Hourly snapshot, expires after 1 day
- Configured maximum = 20,000 snapshots per cluster
 - Maximum 1,024 snapshots per directory



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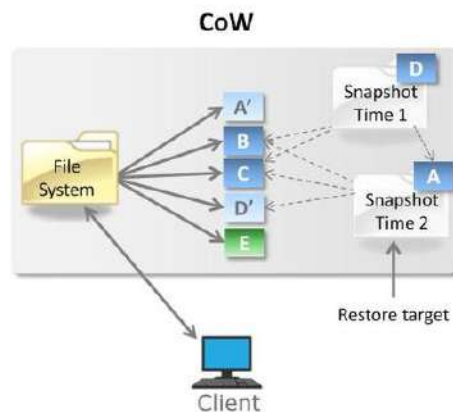
You can create snapshots either by configuring a snapshot schedule or manually generating an individual snapshot. Manual snapshots are useful if you want to create a snapshot immediately, or at a time that is not specified in a snapshot schedule. For example, if you plan to make changes to your file system, but are unsure of the consequences, you can capture the current state of the file system in a snapshot before you make the change.

The most common method is to use schedules to generate the snapshots. A snapshot schedule generates snapshots of a directory according to a schedule. The benefits of scheduled snapshots is not having to manually create a snapshot every time you would like one taken. You can also assign an expiration period to the snapshots that are generated, automating the deletion of snapshots after the expiration period. It is often advantageous to create more than one snapshot per directory, with shorter expiration periods assigned to snapshots that are generated more frequently, and longer expiration periods assigned to snapshots that are generated less frequently.

The default cluster limit is 20,000 snapshots. The default maximum number of snapshots is 1,024 per directory path.

Restoring Snapshots

- Shadow copy emulation
 - Empowers users to recover their own files
- Enabled by default in OneFS
- Users with Windows Shadow Copy Client installed locally
 - Right-click file on folder that contained file
 - Click **Restore Previous Version**
 - Identify file



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If data is accidentally erased, lost, or otherwise corrupted or compromised, any user with Windows Shadow Copy Client installed locally on their computer can restore the data from the snapshot file. To recover an accidentally deleted file, right-click the folder that previously contained the file, click **Restore Previous Version**, and then identify the specific file you want to recover. To restore a corrupted or overwritten file, right-click the file itself, instead of the folder that contains file, and then click **Restore Previous Version**. This functionality is enabled by default starting in OneFS 7.0.

Let's take a look at an example. Here is a file system with writes and snapshots at different times:

- **Time 1:** A,B,C,D. This is preserved in Snapshot Time 1.
- **Time 2:** A,B,C,D'. This is preserved in Snapshot Time 2.

More data is written into the file system:

- **Time 3:** A',B,C,D'
- **Time 4:** A',B,C,D',E

Note that since there is no snapshot taken after Time 2, data corruption to A' or E is not restorable from a snapshot.

So, what happens when the user wants to recover **A** that was overwritten in Time 3 with **A'**?

First a few considerations. When restoring the production file from a snap with the RoW method, no additional storage is consumed and the restore is instant. This is different from CoW (i.e., the operation doesn't need to write data back to the source). Here our snaps are using CoW. The illustration shows Snapshot Time 2 has preserved **A**. Before copying A back

to the file system, a backup snapshot is automatically created as a failback or safety mechanism should the restore from the snap be unacceptable and the user then wants to revert back to **A'**.

Lesson 4: Summary

Having completed this lesson, you are now able to:

- Describe snapshot behavior
- Identify type of snaps OneFS completes
- Understand how snapshots are saved to disk
- Configure and manage snapshot functionality



Having completed this lesson, you are now able to describe snapshot behavior, identify type of snaps OneFS completes, understand how snapshots are saved to disk, and configure and manage snapshot functionality.

Lesson 5: SyncIQ

After completing this lesson, you will be able to:

- Examine replication fundamentals
- Understand SyncIQ replication
- Plan and configure replication policy
- Execute failover and failback operations
- Manage SyncIQ performance
- Describe SyncIQ CloudPools support

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After completing this lesson, you will be able to examine replication fundamentals, understand how SyncIQ replication works, plan and configure a SyncIQ replication policy, execute failover and failback operations, manage SyncIQ performance, and describe SyncIQ CloudPools support.

What is Replication?

- Copying or synchronizing data from source to target
- Target data updated when source data changes
- Two storage devices or storage target locations
 - Primary / source
 - Original or source data
 - Active data set
 - Secondary / target
 - Replicated copy of source data
 - Read only data set

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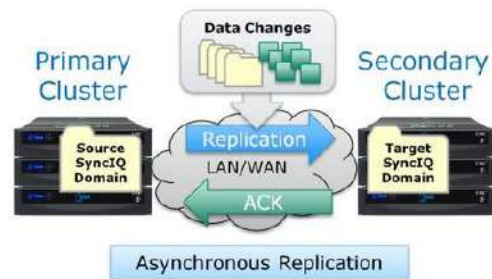
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Replication provides for making additional copies of data, and actively updating those copies as changes are made to the source. While it can be used for many purposes, it is most often implemented as part of a business continuity plan. Replication for business continuity is usually implemented either between block arrays or NAS devices. Most Enterprise NAS products on the market these days offer some type of replication feature. Isilon's replication feature is called SyncIQ.

Replication most often takes place between two storage devices, a primary and a secondary. The primary holding the gold copy of the data which is actively being accessed, updated by clients. The primary is the source of the replication. The secondary being the target of the replication holding a copy of the data. When the source gold data gets updated on the primary, those updates are replicated to the target.

SyncIQ - Isilon Replication Tool

- Asynchronous replication
- SyncIQ license required
- Replicate at directory level
 - Can exclude files or subdirectories
- Metadata replicated
- Automated failover and failback available



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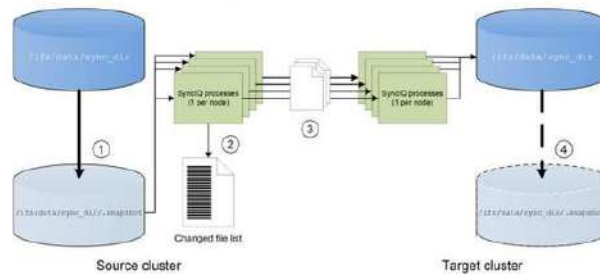
Isilon's replication feature, SyncIQ uses asynchronous replication. Asynchronous replication is similar to an asynchronous file write. The target system passively acknowledges receipt of the data and returns an ACK once the entire file or update is securely received by the target. The data is then passively written to the target. SyncIQ enables you to replicate data from one Isilon cluster to another. You must activate a SyncIQ license on both the primary and the secondary Isilon clusters before you can replicate data between them. You can replicate data at the directory level while optionally excluding specific files and sub-directories from being replicated.

SyncIQ creates and references snapshots to replicate a consistent point-in-time image of a root directory which will be the source of the replication. Metadata, such as access control lists (ACLs) and alternate data streams (ADS), are replicated along with data. SyncIQ enables you to maintain a consistent backup copy of your data on another Isilon cluster.

SyncIQ offers automated failover and failback capabilities that enable you to continue operations on another Isilon cluster if a primary cluster becomes unavailable. In SyncIQ, an administrator configures a policy which details what gets replicated and when. The administrator then starts the replication policy which launches a SyncIQ job. A policy is like an invoice list of what should get replicated and how. A SyncIQ job does the actual work of replicating the data. Policies and jobs are covered in more detail in this lesson.

How Does SyncIQ Work?

- Point in time snapshot of source data
 - Initial: All data (and metadata) replicated to target
 - Subsequent: Only new and changed files are replicated
- SnapshotIQ license not required



SyncIQ uses snapshot technology to take a point in time copy of the data on the source cluster before starting each synchronization or copy job. This source-cluster snapshot does not require a SnapshotIQ license. The first time that a SyncIQ policy is run, a full replication of the data from the source to the target occurs. Subsequently, when the replication policy is run, only new and changed files are replicated. When a SyncIQ job finishes, the system deletes the previous source-cluster snapshot, retaining only the most recent snapshot. The retained snapshot is known as the last known good snapshot. The next incremental replications reference the snapshot tracking file maintained for each SyncIQ domain. When the policy is next run, the changed items are snapshotted, then replicated to the target.

If you require a writeable target, you can break the source/target association. If the sync relationship is broken, a differential or full synchronization job is required to re-establish the relationship. This prevents the inadvertent modification, creation, or deletion of files in the policy's specified target. You can also copy those files to another directory structure for editing.

Each cluster can contain both target and source directories, but a single directory cannot be both a source and a target between the same two clusters (to each other) as this could cause an infinite loop. Only one policy per target path can be configured and each replication set is one way from the source to the target. A snapshot is maintained on the target to facilitate roll back to a previous state. Like the source snapshot, the last known good copy is maintained after a successful replication. You can configure SyncIQ to save historical snapshots on the target, but you must license SnapshotIQ to do this.

In the event that a source becomes unavailable, SyncIQ provides the ability to failover to the target or disaster recovery (DR) cluster. During such a scenario, the administrator makes the decision to redirect client I/O to the DR cluster and initiates SyncIQ failover on the DR cluster. Users will continue to read and write to the DR cluster while the primary cluster is repaired.

SyncIQ Policies

- Governs replication between source and target
- Configured on primary cluster
- Specifies
 - Data source to replicate
 - Target location for replication
 - When and how often data is replicated
- Customizable settings per policy



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With SyncIQ data replication is governed using replication policies. The replication policies are created on the source cluster. The replication policies specify what data is replicated, where the data is replicated from-to, and how often the data is replicated. SyncIQ jobs are the operations that do the work of moving the data from one Isilon cluster to another. SyncIQ generates these jobs according to replication policies.

Two clusters are defined in a SyncIQ policy replication. The primary cluster holds the Source Root Directory and the secondary cluster holds the target directory. The policy is written on the primary cluster. The policy is started on the primary cluster. There are some management capabilities for the policy on both the primary and secondary clusters, though most of the options are on the primary. On the primary these would be accessed under the **Policies** tab in the web administration interface, on the secondary it would be accessed under the **Local Targets** tab. Failover operations are initiated on the secondary cluster.

SyncIQ Jobs

- Does work to move the data
 - Source and target workers or process threads
- Started in accordance with each SyncIQ policy schedule
- Limits
 - One job per SyncIQ policy at a time
 - Up to 50 concurrent jobs per source (others queued)
 - Cluster size and resource dependency
 - Up to 1,000 SyncIQ policies per cluster
 - No cluster limit on quantity of data moved
- SyncIQ policy changes may require replication re-sync



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When a SyncIQ policy is started, SyncIQ generates a SyncIQ job for the policy. A job is started manually or according to the SyncIQ policy schedule.

There is no limit to the number of SyncIQ policies that can exist on a cluster, however the recommended maximum is 1,000 policies. Up to 50 SyncIQ jobs can run at a time, so a maximum of fifty policies can actively replicate at any particular time depending on cluster resources. After a job is started for a SyncIQ policy, another job for the same policy may not be started until the existing job completes, so you cannot have two jobs working on the same policy at the same time. If more than fifty SyncIQ jobs exist on a cluster, the first fifty jobs run while the others are queued to run. While there is no defined limit on how much data can be moved, there are numerous tools which can be used to manage the amount of traffic flowing between the clusters. The number of SyncIQ jobs that a single target cluster can support concurrently is, in part, dependent on the number of workers available on the target cluster. If you modify certain settings of a replication policy after the policy has run, OneFS performs either a full or differential replication the next time the policy runs.

Copy vs. Synchronize Policies

Action

☐ Copy

☒ Synchronize

Run Job

☒ Manually

☐ On a schedule

☐ Whenever the source is modified

☐ Whenever a snapshot of the source directory is taken

Copy Policy	Synchronize Policy
GOAL: File deletion protection - Copy retains deleted source data on target - Files deleted from source are <i>not</i> deleted from target - File deletion protection - Not secure file retention - SmartLock	GOAL: Source cluster protection - Removes deleted source data on target - Files deleted from source are deleted from target - No file deletion protection

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What is the goal or the requirement for replication? Is a mirrored copy of the source the goal? or is the goal to have all source data copied and retain deleted file copies in case they are required later? With SyncIQ you have the option to choose the option to meet your goals for each replication policy. When you create a SyncIQ policy you must choose a replication type of either sync or copy.

Sync maintains a duplicate copy of the source data on the target. Any files deleted on the source are removed from the target. Sync does not provide protection from file deletion, unless the synchronization has not yet taken place.

Copy maintains a duplicate copy of the source data on the target the same as sync. However, files deleted on the source are retained on the target. In this way copy offers file deletion, but not file change protection. This retention is passive and not a secure retention as provided by SmartLock. Copy policies can include file filter criteria not available with the synchronization option.

The view discussed of copy vs. sync is standalone. You can always license SnapshotIQ on the target cluster and retain historic SyncIQ associated snapshots to aid in file deletion and change protection.

SyncIQ Job Process

1. SyncIQ replication job started
2. Point-in-time snapshot of source data
 - Sync job changelist created
3. New or changed directories, files, and metadata replicated
 - Initial: All files and metadata replicated to target
 - Subsequent: Only new and changed files replicated to target
 - Deleted data on source deleted on target (with sync option)
4. Replication successful
 - Target snapshot taken
 - Snapshots become “last known good”



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The SyncIQ process executes the same way each time a SyncIQ job is run. SyncIQ uses snapshot technology to take a point-in-time snapshot copy of the data on the source cluster before starting each replication or copy job; compares the new source snapshot to the last known good source snapshot and creates a changelist based on the differential between the snapshots. The changed directories, files and metadata are replicated at the block level. The initial time a SyncIQ policy is run, a full replication of the data from the source to the target occurs. Subsequently, when the replication policy is run, only new and changed files are replicated. If the SyncIQ policy is a sync policy and not a copy policy, any files deleted on the source cluster are also deleted target cluster.

When a SyncIQ job completes successfully, a snapshot is taken on the target cluster. This snapshot replaces the previous last known good snapshot. The same snapshot starting in SyncIQ 3.5 is taken if a sync job fails and is used to reverse any target cluster modifications to return the target to the last known good state.

On the source cluster when a SyncIQ job completes successfully, the system deletes the previous source cluster snapshot, and retains only the most recent snapshot.

Historical snapshots can be maintained and deleted using the options in the SyncIQ policy. Historical snapshots on the source or target clusters require a SnapshotIQ license.

Target Compare Initial Sync (Diff-Sync)

- Performs differential synchronization (diff-sync)
 - Compares source and target data
 - Restores synchronization relationship
 - Runs on first policy run only
- Required after SyncIQ policy modifications
 - Source directory
 - Included or excluded directories
 - File criteria
 - Target cluster
 - Target directory
- CLI only option
 - Policy option `--target-compare-initial-sync`



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During a full synchronization, SyncIQ transfers all data from the source cluster regardless of what data exists on the target cluster. Full replications consume large amounts of network bandwidth and may take a very long time to complete. A differential synchronization compares the source and target data by doing tree walks on both sides. This is used to re-establish the synchronization relationship between the source and target. Following the tree walks, the changed data is replicated in place of a full data synchronization. The differential synchronization option is only executed during the first time the policy is run.

Some SyncIQ replications issues may require using this option including when a SyncIQ policy is modified. If you modify the source directory, any included or excluded directories, any file criteria, change the target cluster, or target directory, either a full or differential synchronization is required.

Before running the replication policy again, you must enable a target compare initial sync, using the command on the primary **isi sync policies modify <policy name> --target-compare-initial-sync on**. With **target-compare-initial-sync** on for a policy, the next time the policy runs the primary and secondary clusters will do a directory tree walk of the source and target directory to determine what is different. It will then only replicate just those differences from the source to the target. The **target-compare-initial-sync** option determines whether the full or differential replications are performed for this policy. Full or differential replications are performed the first time a policy is run and after a policy has been reset. If set to on, performs a differential replication. If set to off, performs a full replication. If differential replication is enabled the first time a replication policy is run, the policy will run slower without any benefit. The default value is off.

Creating SyncIQ Policy

* New in OneFS 8.0

Settings

- Policy name and description
- Action and Run Job settings

Source Cluster

- Root Directory (domain)
- Included/excluded directories
- File Matching Criteria
- Source node restriction

Target Cluster

- Target Host
- Target Directory
- SmartConnect zone restriction

Target Snapshots

- Snapshot Alias Name
- Snapshot Naming Pattern
- Snapshot Expiration

Advanced Settings

- Policy priority level *
- Log Level
- File integrity validation
- Accelerated failback prep *
- Report retention
- Deletion log
- Copy for CloudPools *



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There are five areas of configuration information required when creating a policy. Those areas are **Settings**, **Source Cluster**, **Target Cluster**, **Target Snapshots**, and **Advanced Settings**. Each of these areas are covered in detail in this lesson.

Create SyncIQ Policy: Settings

Data Protection → SyncIQ → Policies → Create a SyncIQ Policy

Create SyncIQ Policy

* = Required field

Settings

* Policy Name

Description

☒ Enable this policy

Action

☐ Copy

☒ Synchronize

Run Job

☒ Manually

☐ On a schedule

☐ Whenever the source is modified

☐ Whenever a snapshot of the source directory is taken

Settings

Source Cluster

Target Cluster

Target Snapshots

Advanced Settings

- Policy name
- Description
- Enablement
- Replication type
- Policy start selection

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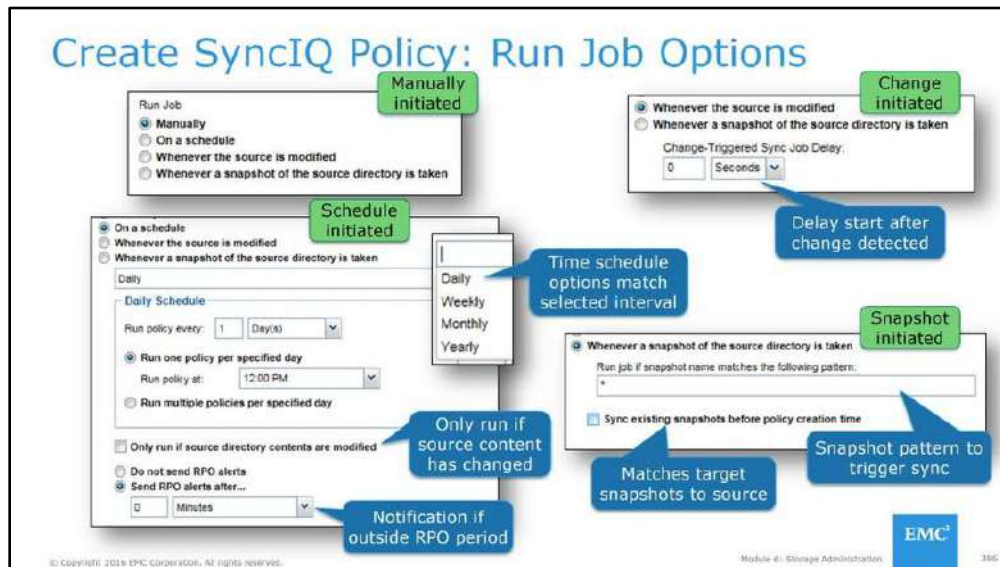
To create a policy in the web administration interface, navigate to **Data Protection > SyncIQ > Policies**. Click the **Create a SyncIQ Policy** button and a **Create SyncIQ Policy** configuration window opens.

In the **Settings** section, you need to assign a unique name to the policy. Optionally you can add a description of the policy. The **Enable this policy** box is checked by default. If you unchecked the box, it would disable the policy and stop the policy from being run. You can always enable the policy later to run the SyncIQ job.

Next you must designate whether this is a **Copy** policy or a **Synchronize** policy.

- A **Copy** policy makes a one time full copy of the source directory to the target directory. **Copy** polices are usually run **Manually**.
- A **Synchronize** policy makes a one time full copy of the source directory to the target directory. It then continues to make incremental copies of the changes in the source directory to the target directory.

The replication policy can be started using one of four different run job options: **Manually**, **On a Schedule**, **Whenever the source is modified**, or **Whenever a snapshot of the source directory is taken**.



Each **Run Job** option produces a different replication behavior and has different associated options.

- **Manually** allows the synchronization to occur on demand. Each time the policy must be manually initiated. The first run of the policy initiates a full copy of the data the same as any other policy based on the copy or sync option chosen.
- **On a schedule** provides a time-based schedule for the SyncIQ policy execution. When selected the time schedule options change to match the elected interval. An option is available to not run the policy if no changes to the data have occurred since the last time the policy was run. This option saves system resources when replication is not required. An option was added in OneFS 8.0 to monitor the recovery point objectives or RPO. Any delayed or failed SyncIQ job sends an alert notification after the selected time period.
- **Whenever the source is modified** is available and intended for select use cases. Content distribution and EDA are the primary select use cases. The SyncIQ domain is checked every 10 seconds for changes. If a change is detected in the data or metadata, the replication is initiated. An option to delay the start of the replication is available to allow new writes to the source to complete prior to triggering the replication. This delay allows fewer and more larger complete file replication rather than short many cycled triggered replication runs.
- **Whenever a snapshot of the source directory is taken** is used to keep historic snapshots on the source and target cluster in sync. The policy is initiated when the snapshot matching the specified pattern is run. You can select during the first time the policy is run to replicate data based on historic snapshots of the source SyncIQ domain. This creates a mirrored image of the snapshots on the target from the source and is particularly useful for snapshot protection for file deletions.

Create SyncIQ Policy: Source Cluster

- SyncIQ domain path
- Included / excluded directory paths
- File filtering
- Select Source Nodes

* Copy policies only

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In the next section of the **Create SyncIQ Policy** window, you define the **Source Cluster** criteria. The **Source Root Directory** is the directory which will be the root of the replication. This is the data which you want to protect by replicating it to another cluster. This is the data to be replicated to the target directory on the secondary cluster. Unless otherwise filtered, everything in the directory structure from the source root directory and below will be replicated to the target directory on the secondary cluster.

The **Included Directories** field allows you to add one or more directory paths below the root which should be included in the replication. Once an include path is listed that means that only paths listed in the include path will be replicated to the target. Without include paths all directories below the root would be included. You can also exclude specific paths from being replicated.

The **Exclude Directories** field lists directories below the root you wish to be explicitly excluded from the replication process. You cannot fail back replication policies that specify includes or exclude settings.

Another filter option is **File Matching Criteria**. File matching allows for the creation of one or more filter rules to filter which files do and do not get replicated. If multiple rules are created they are connected together with Boolean AND or OR statements. When adding a new filter rule, click either the **Add an "And" condition** or **Add an "Or" condition** links. File Matching Criteria says if the file matches these rules then replicate it. If it does not match the rules then do not replicate it. File criteria can be based on several file attributes: **Filename**; includes or excludes files based on the file name. **Path**; Includes or excludes files based on the file path. Paths can also use wildcards. **File Type**; Includes or excludes files based on one of the following file-system object types, soft link, regular file, or directory. **Modified**; Includes or excludes files based on when the file was last modified. **Accessed**; Includes or excludes files based on when the file was last accessed. This option is available only if the global access-time-tracking option of the cluster is enabled. **Created**; Includes or excludes files based on when the file was created. **Size**; Includes or excludes files based on their size. File sizes are represented in multiples of 1024, not 1000.

Restrict Source Nodes - Selecting run on all nodes means that the cluster can use any of its external interfaces to replicate the data to the secondary cluster. Selecting run on only the nodes in the specified subnet and pool, means that only those interfaces which are members of that specific pool will move the replication traffic. This option is effectively selecting a SmartConnect zone over which the replication traffic will transferred. You would choose the appropriate subnet and pool from the drop-down list. The list has all the subnets and pools on the primary cluster. Be aware SyncIQ only supports static IP address pools. Only static address pools should be used. If a replication job connects to a dynamically allocated IP address, SmartConnect might reassign the address while a replication job is running, which would disconnect the job and cause it to fail. In the policy-configuration content, specifying file criteria in

a SyncIQ policy slows down a copy or synchronization job. Using includes or excludes for directory paths does not affect performance, but specifying file criteria does.

Create SyncIQ Policy: Target Cluster

- Target cluster host
- Target cluster root directory
- Only connect to specific nodes
 - Only nodes in a specific SmartConnect zone

Settings
Source Cluster
Target Cluster
Target Snapshots
Advanced Settings

The screenshot shows the 'Target Cluster' configuration window. It has two main input fields: 'Target Host' and 'Target Directory'. The 'Target Host' field has a callout box that says 'Target SmartConnect IP address or FQDN'. The 'Target Directory' field has a callout box that says 'Target SyncIQ domain root path'. Below these fields is a checkbox labeled 'Connect only to nodes within the SmartConnect zone on the target cluster'. A callout box points to this checkbox with the text 'Restrict target nodes'. Another callout box points to the 'Target Host' field with the text 'Local host allowed for same cluster replication'. The window title is '- Target Cluster'.

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The target cluster identification is required for each policy. You specify the target host using the target SmartConnect zone IP address, the fully qualified domain name (FQDN), or local host. Local host is used for replication to the same cluster. You also specify the target SyncIQ domain root path. Best practices suggest the source target name, the access zone name are included in the target directory path. An option is provided to restrict the target nodes used to process to only those node connected within the SmartConnect zone.

Create SyncIQ Policy: Target Snapshots

Target Snapshots

☒ Enable capture of snapshots on the target cluster

Snapshot Alias Name

Default name: SIQ-\$(SrcCluster)-\$(PolicyName)-latest

Snapshot Naming Pattern

Default pattern: SIQ-\$(SrcCluster)-\$(PolicyName)-%Y-%m-%d_%H-%M-%S

Snapshot Expiration
☐ Snapshots do not expire
☒ Snapshots expire after...

Years
 Days
 Weeks
 Months
 Years

Advanced

Priority

Years

Target archival snapshots

- Capture archival snapshots
- Snapshot Alias Name
- Snapshot Naming Pattern
- Snapshot Expiration policy
 - Expiration suggested

Settings
 Source Cluster
 Target Cluster
 Target Snapshots
 Advanced Settings

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Snapshots are used on the target directory on the secondary cluster to retain one or more consistent recover points for the replication data. You can specify if and how these snapshots will be generated on the secondary cluster. If you want to retain the snapshots SyncIQ takes then you should check the box **Capture snapshots on the target cluster**. SyncIQ always retains one snapshot of the most recently replicated delta set on the secondary cluster to facilitate failover, regardless of this setting. Capture snapshots will retain them beyond the time period in which SyncIQ needs them. The snapshots provide you with additional recover points for the data on the secondary cluster.

The **Snapshot Alias Name** is the default alias name is an alias for the most recently taken snapshot. The name alias name pattern is **SIQ_\$(SrcCluster)_\$(PolicyName)**. If this snapshot alias were taken on a cluster called "cluster1" for a policy called "policy2" it would have the alias "SIQ_cluster1_policy2". You can modify this default alias by editing the **Snapshot Alias Name** field. to modify the default alias of the last snapshot created by this replication policy, in the **Snapshot Alias Name** box, type a new alias. You can specify the alias name as a snapshot naming pattern. For example, the following naming pattern is valid:

%{PolicyName}-on-%{SrcCluster}-latest

The previous example produces names similar to the following:

newPolicy-on-Cluster1-latest

The **Snapshot Naming Pattern** field shows the default naming pattern for all snapshots. You can modify this default name by editing the **Snapshot Naming Pattern** field. Then select a snapshot expiration setting. Either snapshots do not expire, or snapshots expire after and then stipulate the time period, the options are days, weeks, months, and years. To modify the snapshot naming pattern, in the **Snapshot Naming Pattern** box, type a naming pattern. Each snapshot generated for this replication policy is assigned a name based on this pattern. For example, the following naming pattern is valid:

%{PolicyName}-from-%{SrcCluster}-at-%H:%M-on-%m-%d-%Y

The example produces names similar to the following:

newPolicy-from-Cluster1-at-10:30-on-7-12-2012

In the **Snapshot Expiration** section, specify whether you want SnapshotIQ to automatically delete snapshots generated according to this policy and or how long to retain the snapshots. It is suggested to always select a snapshot expiration period.

Create SyncIQ Policy: Advanced Settings

- Policy priority level *
- Log Level
- File integrity validation
- Accelerated failback prep *
- Report retention
- Deletion logging
- Copy policy for CloudPools *

* New in OneFS 8.0

Settings
 Source Cluster
 Target Cluster
 Target Snapshots
 Advanced Settings

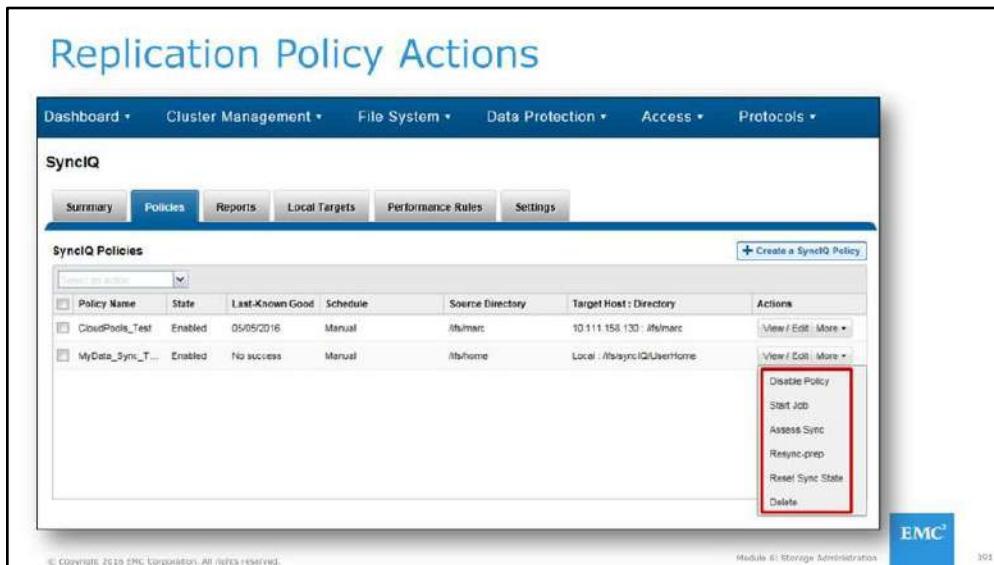
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The **Advanced Settings** section is expanded over the previous versions of OneFS. SyncIQ allows policies to be prioritized. If more than 50 concurrent SyncIQ policies are running at a time, policies with a higher priority take precedent over normal policies. The log level can be selected based on requirements. Replicated file integrity is validated by default, however may be disable if required.

If the SyncIQ replication is intended for failover / failback disaster recovery scenario, you can prepare the domain mark for the failback performance. The original source SyncIQ domain requires a domain mark to be performed. Running a domain mark during the failback process can take a long period to complete.

You have an option to retain SyncIQ job reports for a specified period of time. With an increased number of SyncIQ jobs in OneFS 8.0, the report retention period could be an important consideration. If tracking file and directory deletions performed during synchronization on the target, you can select to record the deletions.

The deep copy setting applies to those policies that have files that are contained in CloudPools target. More details on this feature are included in the CloudPools section of this training. **Deny** is the default. **Deny** allows only stub file replication. The source and target clusters must be at least OneFS 8.0 to support this. **Allow** lets the SyncIQ policy determine if a deep copy should be performed. **Force** automatically enforces a deep copy for all CloudPools data contained within the SyncIQ domain. **Allow** or **Force** are required for target clusters that are not CloudPools aware.



After the policy configuration has been completed, the policy is listed on the **Policies** tab on the source cluster. In the **Actions** column for a policy, the drop-down list has management options. A policy can be enabled or disabled depending on the current state. The policy can be ran manually. For a new policy, an assessment can be run. Failover/failback steps can be performed. The synchronization state can be reset to force a full sync to be performed. Finally, the policy can also be deleted.

After a policy is started, a SyncIQ job then starts based on the **Run Job** setting in the policy. Currently running jobs can be viewed on the **Summary** tab. If the policy was created with a schedule, once the policy has been enabled, the schedule will be able to start the SyncIQ job.

Policy Assessment

- SyncIQ policy assessment actions
 - Only new policies prior to first policy run
 - Does not start replication
 - Run in web administration interface or CLI
- Policy assessment:
 - Provides details about what to replicate
 - Checks validity of policy syntax
 - Verifies communication between clusters
- View assessment information in SyncIQ report



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You can conduct a trial run of a new SyncIQ policy without actually transferring any file data between the source and target clusters. This is called a SyncIQ policy assessment. A SyncIQ policy assessment scans the dataset and provides a detailed report of how many files and directories were scanned. This can be useful if you want to preview the size of the data set that will be transferred if you run the policy. It also verifies that the policy will work and that communication between the source and target clusters is functioning properly. The benefit of an assessment is it can tell you whether your policy will work and how much data will be transferred before you've run the policy. This can be useful when the policy will initially replicate a large amount of data. If there is a problem, with your policy it would be better to know that before you start moving a large amount of data across your network. This functionality is available only after you create a new policy and before you run the SyncIQ policy for the first time. You can assess only replication policies that have never been run before. This can be done in the web administration interface or from the CLI. A SyncIQ report is generated by an assessment run. You can view the assessment information in the SyncIQ report. The report displays the total amount of data that would have been transferred in the **Total Data Bytes** field.

Policy Assessment (cont'd)

```

# iscsi sync reports view archive-data-policy 1
Policy Name: archive-data-policy
Job ID: 1
Start Time: 2014-11-12T22:29:32
End Time: 2014-11-12T23:36:57
Action: run
Status: finished
ID: 1-archiwe-data-policy
Policy ID: 638fbb4c031a75a7c04e7276d8362330
Sync Type: archival
Duration: 1H7m25s
Errors: -
Source Directories Visited: 1
Source Directories Deleted: 0
Run Retransmitted Files: 0
Retransmitted Files: -
Total Files: 7
New Files Registered: 7
Source Files Deleted: 0
Files Changed: 0
Target Files Deleted: 0
Up To Date Files Skipped: 0
User Conflict Files Skipped: 0
Error In Files Skipped: 0
Error New Files Skipped: 0
Error Checksum Files Skipped: 0
Bytes Transferred: 22785485
Total Source Bytes: 0
Total Data Bytes: 22785485
File Data Bytes: 22785485
Sparse Data Bytes: 0
Target Snapshot: S2Q-Follower-archive-data-policy-2014-11-12_22-18-11
Total Phases: 2
Phase:
Phase: STT_PHASE_IN
Start Time: 2014-11-12T23:36:56
End Time: 2014-11-12T23:36:56
Phase: STT_PHASE_IMGOP_SEND
Start Time: 2014-11-12T23:36:56
End Time: 2014-11-12T23:36:56

```

Total data that would have been transferred by this policy

Subreports:	Policy Name	Job ID	Subreport ID	Start Time	End Time	Action	State
	archive-data-policy 1	1		2014-11-12T22:29:32	2014-11-12T23:36:57	test	finished
	Total: 1						



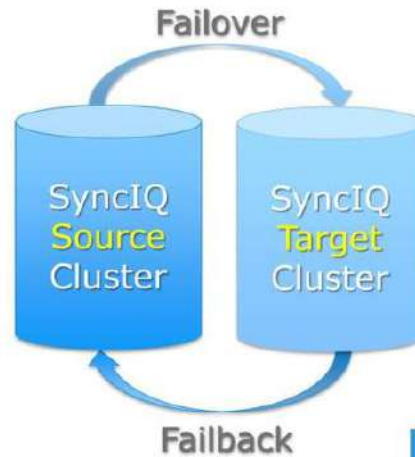
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The results of the assessment can be viewed in the web administration interface by navigating to **Data Protection > SyncIQ > Reports**, and then click view details for the policy you ran. The report can also be viewed from the CLI using the command **isi sync reports view <policy name> <job id>**. In this example, the policy was called *archive-data-policy* and the job id was *1*, which means this was the first instance of a job for this policy. In the **Total Data Bytes** field of the report, the total amount of data that would have been transferred is shown, which in this example was 22785485 bytes.

Failover and Failback Overview

- Failover
- Failback
- Failover revert



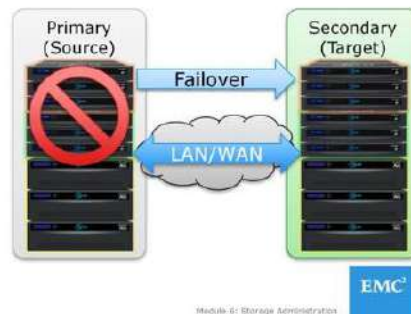
SyncIQ enables you to perform automated data failover and failback operations between source and target Isilon clusters. If a source cluster has gone offline or is rendered unusable, you can fail over to the target cluster, enabling clients to access their data on the target cluster. Failover is the process of allowing clients to modify data on a target cluster. If the offline source cluster later becomes accessible again, you can fail back to the original source cluster. Failback is the process of copying changes that occurred on the original target while failed over back to the original source. This allows clients to access data on the source cluster again, and resuming the normal direction of replication of data back from the source to target.

Failback is the process of returning to normal operations. Normal being the data is read-write on the source cluster and read/only on the target, and the direction of replication is from the source to the target. Failback would be possible if the cause of the outage on the source could be resolved, and the source was then back in operational condition. While the policy was failed over the active copy of the data would reside on the target cluster. Failing back copies the changes that occurred while failed over to the target, back to the source.

Failover revert is a process useful for instances when the source becomes available sooner than expected. Failover revert allows administrators to quickly return access to the source cluster, and restore replication to the target.

Failover with SyncIQ

- Changing SyncIQ target directories from read-only to read/write
 - Assumes source cluster no longer available
- Isilon only manages directory behavior
 - Managed per SyncIQ policy
- Other processes not managed
 - Client / application connectivity
 - Network routing / DNS
 - Authentication sources
 - Shares and exports



We will now discuss the Failover process. Failover is the process of changing the role of the target replication directories into the role of the source directories for assuming new client read, write, and modify data activities.

As part of modification, the site preparation activities must occur.

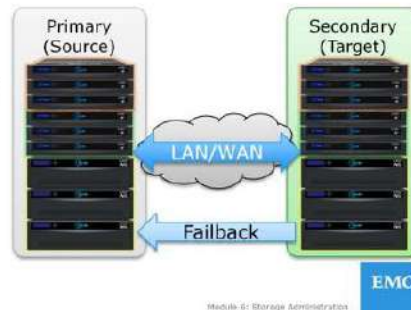
Failovers can happen when the source cluster no longer being available for client activities. The reason could be from any number of circumstances including natural disasters, site communication outages or power outages.

The reason could also be a planned event, such as testing a disaster recovery plan or as a result of upgrade or other schedule maintenance activities.

Failover changes the target directory from read-only to a read-write status. Failover is managed per SyncIQ policy. Only those policies failed over are modified. SyncIQ only changes the directory status and does not change other required operations for client access to the data. Network routing and DNS must be redirected to the target cluster. Any authentication resources such as AD or LDAP must be available to the target cluster. All shares and exports must be available on the target cluster or be created as part of the failover process.

Failback with OneFS

- Restoring original source and target sync relationship
 - Assumes source cluster availability restored
- Isilon only manages directory behavior
 - Managed per SyncIQ policy
- Other processes not managed
 - Client / application connectivity
 - Network routing / DNS



What is failback? A failback is the process of restoring the source-to-target cluster relationship to the original operations where new client activity is again on the source cluster. A failback can happen when the source cluster is available once again for client activities. The reason could be from any number of circumstances including the natural disasters are no longer impacting operations, or site communication or power outages have been restored to normal, or the testing and maintenance activities are finished. Each SyncIQ policy must be failed back. Like failover, failback must be selected for each policy. The same network changes must be made to restore access to direct clients to the source cluster.

Failover Revert

- Use if primary cluster becomes available
 - Enables replication to continue again without fail back
 - Occurs prior to data modification on target cluster
 - If data modified, fail back process may be required to preserve data changes
- Semi-automated
- Not supported for SmartLock directories



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So what is Failover Revert? It is undoing a failover job in process. You would use Failover Revert if the primary cluster or original source cluster once again became available. This could result from a temporary communications outage or if you were doing a failover test scenario. Failover revert stops the fail over job and restores the cluster to a sync ready state and enables replication to the target cluster to once again continue without performing a fail back.

Failover revert may occur even if data modification has occurred to the target directories. If data has been modified on the original target cluster, then either a failback operation must be performed to preserve those changes, otherwise any changes to the target cluster data will be lost.

Before a fail revert can take place, a failover of a replication policy must have occurred.

A Failover Revert is not supported for SmartLock directories.

Managing SyncIQ Performance

- SyncIQ job planning
 - Manage and plan job schedules
 - File control
 - Specifying source nodes
 - Specifying target nodes
- Built-in capabilities
 - File Splitting
- Global throttling control
 - Network bandwidth limits
 - Kb/sec
 - File count limits
 - Files/sec
 - CPU utilization limits
 - Maximum percentage
 - Worker limits
 - Maximum percentage



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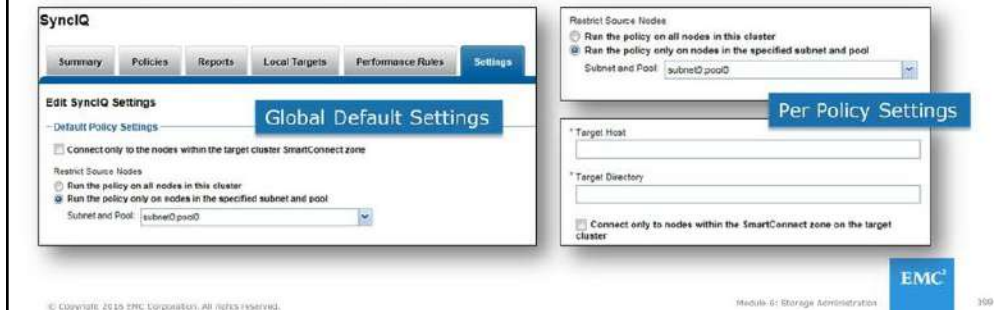
One of the simplest ways to manage resource consumption on the source and target clusters is with proper planning of job scheduling. If the business has certain periods when response time for clients is critical, then replication can be scheduled around these times. If a cluster is a target for multiple source clusters, then modifying schedules to evenly distribute jobs throughout the day is also possible. Another way to maintain performance at either the source or target cluster is to use a more specific directory selection in the SyncIQ policy. This can be useful in excluding unnecessary data from replication and making the entire process run faster, but it does add to the administrative overhead of maintaining policies. However, when required recovery time objectives (RTOs) and recovery point objective (RPOs) dictate that replication schedules be more aggressive or datasets be more complete, there are other features of SyncIQ that help address this.

SyncIQ offers administrators the ability to control the number of workers that are spawned when a SyncIQ job is run. This can improve performance when required or limit resource load if necessary. Administrators can also specify which source and target nodes are used for replication jobs on a per policy basis. This allows for the distribution of workload across specific nodes to avoid using resources on other nodes that are performing more critical functions.

Replication bandwidth between the source and target cluster can be limited to preserve network performance. This is useful when the link between the clusters has limited bandwidth or to maintain performance on the local network. To limit node resource load, administrators can also use file operation rules to limit the number of files that are processed in a given time period, this feature though would only be practical if the majority of the files were close in size.

Performance: Source & Target Nodes

- Replication sends data via any node
- SmartConnect zones limits node participation
- Configured globally or per Policy



If no source subnet:pool is specified then the replication job could potentially use any of the external interfaces on the cluster. SyncIQ attempts to use all available resources across the source cluster to maximize performance. This additional load may have an undesirable effect on other source cluster operations or on client performance. You can control which interfaces, and therefore which nodes, SyncIQ uses by specifying a source subnet:pool. You can specify a source subnet:pool globally on the **Settings** tab or Per Policy when creating a new SyncIQ policy. Specifying a subnet:pool is effectively specifying a SmartConnect zone. You can isolate source node replication resources by defining a SmartConnect zone. The SmartConnect zone can define a subset of nodes in a cluster to be used for replication. It can also be used to define specific subnets or interfaces on each node to isolate replication traffic from client traffic.

When configuring a SyncIQ policy you select a target host. If this hostname is a SmartConnect zone on the secondary cluster, then you have the same ability to control which nodes or interfaces the replication traffic goes through on the secondary. This would, of course, require pre-configuring the SmartConnect zone on the secondary cluster.

Transfer Performance Rules

- Limit bandwidth
- Limit files transferred
- Schedule

You can manage the impact of replication on cluster performance by creating rules that limit the network traffic created and the rate at which files are sent by replication jobs. For a rule to be in effect, it must be enabled. When the **Rule Type** is **Bandwidth**, the limit field is KB/sec. When the **Rule Type** is **File Count**, then the **Limit** field is files/sec.

Using performance rules, you can set network and file processing threshold limits to limit resource utilization. These limits are cluster-wide, they affect all SyncIQ policies, and are shared across jobs running simultaneously. You can configure network-usage rules that limit the bandwidth used by SyncIQ replication processes. This may be useful during peak usage times to preserve the network bandwidth for client response. Limits can also be applied to minimize network consumption on a low bandwidth WAN link that exists between source and target. Multiple network rules can be configured to allow for different bandwidth limits at different times. These rules are configured globally under the performance tab of SyncIQ and apply to all replication jobs running during the defined timeframe on that source cluster.

System resource load can also be modified by using file operation rules. File operation rules are also global. They can limit the total number of files per second that will be processed during replication. You can schedule when the limits will be in effect.

CPU and Worker Throttling

- Maximum CPU utilization
- Maximum source worker percentage (pworkers)
 - Percentage of calculated available workers
- Scheduled

The diagram illustrates the configuration of performance rules for CPU and worker throttling. It shows two side-by-side 'Performance Rule' configuration windows. The left window is for 'CPU' throttling, with a callout 'Limits CPU utilization (% of available)'. The right window is for 'Workers' throttling, with a callout 'Limits percentage of available workers'. Both windows have a 'Set schedule for hours to apply rule' button. Arrows point from the list items to the respective windows.

Performance Rule (Left):

- ☒ Enable this rule
- Rule Type: CPU
- Limit: 100 %
- Description:
- Schedule: This rule will be in effect on the specified days during the specified time interval.
Days: ☒ Sun ☒ Mon ☒ Tue ☒ Wed ☒ Thu ☒ Fri ☒ Sat
Time: 00 to 23:59

Performance Rule (Right):

- ☒ Enable this rule
- Rule Type: Workers
- Limit: 100 % (max)
- Description:
- Schedule: This rule will be in effect on the specified days during the specified time interval.
Days: ☒ Sun ☒ Mon ☒ Tue ☒ Wed ☒ Thu ☒ Fri ☒ Sat
Time: 00 to 23:59

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In OneFS 8.0, additional performance throttling rules can be set; CPU throttling and source worker throttling. CPU throttling allows you to set a maximum CPU threshold for SyncIQ processes, limiting the CPU utilization to a percentage of the total available. You select the maximum percentage out of 100%. The worker throttling is used to limit the maximum calculated workers that can be used to process SyncIQ jobs.

These performance rules will apply to all policies executing during the specified time interval. An individual policy can also have a limit on the number of workers per node. The number of worker calculations is discussed in the next section. The same scheduling rules can be created as with bandwidth and file count throttling options.

SyncIQ Worker Pools Overview

- **Maximum number of source workers**
 - Grows with cluster size
 - Workers per policy grows with cluster size
- **Worker per SyncIQ job allocation**
 - Dynamically allocates workers to policies
 - Based on cluster size and number of running policies
- **Scheduled resource management rules**
 - Bandwidth throttling retained
 - New worker throttling rule added
 - New CPU throttling rule added

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The concept of the SyncIQ worker pool is introduced in OneFS 8.0. As the cluster grows, more workers are available for allocation to all running policies. Workers are then dynamically allocated equally to all running policies. To help manage resource utilization during scheduled events, the bandwidth throttling option is retained and two new throttling options are added, worker throttling and CPU utilization throttling.

SyncIQ Scalability - Limits

- OneFS 8.0 new limits:
 - Defined policies per cluster = **1,000** (10X increase)
 - Concurrent running policies per cluster = **50** (10X increase)
 - Maximum workers (sworkers) per target node = 100 (no change)
- Maximum workers (pworkers) per cluster
 - Determined by number of CPU cores
 - Default is **4 * [# of CPU Cores]**
 - e.g. 20 nodes with 1 x 4-core CPU per node = $(20 * 1 * 4 * 4) = 240$ pworkers
 - e.g. 15 nodes with 2 x 4-core CPU per node = $(15 * 2 * 4 * 4) = 480$ pworkers
- Maximum workers (pworkers) per SyncIQ policy
 - Determined by number of nodes
 - Default is **8 * [# of nodes]**
 - e.g. 20 nodes * 8 = 160 maximum pworkers per policy



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With OneFS 8.0, new limits are defined. The number of active SyncIQ policies is increased from 100 to 1,000, which is a 10 fold increase. The number of running SyncIQ jobs is increased from 5 to 50, also a 10 fold increase. The maximum sworkers or target workers remain at 100 workers per node.

The number of workers on the source cluster is now variable based on the number of CPU cores and the number of nodes. For every CPU core in the cluster, 4 workers are available to the worker pool. So for every CPU with 4-cores, 16 workers are added to the worker pool. If a node has two 4-core CPUs, each node adds 32 workers. As an example to calculate the number of available workers, if the cluster has 20 nodes with 1, 4-core CPU per node, you would have 240 source cluster workers or pworkers available in the pool. If the cluster has 15 nodes with 2, 4-core CPUs per node there are 480 pworkers available to the pool.

The maximum number of workers allocated per SyncIQ job is determined by the multiplying the number of nodes by 8. As an example for a 20 node cluster, we multiple 20 by 8 and the maximum number of workers per job is 160 pworkers.

Dynamic SyncIQ Worker Allocation

- Workers dynamically allocated between policies
 - All running policies get equal share of workers
 - (+/- 1 worker due to rounding)
 - Applied as sync jobs start and stop
 - Workers reallocated slowly
 - Allows smooth worker transition as policies begin and end
- Worker process model remains the same
 - Individual processes
 - Workers created and ended as required
 - Started or stopped when switching policies



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The number of workers per job is a maximum of 8. Why a maximum? Workers are dynamically allocated between running SyncIQ policy jobs. All running policies get an equal share of workers, plus or minus 1 due to rounding. Workers are determined as sync jobs start and stop. So as a job finishes, the job may only have work for a few workers and its allocated workers are released back into the pool. As a new job starts, workers may be allocated from other running jobs to provides resources for the policy to execute its tasks. Workers are allocated slowly and smoothly between jobs as required to eliminate any contention or resource thrashing.

The worker process model remains the same as before. Each worker is an individual process working on an individual task. The workers are created or ended as they are required. Workers are started or stopped when switching between tasks.

Example: Available Workers Calculation

- Source cluster worker calculations
 - Cluster resource calculations
 - 3 node cluster with 4 CPU cores per node
 - Workers
 - Workers per CPU core = 4
 - Workers per node per job policy = 8
 - Maximum workers on the cluster =
 - 3 nodes * 4 CPU cores * 4 workers per core = 48 workers
 - Maximum workers per policy =
 - 3 nodes * 8 workers per node per job = 24 workers per policy

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To illustrate dynamic worker allocation we start with our example cluster. The cluster consists of 3 nodes and has a single 4-core CPU per node. We use the default configuration numbers of 4 workers per CPU core, and 8 workers per node per job limit maximum. The calculations mean we have a total of 48 workers available in the worker pool, and each running policy or job can be assigned up to 24 workers maximum.

Example: Dynamic Worker Allocation

1. A single policy starts

- Maximum workers per policy allocated
 - Policy 1 allocated 24 workers

2. A second policy starts

- Policy 1 retains allocation of 24 workers
 - No reduction required, enough available workers for Policy 2
- Policy 2 allocated 24 workers from available worker pool

3. A third policy starts

- Daemon determines even split of workers per policy
 - Policy 1 smoothly reduced to 16 workers
 - Policy 2 smoothly reduced to 16 workers
 - Policy 3 smoothly allocated 16 workers from Policies 1 and 2

From Example

- Available Workers in Worker Pool = 48
- Maximum Workers per Policy = 24

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When the first SyncIQ policy starts the job and is the only running job, 24 workers are allocated to the running policy because that is the maximum based on the cluster size.

When the second SyncIQ job begins, the remaining 24 workers in the pool are allocated to policy 2. The maximum of 24 workers per job are available in the worker pool, and the workers are evenly distributed between jobs.

Now when a third job begins, no more workers exist in the worker pool. The daemon examines the other running jobs and determines how to reallocate some of their workers to the new job. Each job is evenly allocated 16 workers. The number of workers are smoothly reduced from policies 1 and 2 and allocated to policy 3.

You can carry on this example adding additional jobs and reallocating workers. If the example was of a 100 node cluster, you can quickly calculate the number of workers in the worker pool and maximum workers per job. SyncIQ truly scales with the cluster and available node CPU resources.

Worker Efficiency Improvements

- One worker per file for replication
- File splitting allows multiple workers for large files
 - Initially 1 worker per file
 - Free up workers help with remaining larger files
 - Larger files locked at sub-range for files > 20MB
 - Multiple workers transfer 20MB portion of file
 - Improves sync job time to completion
- File splitting enabled by default



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For most operations, the number of SyncIQ workers per file is fixed as one worker per file on both the primary or source cluster, and the secondary or target cluster. The work is divided amongst the threads or workers at a file level granularity. Each worker “locks” a single file then works to transfer it. That means one worker per file. As the SyncIQ job runs the number of remaining files to replicate decreases and the number of active workers decreases. In many cases the last portion of a SyncIQ job involves a single worker completing a file sync on a large file. Until the SyncIQ job completes, another new or queued SyncIQ job cannot start as part of the five concurrent running SyncIQ jobs.

However, large file synchronization work is divided at the file sub-range and distributed across threads. A sub-range is a given portion of the file. Instead of locking at a file level, locking occurs on the sub-range. The replication state, or repstate, is also tracked based on the file sub-range. This implementation enables multiple workers or threads per file. Dividing of files is necessary when the remaining file replication work is greater than or equal to 20 MB in size. The number of file splits is limited only by the maximum of 40 SyncIQ workers per job. File splitting avoids SyncIQ jobs dropping to single-threaded behavior if the remaining work is a large file. The resultant behavior is overall SyncIQ job performance by providing greater efficiency for large files and a decreased time to job completion.

File splitting is enabled by default, but only when both the source and target cluster are at a minimum. It can be disabled or enabled on a per policy basis using the command **isi sync policies modify <policy_name> --disabled-file-split true or false**. **True** to disable, **false** to re-enable if it had been disabled.

File splitting is enabled by default at the time the replication policy is created. File splitting can be disabled manually using the CLI. Use the **isi sync policies modify** command with the **policy_name** and the **--disable-file-split** option followed by **true** or **false** to set the policy state. Note that the **--disable-file-split** option is hidden and not listed using the **-h** or **--help** options.

Both the source and target clusters must be running OneFS 7.1.1 or newer to enable file splitting. If either the source or the target cluster is pre-OneFS 7.1.1, file splitting cannot be enabled.

SyncIQ CloudPools Support

- Added to support new CloudPools feature
 - Enables replication of CloudPools data from source to target
 - Extends SyncIQ data protection to CloudPools data
 - Uses CloudPools APIs
- Extended SyncIQ capabilities
 - Replicates CloudPools data *
 - Stub files
 - Reference to file data on cloud
 - Cached file data
 - File data retained for CloudPools synchronization
 - Supports all other SyncIQ capabilities *

* Process and capabilities vary by target OneFS version



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To finish the SyncIQ discussion, we'll turn back to SyncIQ's support with CloudPools. SyncIQ is enhanced with new features to support CloudPools. SyncIQ can synchronize CloudPools data from the Isilon CloudPools aware source cluster to an Isilon target cluster. The enhancements extend existing SyncIQ data protection for CloudPools data and provides failover and failback capabilities. SyncIQ leverages the CloudPools application programming interface (API) tools to enable support.

The enhancements extend previous SyncIQ capabilities enabling replication of CloudPools data, including stub files. SyncIQ continues to support all other SyncIQ capabilities during the process including failover and failback for disaster recovery. The processes and capabilities of SyncIQ features are based on the OneFS version relationship between the source cluster and the target cluster. This relationship determines the capabilities and behaviors available for SyncIQ policy replication.

Stub Files and Deep Copy

- Stub file

- Contains file metadata and cloud storage location information
- Includes cached CloudPools file transactional data
- Only applicable to CloudPools stored files



- Deep copy

- Pertains to SyncIQ files for CloudPools data
- CloudPools stub file information plus full file data
- Used when target is not CloudPools aware



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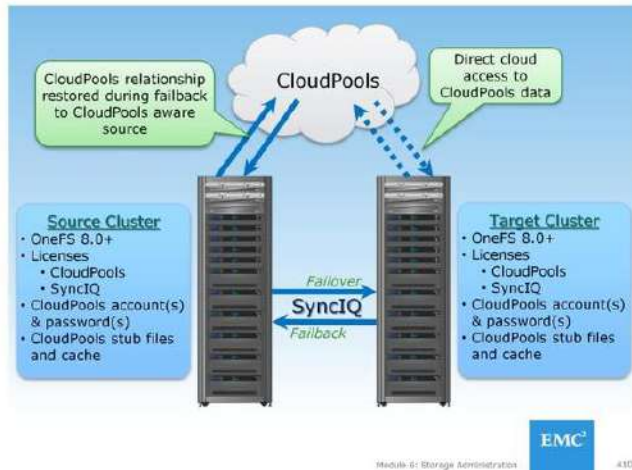
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As discussed in the CloudPools lesson, when a file is saved to the cloud storage location, the file structure changes on the cluster for the file. This is called a SmartLink file or stub file. The stub file contains the file metadata, the cloud storage location and any cached CloudPools transactional data for the file. Stub files are only applicable for CloudPools stored files. The illustration represents what is contained in a stub file.

With SyncIQ we have the option to synchronize the stub files to the target cluster, or we have the option to copy the stub file data and the actual file data. If we synchronize the full file data with the stub file data, it is called a deep copy. Deep copy preserves the entire file to the target. The primary use is with SyncIQ when the target is not CloudPools aware. An example of a non-CloudPools aware target is a cluster running pre-OneFS 8.0, or a cluster without access to the cloud location storage provider. The lower illustration represents the data stored during a deep copy.

SyncIQ with CloudPools: 8.0+ → 8.0+

- OneFS 8.0+ on source and target
- Full stub file and file data cache supported in syncs
- Target access to CloudPools data
- Seamless DR failover to target
- Intact failback to source

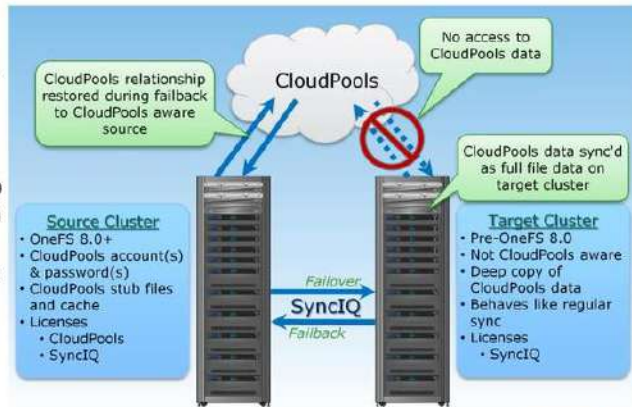


We now take a look at how SyncIQ works with CloudPools data when we have OneFS 8.0 or later on both the source and target clusters. In this case SyncIQ can replicate and understand the CloudPools data natively. The CloudPools data contains the stub file and the cached CloudPools synchronization data. SyncIQ replicates and synchronizes both data components to the target cluster.

Both the source cluster and target cluster are CloudPools aware. The target cluster supports direct access to CloudPools data if the CloudPools license is purchased and enabled by adding the CloudPools account and password information on the target cluster. This enables seamless failover for disaster recovery with utilizing the standard SyncIQ failover processes. Failback to the original source cluster updates the stub file information and current cached CloudPools data as part of the process.

SyncIQ with CloudPools: 8.0+ → pre 8.0

- OneFS 8.0+ on source
OneFS 6.5 – OneFS 7.2.1 on target
- Target not CloudPools aware
- Deep copy populates stub file with full file data from CloudPools data
- DR failover to full data on target
 - Target access to local data only
- Failback to source
 - CloudPools restored



How does SyncIQ differ when the source cluster is CloudPools aware and the target cluster is not? SyncIQ has been updated to support target clusters with OneFS 6.5 through OneFS 7.2.1. These OneFS versions are pre-CloudPools and are not aware of CloudPools stub files. When this occurs, SyncIQ initiates a deep copy of the CloudPools data to the target. The files synchronized contain the CloudPools information stored as part of the file along with a full copy of the file data. The target cluster cannot connect directly to the CloudPools and relies on the deep copy data stored locally on the cluster. The synchronization behaves like any standard SyncIQ job updating the target data. In the event of a failover or a failback, the target relies on the local copy of the data. During failback, the source cluster recognizes when a file has been tiered to the cloud and updates the cloud with data from the target appropriately. Any changes made to the target file data is saved as a new file version on the cloud.

Deep Copy Configuration

- Configured per SyncIQ policy
 - Affects sync to target behavior
 - Applies to OneFS 8.0+ to any supported OneFS version
 - Enables customer desired control of CloudPools data sync behavior
- Options include **Deny**, **Allow**, or **Force**
 - **Deny**: Never deep copy, fail policy if SyncIQ needs to deep copy
 - CloudPools data not replicated to target
 - Default behavior
 - **Allow**: Do stub sync, but if SyncIQ needs to, do deep copy
 - **Force**: Always deep copy, never stub sync
 - Generates full data sync to target

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In addition to the default SyncIQ behavior, options are provided to control the how CloudPools file data is synchronized. Customers may desire different replication behavior based on their policies for different data sets. As an example, low importance data stored on the cloud may not merit the storage space required for a deep copy to a non-CloudPools aware cluster. Or they have decided to keep a local copy of all CloudPools data for archive or as a backup to the services provided through the cloud storage provider.

Three options are available to configure with each SyncIQ policy: Deny, Allow, and Force.

- **Deny** never deep copies CloudPools data to a target cluster and fails the SyncIQ policy if a deep copy is required. Deny is the default behavior.
- **Allow** copies stub file and cached file data when it can, and does a deep copy of the data when it needs to.
- **Force** deep copies all data and never the stub file data to the target.

Lesson 5: Summary

Having completed this lesson, you can now:

- Know replication fundamentals
- Understand SyncIQ replication
- Plan and configure replication policy
- Execute failover and failback operations
- Manage SyncIQ performance
- Describe SyncIQ CloudPools support

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Having completed this lesson, you now know replication fundamentals, understand how SyncIQ replication works, can plan and configure a SyncIQ replication policy, execute failover and failback operations, manage SyncIQ performance, and describe SyncIQ CloudPools support.

Lesson 6: Deduplication (SmartDedupe)

After completing this lesson, you will be able to:

- Understand deduplication
- Describe Isilon's deduplication (SmartDedupe)
- Configure deduplication on OneFS

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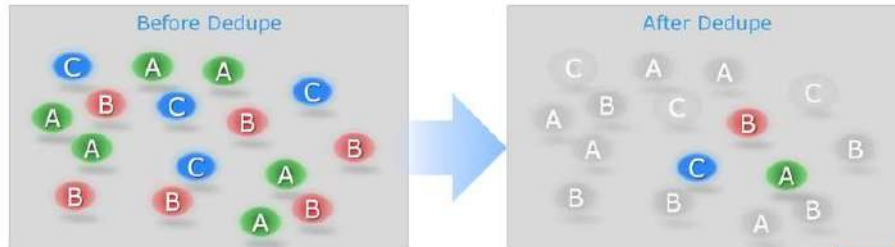


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After completing this lesson, you will be able to explain what deduplication is, and then describe how it is handled and configured on Isilon clusters.

Deduplication Overview

- Store duplicate blocks once
- Runs as job
- Does not affect user metadata
- Configurable target directories



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Deduplication saves a single instance of data when multiple identical instances of that data exist, in effect, reducing storage consumption. This can be done in a few ways – you can look for duplicate files, duplicate blocks in files, or identical extents of data within files.

The OneFS deduplication (SmartDedupe) functionality deduplicates at the block level.

Deduplication on Isilon is an asynchronous batch job that identifies identical blocks of storage across the pool. The job is transparent to the user. Stored data on the cluster is inspected, block by block, and one copy of duplicate blocks is saved, thus reducing storage expenses by reducing storage consumption. File records point to the shared blocks, but file metadata is not deduplicated. The user should not experience any difference except for greater efficiency in data storage on the cluster, because the user visible metadata remains untouched – only internal metadata is altered. Storage administrators can designate which directories are to go through deduplication, so as to manage the cluster's resources to best advantage, not all workflows are right for every cluster.

Dedupe Considerations

- SmartDedupe License
- Post process
 - Not immediate
 - Eventual
- Asynchronous
 - Does not block writes
- Per disc pool
- File metadata is not deduplicated
- Files less than 32KB
- Shadow store
 - 2GB default size
 - Up to 256,000 blocks storable
- Replication and backup behavior
- Snapshots



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A SmartDedupe license is required to deduplicate data on a cluster. Deduplication on Isilon is a relatively nonintrusive process. Rather than increasing the latency of write operations by deduplicating data on the fly, it is done after the fact. This means that the data starts out at the full literal size on the cluster's drives, and might only reduce to its deduplicated, more efficient representation hours or days later.

Because the amount of time that deduplication takes is heavily dependent on the size and usage level of the cluster, a large and complex environment would benefit not only from using the dry run procedure, but also from consultation with high-level support or engineering. Deduplicating petabytes is harder than deduplicating gigabytes.

Another limitation is that the deduplication does not occur across the length and breadth of the entire cluster, but only on each disc pool individually. This means that some opportunities for deduplication may be missed if the identical blocks are on different disc pools. This means that data which is moved between node pools may change what level of deduplication is available for it.

An example would be data that was moved owing to SmartPools configurations from a high performance node pool to nearline storage. That data would no longer be available for deduplication with respect to the other data on the high performance node pool, but would be newly available for deduplication on nearline storage.

SmartDedupe does not deduplicate files that are 32 KB or smaller, because doing so would consume more cluster resources than the storage savings are worth.

The default size of a shadow store is 2 GB, and each shadow store can contain up to 256,000 blocks. Each block in a shadow store can be referenced up to 32,000 times.

When deduplicated files are replicated to another Isilon cluster or backed up to a tape device, the deduplicated files no longer share blocks on the target Isilon cluster or backup device. However, although you can deduplicate data on a target Isilon cluster, you cannot deduplicate data on an NDMP backup device.

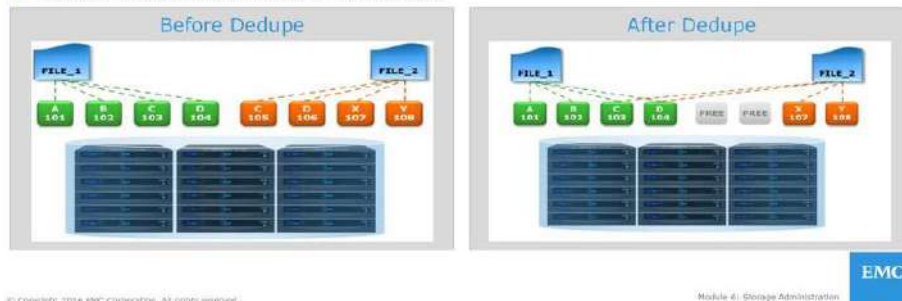
Shadow stores are not transferred to target clusters or backup devices. Because of this, deduplicated files do not consume less space than non-deduplicated files when they are replicated or backed up. To avoid running out of space, you must ensure that target clusters and tape devices have enough free space to store deduplicated data as if the data had not been deduplicated.

You cannot deduplicate the data stored in a snapshot. However, you can create snapshots of

deduplicated data.

How Deduplication Works on OneFS

1. Compares 8KB blocks in `/ifs`
2. Finds matching blocks with help of hash values
3. Moves matching blocks to shadow store
4. Saves block references in metadata



A job in the OneFS Job Engine runs through blocks saved in every disc pool, and compares the block hash values. If a match is found, and confirmed to be a true copy, the block is moved to the shadow store, and the file block references are updated in the metadata.

This job has a few of phases. Under the hood, the job first builds an index of blocks, against which comparisons are done in a later phase, and ultimately confirmations and copies take place. This does not happen immediately when a file is written, but after the fact, behind the scenes. The actual deduplication job can be a very time consuming one, but because it happens as a job which is throttled by the load on the system, the actual customer experience is fairly seamless. Customers find that their cluster space usage has dropped once the job has run.

Because this is a post process form of deduplication, data has to be written to the system before it is inspected. This has the benefit that cluster writes happen faster, but the disadvantage is that the Isilon cluster may have duplicate data written to it before it is picked up and reorganized to eliminate the duplicates.

Dedupe Phases



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The process of deduplication consists of four phases. The first phase is sampling, in which blocks in files are taken for measurement, and hash values calculated. In the second phase, blocks are compared with each other using the sampled data. In the sharing phase, matching blocks are written to shared locations. Finally the index of blocks is updated to reflect what has changed. The deduplication job is potentially very time consuming. It is heavily dependent on the cluster size and the cluster's usage level.

The deduplication dry run has three phases – the sharing phase is missing compared to the full deduplication job. Because this is the slowest phase, it allows customers to get a fairly quick overview of how much data storage they are likely to reclaim through deduplication. The dry run has no licensing requirement, so customers can run it before they pay for deduplication.

The deduplication job runs with the parameters as set by Isilon's engineering department. The only factors that are open to customer alteration are scheduling, job impact policy, and which directories on the cluster will be deduplicated.

Use Cases and Considerations for Dedupe

Use Cases

- Home directories
 - Many users with many of same files
- Archival files
 - Relatively static data
- Uncompressed virtual machine images

Considerations

- Compressed vs. uncompressed data
- Unique vs. replica files
- Rapid change vs. near-static
- Do a dry run



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A good use case for deduplication is home directories. A home directory scenario in which many users may be saving copies of the same file can offer excellent opportunities for deduplication. Static, archival files are another example. Typically this data is seldom changing and therefore the storage saved may far outweigh the load dedupe places on a cluster. Deduplication is more justifiable when the data is relatively static. Workflows that create many copies of uncompressed virtual machine images can benefit from deduplication.

Deduplication by its nature does not deal well with compressed data because the compression process tends to rearrange data to the point that identical files in separate archives are not identified as such. Environments with many unique files don't duplicate each other, so the chances of blocks being found which are identical are very low. Rapid changes in the file system tend to undo deduplication, so that the net savings achieved at any one time are low.

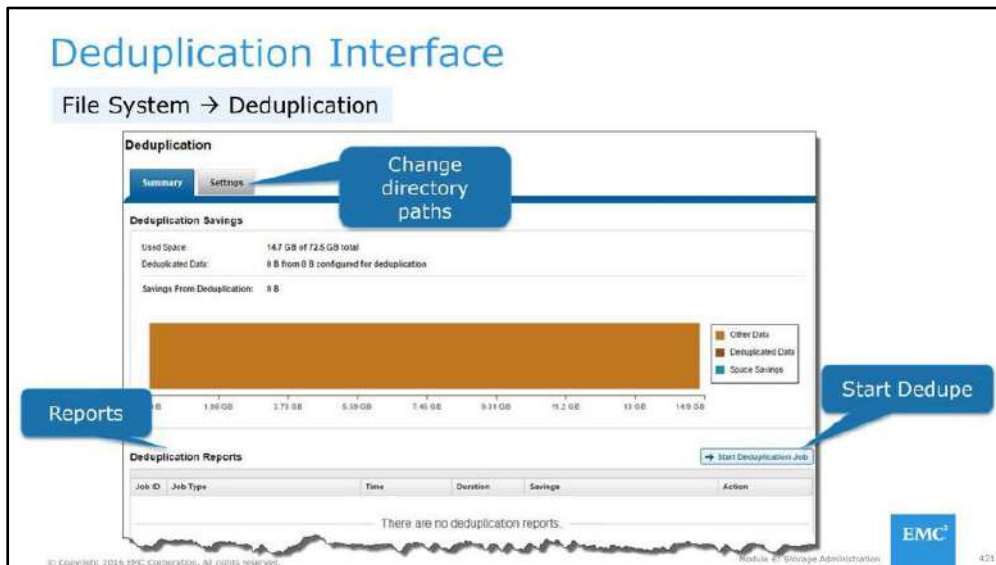
If in doubt, or attempting to establish the viability of deduplication, a good and relatively nonintrusive way of identifying the practicality of deduplication is to perform a dry run.



Because the sharing phase is the slowest deduplication phase, a dry run (or deduplication assessment) returns an estimate of capacity savings. The dry run places minor load on the cluster and completes more quickly than a full deduplication run. This enables a customer to decide whether or not the savings offered by deduplication are worth the effort, load, and cost.

Shown in the screen capture are the jobs associated with deduplication: Dedupe and DedupeAssessment. The administrator can start the dry run as well as edit the job type. Editing the Dedupe or DedupeAssessment jobs allows the administrator to change the:

- **Default Priority** – the job priority as compared to other system maintenance jobs running at the same time.
- **Default Impact Policy** – the amount of system resources that the job uses compared to other system maintenance jobs running at the same time.
- **Schedule** – start the job manually or set to run on a regularly scheduled basis.



After enabling the SmartDedupe license, you can find the **Deduplication** under the **File System** tab. From this screen you can start a deduplication job and view any reports that have been generated. On the **Settings** tab, you can also make alterations to in terms of which paths are deduplicated. This is so that a storage administrator can avoid attempting to deduplicate data where no duplicate blocks are expected, like large collections of compressed data. Deduplicating an entire cluster without considering the nature of the data is unlikely to be efficient.

Lesson 6: Summary

Having completed this lesson, you can now:

- Understand deduplication
- Describe Isilon's deduplication (SmartDedupe)
- Configure deduplication on OneFS

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Module 6: Storage Administration



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Having completed this lesson, you can now explain what deduplication is, and then describe how it is handled and configured on Isilon clusters.

Module 6: Summary

Having completed this module, you can now:

- Implement SmartPools and file pool policies
- Deploy CloudPools
- Configure SmartQuotas functionality
- Apply SnapshotIQ
- Execute SyncIQ policies and actions
- Accomplish data deduplication

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Module 6: Storage Administration



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Having completed this module, you can now implement SmartPools and file pool policies, deploy CloudPools, configure SmartQuotas, apply SnapshotIQ, execute SyncIQ policies, and accomplish data deduplication.

Lab 6: Storage Administration

- Working with SmartPools and file pool policies
- Working with SmartQuotas
- Working with SnapshotIQ
- Working with SyncIQ
- Working with Dedupe



Module 6: Storage Administration

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In Lab 6, you will validate SmartPools settings and default file pool policies, and then create a new file pool policy, configure a SmartQuotas directory, and then add and test a user's quota, create a SnapshotIQ schedule, and then view the generated snapshots, create, view, and modify a SyncIQ policy, and assess the amount of disk space saved by deduping the cluster.

Module 7: Job Engine

Upon completion of this module, you should be able to:

- Describe cluster Job Engine
- Identify critical cluster jobs
- Establish job roles
- Understand Job Engine risks

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Module 7: Job Engine



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Upon completion of this module, you will be able to describe the Job Engine and how jobs work. You will be able to identify the jobs that run on the cluster and describe the role of jobs as part of cluster operations and understand some of the risks surrounding the job system. Finally, you will understand how to manage jobs using the web administration interface and the CLI.

Lesson 1: Job Engine Architecture

After completing this lesson, you will be able to:

- Describe cluster Job Engine
- Define job tasks
- Explain Job Engine functionality
- Characterize coordinator role
- Identify jobs and job threads

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Module 7: Job Engine



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After completing this lesson, you will be able to describe the cluster's Job Engine and define the job tasks, explain the Job Engine functionality, know how to characterize the coordinator role, and identify jobs and job threads.

Job Engine in OneFS

- Meets need for cluster-wide automation
- Runs as daemon on each cluster node
- Starts and runs jobs
- Coordinates between nodes
- Keeps track of jobs and priorities
- Reports results

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Module 7: Job Engine



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The Job Engine performs cluster-wide automation of tasks on the cluster. The Job Engine is a daemon that runs on each node. The daemon manages the separate jobs that are run on the cluster. The daemons run continuously, and spawn off processes to perform jobs as necessary. Individual jobs are procedures that are run until complete. Individual jobs are scheduled to run at certain times, are started by an event, such as a drive failure, or manually started by the administrator. Jobs do not run on a continuous basis.

The **isi_job_d** daemons on each node communicate with each other to confirm actions are coordinated across the cluster. This communication ensures that jobs are shared between nodes to keep the work load as evenly distributed as possible. Each job is broken down into work units. The work units are handed off to nodes based on node speed and workload. Every unit of work is tracked. That way, if a job needs to be paused it can be restarted from where it last stopped.

- Jobs are tasks
 - Perform maintenance activities
- Have priorities
 - Most Important higher priority
- Interruptible
- Have impact description
 - Node resource impact
- Not user definable

[illegible]

Module 7: Job Engine

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OneFS does not enable administrators to define custom jobs. It does permit administrators to change the configured priority and impact policies for existing jobs. Changing the configured priority and impact policies can impact cluster operations.

Job Engine Complex Relationships

- System resources utilization
- Interdependencies between jobs
- Default job settings
 - Balance of requirements
 - Most important jobs first
 - Data integrity and protection priority
 - Recommendation: Do not change default settings
 - Can impact system balance
 - Can put data at risk



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Module 7: Job Engine

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The Job Engine can run up to three jobs at a time. The relationship between the running jobs and the system resources is complex. Several dependencies exist between the category of the different jobs and the amount of system resources consumed before resource throttling begins. The default job settings, job priorities, and impact policies are designed to balance the job requirements to optimize job system. The most important jobs have the highest job priority and should not be modified. **FlexProtect** and **FlexProtectLin** are the top-priority jobs in OneFS and are responsible for reprotecting data in the event of a drive failure. Do not ever change the priority of these jobs. Changing the job priority can impact the systems ability to maintain data protection and integrity. The recommendation is to not change the default impact policies or job priorities without consulting qualified EMC Isilon engineers. Changing the settings can impact the system balance and potentially put data at risk.

Job Engine Terminology

- Job
- Phase
- Task
- Task result
- Item
- Item result
- Checkpoints

```

asandbox1-1$ isi job engine list
Time      Message
-----
2014-11-10T07:09:38 SmartPools[9636] Phase 2: end sin policy update
2014-11-10T07:09:38 SmartPools[9636] Succeeded
2014-11-10T07:10:10 SmartPools[9637] Waiting
2014-11-10T07:10:10 SmartPools[9637] Running
2014-11-10T07:10:10 SmartPools[9637] Phase 1: begin lin policy update
2014-11-10T07:10:41 SmartPools[9637] Phase 1: end lin policy update
2014-11-10T07:10:41 SmartPools[9637] Phase 2: begin sin policy update
2014-11-10T07:10:42 SmartPools[9637] Phase 2: end sin policy update
2014-11-10T07:10:42 SmartPools[9637] Succeeded
2014-11-10T07:15:13 SmartPools[9638] Waiting
2014-11-10T07:15:13 SmartPools[9638] Running
2014-11-10T07:15:13 SmartPools[9638] Phase 1: begin lin policy update
2014-11-10T07:15:48 SmartPools[9638] Phase 1: end lin policy update
2014-11-10T07:15:48 SmartPools[9638] Phase 2: begin sin policy update
2014-11-10T07:15:48 SmartPools[9638] Phase 2: end sin policy update
2014-11-10T07:15:48 SmartPools[9638] Succeeded
2014-11-10T07:20:19 SmartPools[9639] Waiting
2014-11-10T07:20:19 SmartPools[9639] Running
2014-11-10T07:20:20 SmartPools[9639] Phase 1: begin lin policy update
2014-11-10T07:20:53 SmartPools[9639] Phase 1: end lin policy update
2014-11-10T07:20:53 SmartPools[9639] Phase 2: begin sin policy update
2014-11-10T07:20:53 SmartPools[9639] Phase 2: end sin policy update
2014-11-10T07:20:53 SmartPools[9639] Succeeded
2014-11-10T07:25:25 SmartPools[9640] Waiting
2014-11-10T07:25:25 SmartPools[9640] Running
2014-11-10T07:25:25 SmartPools[9640] Phase 1: begin lin policy update
2014-11-10T07:25:59 SmartPools[9640] Phase 1: end lin policy update
    
```



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Module 7: Job Engine

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Job - An application built on the distributed work system of the Job Engine. A specific instance of a job, often just called a job, is controlled primarily through its job ID that is returned using the **isi job jobs start** command.

Phase - One complete stage of a job. Some jobs have only one phase, while others, like **MediaScan**, have as many as seven. If an error occurs in a phase, the job is marked failed at the end of the phase and does not progress. Each phase of a job must complete successfully before advancing to the next stage or being marked complete returning a job state Succeeded message.

Task - A task is a division of work. A phase is started with one or more tasks created during job startup. All remaining tasks are derived from those original tasks similar to the way a cell divides. A single task will not split if one of the halves reduces to a unit less than whatever makes up an item for the job. At this point, this task reduces to a single item. For example, if a task derived from a restripe job has the configuration setting to a minimum of 100 logical inode number (LINS), then that task will not split further if it derives two tasks, one of which produces an item with less than 100 LINS. A LIN is the indexed information associated with specific data.

Task result - A task result is a usually small set of statistics about the work done by a task up to that point. A task will produce one or more results; usually several, sometimes hundreds. Task results are producing by merging item results, usually on the order of 500 or 1000 item results in one task result. The task results are themselves accumulated and merged by the coordinator. Each task result received on the coordinator updates the status of the job phase seen in the **isi job status** command.

Item - An item is an individual work item, produced by a task. For instance, in **QuotaScan** an item is a file, with its path, statistics, and directory information.

Item result - An accumulated accounting of work on a single item; for instance, it might contain a count of the number of retries required to repair a file, plus any error found during processing.

Checkpoints - Tasks and task results are written to disk, along with some details about the job and phase, in order to provide a restart point.

How the Job Engine Works

- **Jobs**
 - Hierarchical structure
 - Managed by coordinator
 - Coordinator chosen by race between nodes
- Job can have multiple phases
- Phases split into tasks
- Tasks broken down into items



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The Job Engine consists of all the job daemons across the whole cluster. The job daemons elect a job coordinator. The election is by the first daemon to respond when a job is started.

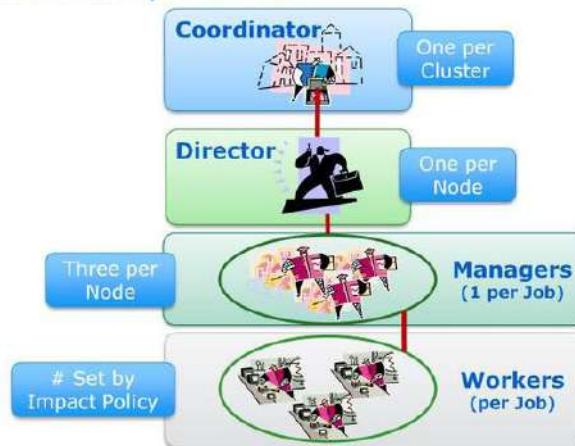
Jobs can have a number of phases. There might be only one phase, for simpler jobs, but more complex ones can have multiple phases. Each phase is executed in turn, but the job is not finished until all the phases are complete.

Each phase is broken down into tasks. These tasks are distributed to the nodes by the coordinator, and the job is executed across the entire cluster.

Each task consists of a list of items. The result of each item's execution is logged, so that if there is an interruption the job can restart from where it stopped.

Job Engine v2 Components

- Coordinator
- Director
- Managers
- Workers



Job Engine v2 is comprised of four main functional components; the coordinator, the directors, the managers, and the workers.

- The coordinator is the executive of the Job Engine, this thread starts and stops jobs and processes work results as they are returned during the execution of the job.
- The director runs on each node and communicates with the job coordinator for the cluster and coordinates tasks with the three managers.
- Each manager manages a single job at a time on the node. The three managers on each node coordinate and manage the tasks with the workers on their respective node. Each node has a manager, responsible for managing the flow of tasks and task results throughout the node. Managers request and exchange work with each other and supervise the worker processes they assign.
- Each worker is given a task, if any task is available. The worker then processes the task item by item until the task is complete or the manager removes the task from the worker. The number of workers assigned to a task is set by the job's impact policy. The impact policy applied to the cluster is based on the highest impact policy for all current running jobs.

Job Coordinator

- Temporarily elected position
- Manages
 - Job scheduling
 - Job phase progression
 - Task sharing
 - Pausing and restarting jobs
- Upon failure, cluster elects new coordinator

```
Identify Job Coordinator
Cloud9-1# isi_job_d status
isi_job_d is running, pid is 2180
Coordinator is on node 3
```



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The job daemons elect a coordinator by racing to lock a file. The node that first locks the file becomes the coordinator. This is an approximate way of choosing the least busy node as the coordinator. If the coordinator's node goes offline and the lock is released, the next node in line becomes the new coordinator. The coordinator then coordinates the execution of each job, and shares out the parts of each job.

To find the coordinator node, run **isi_job_d status** from the CLI. The node number displayed is the node array ID.

Job Workers

- Job daemon uses threaded model
- Threads run simultaneously on each node
- Number of threads depends on impact policy level
- Worker has number of threads
- More threads not always better



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The job daemon uses threads to enable it to run multiple tasks at the same time. A thread is the processing of a single command by the CPU. The coordinator tells the job daemon on each node what the impact policy of the job is, and consequently, how many threads should be started to get the job done.

Each thread handles its task one item at a time and the threads operate in parallel. A number of items are being processed at any time. The number of items being processed is determined by the number of threads. The defined impact level and the actual load placed on any one node is managed by the maximum number of assigned threads.

It is possible to run enough threads on a node that they can conflict with each other. An example would be five threads all trying to read data off the same hard drive. Each thread cannot be served at once and are queued and wait for each other to complete. The disk can thrash from over access reducing efficiency. A threshold exists to the useful degree of parallelism available depending upon the job.

Increasing the impact policy for a job is not usually advisable. You need to understand what each job is doing to assess the costs and benefits before changing the impact policy. As a general recommendation, all impact policy settings should remain as the default settings.

Exclusion Sets – Job Engine

- Three categories of job activity
 - Restripe phases
 - Mark phases
 - All other phases
- Maximum one phase at a time from:
 - Restripe category
 - Mark category
 - Both restripe and mark
- Up to 3 other jobs at a time
- One instance per individual job



Job Engine v2 includes the concept of job exclusions sets. Job phases are grouped into three categories: restripe, mark and all other job phase activities. Two categories of job phase activity, restripe and mark, modify core data and metadata. Up to three jobs can run at the same time with Job Engine v2. Although multiple restripe or mark job phases cannot safely and securely run at the same time without either interfering with each other or the risk of data corruption. Job Engine restricts the simultaneous jobs to include only one restripe category job phase and one mark category job phase at the same time. There is one job that is both a restripe job and a mark job. When this job runs, no additional restripe or mark job phases are permitted to run. Up to three other jobs can run at the same time and run simultaneous with the running restripe or mark job phases. Only one instance of any job may run at the same time.

The valid simultaneous job combinations include:

- One restripe job phase, one mark job phase, and one all other phases
- One restripe job phase and two all other phases
- One mark job phase and two all other phases
- One combined mark/restripe job activity and two all other phases
- Or three all other job phases

Lesson 1: Summary

Having completed this lesson, you can now:

- Describe cluster Job Engine
- Define job tasks
- Explain Job Engine functionality
- Characterize coordinator role
- Identify jobs and job threads

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Having completed this lesson, you can now describe the cluster's Job Engine and define the job tasks, explain the Job Engine functionality, know how to characterize the coordinator role, and identify jobs and job threads.

Lesson 2: Jobs and Job Configuration Settings

After completing this lesson, you will be able to:

- Differentiate feature-related and specific-use jobs
- Understand data restriping and balancing jobs
- Describe impact policies
- Establish how job priority is applied
- Explain job exclusion sets

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Upon completion of this lesson, you will be able to differentiate between feature-related and specific-use jobs, explain the different jobs and their usage relationship. You will also be able to discuss job priorities and impact policies. You will also be able to explain what exclusion sets are and how they are used in determining which jobs are run.

Jobs in Context

- Integral part of OneFS
 - Jobs perform system maintenance tasks
 - Jobs key role in data protection
 - Jobs involved in major features
- Job errors affect cluster
 - Which job?
 - When?
 - Priority?
 - Impact policy?



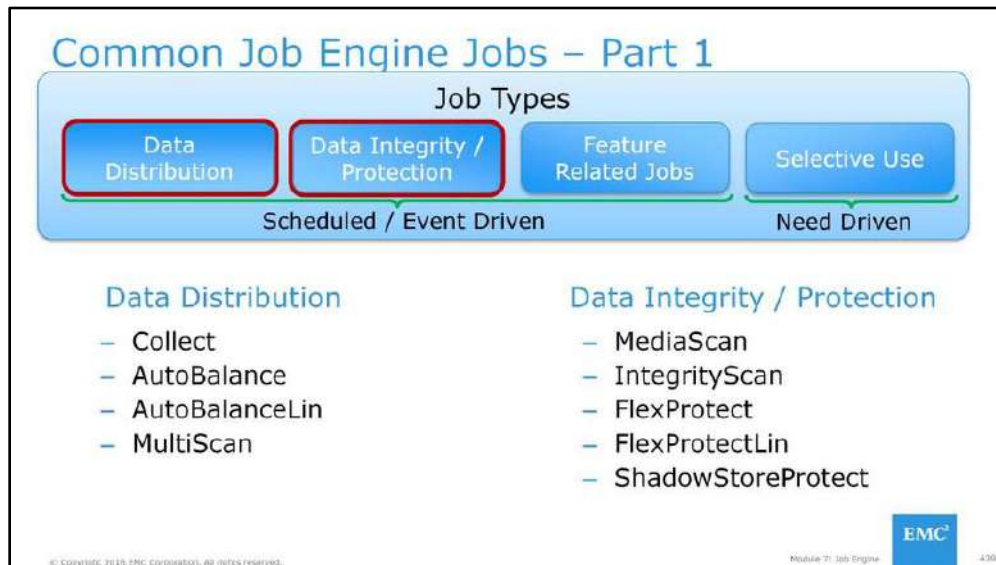
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A lot of the functions and features of an Isilon cluster depend on jobs, which means that the Job Engine and the jobs that run through it are critical to cluster health. Jobs play a key role in data reProtection and balancing data across the cluster, especially in the event of hardware failure or cluster reconfiguration. Features such as antivirus scanning and quota calculation also involve jobs.

Up to three jobs can run at a time. Additional jobs or job phases limited by exclusion sets are queued and run sequentially. Higher priority jobs are run before of lower priority jobs, and jobs with the same priority run in the order that the job start request is made, a first-in-queue, first-to-run order.

Jobs run sequentially, one job that holds up other jobs can affect cluster operations. If this occurs, you should examine which jobs are running, which jobs are queued, when the jobs started, the job priority and impact policies for the jobs. Some jobs can take a long time to complete. This is a normal condition. However if those jobs should get paused so jobs of higher immediate importance can complete. MediaScan can take days to complete and why the default priority is set to 8 as the lowest priority job in OneFS. All other jobs may interrupt MediaScan. This is an example of the balance for job priorities taken into consideration when the default setting were determined.



The most common Job Engine jobs can be broken into different types of use; jobs related to the distribution of the data on the cluster, jobs related to testing the data integrity and protection, jobs associated with specific feature functionality, and other jobs which are used selectively for particular needs. Jobs are not exclusive to themselves and often work in conjunction calling other jobs to complete their task.

Looking at the data distribution jobs, four of the most common jobs are used to help distribute data across the cluster.

- **Collect** - Runs a mark-and-sweep looking for orphaned or leaked inodes or blocks.
- **AutoBalance** - Scans drives of an imbalanced cluster, balances the distribution of files across the node pools and tiers.
- **AutoBalanceLin** - Logical inode number (LIN) based version of AutoBalance
- **MultiScan** - A combination of AutoBalance and Collect, it is triggered after every group change. Collect is run if it hasn't been run recently, the default is within the last 2 weeks.

Data integrity and protection jobs are regularly run on the cluster. These jobs can be further broken down into proactive error detection and re-protection of the data. The proactive error detection includes jobs that will often be found running for long periods of time. These run when no other jobs are active and look primarily for errors on the drives or within the files.

- **MediaScan** - Scans the drives looking for error correction code (ECC)-detected error entries. It has many phases, with the general purpose of moving any file system information off ECC-producing areas and repairing any damage.
- **IntegrityScan** - Like the first phase of collect, identifies everything valid in the file system. Nothing is changed; the inspection process itself is meant to catch invalid file system elements.

The re-protection jobs focus on returning data to a fully protected state. These jobs are usually triggered by events such as a drive failure.

- **FlexProtect** - Restores the protection level of individual files. Without getting into too much detail, this makes sure that a file which is supposed to be protected at, say, 3x, is still protected at 3x. It is run automatically after a drive or node removal (or failure).
- **FlexProtectLin** - LIN based version of FlexProtect.
- **ShadowStoreProtect** - Re-protects data in shadow stores to a higher protection level that are referenced by a LIN with a higher protection level.

Common Job Engine Jobs – Part 2



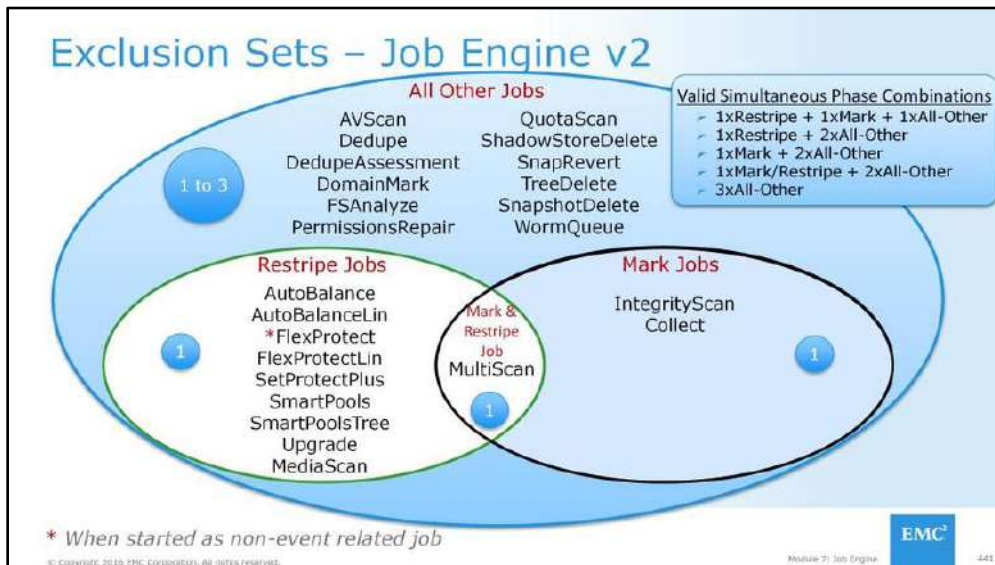
Feature related jobs are jobs that run as part of specific features scheduled in OneFS.

- **SetProtectPlus** - The unlicensed version of SmartPools; it enforces the default system pool policies but does not enforce user pool policies. SetProtectPlus is disabled when a SmartPools license is activated on the cluster.
- **SmartPools** - Responsible for maintaining layout of files in the node or file pools according to file pool policies. Requires a SmartPools license.
- **SmartPoolsTree** - Allows an administrator to run SmartPools on a particular directory tree, rather than the whole file system at once.
- **QuotaScan** - Scans modified quota domains to incorporate existing data into new quotas. QuotaScan is automatically triggered by quota creation. Requires a SmartQuotas license.
- **SnapshotDelete** - In order from the oldest to newest deleted snapshot, deletes the file reference in the snapshot, and then deletes the snapshot itself.
- **SnapRevert** - Revert an entire snapshot back to the original version. Requires a SnapshotIQ license.
- **AVScan** - Scans the filesystem for viruses. Uses an external antivirus server. Scheduled independently by the AV system.
- **FSAnalyze** - FSAnalyze is the Data gatherer for InsightIQ, or filesystem analytics to provide cluster data such as file counts, a heat mapping, and usage by user. Requires an InsightIQ license.
- **ChangelistCreate** - Create a list of changes between two consecutive SyncIQ snapshots
- **Dedupe** - Scan a directory for redundant data blocks and deduplicates the redundant data stored in the directory. Requires a SmartDedupe license.
- **DedupeAssessment** - Scan directory for redundant data blocks and reports and estimate of the amount of space that could be saved by deduplicating the directory. No license is required.
- **WormQueue** - Scans the SmartLock directories for uncommitted files for retention, and commits the appropriate files to WORM state.

The last category of jobs contains the jobs selectively run for specific purposes. These jobs may be scheduled, however, they are generally run by the administrator only when they are required.

- **PermissionsRepair** - Correct permissions of files and directories in **/ifs**.
- **DomainMark** - Associate a path and its contents with a SyncIQ or SnapRevert domain.
- **TreeDelete** - Deletes complete directories with speed by splitting up the work of deleting the potentially large directory.
- **ShadowStoreDelete** - Free space associated with a shadow store. Removes shadow stores that are no longer referenced and have 0 refcounts associated with them. This is a good thing to run before IntegrityScan.
- **Upgrade** - The exact content of this job varies from release to release, but always runs exactly once on upgrade from a previous OneFS version. The job ensures that whatever filesystem changes are in the new version are applied to the old data. It has no responsibility for the rest of upgrade (new daemons, functionality,

command-line tools, configuration, etc.).



Earlier exclusion sets were discussed. In the diagram, the jobs are displayed in their exclusion set categories, as determined by the needs of their individual phases. Just because a job is in an exclusion set does not mean that all its phases fit into the same exclusion set, so OneFS makes the exclusion determination at the outset of a phase, not the entire job. **FlexProtect** can be part of an exclusion set when run proactively. **FlexProtect** will override and pause all other jobs when run as a event triggered job.

FlexProtect Job

- Started two ways
- Non-event job - Started through Job Engine request
 - Manually started
 - Restripe job + other jobs can run
- Event related job - Automatically started with disk failure
 - SmartFail process
 - Restripe job, all other jobs suspended

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FlexProtect is the highest priority job on the cluster. **FlexProtect** can be run manually as a non-event triggered job to and coexist with other Job Engine jobs on the cluster. An example would be when there is proactive action to SmartFail a drive out to replace it with an SSD during a hardware upgrade activity. If the **FlexProtect** job is triggered by a drive failure, **FlexProtect** takes exclusive ownership of the Job Engine. All other jobs are paused or suspended until the **FlexProtect** job completes. This is normal behavior and is intended to reprotect the data as quickly as possible to minimize any potential risk of data loss. Do not change the priority or impact policy of the **FlexProtect** job.

Job Priority

- Order of importance for job
- Priorities values: 1 to 10
 - 1 is highest priority
 - 10 is lowest priority
- Higher priority take precedence over lower priority
- Interrupt and pause lower priority jobs
- Same priority jobs are queued
- Priority can be changed temporarily or permanently



IMPORTANT!

Before changing job priority or impact policy, consider possible impact carefully. Historically, this has caused many cluster issues

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Every job is assigned a priority that determines the order of precedence relative to other jobs. The lower the number assigned, the higher the priority of the job. As an example, **FlexProtect**, the job to reprotect data from a failed drive and restore the protection level of individual files, is assigned a priority of 1, which is the top job priority.

When multiple jobs attempt to run at the same time, the job with the highest priority takes precedence over the lower priority jobs. If a job of a lower priority is currently running and a higher priority job is called to run, the lower priority job is interrupted and paused until the higher priority job completes its task. The paused job restarts from the point at which it was interrupted.

New jobs of the same or lower priority of a currently running job are queued and then started after current job completes.

Job priority can be changed either permanently or during a manual execution of a job. If a job is set to the same priority as the running job, the running job will not be interrupted by the new job. It is possible to have a low impact, high priority job, or a high impact, low priority job.

In Job Engine, jobs from similar exclusion sets are queued when conflicting phases may run. If there is a queued job or new job phase ready to start from another exclusion set or from the all other jobs category, the job will also be run.

Changing the priority of a job can have negative effect on the cluster. Jobs priority is a tradeoff of importance. Historically, many issues have been created by changing job priorities. Job priorities should remain at their default unless instructed to be changed by a senior level support engineer.

Job Impact Policies

- Impact policy determines cluster resource allocations
- Increases number of workers
- Default impact policies
 - LOW
 - MEDIUM
 - HIGH
 - OFF_HOURS
- Running priority in Job Engine
 - Highest running job impact policy
 - Share resources



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In addition to being assigned a priority, every job is assigned an impact policy that determines the amount of cluster or node resources assigned to the job. The determination of what is more important must be made, the use of system resources to complete the job or to have the resources available for processing workflow requirements.

A default impact policy has been set for each job based on how much of a load the job places on the system. Very complex calculations are used in determining how cluster resources are allocated.

By default, the system includes default impact profiles with varying impact levels assigned—low, medium, high; and the ability to create custom schedule policies if required. Increasing or lowering an impact level from its default results in increasing or lowering the number of workers assigned to the job. The number of workers assigned to the job impacts the time required to complete the job and the impact on cluster resources.

By default, the majority of jobs have the LOW impact policy, which has a minimum impact on the cluster resources.

More time-sensitive jobs have a MEDIUM impact policy. These jobs have a higher urgency of completion usually related to data protection or data integrity concerns.

The use of the HIGH impact policy is discouraged because it can affect cluster stability. This has not been found to be a problem with **TreeDelete**, but is known to be a problem with other jobs. The HIGH impact policy should not be assigned to other jobs. HIGH impact policy use can cause contention for cluster resources and locks that can result in higher error rates and negatively impact job performance.

The OFF_HOURS impact policy allows greater control of when jobs run in order to minimize impact on the cluster and provide the maximum amount of resources to handle customer workflows.

Impact policies in Job Engine v2 are based on the highest impact policy for any currently running job. Impact policies are not cumulative between jobs but set the resource levels and number of workers shared between the jobs.

Significant issues are caused when cluster resources are modified in the job impact settings. Lowering the number of workers for a job can cause jobs to never complete. Raising the impact level can generate errors or disrupt production workflows. Use the default impact policies for the jobs whenever possible. If customer workflows require reduced impact levels, create a custom schedule based on the OFF_HOURS impact policy.

System Jobs - Priority and Impact Policies

Job	Priority	Impact	Job	Priority	Impact
AutoBalance	4	LOW	MultiScan	4	LOW
AutoBalanceLin	4	LOW	PermissionRepair	5	LOW
AVScan	6	LOW	QuotaScan	6	LOW
ChangelistCreate	5	LOW	SetProtectPlus	6	LOW
Collect	4	LOW	ShadowStoreDelete	2	MEDIUM
Dedupe	4	LOW	ShadowStoreProtect	6	LOW
DedupeAssessment	6	LOW	SmartPools	6	LOW
DomainMark *	5	LOW	SmartPoolsTree	5	MEDIUM
FlexProtect	1	MEDIUM	SnapRevert	5	LOW
FlexProtectLin	1	MEDIUM	SnapshotDelete *	2	MEDIUM
FSAnalyze	1	LOW	TreeDelete	4	MEDIUM
IntegrityScan	1	MEDIUM	Upgrade	3	MEDIUM
MediaScan	8	LOW	WormQueue	6	LOW

* Started by Job Engine, runs in SyncIQ framework

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This chart displays the default job priority and impact policy for each of the system jobs.

Only a few jobs are priority 1 and these have the MEDIUM impact policy. All three of these jobs are related to data protection and data integrity.

Two jobs have a priority of 2 with the MEDIUM impact policy. These jobs need to be completed quickly to ensure no disruption to the system processes.

No jobs have the HIGH impact policy. Very few workflows can tolerate disruption in cluster responsiveness when HIGH impact policy is used.

The **DomainMark** and **SnapshotDelete** jobs are started by the Job Engine, but run under the SyncIQ framework. The SyncIQ framework utilizes a different mechanism to perform tasks.

Lesson 2: Summary

Having completed this lesson, you can now:

- Differentiate feature-related and specific-use jobs
- Understand data restriping and balancing jobs
- Describe impact policies
- Establish how job priority is applied
- Explain job exclusion sets

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Having completed this lesson, you can now differentiate between feature-related and specific-use jobs, explain the different jobs and their usage relationship. You can also discuss job priorities and impact policies, and can now explain what exclusion sets are and how they are used in determining which jobs are run.

Lesson 3: Managing Jobs

After completing this lesson, you will be able to:

- Access Job Engine
- Understand operations, jobs, and impact policies
- Edit job defaults
- Execute manual jobs
- Troubleshoot job implications for cluster features, performance, and stability

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Upon completion of this lesson, you will be able to access the Job Engine using the web administration interface and the CLI, understand different job operations and settings, and be able to edit job settings and return them to their defaults. You will also be able to manually run a job and customize the job settings. Finally, you will understand the importance and apply troubleshooting to job-related cluster issues, including cluster performance and stability.

- Web administration interface
- Command-line interface (CLI)
- Scheduled through features

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Management Capabilities

- View jobs
 - Current job(s)
 - Job history
 - Job events
 - Job settings
- Manipulate jobs
 - Pause or stop jobs
 - Start individual jobs manually
- No custom jobs or impact levels
- Modify jobs
 - Change priorities
 - Change impact policies
 - Change schedules



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The cluster health depends on the Job Engine and the configuration of jobs in relationship to each other. Many customers modify the relationship by altering the job priorities, impact policies, and job schedules to meet their workflow requirements. While there are appropriate reasons to allow changes to the jobs, many customers have also suffered consequences as a result of these modifications. The system is engineered to maintain a delicate balance between cluster maintenance and cluster performance.

Many capabilities are available through the web administration interface and using the CLI. As of OneFS 7.2, job management is also available using PAPI. Job status and history can be easily viewed. Failed jobs or jobs with frequent starts or restarts can easily be identified.

Administrators can view and modify job settings. They can change the job priorities, impact policies and schedules for jobs.

Administrators can also manipulate currently running jobs. Jobs can be paused or stopped at any time. Jobs can also be run manually. If necessary to run a job with a modified priority or impact level from the default, it is recommended to manually run the job. Both setting can be set and only in place during the manually run job.

OneFS does not allow the capability to create custom jobs or custom impact levels. If it is required to adjust the impact level for a job, it is recommended to create a custom schedule using the OFF_HOURS impact policy and adjust the impact levels based on the time of day and day of the week.

Management - Web Admin Interface

- Job(s) status
- Job summary
- Job history
- Jobs settings
 - Priority
 - Impact policy
- Running jobs manually

The screenshot shows the 'Job Operations' section of the EMC Job Engine web interface. It includes tabs for 'Job Summary', 'Job Types', 'Job Reports', 'Job Events', and 'Impact Policies'. The 'Job Types' tab is active, displaying a table with columns: Name, State, Priority, Impact, Schedule, and Actions. The table lists several job types, all with a state of 'Enabled' and a priority of 4.

Name	State	Priority	Impact	Schedule	Actions
AutoRebuild Rebuild files across in a cluster, substituting a more efficient scheduling that is more efficient.	Enabled	4	LOW	Manual	Start Job View Job
AutoRebuildIn Rebuild files across in a cluster, substituting a more efficient if the system metadata is stored on SSDs.	Enabled	4	LOW	Manual	Start Job View Job
ChangeMetadata Create a set of changes between two snapshots, with metadata and policy.	Enabled	5	LOW	Manual	Start Job View Job
Copy Replicate files from previously unavailable media or disks.	Enabled	4	LOW	Manual	Start Job View Job
DeDup Create a directory for redundant data blocks and duplicates of redundant data stored in the directory. This job requires a SnapDrive license.	Enabled	4	LOW	Manual	Start Job View Job
DeDupRebuild Rebuild a directory for redundant data blocks and report an estimate of the error rate of data that is not covered by the directory. This job does not require a SnapDrive license.	Enabled	5	LOW	Manual	Start Job View Job
DomainJoin Associate a path and its contents with a domain.	Enabled	5	LOW	Manual	Start Job View Job
FSMigrate Perform a full migration of the system.	Enabled	6	LOW	Every day at 22:00	Start Job View Job

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The web administration interface is the primary customer interface into the Job Engine. You can view job status, job histories, view and change current job schedules, view and manage job priorities and impact policies, and run jobs manually.

Job management in the web administration interface can vary in different versions of OneFS. Although the information may be organized differently and displayed in different formats, the operational functionality remains similar. You should familiarize yourself with the web administration interface for your version of OneFS.

Job Engine Access

Cluster Management → Job Operations

The screenshot displays the EMC Job Engine web interface. The top navigation bar includes 'Dashboard', 'Cluster Management', 'File System', 'Data Protection', 'Access', and 'Protocols'. The 'Cluster Management' menu is expanded, showing options like 'General Settings', 'Events and Alerts', 'Network Configuration', 'Hardware Configuration', 'Job Operations', 'Auditing', 'Diagnostics', 'Licensing', and 'Upgrade'. The 'Job Operations' option is highlighted. On the left sidebar, the 'Job Operation' section is visible, with 'Job Summary' and 'Job Types' tabs. The 'Job Types' tab is active, showing a table of job types. The table has columns for Name, State, Priority, Impact, Schedule, and Actions. Two job types are listed: 'AutoBalance' and 'AutoBalanceIn'. The 'AutoBalance' job type is enabled, has a priority of 4, a low impact, and a manual schedule. The 'AutoBalanceIn' job type is also enabled, has a priority of 4, a low impact, and a manual schedule. The 'Actions' column for each job type contains 'Start Job' and 'View / Edit' buttons. The EMC logo is in the bottom right corner, and the footer text reads '© Copyright 2015 EMC Corporation. All rights reserved. Module 7: Job Engine 451'.

Name	State	Priority	Impact	Schedule	Actions
AutoBalance Balance free space in contain only HDDs.	Enabled	4	LOW	Manual	Start Job View / Edit
AutoBalanceIn Balance free space in metadata is stored on	Enabled	4	LOW	Manual	Start Job View / Edit

To get to the Job Engine information, click the **Cluster Management** menu, and then click **Job Operations**. The available tabs are **Job Summary**, **Job Types**, **Job Reports**, **Job Events**, and **Impact Policies**, which we'll cover in more detail on the following slides.

Job Summary

Cluster Management → Job Operations → Job Summary

The screenshot displays the 'Job Operations' section with the 'Job Summary' tab selected. It shows a table of active jobs with columns for Status, ID, Type, Priority, Impact Policy, Elapsed, Phase, Progress, and Actions. A red box highlights the 'Actions' column for a running job, showing options like 'View / Edit / More...'. Another red box highlights a bulk action dropdown menu with options: 'Cancel Selection', 'Pause Selection', and 'Resume Selection'. A blue arrow points from the bulk action menu to the 'Actions' column of the job table.

Status	ID	Type	Priority	Impact Policy	Elapsed	Phase	Progress	Actions
Running	225	Correct	4	LOW (Low)	1.4 m	2 of 2	Started	View / Edit / More...
Waiting	226	MediaScan	5	LOW (Low)	23 s	1 of 1	Found 6 ECCs on 22 drives, 1...	View / Edit / More...
Running	227	AutoBalance	4	LOW (Low)	20 s	1 of 5	Drives: 101 done, 2 in progr...	View / Edit / More...

The **Job Summary** tab displays all currently active jobs and the capability to manage the job. The information provided includes the job status, job ID, job type, priority, impact policy, elapsed time, job phase and progress. The actions include the ability to modify the settings for a running job. Individual jobs can also be cancelled, paused, or restarted depending upon their status.

A bulk action capability is also provided. Select the desired jobs, and then from the drop-down list, select the desired action.

Bulk cancellation or pausing of jobs is useful when troubleshooting job related issues, such as high CPU or memory utilization. Pausing or cancelling the job can confirm or eliminate the jobs from the list of possible causes.

Job Types

Job Operations

Job Summary **Job Types** Job Reports Job Events Impact Policies

Job Types

Name	State	Priority	Impact	Schedule	Actions
AVScan Perform an antivirus scan on all files.	Enabled	8	LOW	Manual	View / Edit / More ▾
AutoBalance Balance free space in a cluster. AutoBalance is most efficient in clusters that contain only HDDs.	Enabled	4	LOW	Manual	View / Edit / More ▾
AutoBalanceLin Balance free space in a cluster. AutoBalanceLin is most efficient if file system metadata is stored on SSDs.	Enabled	4	LOW	Manual	View / Edit / More ▾
Collect Reclaim free space from previously unavailable nodes or drives.	Enabled	4	LOW	Manual	View / Edit / More ▾
DeDup Scan directory for redundant data blocks and deduplicate all redundant data stored in the directory. This job requires a SmartCloud license.	Enabled	4	LOW	Every 1 days at 12:00 PM	View / Edit / More ▾
DeDupAssessment Scan a directory for redundant data blocks and report an estimate of the amount of space that could be saved by deduplicating the directory. This job does not require a SmartCloud license.	Enabled	8	LOW	Manual	View / Edit / More ▾
DomainTask Associate a path and its contents with a domain.	Enabled	5	LOW	Manual	View / Edit / More ▾
FSAnalyze Gather information about the file system.	Enabled	8	LOW	Every day at 22:00	View / Edit / More ▾
FileProtect Scan the file system after a device failure to ensure that all files remain protected. FileProtect is most efficient in clusters that contain only HDDs.	Enabled	1	MEDIUM	Manual	View / Edit / More ▾

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Job Types is the tab to examine and modify the current job settings. You use the **View / Edit** button to modify the job settings and the **More** button to manually start a job. For troubleshooting, you use this page to verify current job settings, and then modify the settings back to default as necessary.

Job Reports

The screenshot shows the 'Job Reports' tab in the Job Operations section. A blue callout box points to a filter dropdown menu on the left, labeled 'Filter by Job Type'. The dropdown lists various job types: AVScan, AutoBalance, AutoBalanceLin, ChangelistCreate, Collect, Dedupe, DedupeAssessment, DomainMark, FSAnalyze, FlexProtect, FlexProtectLin, IntegrityScan, MediaScan, MultiScan, PermissionRepair, and QuotaScan. The main table displays job history with columns: Time, Event ID, Job ID, Type, Phase, and Action. The 'Action' column contains 'View Details' buttons for each row. A red box highlights the 'View Details' button for the first row. The table shows 10 rows of data, with the last row being 'FSAnalyze' at 2016-01-18 19:01:52. The bottom of the interface shows 'Page 1' and 'Showing 1 - 10'.

Time	Event ID	Job ID	Type	Phase	Action
2016-01-20 05:17:40	1.1991	257	AutoBalance	1	View Details
2016-01-20 05:17:43	1.1609	285	Collect	2	View Details
2016-01-20 05:17:38	1.1907	285	Collect	1	View Details
2016-01-20 05:18:33	1.1970	204	DedupeAssessment	1	View Details
2016-01-19 19:01:38	1.1970	282	FSAnalyze	2	View Details
2016-01-19 19:02:42	1.1968	283	FSAnalyze	1	View Details
2016-01-19 09:02:30	1.1963	282	Dedupe	1	View Details
2016-01-19 21:09:16	1.1958	261	Shadow-Snapshot	1	View Details
2016-01-18 19:01:45	1.1953	280	FSAnalyze	2	View Details
2016-01-18 19:01:52	1.1951	280	FSAnalyze	1	View Details

The **Job Reports** tab displays the job history, including the associated **Event ID** and **Job ID**. On the main page, the phase information is displayed but not the overall job state. Click the **View Details** button for detailed information about the job. A filter mechanism has been provided to narrow the displayed information based on the selected job listed in the drop-down list.

Job Events

The screenshot shows the 'Job Operations' tab with the 'Job Events' sub-tab selected. A 'Filter by Job Type' dropdown is on the left, and a 'Filter by Job ID' dropdown is on the right. The main area displays a table of job events with columns: Time, Event ID, Job ID, Type, State, and Message. An 'Action' column on the right contains 'View Details' buttons for each row. The table lists several events for SmartPools, including 'Succeeded', 'Phase', and 'Waiting' states.

Time	Event ID	Job ID	Type	State	Message	Action
2014-11-10 09:16:04	340015	9646	SmartPools	Succeeded	SmartPools(9646) Succeeded	View Details
2014-11-10 09:16:04	340014	9646	SmartPools	Phase	SmartPools(9646) Phase 2: end sin policy update	View Details
2014-11-10 09:16:04	340013	9646	SmartPools	Phase	SmartPools(9646) Phase 2: begin sin policy update	View Details
2014-11-10 09:16:04	340012	9646	SmartPools	Phase	SmartPools(9646) Phase 1: end sin policy update	View Details
2014-11-10 09:16:25	340011	9646	SmartPools	Phase	SmartPools(9646) Phase 1: begin sin policy update	View Details
2014-11-10 09:16:26	340010	9646	SmartPools	Running	SmartPools(9646) Running	View Details
2014-11-10 09:16:26	340009	9646	SmartPools	Waiting	SmartPools(9646) Waiting	View Details
2014-11-10 09:16:58	340008	9645	SmartPools	Succeeded	SmartPools(9645) Succeeded	View Details
2014-11-10 09:16:58	340007	9645	SmartPools	Phase	SmartPools(9645) Phase 2: end sin policy update	View Details
2014-11-10 09:16:57	340006	9645	SmartPools	Phase	SmartPools(9645) Phase 2: begin sin policy update	View Details

The **Job Events** tab provides both the phase and state for each job. Successful, failed, running and waiting jobs appear in the job state. As a job completes a phase, the phase information is updated and displayed. Additional information is provided in the message column. The same capability is provided as with job reports to filter by job. Additional information can be viewed by clicking the **View Details** button. The **Job Events** tab is the primary source for job status information. Use this page to quickly view and identify jobs running during specific times and current phase information.

Impact Policies

Job Operations

Job Summary | Job Types | Job Reports | Job Events | **Impact Policies**

Impact Policies

+ Add an Impact Policy

Delete Selection

Actions

View Details, Edit, View Details, More, View Details, More, View Details, More

Name	Description	Impact Schedule
HIGH	job template: high impact at all times	Sunday 00:00 - Sunday 00:00: High
LOW	job template: low impact at all times	Sunday 00:00 - Sunday 00:00: Low
MEDIUM	job template: medium impact at all times	Sunday 00:00 - Sunday 00:00: Medium
OFF_HOURS	job template: paused at P 9-5, low impact at other times	Sunday 00:00 - Monday 09:00: Low Monday 09:00 - Monday 17:00: Paused Monday 17:00 - Tuesday 09:00: Low Tuesday 09:00 - Tuesday 17:00: Paused Tuesday 17:00 - Wednesday 09:00: Low Wednesday 09:00 - Wednesday 17:00: Paused Wednesday 17:00 - Thursday 09:00: Low Thursday 09:00 - Thursday 17:00: Paused Thursday 17:00 - Friday 09:00: Low Friday 09:00 - Friday 17:00: Paused Friday 17:00 - Sunday 00:00: Low

Custom Impact Policies are variation of OFF_HOURS policy

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Impact Policies are on their own tab. You can view the information from the **Actions** column or copy the policy from the **More** selection. Copying and modifying an impact policy modifies the schedule of when the policy is allowed to run. Creating custom schedules is the appropriate means to adjust impact policies to meet customer workflow demands. A drop-down list allows you to select from custom impact policies for bulk.

[illegible]

Run Manual Job

Start Job

Job Type Details

Name: SmartPools

Description: End of SmartPools file process. This job requires a SmartPools license.

☐ Allow Duplicate Jobs

Priority: 8

Impact Policy: LOW

Cancel
Start Job

eFS 7.2, jobs are manually started from the **Job Types** tab. Click the **More** button on a job manually. A new window is displayed providing the capability to set the job and the impact policy for the manual job.

isi job status Command

- Syntax
 - `isi job status -v`
 - `--verbose` | `-v`
- Running jobs
- Recently finished jobs
- Queued jobs
- Coordinator node number
- Job related cluster status

```
se-sandbox2-cluster-3$ isi job status -v
The job engine is running.
  Coordinator: 1
    Connected: True
  Disconnected Nodes: -
Down or Read-Only Nodes: False
  Statistics Ready: True
  Cluster Is Degraded: False
  Run Jobs When Degraded: False

No running or queued jobs.

Recent finished jobs:
ID   Type                               State      Time
-----
11   SmartPools                         Succeeded  2013-10-29T22:00:37
282  Dedupe                             Succeeded  2014-01-19T12:00:55
283  FSAnalyze                          Succeeded  2014-01-19T22:01:28
284  DedupeAssessment                   Succeeded  2014-01-20T08:16:33
285  Collect                            Succeeded  2014-01-20T08:17:43
288  FlexProtect                        Succeeded  2014-01-20T09:00:23
289  FlexProtect                        Succeeded  2014-01-20T09:23:07
287  AutoBalance                        Succeeded  2014-01-20T09:25:41
286  MediaScan                          Failed     2014-01-20T09:37:06
290  MediaScan                          Succeeded  2014-01-20T09:53:40
-----
Total: 10
```



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The **isi job status** command is used to view currently running, paused, or queued jobs, and the status of the most recent jobs. Use this command to view running and most recent jobs quickly. Failed jobs are clearly indicated with messages.

The output provides job-related cluster information, including identifying the coordinator node and if any nodes are disconnected from the cluster.

Syntax:

- **isi job status**
 - `[--verbose | -v]`

isi job statistics Command

- `isi job statistics`
- Actions
 - `list`
 - `view`

```

Sterling-375-18 isi job statistics -h
Description:
  View job engine workers and statistics.
Usage:
  isi job statistics <action>
  [--timeout <integer>]
  [--help | -h]

Actions:
  list    List job engine workers and statistics.
  view    View job engine workers and statistics.

Options:
  Display Options:
    --timeout <integer>
      Number of seconds for a command timeout.
    --help | -h
      Display help for this command.

See 'isi job statistics <subcommand> --help' for more information on a specific
subcommand.
  
```

```

se-sandbox2-cluster-39 isi job statistics list -v
Job ID: 289
Phase: 2
Nodes:

Node 1: 1
  STD: 13240
  CPU: 11.96% (0.00% min, 11.96% max, 0.31% avg)
  Memory:
    Virtual: 100.23M (100.23M min, 100.23M max, 100.23M avg)
    Physical: 11.43M (11.33M min, 11.43M max, 11.39M avg)
  I/O:
    Read: 106993 ops, 835.50M
    Write: 156813 ops, 970.06M
    Workers: 1 (0.00 SIW avg.)

Node 2: 2
  STD: 73875
  CPU: 13.96% (0.00% min, 10.97% max, 10.75% avg)
  Memory:
    Virtual: 100.23M (100.23M min, 100.23M max, 100.23M avg)
    Physical: 10.84M (10.83M min, 10.89M max, 10.84M avg)
  I/O:
    Read: 107695 ops, 841.36M
    Write: 145542 ops, 949.60M
    Workers: 1 (0.00 SIW avg.)

Node 3: 3
  STD: 2491
  CPU: 22.99% (12.79% min, 29.90% max, 19.65% avg)
  Memory:
    Virtual: 100.23M (100.23M min, 100.23M max, 100.23M avg)
    Physical: 10.72M (10.55M min, 10.72M max, 10.70M avg)
  I/O:
    Read: 122409 ops, 956.81M
    Write: 142440 ops, 1032.77M
    Workers: 1 (0.00 SIW avg.)
  
```



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The **isi job statistics** command includes the options of **list** and **view**. The **verbose** option is provided to provide detail information about the job operations. To get the most information about all current jobs, use the **isi job statistics list -v** command. To limit the information to a specific job, use the **isi job statistics view <jobID> -v** command. For troubleshooting, this provides the most granular real-time information available for running jobs.

Troubleshooting Job Issues

- Key indicators
 - Failed jobs
 - Jobs that have not completed after a long time
- Where to check in web administration interface
 - Events and summary
 - Job history and reports
- Check for changes
 - Changed from defaults
 - Priorities
 - Impact policy
 - Custom schedules



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Misconfigured jobs can affect cluster operations. The vast majority of these failures can be observed by examining how the jobs have been configured to run, and how they have actually been running and if jobs are failing. Failed jobs can also be an indicator of other cluster issues. For example, if the **MultiScan** or **Collect** jobs have many starts and restarts, this is an indicator of group changes. Group changes occur when drives or nodes leave or join the cluster.

The job events and operations summary either from the web administration interface or using the CLI are useful for immediate history, to view recent failures, but often an issue is reoccurring over time and can be more easily spotted from the job history or job reports. For example, if a high priority job is constantly pushing other jobs aside, that is easy enough to see from the Operations Summary, but a less consistent queue backup can still prevent features from properly operating. This can require much deeper dives into the job history to see what isn't running, or is running only infrequently.

A common way in which customers effect performance is in misconfigurations of the Job Engine. Changing the priority of a job and when a job is scheduled to run can interfere with the another job to run on schedule. As an example, a customer changed the priority of the **SmartPools** job to a 2 and changed the priority of the **SnapshotDelete** job to an 8 and scheduled both jobs at the same time. Almost all other jobs took priority and the **SnapshotDelete** job only would run about twice a month. The result was the customer's snapshots filled the available space on the cluster frequently and when the job did run, it usually ran during peak workflow hours and impacted the cluster performance. If a priority of a job was changed by the customer, continue to investigate why the change was made. If a customer changed a job priority, there is a good probability that it was done with some goal in mind. Look for alternative configuration options to achieve the goal.

Impact level changes have been referred to throughout this module. These directly affect the time to complete and the cluster resources utilized for job execution. One customer example was to modify the LOW impact policy to have 0.1 maximum workers or threads per storage unit. The result was no low impact job ever completed. The customer then changed all of the jobs with LOW impact policies to use the MEDIUM impact policy. When the jobs ran, cluster performance was noticeably negatively impacted. After investigation, the reason the customer made the changes was to limit impact during their peak workflow hours. To fix the issue, all settings were first restored to the system defaults. The use of a custom schedule was implemented using a modification of the OFF_HOURS policy. The customer's goal was obtained.

Lesson 3: Summary

Having completed this lesson, you can now:

- Access Job Engine
- Understand operations, jobs, and impact policies
- Edit job defaults and schedules
- Execute manual jobs
- Troubleshoot job implications for cluster features, performance, and stability

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Module 7: Job Engine



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Having completed this lesson, you can now access the Job Engine using the web administration interface and the CLI, understand different job operations and settings, and can now edit job settings and return them to their defaults. You can also manually run a job and customize the job settings. Finally, you now understand the importance and apply troubleshooting to job-related cluster issues, including cluster performance and stability.

Lesson 4: Upgrades

After completing this lesson, you will be able to:

- Distinguish between upgrade types
- Understand supported upgrade paths
- Explain commit and rollback
- Define non-disruptive operations

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Module 7: Job Engine



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Upon completion of this lesson, you will be able to distinguish between upgrade types, understand the supported upgrade paths, explain commit and rollback, and define non-disruptive operations.

Non-Disruptive Upgrades (NDU)

- Rolling versus simultaneous upgrades
- To OneFS 8.0 → simultaneous reboots
- Transparent failover for NFSv3, NFSv4*, & SMB 3.0* clients
- Improved upgrade reliability
- Improved client failover between nodes
- New upgrade control features
- Monitor upgrade progress via web administration interface

* New in OneFS 8.0



Module 7: Job Engine

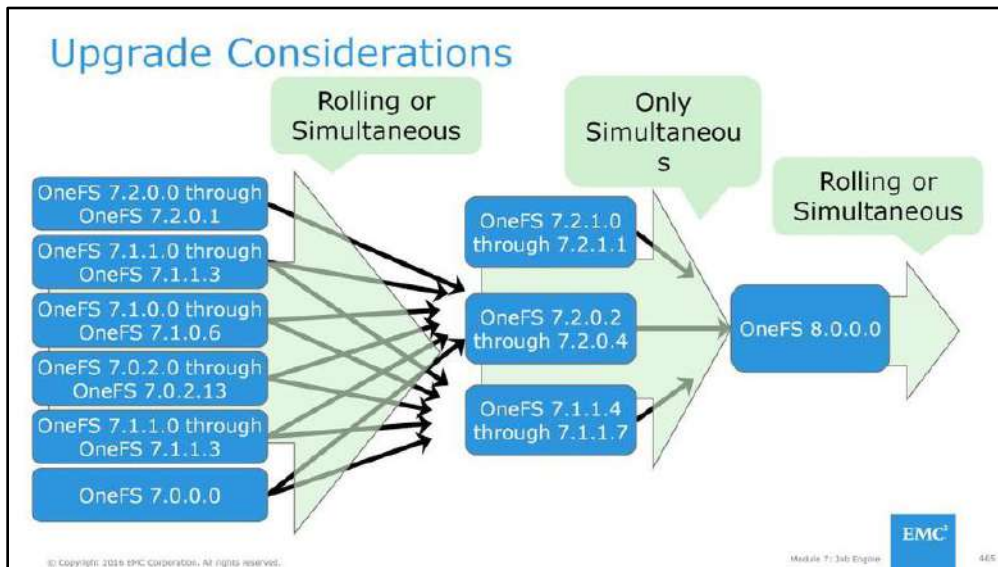
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A full operating system upgrade is done when upgrading OneFS, requiring a cluster reboot. Two types of upgrade can be done, rolling and simultaneous. A rolling upgrade is non-disruptive, upgrading and rebooting cluster nodes one at a time. Only one node is offline at a time. Nodes are upgraded and restarted sequentially. Hosts connected to a restarting node are disconnected and reconnected. Rolling upgrades are not available between all OneFS versions. A simultaneous upgrades are faster than rolling upgrades, but reboot all nodes at the same time, thus incurring an interruption in data access. Isilon has re-designed and re-built the architecture surrounding upgrades to ensure all supported upgrades can be performed in a rolling fashion.

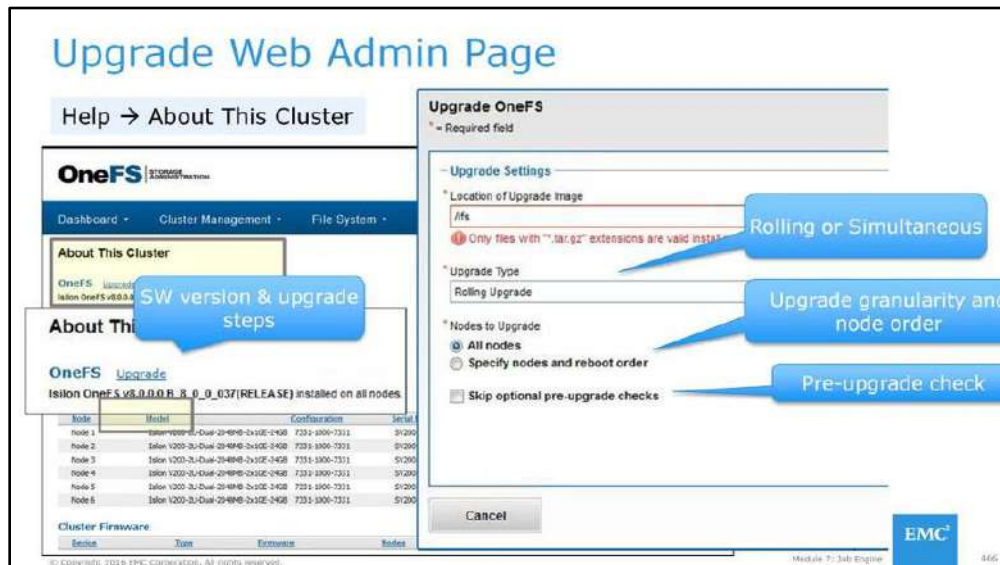
The upgrade to OneFS 8.0 requires a simultaneous reboot to implement the new upgrade infrastructure.

Rolling upgrades are non-disruptive to clients that can seamlessly failover their connections between nodes. These clients include NFSv2, NFSv3, and SMB 3.0's continuous availability shares and witness protocol features. SMB 1.0 and SMB 2.0 are stateful protocols and do not support transparent failover of their connections. Those customers will see a brief disruption when a node is rebooted into the new code.

It is important to note these NDU features are being added *in* the OneFS 8.0 release, therefore, only upgrades *from* OneFS 8.0 and beyond will have the features available.



Noted here are the supported upgrade paths. Note that all upgrades to OneFS 8.0 are only simultaneous. Using the supported upgrade paths ensures all bug fixes and enhancements are included. If the cluster's version of OneFS is not supported and an upgrade to a supported version cannot be done, EMC Isilon Technical Support should be contacted.



Shown here is the web administration page for upgrades. Navigation is via the **Help** menu. A pre-upgrade check can be run well in advance of the actual upgrade to assist in upgrade planning and address issues that may impact the upgrade before it happens. The pre-upgrade check is also run automatically as the first step of any upgrade. Selecting **Upgrade** launches the upgrade settings window. The upgrade settings allow the ability to specify the upgrade type, rolling or simultaneous, the option to select a group of nodes to upgrade, and the ability to set an upgrade order within any group of nodes. Upgrade progress can be monitor via the web administration interface and the command-line, and will list alerts on upgrade success or failure.

Commit and Rollback



- Rollback allowed until commit
- Pre-commit allows functionality validation
- Rollback to previously installed release
- 10 day rollback window
- Only new version lost after rollback



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Any good change management process includes planning for backing out changes. Rollback to the previously installed OS can be achieved with all cluster data fully intact, giving organizations the ability to halt or back out of an upgrade plan. A rollback can be done any time before the release is committed. The upgrade type will not impact the ability to rollback. Customers can remain in an upgraded, uncommitted state for 10 days, after which they will be prompted to commit to the upgrade. A rollback can be initiated through the web administration interface, or CLI, at any time and will initiate a cluster-wide reboot to return the cluster to the prior state. Any data written after the initiation of the upgrade will remain in tact with any applicable user changes during that time. However, configuration changes specific to features in the upgraded version that are not supported by the prior version will be lost upon rollback to that version.

If no issues are found, the administrator can “commit” the release. Once the commit is initiated, any post upgrade jobs that could not be rolled back safely will be initiated and the entire upgrade process will complete.

***Note:** Rollback is available only after upgrade FROM OneFS 8.0. A rollback cannot be done to a release prior.

Implementation Considerations

- Non-disruptive Operations (NDO)
- Software patch
 - Standard and Rollup patches
 - Requires restart to targeted service or processes
 - Performed as cluster-wide simultaneous update
- Firmware update installation
 - Performed as a cluster-wide rolling update
 - One single node at a time updates firmware
 - Requires reboot to initialize updated hardware
 - No rollback feature available for firmware

Note: Customers can receive new patch notifications by subscribing to the OneFS Support by Product page via support.emc.com

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Module 7: Job Engine



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The non-disruptive features enabled for rolling upgrades extend to patches and firmware updates as well. The intention is to eliminate maintenance disruptions wherever possible. This means if reboots or service restarts are required, they can be controlled, monitored, and performed in a rolling fashion to minimize any disruption. In addition, new features are enabled to support protocols, such as improving handling of connection transition from one node to the next.

All recommended patches, and any other patches that could affect the workflow, should be installed. There are two types of patches, a standard patch and a rollup patch. A standard patch addresses known issues for a major, minor, or MR release of OneFS. Some patches contain minor enhancements or additional logging functionality that can help EMC Isilon Technical Support troubleshoot issues with your cluster. Rollup patches address multiple issues related to one component of OneFS functionality, such as SMB. It might also contain fixes from previous, standard patches that addressed issues related to that component.

Similar to OneFS upgrades, firmware updates and even some patches may require services to go down across the cluster and cause outages. Due to these interruptions, it's recommended to stay current with the latest patch and firmware updates.

SyncIQ NDU Support

- SyncIQ supports rolling upgrades OneFS 8.0+
- Requires upgrade commit before new feature availability
 - New feature testing before commit, not supported
- SyncIQ jobs track version originally started as
 - Executes that version until job completes successfully
 - New features not utilized until next job started

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Module 7: Job Engine



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SyncIQ supports rolling non-disruptive upgrades, or NDU, in OneFS 8.0. New features in SyncIQ become available only after the upgrade commit process is completed. This means that new features may not be tested or used before the commit to the upgrade is completed. SyncIQ tracks the OneFS version used for every running job during the upgrade process. Any running job completes on the OneFS features available at the start of the job. New features are implemented only after successful completion of the existing job, when the job is next executed.



Shown here are source material that should be consulted before an upgrade. The OneFS upgrade process flowchart provides an end-to-end process in the form of a decision tree for a OneFS code upgrade. Release notes will provide the most current enhancements for a given release. The Upgrade Planning and Process Guide provides comprehensive planning information for upgrading to a later version of OneFS and includes instructions for assessing a cluster to ensure its readiness.

Lesson 4: Summary

Having completed this lesson, you can now:

- Distinguish between upgrade types
- Understand the supported upgrade paths
- Explain commit and rollback
- Define non-disruptive operations

Having completed this lesson, you can distinguish between upgrade types, understand the supported upgrade paths, explain commit and rollback, and define non-disruptive operations.

Module 7: Summary

Having completed this module, you can now:

- Describe cluster Job Engine
- Identify critical cluster jobs
- Establish job roles
- Understand Job Engine risks

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Module 7: Job Engine



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Having completed this module, you can now describe the Job Engine and how jobs work. You can identify the jobs that run on the cluster, describe the role of jobs as part of cluster operations and understand some of the risks surrounding the job system, and, finally, understand how to manage jobs using the web administration interface and the CLI.

Lab 7: Job Engine

- Check job status
- Modify impact policies
- Configure job priorities

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Module 7: Job Engine



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In this lab, you will modify impact policies and job priorities, and then you will analyze and troubleshoot jobs.

Module 8: Monitoring

Upon completion of this module, you will be able to:

- Understand cluster events
- Use InsightIQ monitoring tool
- Use `isi statistics` command
- Configure ESRS

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Module 8: Monitoring



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Upon completion of this module, you will be able to use the InsightIQ graphical monitoring tool and the **isi statistics** command. Additionally, you will be able to understand cluster events and configure ESRS.

Lesson 1: Cluster Event Architecture

Upon completion of this lesson, you will be able to:

- Describe purpose of cluster event system
- Explain event groups
- Configure alert channels

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Module 8: Monitoring



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Upon completion of this lesson, you will be able to describe the purpose of the cluster event system, explain event groups, and configure alert channels.

Cluster Events Log (CELOG) System

- Monitors and logs events
- Processes
 1. Receives event messages
 2. Coalesces events into event groups
 - Prevents over-notification
 - Presents events through CLI or web administration interface
 3. Sends event notifications based on rules
 - Configurable settings
 - Conditional alerting

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Module 8: Monitoring



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The Isilon's cluster events log (CELOG) monitors, logs and reports important activities and error conditions on the nodes and cluster. Different processes that monitor cluster conditions, or that have a need to log important events during the course of their operation, will communicate with the CELOG system. The CELOG system is designed to provide a single location for the logging of events. CELOG provides a single point from which notifications are generated, including sending alert emails and SNMP traps. SNMP Version 3 (SNMPv3) is supported, providing authentication, adding greater security than previous versions. Note that in OneFS 8.0, SNMP has moved to the FreeBSD SNMP software: `bsnmpd`. This is a faster, more stable solution than the `net-snmpd` that OneFS had used in previous versions. This means better scalability, and better stability.

The CELOG system receives event messages from other processes in the system. Multiple related or duplicate event occurrences are grouped, or coalesced, into one event group by the OneFS system. Combining events into groups prevents over notification and prevents spamming the user in the user interfaces and over email. You can view individual events and event groups and details through the web administration interface or the command-line interface.

The administrator can configure conditions for alert delivery, to best reflect the needs of the organization.

What are System Events?

- Report items created by CELOG
- Two types of system events
 - Task management events
 - Error events
- Provides single location for information
 - Outstanding issues
 - Current file system state
 - Reduced message clutter
 - Per node and debug level data, if needed
 - History of issues
 - Sort and filter capabilities
 - Notifications for issue

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Module 8: Monitoring

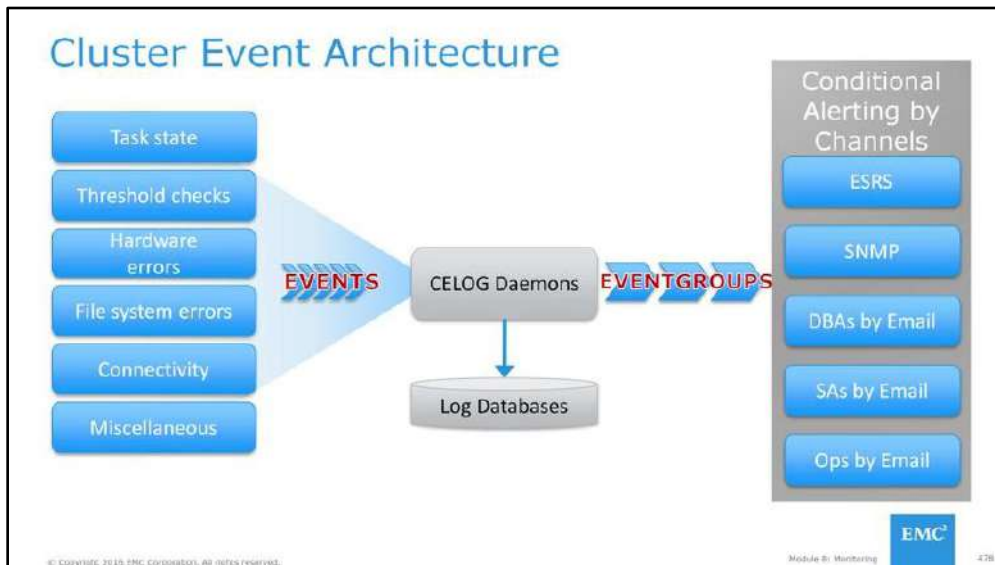


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OneFS uses events and event notifications to alert you to potential problems with cluster health and performance. Events and event notifications enable you to receive information about the health and performance of the cluster, including drives, nodes, snapshots, network traffic, and hardware.

The main goals to a system events feature is provide a mechanism for customers and support to view the status of the cluster. Events provide notifications for any ongoing issues and displays the history of an issue. This information can be sorted and filtered by date, type/module, and criticality of the event.

CELOG is designed to support the task-management systems, such as the Job Engine. The task-management systems notify CELOG of major task changes, such as starting and stopping a job. However, the task-management system does not notify CELOG of internal substates, such as what files are being worked on and what percentage of completion the job has reached. The other type of system events that are generated are a result of errors such as file system errors, threshold violations, system messages, and Simple Network Management Protocol (SNMP) traps.



An event is a notification that provides important information about the health or performance of the cluster. Some of the areas include the task state, threshold checks, hardware errors, file system errors, connectivity state and a variety of other miscellaneous states and errors.

The raw events are processed by the CELOG coalescers and are stored in log databases, and coalesced into event groups. Events themselves are not reported, but CELOG reports on event groups. Reporting on event groups is not uniform, but depends on conditions, and defined reporting channels. For example, networking issues would be reported to a channel that includes network administrators, but database administrators would probably not benefit much from the information, so their reporting channel need not be on the list for networking related issues.

What are Event Groups?

Event Groups	
Unified Reporting	One alert per group condition vs. alerts on every event for group.
Resolved as Group	Resolve all related events in one action vs. chasing down hundreds of linked events. New events go into new groups.
Ignored as Group	Ignore whole group as unit, if you have managed underlying cause.

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CELOG combines events into groups for ease of management. Very often a single underlying cause gives rise to a large number of events. For example, a single node failure can result in networking changes (routes changing, dynamically assigned IP addresses moving), storage management occurrences (pool usage thresholds being reached, protection levels changing) and other system activities (jobs failing, **FlexProtect** running) resulting in a huge number of possible alerts. In actual fact, there is one key alert that needs to cut through the noise: the fact that a node is unreachable. Everything else that results from that is a natural outflow of the fact that the node became unreachable.

CELOG does not make the related events go away. The interface lets the storage administrator dive down into each individual event and inspect it. What CELOG does by combining events into groups, is to cut down on the confusion of events precisely when confusion is most detrimental: when something serious has just happened in the storage environment.

Similarly, a whole group can be ignored at a time. For example, if an event group was created in the context of a drive failing, and the administrator has already called in the issue and arranged for a replacement drive, further alerts on that event will not add much useful to the understanding of the situation. The event group should stay open – after all, the situation is not yet resolved – but there is no real reason for alert delivery on it while the administrator waits for the replacement to arrive. It can be ignored.

Event Group Example

Cluster Management → Events and Alerts

Events
Alerts
Settings

Event Groups

Resolved event groups Shows

Ignored event groups Shows

ID	Time Noticed	Event Group Causes	Severity	Event Count	Actions
196628	2016-04-25 01:00:26	R Job state	Information	1	View Details
196627	2016-04-25 01:00:20	R Job phase end	Information	1	View Details
196626	2016-04-25 01:00:16	R Job phase begin	Information	1	View Details

View Event Group Details

Event Details

Event Group Causes:
SWR_JOBENG_JOB_STATE: Job state

Event Group ID:
196628

Severity:
Information

Alert Channels:
No value

Event Count:
1

Time Noticed:
2016-04-25 01:00:26

Marked Resolved:
Yes

Resolver Name:
70260744309910

Resolve Time:
2016-04-25 01:00:26

Ignored:
Yes

Event Details

Events

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To display the event details, on the **Events and Alerts** page, in the **Actions** column, click **View Details**.

Key information is displayed about the event:

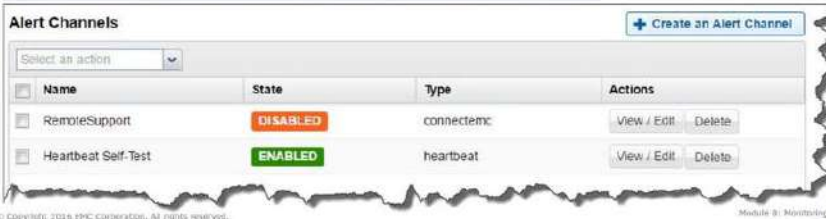
- **Event Group ID** – The unique event identifier
- **Severity** – The level of the event group's severity
- **Time Noticed** – When the cluster logged the initiating event of the group
- **Resolve Time** – When the event group was resolved, if applicable
- **Ignored** – Whether or not the event group was marked ignored

Events within the event group are displayed below the event group's summary information in case the administrator wants to inspect the details.

What are Channels?

- Communication channels for alerts
- Each channel:
 - One way for alerts to travel
 - Preconfigured or custom
 - Defines destination and connection method

Cluster Management → Events and Alerts → Alerts



Name	State	Type	Actions
RemoteSupport	DISABLED	connectemc	View / Edit / Delete
Heartbeat Self-Test	ENABLED	heartbeat	View / Edit / Delete

CELOG manages alerting on event group state changes through channels. Some channels, such as the Heartbeat Self-Test channel, are created automatically. Heartbeat events are test events that are sent every day, one event from each node in your cluster. The RemoteSupport is used for connectemc and as the channel name implies, to alert remote service. Other channels are up to the storage administrator to create and manage. This allows for very flexible alert management, by controlling who receives which alerts. A typical commercial configuration would probably include alerting channels for system administration, storage administration, network administration, ESRS as well as SNMP servers and possibly other groups such as auditing, development and database administration.

Each channel is one destination, but an alert can travel to multiple destinations. The channel is a convenient way of managing alerting configurations, such as SNMP hosts, and lists of email addresses.

Creating an Alert Channel

Cluster Management → Events and Alerts → Alerts

Create an Alert Channel
* Required field

Alert Settings
* Name:
☐ Enable this Channel
Type:
SMTP Settings
Send to:

☒ Use global SMTP server settings [View SMTP Server Settings](#)
☐ Manually configured SMTP server settings
Master Node Settings
Advanced Topics:

Excluded Nodes:

Type
SMTP
SMTP
ConnectEmc
SNMP

EMC
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An Alert Channel can be created by going to **Cluster Management > Events and Alerts > Alerts**. Here you can configure the channel for sending notifications. The types of channels are SMTP, ConnectEmc, and SNMP.

With SMTP, email messages are sent to a distribution lists. You can also specify SMTP, authorization, and security settings. ConnectEmc enables you to receive alerts from ESRS regarding cluster health. It allows support personnel to run scripts to gather data for troubleshooting the cluster. Configuring SNMP enables sending SNMP traps to one or more network monitoring stations. The management information base files (MIBs) for SNMP can be downloaded from the cluster at **/usr/local/share/snmp/mibs/**.

View Events in CLI

View System Events

- `isi event`

List and View Events

- `isi event events list`
- `isi event events view`

Ignore or Resolve Event Groups

- `isi event groups modify --ignore`
- `isi event groups modify --resolved`

View Notifications, Settings, and Test Notifications

- `isi event alerts`
- `isi event channels`
- `isi event settings`
- `isi event test`



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Through the CLI, you can list system events, view details for a specified event, ignore or resolve event groups, send test events, and view event log files. Use the **isi event** command to display and manage events through the CLI. You can access and configure OneFS events and notification rules settings using the **isi event** command. Use **isi event -h** to list available command actions and options.

isi event events list – List events either by default or using available options to refine output; including specific node, event types, severity and date ranges.

isi event events view – Displays event details associated with a specific event.

isi event groups modify --ignore – Ignores alerts from events relating to a particular event group.

isi event groups modify --resolved – Changes event group to resolved. Any events that would have joined this group now form new event groups.

isi event alerts – Used to create, delete or manage alerts.

isi event channels – Used to set up channels for sending alert notifications.

isi event settings – Used to change or view event settings.

isi event test – Sends test notifications.

Isilon Advisor Tool

- Replaces Isilon Self Service Platform
 - Same tool used by Technical and Field Support
- Features:
 - Proactively identifies issues and potential solutions
 - Runs reports on 50+ different checks
 - Takes preventative action on known issues
 - Gains visibility into the health of environments
- Tool download and additional information:
 - <https://help.psapps.emc.com/pages/viewpage.action?pageId=2853972>



The EMC Isilon Advisor (IA) is a smart, fast and effective application that enables customers to self-support common Isilon issues and accelerate time-to-resolution. This is the same application used by Technical Support Engineers and Field Representatives to resolve service requests. You can use IA to diagnose, troubleshoot and proactively avoid issues by analyzing the current health of your cluster and listing items that require attention.

Items that require attention can range from simple checks to critical alerts. They are listed in order of importance and color-coded for quick visual reference:

- **red**: critical
- **yellow**: needs attention
- **green**: no problem found

More importantly, IA provides links to documentation about how to resolve the issues identified and if further assistance by Technical or Field Support is needed, a summary of each check can be extracted to a flat file and attached to a Service Request (SR). This information will aid the Support team in resolving your issue faster.

For more information and to download the IA tool, following the link on the slide.

Lesson 1: Summary

Having completed this lesson, you are able to:

- Describe the purpose of the cluster event system
- Explain event groups
- Configure alert channels

Having completed this lesson, you can now describe the purpose of the cluster event system, explain event groups, and configure alert channels.

Lesson 2: InsightIQ Overview

Upon completion of this lesson, you will be able to:

- Understand InsightIQ architecture
- Plan InsightIQ installation
- Install InsightIQ

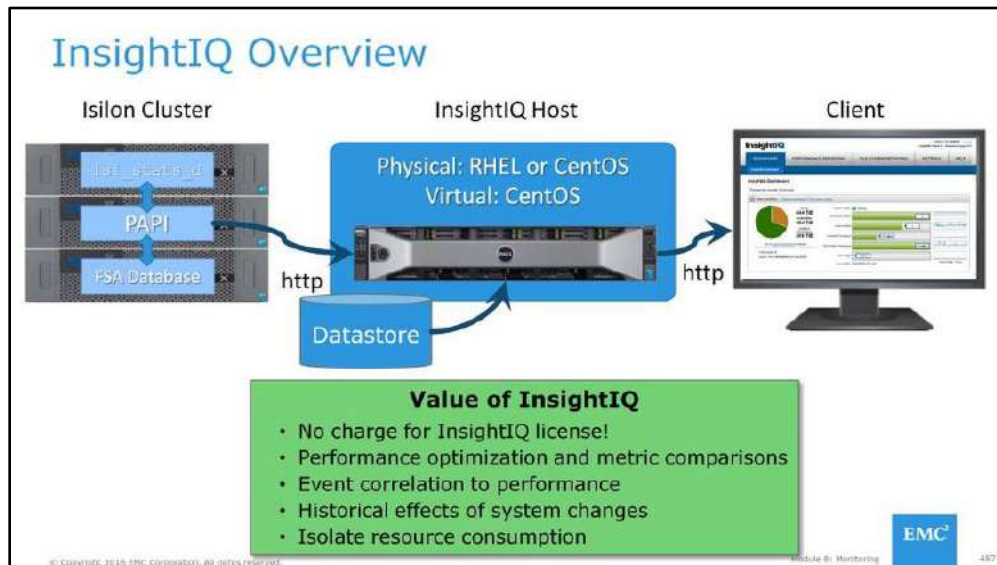
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Upon completion of this lesson, you will be able to install InsightIQ and understand the InsightIQ environment.



Whereas the CELOG monitors, logs and reports important activities and error conditions on the nodes and cluster, InsightIQ focuses on the Isilon data and performance. InsightIQ is available for no charge and provides advanced analytics to optimize applications, correlate workflow and network events. It provides tools to monitor and analyze a cluster's performance and file systems. Cluster monitoring includes performance, capacity, activity, trending, and analysis. InsightIQ runs on separate hardware from the clusters it monitors and provides a graphical output for easy trend observation and analysis. It does not take cluster resources beyond the data collection process.

InsightIQ has a straightforward layout of independent components. Inside the Isilon cluster, monitoring information is generated and statistical data collected by **isi_stat_d**, and presented through **isi_api_d**, which handles PAPI calls, over HTTP. The InsightIQ datastore can be local to the host or external via an NFS mount from the Isilon cluster, or any NFS-mounted server. The datastore must have at least 70GB of free disk space. File System Analytics (FSA) data is kept in a database on the cluster. InsightIQ accesses the cluster through PAPI rather than as an NFS mount. Previous releases stored FSA data externally, which was inefficient for a number of reasons.

InsightIQ is accessed through any modern web browser, such as Microsoft Edge, Internet Explorer, Mozilla Firefox, Apple Safari, and Google Chrome. If InsightIQ is to be loaded on a Red Hat or CentOS Linux system, EMC provides it in the form of an rpm package.

Some of the value InsightIQ offers is its ability to:

- Determine whether a storage cluster is performing optimally
- Compare changes in performance across multiple metrics, such as CPU usage, network traffic, protocol operations, and client activity
- Correlate critical storage cluster events with performance changes
- Determine the effect of workflows, software, and systems on storage cluster performance over time
- View and compare properties of the data on the file system
- Pinpoint users who are using the most system resources and identify their activity

Customizable Reports

- Isolate problems
 - Performing optimally
- Measure change effects
 - Correlate past vs. present performance
- Analysis
 - Performance over time
- Trending
 - Forecast capacity needs



InsightIQ's reporting allows monitoring and analysis of cluster activity in the InsightIQ web-based application. Reports are customizable, and can provide cluster hardware, software, and protocol operations information. InsightIQ data can highlight performance outliers, helping to diagnose bottlenecks and optimize workflows. Use cases include:

- **Problem isolation:** Report to isolate the cause of performance or efficiency related issues
- **Measurable effects of configuration changes:** Report comparing past performance to present performance
- **Application optimization:** Report to identify performance bottlenecks or inefficiencies
- **Analyze real-time and historical data:** Report on cluster information such as individual component performance
- **Forecasting:** Report on the past cluster capacity consumption to forecast future needs

- Provides detailed information
 - Files
 - Directories
 - Stored in `/ifs/.ifsvar/modules/fsa`
- Information available in InsightIQ
- Runs through Job Engine
 - Optional process



The OneFS Job Engine runs the **FSAalyze** job daily, which then collects all the information across the cluster such as the number of files per location or path, the file sizes, and the directory activity tracking. InsightIQ collects the FSA data from the cluster for display to the storage administrator.

FSA Enhancement

- Uses changelist API
 - Tracking list based on snapshot comparisons
 - SnapshotIQ license not required
- FSA database managed by cluster
- Uses at least one thread per node
- Reporting database files split out
 - Parallel database merge
 - Removes point of contention for processes

OneFS 8.0 and
InsightIQ 4.0
Enhancements



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Prior to OneFS 8.0, when the **FSA** job ran, it performed a complete LIN scan of the file system and the **FSA** job could take a long time to complete. Using the changelist API, snapshots are taken for tracking changes to the file system on the entire **/ifs** tree. A comparison between the last snapshot and the new snapshot creates a tracking list that you can examine for changes. This results in a much quicker completion for the **FSA** job than previous releases. The snapshots are system snapshots and no SnapshotIQ license is required.

The FSA data and the database remains on the cluster and is managed by the cluster, making it more efficient than in versions before InsightIQ 4.0 where the database was hosted to InsightIQ via an NFS mount. InsightIQ merely queries the database directly through PAPI.

In addition, the **FSA** job is now multi-threaded, with at least one thread or process per node is used to generate and update the results. The results are reported per node in parallel, and the final results are combined to produce the results. Any point of database access contention is removed by the new design.

Considerations

- **Compatibility**
 - **OneFS 8.0 FSA Statistics:** Only compatible with InsightIQ 4.0 and later
 - **InsightIQ 2.5 – 4.0:** Compatible with OneFS 7.1 through 8.0
 - **InsightIQ 3.2.0 and above:** Upgrade direct to InsightIQ 4.0
 - **InsightIQ pre-3.2.0:** First upgrade to InsightIQ 3.2.0
- **IsilonSD Edge support**
- **Default access via HTTPS port 443 or HTTP port 80**
- **Snapshots of InsightIQ datastore not recommended**
- **Single instance of InsightIQ recommended maximum is monitoring 8 clusters or 150 nodes**

Ref: https://support.emc.com/docu44518_Isilon-Supportability-and-Compatibility-Guide.pdf?language=en_US

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OneFS 8.0.0 FSA statistics are compatible only with InsightIQ 4.0 and later releases. InsightIQ 4.0 is capable of dealing with all versions of OneFS from 7.0 forward. This includes both Isilon SD Edge and OneFS 8.0. InsightIQ 4.0 can differentiate between the FSA database handling of OneFS 8.0 and earlier versions of OneFS, and will handle each one correctly. InsightIQ can be directly upgraded from any earlier version at least as recent as InsightIQ 3.2. Versions before that will need an interim upgrade step. Reference the Isilon Supportability and Compatibility Guide on support.emc.com for a comprehensive list of upgrade paths and version support.

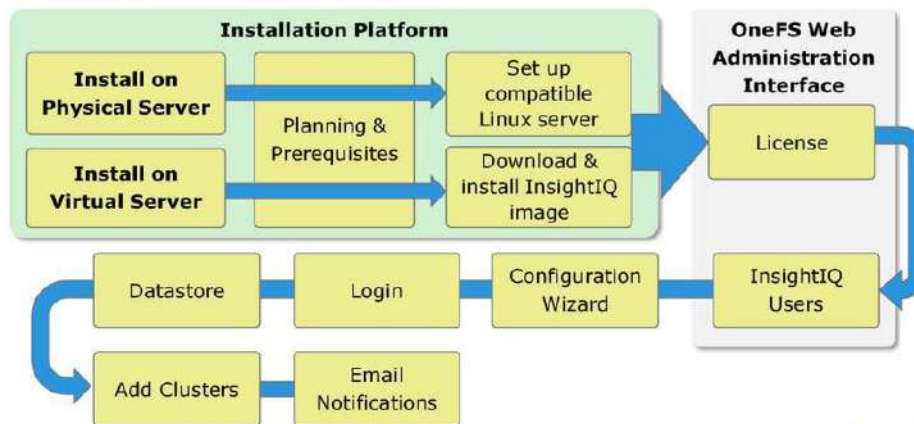
The community support network is available for the free version of IsilonSD Edge.

By default, web browsers connect to InsightIQ over HTTPS or HTTP via port 443 for HTTPS and port 80 for HTTP.

A revert to a snapshot or modifications of the InsightIQ datastore can cause the datastore corruption. Snapshots should not be used for the datastore.

The maximum number of clusters that you can simultaneously monitor is based on the system resources available to the Linux computer or virtual machine. It is recommended that you monitor no more than 8 storage clusters or 150 nodes with a single instance of InsightIQ.

Installation Overview



https://support.emc.com/docu65874_InsightIQ-4.0-Installation-Guide.pdf?language=en_US

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This page provides an overview of the InsightIQ installation steps. Details for each step are shown on the slides that follow. For comprehensive system requirements, see the InsightIQ Installation Guide on **support.emc.com**.

Prerequisites

If Virtual Server:

- VMware ESXi version 4.x, 5.x
- VMware Workstation
- VMware Fusion (Mac OS)
- 8 GB RAM

70 GB free disk space
(local or NFS datastore)

Datastore

InsightIQ Host

If Physical Server:

- CentOS 6.7 or higher,
or RHEL 6.7 or higher
- x86_64
- 8 GB RAM
- 5 GB disk space on OS
partition

Sizing Formula for Datastore

$\text{<minimum_datastore_size> = <number_of_nodes> * 1 + 10}$

$\text{<larger_datastore_size> = <number_of_nodes> * <number_of_months> * 2 + 10}$



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Shown here are the virtual and physical requirements to install InsightIQ. Note that a license is required. Consideration needs to be given to the NFS datastore size. On average, InsightIQ creates 1 GB of data per monitored node every 2 weeks. To retain more than 2 weeks of data, the size of the InsightIQ datastore should be increased by 2 GB per node per month. Also, it's recommended that the disk space includes at least 10 GB of free space. If the datastore has less than 3 GB of free space available, InsightIQ begins to delete older data to create room for new data. If InsightIQ is unable to free at least 5 GB of disk space, monitoring stops until more available free disk space is available. There are two basic formulas for sizing the datastore. The first is the capacity in GBs needed for the default retention. The second is if needing more than two months of data. For example, if monitoring 12 nodes, it is recommended reserving at least 22 GBs of disk space. Retaining 3 months of data, 82 GB of disk space is recommended.

InsightIQ Installation Video

- Virtual server
- Demonstration steps:
 - Installing/verifying license
 - Enabling InsightIQ user
 - Stepping through configurations wizard
 - Logging in to InsightIQ web administration interface
 - Adding a cluster



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This video steps through an InsightIQ installation on a virtual server. Demonstrated is installing/verifying the license, enabling the InsightIQ user, stepping through the configurations wizard, logging in to the InsightIQ web administration interface, and adding a cluster. Click the “clip” icon to launch the video.

Installation & Configuration Wizard

Install on Physical Server

1. Get installation script in *.sh format
2. Log in using account with **sudo** access or **root** user
3. Run `sudo sh <path>` (path of *.sh installation script)
4. Create user account:
`sudo useradd <username>`
`sudo passwd <username>`
5. Enable access to HTTP port 80 or HTTPS port 443

Install on Virtual Server

1. Get *.ova and extract files
2. Add InsightIQ to VM inventory
3. Open InsightIQ console
4. Create administrator password
5. Configure InsightIQ IP address
6. Set time zones



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On the slide are the high-level steps to install InsightIQ on a physical and virtual server. The installation files are located at support.emc.com. On a physical system, log on with either a **sudo** user or **root** user account. Run **sudo sh <path>** to install InsightIQ, where **<path>** is the file path of the *.sh installation script. An optional user account can be created to access the InsightIQ web application. Note that **root** user cannot log in to the InsightIQ web application access to HTTP port 80 or HTTPS port 443 needs to be enabled.

For a virtual machine install, extract the *.ova and add InsightIQ to the virtual machine inventory. In the InsightIQ VM console, create a password for the administrator account. Configure an IP address either DHCP or static. The last step of the configuration wizard is to set the time zone.

Datastore for InsightIQ

SETTINGS → Datastore

- Included with VM *.ova
- Local or NFS mounted to physical server
- InsightIQ data stored in datastore

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If InsightIQ is installed on a VM, the datastore will be a virtual hard drive configured in the image. For a physical system, the datastore can be local or NFS mounted. The NFS datastore can be either an Isilon cluster or another NFS-mounted server. Shown is the **SETTINGS** page, **Datastore** submenu where the configuration is done.

Lesson 2: Summary

Having completed this lesson, you are able to:

- Understand InsightIQ architecture
- Plan InsightIQ installation
- Install InsightIQ

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Having completed this lesson, you can now install InsightIQ and understand the InsightIQ environment.

Lesson 3: Using InsightIQ

Upon completion of this lesson, you will be able to:

- Use InsightIQ interface
- Configure InsightIQ
- Troubleshoot InsightIQ

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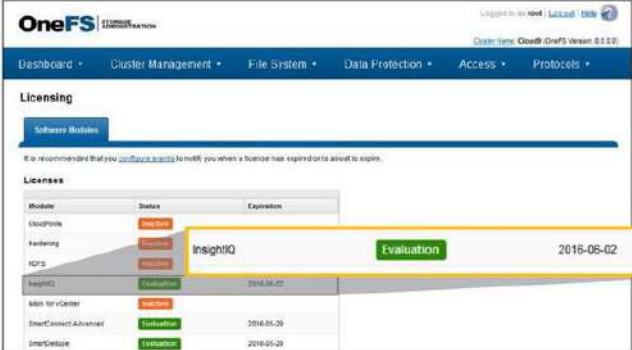


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Upon completion of this lesson, you will be able to use, configure, and troubleshoot InsightIQ.

InsightIQ License

Cluster Management → Licensing



The screenshot shows the OneFS web administration interface. The top navigation bar includes links for Dashboard, Cluster Management, File System, Data Protection, Access, and Protocols. The 'Cluster Management' section is expanded, and the 'Licensing' page is selected. A table lists various software modules and their license status. The 'InsightIQ' module is highlighted with a yellow box, showing a status of 'Evaluation' and an expiration date of '2016-05-02'.

Module	Status	Expiration
CloudPools	Enabled	
Hardening	Enabled	
NDFS	Enabled	
InsightIQ	Evaluation	2016-05-02
Admin for vCenter	Enabled	
SmartConnect Advanced	Evaluation	2016-05-29
SmartSense	Evaluation	2016-05-29

In the OneFS web administration interface, go to **Cluster Management > Licenses**. Verify that a valid InsightIQ license is enabled on the monitored cluster and that the local InsightIQ user is enabled and configured with a password on the monitored cluster.

Local InsightIQ User

Access → Membership & Roles → Users

The screenshot shows the OneFS Management Console interface. The top navigation bar includes 'Dashboard', 'Cluster Management', 'File System', 'Data Protection', 'Access', and 'Protocols'. The 'Access' menu is expanded, showing 'Membership & Roles' and 'Users'. The 'Users' page is active, displaying a table of users. The 'Current Access Zone' is set to 'System'. The 'Providers' dropdown is set to 'File: System'. The 'insightiq' user is highlighted in a yellow box, showing it is enabled.

Short Name	Full Name	Primary Group	Account Status	Actions
root	Root	admin	Enabled	View / Edit
admin	Admin User	admin	Enabled	View / Edit
compliance	Admin InsightIQ Compliance Administrator	admin	Enabled	View / Edit
insightiq	InsightIQ User	insightiq	Enabled	View / Edit
remote-support	Remote Support User	admin	Enabled	View / Edit

Next, verify that a local InsightIQ user is created and active by going to **Access > Membership & Roles > Users**. Ensure the **Current Access Zone** is **System**. From the **Providers** drop-down list, select **File: System**. There should be a user named **insightiq**. If not enabled, select **View/Edit** and then assign a password and check the enable user checkbox and save.



In a supported web browser, connect to the InsightIQ application at `http://<IPAddressOrHostName>`, where `<IPAddressOrHostName>` is the IP address or the host name of the InsightIQ appliance. The InsightIQ application login page displays.

In the **Username** box, type a valid user name that has been configured for this instance of the InsightIQ application. The user name for the administrator account is **administrator**. The user names for read-only accounts are configured by the administrator. In the **Password** box, type the password that is associated with the user name that you entered in the **Username** box.



Shown here is the **Dashboard** page you see after logging in. There are five tabs to view data and configure settings. The **DASHBOARD** provides an aggregated cluster overview and a cluster-by-cluster overview. This graphic shows that InsightIQ is already configured and monitoring clusters. The view can be modified to represent any period of time for which InsightIQ has collected data. Also, breakouts and filters can be applied to the data.

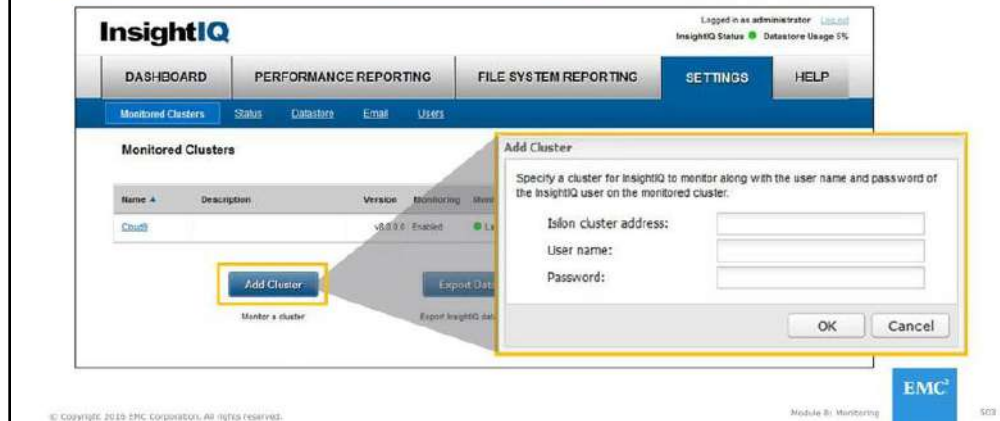
In the **Aggregated Cluster Overview** section, you can view the status of all monitored clusters as a whole. There is a list of all the clusters and nodes that are monitored. **Total capacity, data usage, and remaining capacity** are shown. Overall **health** of the clusters is displayed. There are graphical and numeral indicators for **Connected Clients, Active Clients, Network Throughput, File System Throughput, and Average CPU Usage**. There is also a **Cluster-by-Cluster Overview** section that can be expanded.

Depending on the chart type, the data can be broken out and viewed by pre-set filters. For example, In/Out displays data by inbound traffic versus outbound traffic. You can also view data by file access protocol, individual node, disk, network interface, and individual file or directory name. If the data is displayed by client only, the most active clients are represented in the displayed data. Displaying data by event can include an individual file system event, such as read, write, or lookup. Filtering by Operation Class displays data by the type of operation being performed.

If File System Analytics is enabled, data can be viewed by when a file was last accessed, by when a file was last modified, by the size of files in each disk pool, and by file extension. You can also view data by a user-defined attribute. To do this you must first define the attributes through the command-line interface. If you want to view data by logical file size or physical data size, note that logical file size calculations include only data and do not include data-protection overhead, while physical file size calculations include data-protection overhead.

Add Cluster in InsightIQ

SETTINGS → Monitored Clusters → Add Cluster



Adding clusters to monitor is done with the InsightIQ web interface. Go to **Settings > Monitored Clusters**, and then on the **Monitored Clusters** page, click **Add Cluster**. In the **Add Cluster** dialog box, type the name of an Isilon SmartConnect zone for the cluster to be monitored. In the **Username** box, type **insightiq**. In the **Password** box, type the local InsightIQ user's password exactly as it is configured on the monitored cluster, and then click **OK**. InsightIQ begins monitoring the cluster.

InsightIQ Email Notification Settings

SETTINGS → Email

InsightIQ

Logged in as administrator | Logout
InsightIQ Status: ● Defective Stage: 5%

DASHBOARD PERFORMANCE REPORTING FILE SYSTEM REPORTING **SETTINGS** HELP

Business Overview Status Alerts **Email** Logs

Configure Email Settings (SMTP)

InsightIQ uses email to deliver configuration, status, and alert notifications. Configure the email settings below to enable email notifications.

SMTP server:		Enter host name or IP address.
SMTP port:		Enter port used to connect to SMTP server (if left blank then the default is 25, or 587 if using TLS).
Username:		Specify the username to authenticate to the SMTP server (if required).
Password:		Specify the password to authenticate to the SMTP server (if required).
From Email:		By default, emails will be sent from ns_reply@isilon-insightiq. If your server requires a valid email address in the from field, specify it here.
TLS Connection:	☐	Check this box to specify that TLS (SSL) is required to connect to this SMTP server.

Send a test email:

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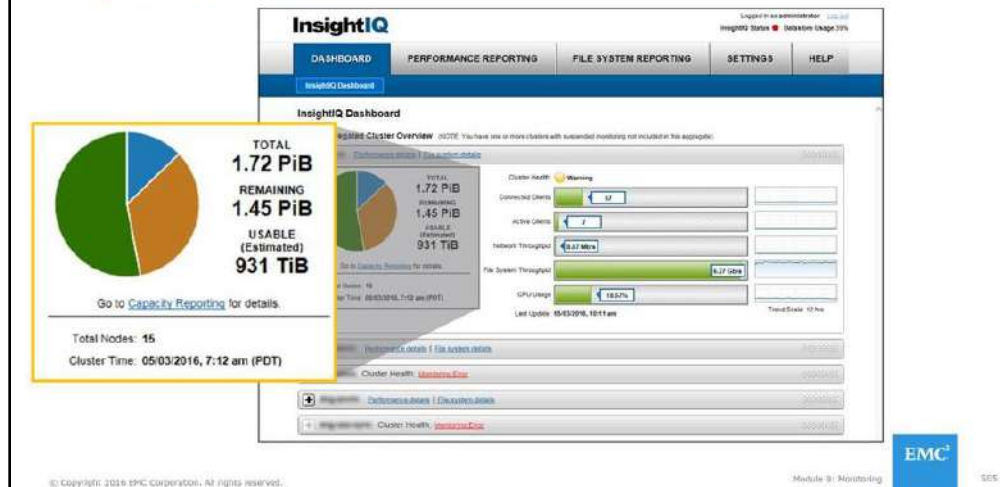
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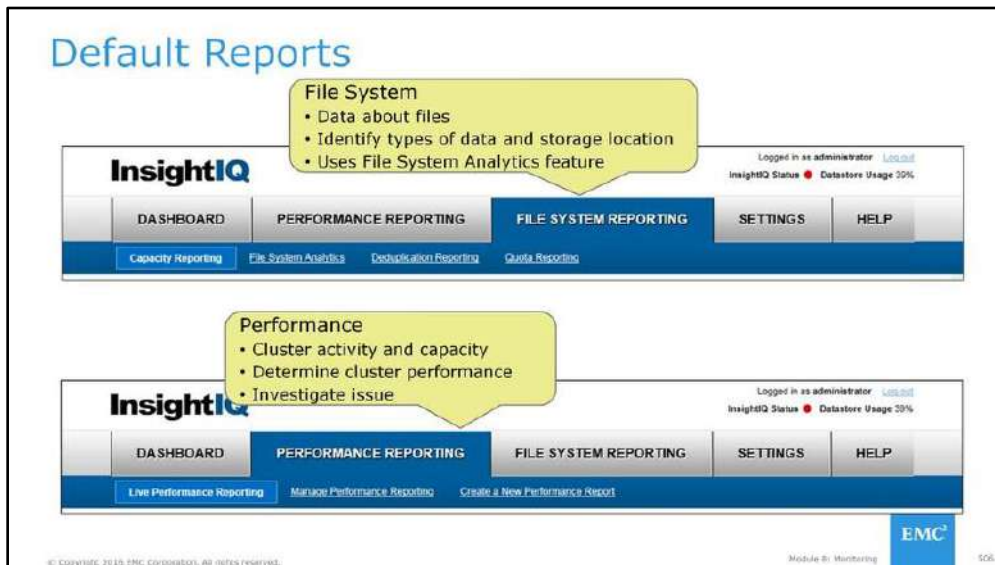
504

A protocol for sending email messages over networks is called Simple Mail Transport Protocol (SMTP), which Isilon clusters support. If the customer wants to email scheduled PDF reports, you must enable and configure InsightIQ to send outbound email through a specified email server. Click **Settings > Email**. The **Configure Email Settings (SMTP)** page appears. In the **SMTP server** box, type the host name or IP address of an SMTP server that handles email for the customer's organization. In the **SMTP port** box, type the port number used to connect to the SMTP server that you specified. If the SMTP server requires a username and password for authentication, specify a username and password. In the **Username** box, type the name of a valid user on the server. In the **Password** box, type the password of the user you specified. If the SMTP server you specified accepts email only from valid email addresses, type a valid email address in the **From Email** box. The address that you type will appear in the **From** field of email messages sent by InsightIQ. If either the Transport Layer Security, or TLS, or the Secure Sockets Layer, or SSL, protocol is required to connect to the SMTP server that you specified, select the **TLS Connection** box, and then click **Submit**.

Capacity Analysis



The InsightIQ dashboard includes a capacity analysis pie chart. The estimate of usable capacity is based on the existing ratio of user data to overhead. This does mean that there is an assumption that data usage factors will remain fairly constant over additional use. If a customer has been using the Isilon cluster for many small files and then wants to add some large files, or vice versa, the result will not be precisely what is predicted by the system.



You can monitor clusters through customizable reports that display detailed data about clusters over specific periods of time. InsightIQ enables you to view two general types of reports: performance reports and file system reports.

Performance reports have information about cluster activity and capacity. Performance reports can be useful if, for example, you want to determine whether clusters are performing as expected or you want to investigate the specific cause of a performance issue. File system reports include data about the files that are stored on a cluster and can be useful if, for example, you want to identify the types of data being stored and where on a cluster that data is stored. Before you can apply a file system report to a cluster, you must enable the InsightIQ File System Analytics feature for that cluster.

InsightIQ supports live versions of reports that are available through the InsightIQ web application. You can create live versions of both performance and file system reports and can modify certain attributes as you are viewing the reports, including the time period, breakouts, and filters.

Enable File System Analytics

SETTINGS → Monitored Clusters → Configure

InsightIQ

Logged in as administrator | Logout
InsightIQ Status: ● Datastore Usage: 5%

DASHBOARD PERFORMANCE REPORTING FILE SYSTEM REPORTING **SETTINGS** HELP

Monitored Clusters Status Databases Crawl Alerts

Monitored Clusters

Name	Description	Version	Monitoring	Monitoring Status	Data Size	Actions
Ceph		v8.1.0.0	Enabled	● Latest data: 05/13/2016 8:36 am	53.7 MB	Configure Delete Suspend

Configuration for Cloudf

Cluster Credentials **Enable FSA** FSA Configuration Manage FSA Result Sets Data Set Configuration

Job Control

Filesystem analytics runs as a periodic job on the monitored cluster. The File System Analytics job is scheduled to run on the monitored cluster every day at 22:00 monitored through the command-line interface or the [job administration interface](#).

Warning: creating the FSA job will consume computing resources on the monitored cluster and this may have an impact on cluster performance.

Generate FSA reports on the monitored cluster ☒

View FSA reports in InsightIQ: ☒

Submit

EMC

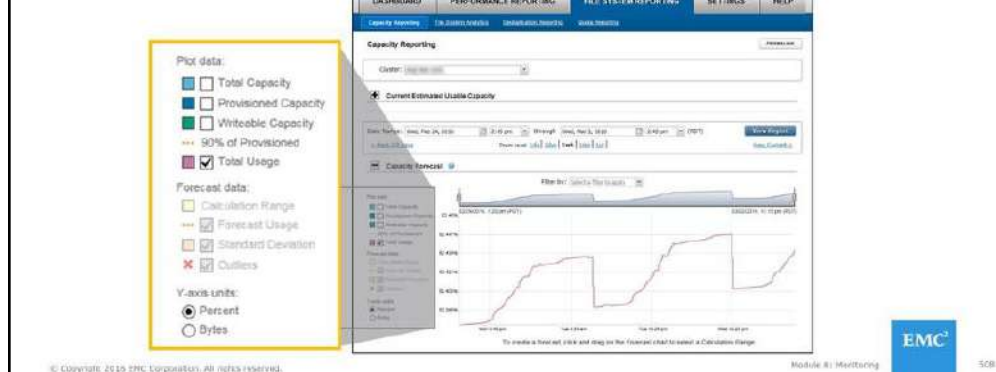
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Before you can view and analyze data usage and properties through InsightIQ, you must enable the File System Analytics feature.

In InsightIQ, click **Settings > Monitored Clusters**. The **Monitored Clusters** page appears. In the **Actions** column for the cluster for which you want to enable or disable File System Analytics, click **Configure**. The **Configuration** page displays. Click the **Enable FSA** tab. The **Enable FSA** tab displays. To enable the **FSA** job, select the **Generate FSA reports on the monitored cluster** box. To enable InsightIQ for File System Analytics reports, select the **View FSA reports in InsightIQ** box.

Capacity Reporting and Forecasting

FILE SYSTEM REPORTING → Capacity Reporting

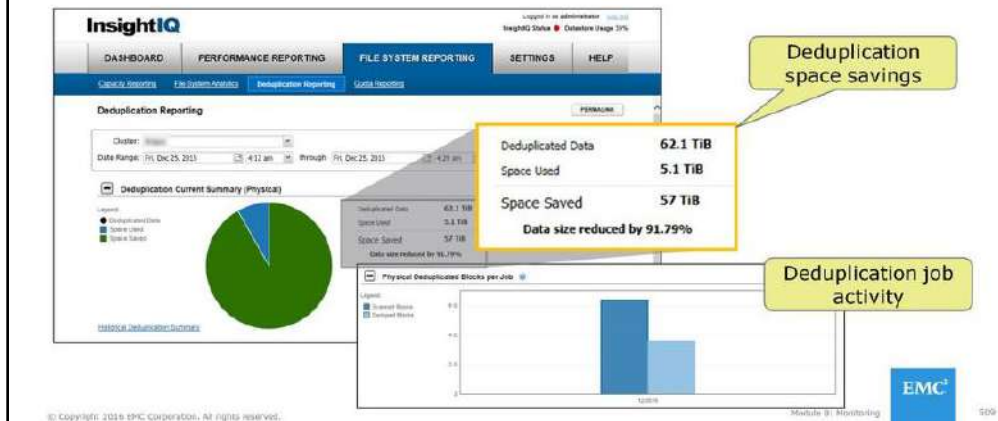


Let's take a look at the file system reports, starting with capacity reporting. The administrator can drill-down to file system reporting to get a capacity reporting interface that displays more detail about usage, overhead and anticipated capacity. InsightIQ 4.0 introduces capacity forecasting. The administrator can select a period of InsightIQ's information on a cluster and use that typical usage profile to estimate when the cluster will be 90% full. This is useful for planning upgrades well ahead of time, so that delays around procurement and order fulfilment do not cause unnecessary difficulties. Capacity Forecast, shown here, displays the amount data that can be added to the cluster before the cluster reaches capacity.

The Plot data metrics show the total amount of storage capacity, the storage capacity of nodes provisioned to node pools, the amount of storage capacity that user data can write to, and the total amount of user data and the associated protection overhead stored on the cluster. Forecast data shows the breakout of information shown in the Forecast chart. This data includes a calculation range highlighting the range of data used to calculate the forecast, the projected forecast total usage over time, the standard deviation of the forecast usage calculation, and outliers that fall outside of the range of the bulk of calculated. Depending on the frequency and amount of variation, outliers can have a major impact on the accuracy of the forecast usage data.

Deduplication Reporting

FILE SYSTEM REPORTING → Deduplication Reporting



The deduplication interface in InsightIQ displays several key metrics. The administrator can clearly see how much space has been saved, in terms of deduplicated data as well as data in general. The run of deduplication jobs is also displayed so that the administrator can correlate cluster activity with deduplication successes.

Quota Reporting

FILE SYSTEM REPORTING → Quota Reporting

Granular reporting period

Configured quota limits and usage

Path #	Type	% Used	Range	Hard Limit	Soft Limit	Advisory Limit	Snapshots Included	Overhead Included
/insightiq	user (101 total)	0.11%	13.4 MB	1 GB	1 GB	1 GB	100	100
/insightiq	default user	0%	1 GB	1 GB	1 GB	1 GB	100	100
/insightiq/backup	user (101 total)	0%	11 GB	200 GB	200 GB	200 GB	100	100
/insightiq/restore	user (101 total)	0%	11 GB	200 GB	200 GB	200 GB	100	100
/insightiq/monitoring	user (101 total)	11.11%	122 MB	200 GB	200 GB	200 GB	100	100
/insightiq/agent	user (101 total)	0.10%	109 MB	200 GB	200 GB	200 GB	100	100
/insightiq/management	user (101 total)	45.05%	462 GB	1 TB	1 TB	1 TB	100	100
/insightiq	user (101 total)	0%	13.4 MB	200 GB	200 GB	200 GB	100	100

The interface for quota monitoring displays which quotas have been defined on the cluster, as well as actual usage rates. The storage administrator can use this as a trending tool to discover where quotas are turning into limiting factors before it happens without necessarily scripting a lot of analysis on the front-end. If SmartQuotas has not been licensed on the cluster, InsightIQ will report this fact.

Create New Performance Report

PERFORMANCE REPORTING → Create a New Performance Report

Create a New Performance Report

Create your new performance report, and optionally schedule it for automatic generation and email delivery. After you save it, you can edit the report.

Build Your New Performance Report

Performance Report Name:

15 character maximum length

Select Your Reporting Type(s): ☐ Live Performance Reporting

☐ Scheduled Performance Report

Specify the performance modules you want to see, then use this report whenever you see cluster over any time range.

Add a schedule and specify up to 10 report recipients, and messages will automatically get sent by email according to your specifications.

Select the Data You Want to See

PERFORMANCE MODULES

Select a Module for this Position:

Performance Modules:

- Active Clients
- Average Cached Data Age
- Average Disk Hardware Latency
- Average Disk Operation Size
- Average Pending Disk Operations C...
- Blocking File System Events Rate
- CPU % Use
- Client Summary
- Cluster Capacity
- Connected Client
- Contended File System Events Rate
- Deadlocked File System Events Rate
- Deduplication Summary (Logical)
- Deduplication Summary (Physical)
- Disk Activity

Cancel Save Cancel

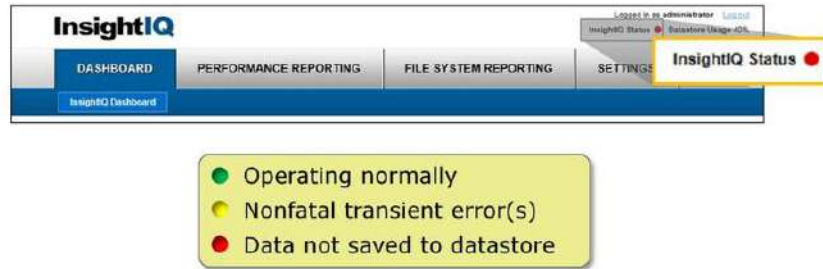
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You can create custom live performance reports by clicking **Performance Reporting > Create a New Performance Report**. On the **Create a New Performance Report** page, specify a template to use for the new report. There are three types of reports: Create a live performance report from a template that is based on the default settings as shown; create a live performance report based on a saved performance report; or select one of the live performance reports based on one of the template reports.

In the **Create a New Performance Report** area, in the **Performance Report Name** box, type a name for the live performance report. Select the **Live Performance Reporting** checkbox. In the **Select the Data You Want to See** area, specify the performance modules that you want to view in the report. There are two options: You can add a new performance module or modify an existing one. Repeat this step for each performance module that you want to include. Save the report.

Status Indicators



The first indication of a problem may be seen by the change of the status indicator in the upper-right corner of the InsightIQ web administration interface.

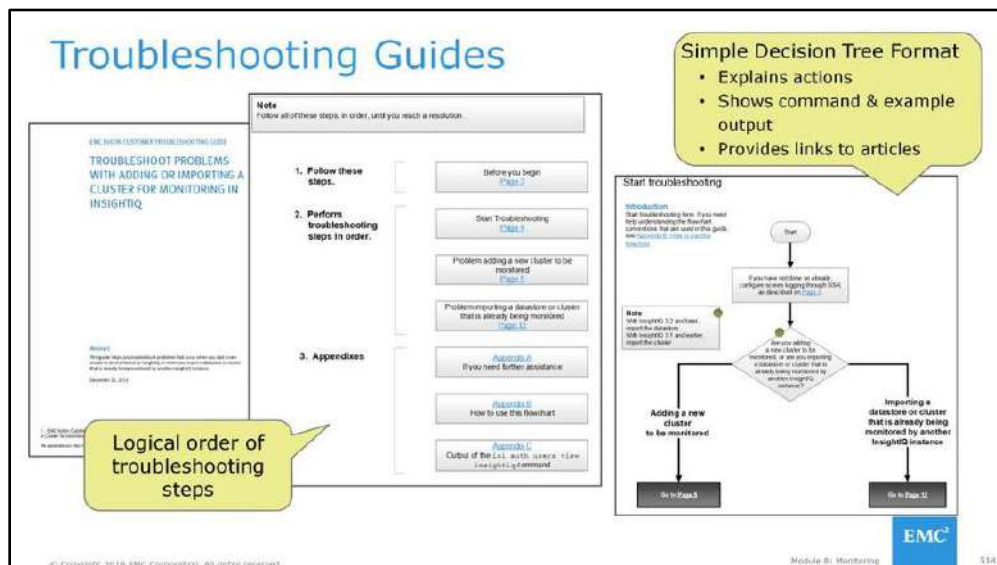
- **Green** indicates InsightIQ is operating normally.
- **Yellow** is an indication at least one transient, nonfatal error has occurred.
- **Red** shows if InsightIQ could not save data to the datastore, such as with instances where the datastore is full. Red may also indicate InsightIQ could not contact the InsightIQ virtual machine. When the status is red, InsightIQ does not collect additional data until the issue is resolved.

Troubleshooting

SETTINGS → Status

The screenshot shows the InsightIQ web interface. The top navigation bar includes 'Dashboard', 'Performance Reporting', 'File System Reporting', 'Settings', and 'Help'. The 'Settings' tab is active, and the 'Status' sub-tab is selected. The 'InsightIQ Status' section shows an 'Overall Status' of 'Error'. Below this, there are configuration options for email alerts. A yellow callout box highlights an error message: 'Monitoring halted: UnicodeDecodeError'. The error details state: 'utf8 codec can't decode byte 0x00 in position 113790: invalid start byte. InsightIQ cannot resolve this error automatically. You must fix the error and restart InsightIQ to restart monitoring. Error occurred on Mar 02 2016, 07:40 PM (UTC)'. A yellow speech bubble next to the callout box contains the text: 'Error identified, explained, and probable fix given'.

By selecting the **InsightIQ Status** link in the upper right of the web administrator interface or by navigating to **SETTINGS > Status** the errors are listed. Some configuration can be done on the page for how email notifications are handled. The graphic highlights an error to one of the monitored clusters. The status page shows which cluster has an error, explains the error, lists a course action for the administrator to take, and logs the time of the error.



To begin, **support.emc.com** has many customer troubleshooting guides as well as troubleshooting guides designed for EMC personnel and SE Partners. The guides are a good starting point for isolating the issue. Shown are excerpts from a troubleshooting guide. Note the logical approach to troubleshooting steps and the simple to follow flow diagram.

The customer troubleshooting guides on community.emc.com can be accessed from the InsightIQ - Isilon Info Hub: **<https://community.emc.com/docs/DOC-42096>**

Lesson 3 Summary

Having completed this lesson, you are able to:

- Use InsightIQ interface
- Configure InsightIQ
- Troubleshoot InsightIQ

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Having completed this lesson, you are able to use, configure and troubleshoot InsightIQ.

Lesson 4: Statistics from Command-line

Upon completion of this lesson, you will be able to:

- Use `isi statistics`
- Understand `isi statistics` options
- Manipulate `isi statistics` output

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Upon completion of this lesson, you will be able to use the **isi statistics** command, understand **isi statistics** options, and manipulate the **isi statistics** output via the CLI.

Cluster Monitoring Commands

isi status	isi devices	isi statistics
General status	Drive status	Protocol statistics
General performance	Hardware condition	Drive statistics
Critical alerts	Node management	Hardware statistics
Job Engine status	Drive replacement	Node statistics



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Three main commands that enable you to view the cluster from the command-line are **isi status**, **isi devices**, and **isi statistics**.

The **isi status** command displays information on the current status of the cluster, alerts, and jobs. To view information on the cluster, critical events, cluster job status, and the basic identification, statistics, and usage, run **isi status** at the CLI prompt.

The **isi devices** command displays information about devices in the cluster and changes their status. There are multiple actions available including adding drives and nodes to your cluster.

The **isi statistics** command has approximately 1,500 combinations of data you can display as statistical output of cluster operations. We will take a closer look at **isi statistics** in the following slides.

Basic isi statistics Functions

- Cluster and node statistics from kernel counters
- Collected and stored in `/ifs/.ifsvvar/modules/stats`
 - Sqlite3 database
- Can be used by higher-level applications using API
 - InsightIQ
 - Web administration interface
 - SNMP
- `isi_stats_d`
 - Majority of data collection
 - Works with InsightIQ



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The **isi statistics** command provides a set of cluster and node statistics. The statistics collected are stored in an **sqlite3** database that is under the **/ifs** folder on the cluster. Additionally, other Isilon services such as InsightIQ, the web administration interface, and SNMP gather needed information using the **isi statistics** command.

The **isi statistics** command enables you to view cluster throughput based on connection type, protocol type, and open files per node. You can also use this information to troubleshoot your cluster as needed.

In the background, **isi_stats_d** is the daemon that performs a lot of the data collection.

To get more information on **isi statistics**, run **man isi statistics** from any node.

To display usage help:

- **isi statistics system --help**
- **isi statistics protocol --help**
- **isi statistics client --help**
- **isi statistics drive --help**
- **isi statistics list keys**

isi statistics can list over 1500 statistics, dumps all collected statistics, and is useful when you want to run the **query** subcommand on a specific statistic. It can be used to build a custom **isi statistics** query that is not included in the provided subcommands (such as, drive, protocol, etc.).

InsightIQ vs. isi statistics

InsightIQ	isi statistics
Licensed	Not licensed
Graphical output	Produces raw CSV output on demand
Not easily scripted	Easily scripted
Not easily automated	Automate with <code>cron</code>
Use from remote host	Use from any node
Web interface	Produces data similar to other UNIX utilities (such as <code>top</code>)
Fixed interval	Flexible sampling interval



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isi statistics gathers the same kind of information as InsightIQ, but presents the information in a different way. The table lists some of the major differences between **isi statistics** and InsightIQ. In situations where InsightIQ is unavailable or malfunctioning, **isi statistics** is still a powerful and flexible way of gathering cluster data.

Some **isi statistics** parameters include the following list:

- **isi statistics protocol --classes read,write,namespace_read,namespace_write**
This format provides a display of statistics organized by protocol, such as NFS3, HTTP, and others. The **--classes** options describe the list of protocol operations to measure.
- **isi statistics client --remote_names "<IP Address>"**
This format provides statistics broken out by users or clients accessing the cluster.

Here are some of the other **isi statistics** subcommands:

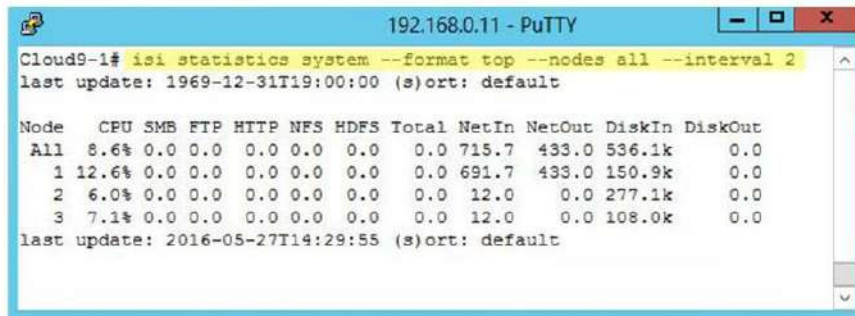
- **query** mode provides highly customizable access to any statistic in the cluster statistics library.
- **query history** mode provides basic access to historical values of statistics which are configured to support history.
- **drive** mode shows performance by drive.
- **heat** mode displays the most active areas of the cluster file system.
- **pstat** mode displays a selection of cluster-wide and protocol data.
- **list** valid arguments to given options.
- **system** mode displays general cluster statistics. This mode displays operation rates for all supported protocols, as well as network and disk traffic (in kB per second).

You can use the **isi statistics** command within a **cron** job to gather raw statistics over a specified time period. A **cron** job can run on UNIX-based systems to schedule periodic jobs. Note that **cron** works differently on an Isilon cluster vs. a UNIX machine so contact support before using it.

InsightIQ retains a configurable amount of historic information with regard to the statistics it collects. To prevent collection of an large backlog of data, InsightIQ retains data sets to

provide trending information over a year, but these settings are also configurable.

Example: Statistics by Most Active Node



```
192.168.0.11 - PuTTY
Cloud9-1# isi statistics system --format top --nodes all --interval 2
last update: 1969-12-31T19:00:00 (s)ort: default

Node   CPU  SMB  FTP  HTTP  NFS  HDFS  Total  NetIn  NetOut  DiskIn  DiskOut
All    8.6% 0.0  0.0  0.0  0.0  0.0    0.0  715.7  433.0  536.1k  0.0
1     12.6% 0.0  0.0  0.0  0.0  0.0    0.0  691.7  433.0  150.9k  0.0
2      6.0% 0.0  0.0  0.0  0.0  0.0    0.0   12.0   0.0  277.1k  0.0
3      7.1% 0.0  0.0  0.0  0.0  0.0    0.0   12.0   0.0  108.0k  0.0
last update: 2016-05-27T14:29:55 (s)ort: default
```

The command shown here gives you the general cluster statistics showing the most active nodes on top, and the output refreshes every two seconds. Data is broken down by protocol and interface.

If the administrator would like a result sorted by node number, one option is to run the following command:

```
while true ; do isi statistics system --nodes all | sort -n ; sleep 2 ; done
```

Example: Statistics for Drive Activity

192.168.0.11 - PuTTY

```
Cloud9-1# isi_for_array "isi statistics drive | head -5"
```

Node	Drive	Type	OpsIn	BytesIn	OpsOut	BytesOut	TimeAvg	TimeInQ	Queued	Busy
Cloud9-2:	2:1	SSD	22.8	216.3k	0.0	0.0	0.0	0.0	0.0	17.3%
Cloud9-2:	2:2	SAS	0.4	1.7k	0.0	0.0	0.0	0.0	0.0	0.2%
Cloud9-2:	2:3	SAS	0.6	6.6k	0.0	0.0	0.0	0.0	0.0	0.3%
Cloud9-1:	1:1	SSD	23.0	217.1k	0.0	0.0	0.0	0.0	0.0	19.1%
Cloud9-1:	1:2	SAS	0.6	9.8k	0.0	0.0	0.0	0.0	0.0	0.2%
Cloud9-1:	1:3	SAS	0.2	3.3k	0.0	0.0	0.0	0.0	0.0	0.1%
Cloud9-3:	3:1	SSD	22.8	216.3k	0.0	0.0	0.0	0.0	0.0	20.2%
Cloud9-3:	3:2	SAS	1.0	9.9k	0.0	0.0	0.0	0.0	0.0	0.3%
Cloud9-3:	3:3	SAS	0.4	3.3k	0.0	0.0	0.0	0.0	0.0	0.2%



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This slide shows example output of **isi statistics drive**, using **isi_for_array** to examine all the nodes on the cluster, and **head -5** to display only the most active results on each node. Each line shows the node providing the data, and each node displays the top three drives and what levels of activity they are displaying. This can be very useful to establish if there is an imbalanced load across the cluster. Specifically the **drive** option makes each node report where its busiest drives are and what their levels of activity are.

Example: Statistics by Most Active Files/Dirs

192.168.0.11 - PuTTY

```
Cloud9-1# isi statistics heat --long | head -20
```

Timestamp	Ops	Node	Event	Class	LIN	Path
1462363983	42.5	1	lock	other	1:0000:0001	/ifs/.ifsvaz
1462363983	41.0	1	lookup	namespace_read	2	/ifs/
1462363983	40.5	1	lock	other	2	/ifs/
1462363983	40.0	1	lookup	namespace_read	1:0000:0001	/ifs/.ifsvaz
1462363983	32.9	3	lookup	namespace_read	2	/ifs/
1462363983	31.9	3	lookup	namespace_read	1:0000:0001	/ifs/.ifsvaz
1462363983	31.5	2	lock	other	2	/ifs/
1462363983	30.5	2	lookup	namespace_read	2	/ifs/
1462363983	30.4	1	lock	other	1:0003:000b	/ifs/.ifsvaz/modules
1462363983	29.5	2	lookup	namespace_read	1:0000:0001	/ifs/.ifsvaz
1462363983	29.2	2	lock	other	1:0000:0001	/ifs/.ifsvaz
1462363983	29.1	1	lookup	namespace_read	1:0003:000b	/ifs/.ifsvaz/modules
1462363983	26.9	3	lock	other	2	/ifs/
1462363983	24.7	3	lock	other	1:0000:0001	/ifs/.ifsvaz
1462363983	23.6	3	lookup	namespace_read	1:0003:000b	/ifs/.ifsvaz/modules
1462363983	19.4	2	lookup	namespace_read	1:0003:000b	/ifs/.ifsvaz/modules
1462363983	19.3	2	lock	other	1:0003:000b	/ifs/.ifsvaz/modules
1462363983	19.0	3	lock	other	1:0003:000b	/ifs/.ifsvaz/modules

Cloud9-1#

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Here is an example of **isi statistics heat**, now using **--long** to include more columns. The **head -20** command only shows the first 20 lines, again allowing you to see what is most active on the cluster. The **heat** option identifies the most accessed files and directories.

Practical Skills

Use InsightIQ to Troubleshoot

- Sometimes not possible - customers may not allow new software
- May be limited time or facilities

`isi statistics` can fill gap

- Equivalent data available
- Also refers to `isi_stats_d`
- Many performance related options

Statistic Keys

- `isi statistics list keys`
- `isi statistics query current --keys node.uptime`

Examine state of data caches

- `isi_cache_stats`

InsightIQ customer troubleshooting guides: <https://community.emc.com/docs/DOC-42096>

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Troubleshooters need to be flexible with respect to the tools at their disposal. Skillful use of **isi statistics** can produce equivalent information to what can be gleaned from InsightIQ, for example. Using these skills to improve diagnostics is a powerful addition to the technician's toolbox. Combining large sets of collected data with log analysis skills can help identify long term trends and sources of trouble.

The **isi statistics** and **isi_stats_d** commands can help isolate or identify issues where InsightIQ may not have visibility. Using **isi statistics keys** can show specific metrics, such as **isi statistics query current --keys node.uptime** displays the node uptime.

Another area to examine may be the cache statistics using the **isi_cache_stats** command. The output shows reads and prefetch stats. The output also shows prefetch hits and misses.

Lesson 4: Summary

Having completed this lesson, you are able to:

- Use `isi statistics`
- Understand `isi statistics` options
- Manipulate `isi statistics` output

Now that you have completed this lesson, you should be able to use **isi statistics**, understand **isi statistics** options and manipulate **isi statistics** output via the CLI.

Lesson 5: ESRS

Upon completion of this lesson, you will be able to:

- Describe ESRS
- Understand ESRS architectural layout
- Configure ESRS on Isilon nodes

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Upon completion of this lesson, you will be able to describe and understand the EMC Secure Report Services (ESRS) environment, and configure ESRS on Isilon nodes.

ESRS Overview

- Remote support tool
- Secure back-end connection
- Encrypted and authenticated communications
- Uploads logs to support
- Communicates events to support



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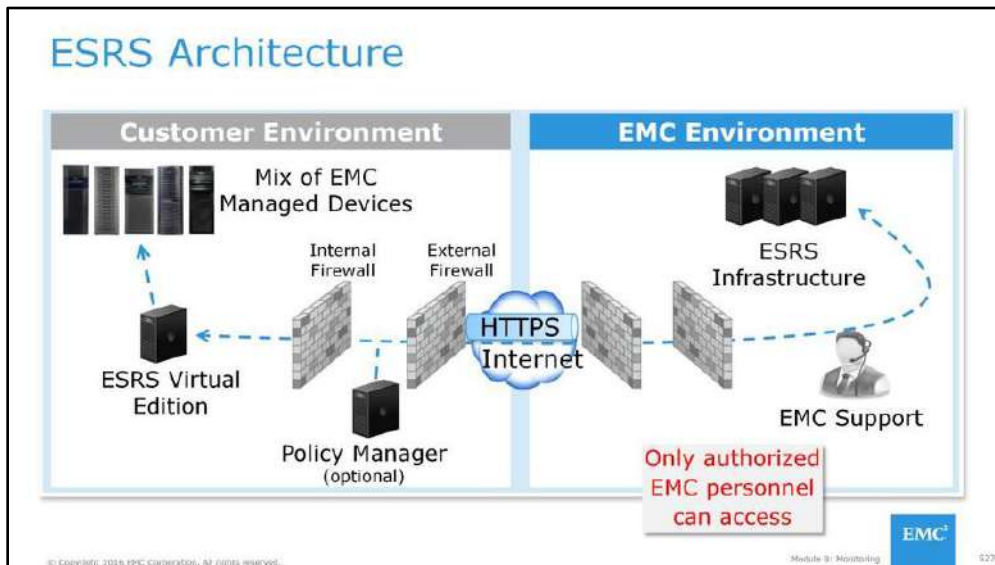
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EMC Secure Report Services (ESRS) is a mature and well-established system that communicates alerts and logs, and enables EMC support staff to remotely perform support and maintenance tasks. ESRS monitors the Isilon cluster on a node-by-node basis, sending alerts regarding the health of your devices. It provides a secure, IP-based customer service support system that features 24x7 remote monitoring, secure authentication with AES 256-bit encryption, and RSA digital certificates. ESRS is included with the OneFS operating system and not licensed separately.

InsightIQ status is monitored through ESRS. Information passed to the cluster is automatic, passing registration information through to ESRS. There is no administrative intervention needed to achieve the registration.



The graphic shows the general architecture of ESRS operation in a heterogeneous EMC environment. ESRS functions as communications brokers between the managed devices, the Policy Manager, and the EMC Enterprise. All communication with EMC initiates from ESRS on port 443 or 8443 outbound from the customer site to EMC support services. EMC does not establish inbound network communications to the systems. This is a security measure which is to the benefit of customers that run secure sites but do permit limited, controlled outbound communications.

Although the Policy Manager is optional, it is required to fulfill requirements for authentication, authorization and auditing. By implementing the optional ESRS Policy Manager, customers can enable monitoring on a node-by-node basis, allow or deny remote support sessions, and review remote customer service activities. The Policy Manager enables permissions to be set for ESRS managed devices. When the ESRS server retrieves a remote access request from the EMC Enterprise, the access is controlled by the policies configured on the Policy Manager and are enforced by the ESRS server.

Communications between the customer site and EMC support flow over an encrypted HTTPS connection, which means that sensitive information does not traverse the internet unprotected.

ESRS can be configured for redundancy with more than one ESRS instance installed, allowing reports through ESRS in the event of hardware or partial data environment failure.

On the EMC support side, only authorized EMC representatives have access to the customer systems or their information at all.

Installation and Design Considerations

- ESRS installation is an EMC service
- Dedicated VM on ESX or Hyper-V
- One ESRS server supports up to 250 devices
 - Each node individually monitored by ESRS
- Multiple subnets configurable
- SupportIQ deprecated and replaced with ESRS in OneFS 8.0

Virtual Server Requirements	
RAM	4 GB of free memory
Storage	64 GB of free disk space
CPU	One 64-bit vCPU 2.0 GHz or higher
Browser	IE 9 or higher, or Google Chrome

Documentation: https://support.emc.com/docu65068_OneFS-8.0.0-Web-Administration-Guide.pdf?language=en_US

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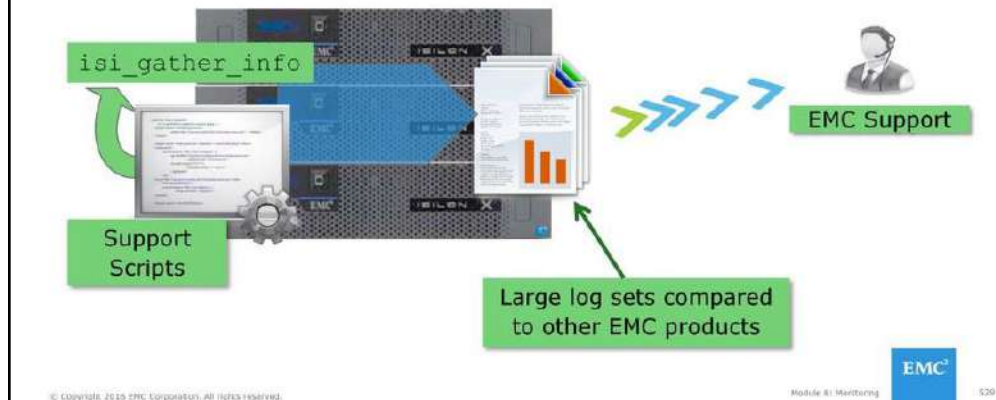
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ESRS has improved over the years, just as OneFS has. The ESRS installation is a service provided by EMC staff. Presently, the configuration and installation is not open for customers to perform. The customer used to have to accept a Windows server (either virtual or physical hardware) in their data center, which some customers were unwilling to do. Now there is a dedicated virtual machine that only runs the ESRS gateway software. This eliminates dependency on a product or operating system, such as Windows. ESRS treats each node as a separate device, and each node is connected to ESRS individually. The cluster is not monitored as a whole.

ESRS can operate through different subnets. By crafting the right set of subnets, a storage administrator can address any set of network interfaces on any set of Isilon cluster nodes.

In OneFS 8.0, SupportIQ is fully deprecated.

Isilon Log Transmission through ESRS



Isilon logs, even compressed, can be many gigabytes of data. There are ways of reducing the log burden, such as gathering incremental logs rather than complete log records or selecting specific logs to gather, but even so, logs on Isilon tend to be large. Uploading logs may require a lot of bandwidth and could take awhile with the risk of timeouts and restarts. The support scripts are based on the **isi_gather_info** tool. The remote support scripts are located in the **ifs/data/Isilon_Support/** directory on each node. The scripts can be run automatically to collect information about your cluster's configuration settings and operations. ESRS uploads the information to a secure Isilon FTP site, so that it is available for Isilon Technical Support personnel to analyze. The remote support scripts do not affect cluster services or the availability of your data.

ESRS on NANON Clusters

- Configure networked nodes for ESRS
- CELOG alerts come through network attached node
 - Includes all CELOG alerts, including for NANON nodes
- Use `isi_gather_info` to gather logs
 - Works for broad queries
 - Does not enable ESRS to query only NANON node
- ESRS does not currently have cluster comprehension
 - Each node handled as separate device

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NANON clusters are clusters where not all the nodes are on the network. This can be a deliberate design choice for a number of reasons. One fairly typical scenario is where nearline or high-density nodes are not externally accessible so as to prevent clients from overloading their limited CPU and RAM resources.

CELOG alerts that go through an ESRS channel are always directed through a network connected node. This means that ESRS won't be inadvertently be blinded to alerts by a random movement of CELOG binaries.

ESRS can also perform a log gather for the whole cluster through a connected node, rather than having to reach each node individually. This way the connected node acts as a proxy for the inaccessible nodes, but it does not allow ESRS to only reach disconnected nodes.

Despite all this, ESRS still recognizes each node as a separate device and has no unified concept of the cluster. The cluster is not semantically accessible to ESRS as a service.

Lesson 5: Summary

Having completed this lesson, you are able to:

- Describe ESRS
- Understand ESRS architectural layout
- Configure ESRS on Isilon nodes

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Module 8: Monitoring



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Having completed this lesson, you are able to understand the ESRS environment and configure ESRS on Isilon nodes.

Module 8: Summary

Having completed this module, you are able to:

- Understand cluster events
- Use InsightIQ graphical monitoring tool
- Use `isi statistics` command
- Configure ESRS

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Module 8: Monitoring



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Having completed this module, you are able to use the InsightIQ graphical monitoring tool and the **isi statistics** command. Additionally, you should now be able to understand cluster events and configure ESRS.

Lab 8: Monitoring

- View cluster events
- Install and configure InsightIQ
- View InsightIQ reports
- Use `isi statistics` command

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Module 8: Monitoring



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In this lab, you'll learn different techniques for monitoring your cluster.

Course Summary

Having completed this course, you can now:

- Explain Isilon cluster functionality
- Implement data protection preferences
- Differentiate internal and external networking configurations
- Utilize access management controls
- Define options for user authentication and file access
- Describe Isilon's backup and disaster recovery methods
- Use Isilon Job Engine
- Monitor your Isilon cluster

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Course Summary



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Having completed this course, you can now explain Isilon cluster functionality, implement data protection preferences, differentiate internal and external networking configurations, utilize access management controls, define options for user authentication and file access, describe Isilon's backup and disaster recovery methods, use the Isilon Job Engine, and monitor your Isilon cluster.



This concludes the Isilon Administration and Management course. Thank you for your participation!